

1. INTRODUCTION

1.1 Preamble

Established in 1958, the Lucinda Bulk Sugar Terminal has a storage capacity of around 230,000 tonnes of raw sugar. The terminal is owned by Sugar Terminals Limited (STL) and is managed exclusively by QSL under an operating arrangement.

Queensland Sugar Ltd (QSL) have an ongoing commitment to maintaining plant and structures at the BSTs that it operates.

The Bulk Sugar Terminal (BST) where the WUC shall be executed is located at:

Lucinda Bulk Sugar Terminal – 2 Dungeness Road, Lucinda, QLD 4850



Figure 1: Lucinda Wharf & Jetty

Works for this Contract include structural repair and corrosion protection. They will be conducted on the 5.7km long Jetty.



Figure 2: A Typical Lucinda Jetty Pilebent consisting of a headstock and multiple piles

1.2 Construction Schedule

The works on the Jetty piles and headstocks are to be carried out over a multi-year period as per the Contract. A listing of WUC for the jetty piles and headstocks is contained in Appendix D.

The works are to be completed according to the Contract Program and it is not permitted to accelerate the works without a written direction to do so by the Superintendent.

Mobilisation to start following tail end works

The Principal is undertaking works to the tail end of the Jetty in early 2025 that will restrict access. These are expected to be completed by June 2025. However, in the event of delay to the recommencement of access to the Jetty the Contractor will not be able to mobilise. Where there is a delay due to this cause, the Contractor will be entitled to an extension of time, but no delay costs or damages as result of the delay will be payable.

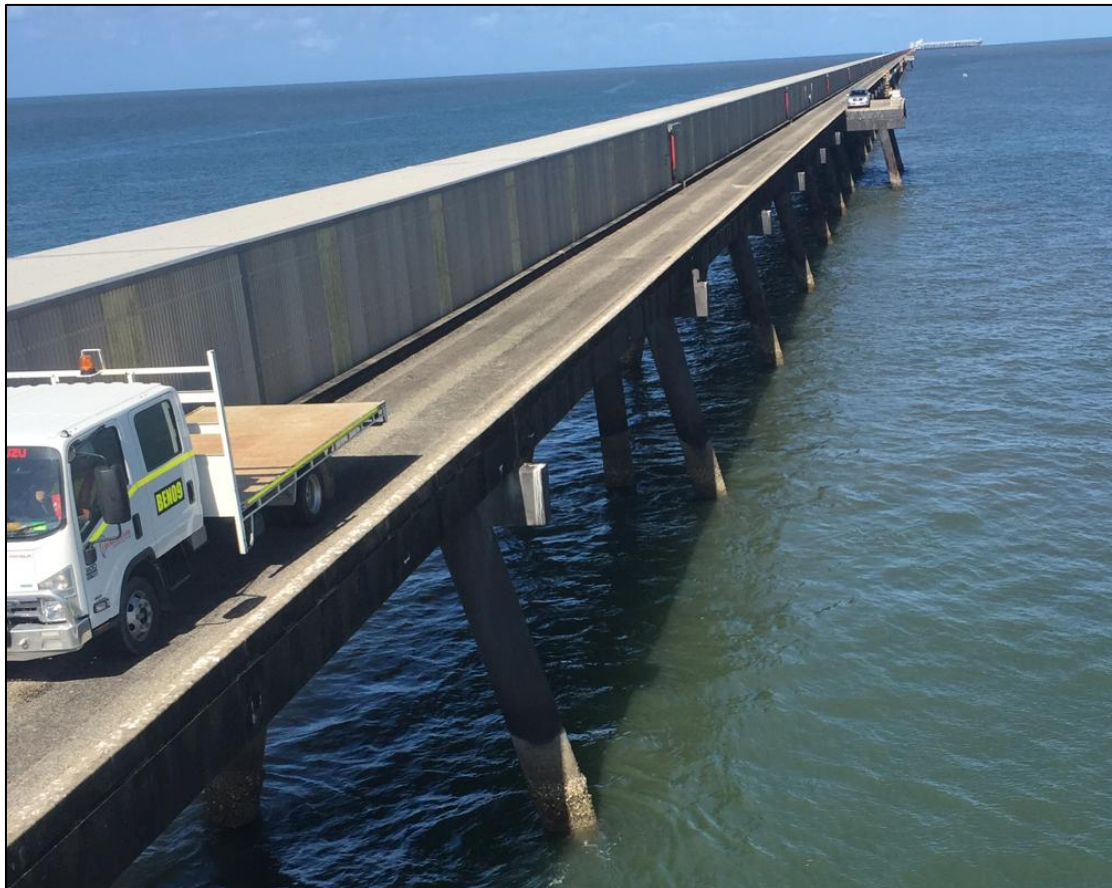


Figure 3: Roadway section

1.3 Interpretation

Capitalised terms used in this Detailed Specification have the meanings assigned to them in the General Conditions of Contract or the General Specification (if any) unless the context otherwise requires.

2. SCOPE OF WORKS

2.1 General

Without limiting any other requirement of the Contract, the scope of works (SOW) for the WUC includes::

1. Site Establishment & Inductions;
2. Administration and Documentation;
3. Development, maintenance and operation of safety, environment and quality management plans.
4. Preparation, treatment, and repairs as required to perform the WUC in accordance with the Specification;
5. Completion of WUC in accordance with the Scope;
6. Clean-up and demobilise.

2.2 WUC on jetty piles and headstocks

Figure 2 shows a typical pilebent on the LBST Jetty consisting of a headstock and multiple piles. The Jetty Pile and Headstocks WUC consists of remediation of nominated headstocks plus nominated piles that form part of the same pilebents. The majority of pilebents on the structure are included.

QSL have been completing a program of repairing piles and applying a petrolatum protective jacket system using the Jetty Maintenance Vehicle (JMV). The piles included in the WUC are those that have not been and are not expected to be wrapped under that separate program using the JMV. The nominated piles have generally been chosen as those that have sustained significant corrosion and require substantial welded repair.

2.2.1 *Nominated piles and headstocks*

Appendix D contains a listing of nominated pilebents and whether the WUC for each pilebent includes only headstock works or includes headstock works plus nominated piles or other items.

The WUC on each pilebent is allocated to a particular separable portion.

2.2.2 *Headstock works*

At each nominated pilebent the WUC includes:

1. Provision of safe access
2. Removal of the petrolatum tape jacket systems and any other tape systems to a minimum 600mm below the headstock as per drawing QSL2501-JPHR-G01 in Appendix A. The remaining petrolatum tape shall be protected by masking with a suitable product during further processes.
3. Removal of petrolatum tape around the gallery feet and masking of sliding surfaces to avoid damage from abrasive blasting and painting as per drawings QSL2501-JPHR-G01 and QSL2501-JPHR-G02 in Appendix A.
4. Cleaning of all surfaces by high pressure water blasting or similar including de-greasing of surfaces as required.
5. Inspection of corroded areas including thickness testing and recording as per section 4.1.2
6. Replacement of deck unit keeper angles as per drawing QSL2501-JPHR-G02 in Appendix A.

7. Welded repairs as approved by the Superintendent in accordance with drawings in Appendix J or similar as updated – see section 4.1 for further details.
8. Abrasive blasting and application of the protective coating system as per requirements of this specification including Appendix A and Appendix E.
9. Reinstatement of petrolatum tape jacket systems.
10. Reinstatement of petrolatum around conveyor gallery feet as per drawing QSL2501-JPHR-G02.
11. Installation of elastomeric block and pouring of epoxy for each deck unit keeper angle as per drawing QSL2501-JPHR-G02, Appendix A.
12. Safe access and pile remediation works as below for nominated piles.
13. Safe access and underpass remediation works as below for nominated underpasses.
14. 2-week hire of scaffold access system for expansion joint pilebents, to allow Principal works (see below)

2.2.3 Pile works

Where piles are nominated as part of the WUC for a pilebent the piles should, subject to the Contractor's methodology, be completed concurrently with the headstock works for that pilebent. The WUC for each nominated pile include:

1. Provision of safe access to the pile to allow installation of jacket down to a level of ~1.7m LAT
2. Removal of marine growth to jacket extent
3. Surface preparation by high pressure water blasting, minimum 5000psi, plus scraping or other means to remove all flaking paint, loose corrosion and marine life.
4. Inspection of corroded areas including thickness testing and recording as per procedure section 4.
5. Welded repairs as approved by the Superintendent in accordance with this Specification and drawings Appendix J or similar as updated – see section 4.1 for further details
6. Application of Denso SeaShield 2000FD system as per Appendix G

2.2.4 Underpass works

For nominated underpasses the works include treatment as per drawing QSL2501-JPHR-G03 in Appendix B inclusive of:

1. Abrasive blasting and protective coating in accordance with the specification Appendix E and requirements Appendix B
2. Application of corrosion protection tapes to structural members as per Appendix B

2.2.5 Expansion joint pilebents

Expansion joint pilebents have been nominated as "optional provisional, if ordered". Refer to clause 3 of the General Conditions of Contract. The timing of these items will be as stated in the applicable Direction (if any) but, if ordered, will be ordered so as to coincide with other WUC on the Jetty.

The expansion joint pilebents require additional works that will be completed by the Principal that includes lifting of the roadway by jacking from the headstock, and installation of new sliding plates. (Note: when the first one or two expansion joints are being completed the Contractor will be invited to view the process and optionally provide a variation price to carry out the remaining expansion joint works).

The WUC for expansion joint pilebent works includes the following:

- Provision of access system with modification if required to allow the roadway to be raised approximately 200mm.

- Cleaning of all surfaces by high pressure water blasting or similar including de-greasing of surfaces as required.
- Inspection of corroded areas including thickness testing and recording as per section 4.1.2
- Hire of scaffold access system for up to 2 weeks
- Provision for allowing Principal's works to occur on the pilebent including allowing for closure of roadway during lifting of roadway girders
- Remainder of works as per standard pilebents (excluding deck unit keeper angle replacement).



Figure 6: Expansion joint lifting to be carried out by others

2.3 C7B conveyor gallery WUC

Corrosion protection wrapping of the steel members of nominated conveyor gallery bays have been included in the WUC. This item provides long-term corrosion protection to the structure but has primarily been included in this contract to help the Contractor to manage the impacts of unsuitable weather such as high humidity.

WUC include:

1. Provision of access as required (platforms, step ladders, access over belt)
2. High pressure water blasting of nominated sections
3. Inspection of nominated sections for holes or deep pitting due to corrosion
4. Supply and application of petrolatum tapes and topcoat materials as per drawing QSL2501-JPHR-G04 in Appendix C and any other requirements noted therein.
5. Cleanup of area including conveyor belt.

Access restrictions apply to the conveyor gallery during shipping activities as detailed in Section 3.

3. SITE, ACCESS AND GENERAL REQUIREMENTS

3.1 Regulations, specifications and standards

The Contractor will be responsible for obtaining a Port of Townsville Blasting and Surface Coating permit available at www.townsville-port.com.au/operations/permits-forms/

3.2 The Site

The Site, in which the Contractor is nominated as Principal Contractor, will vary throughout the length of the contract depending on the stage of the Works. Access to and Possession of the Site shall be granted in accordance with the General Conditions of Contract. Unless otherwise directed, the Contractor shall not carry out WUC in areas where it has not been granted possession in writing.

Where the Contractor demobilises from the Site for a period, for example over the wet season, they shall clean up and relinquish access to the Site areas on the jetty. The onshore storage areas may be retained as part of the Site.

3.2.1 *Jetty pile and headstock works*

The Site includes of:

- i. The section of the Jetty between passing bays where the WUC are being undertaken.
- ii. One or more Jetty passing bays adjacent to the section of the Jetty where the WUC are being undertaken and where the Superintendent has directed these to be included in the Site for storage of the Contractor's materials and equipment.

See Drawing 1122-CTQ-001 for the locations of passing bays.

3.2.2 *C7B conveyor gallery wrapping*

The Site includes of:

- iii. The interior of the C7B conveyor gallery where the WUC are underway, as well as the adjacent galleries as required for access and storage of materials and equipment.

3.2.3 *Onshore storage area*

An onshore storage area may be granted as part of the Site upon request by the Contractor. The Contractor will be responsible for establishing this area including provision of security, vehicle access and any other requirements. The location of this area is seen in the blue rectangle Figure 4 but does not include the entire area. The Contractor will be given access to a sufficient portion of this area as required to support the WUC. The area must be kept neat and tidy. Note that the area does not have a power connection.



Figure 5: Onshore storage area location

3.2.4 Area adjacent to onshore crib room

An area adjacent to the onshore crib room will be granted as part of the Site upon request by the Contractor. The Contractor will be responsible for establishing this area including provision of security, vehicle access, services and any other requirements. The location of this area is seen in Figure 5. The area must be kept neat and tidy.



Figure 6: Area adjacent to onshore crib room

3.3 Crib Room and office facilities

There is a crib room available for use both onshore and offshore. The onshore crib room facility is used by all contractors working on the site. The offshore crib room facility is used by both contractors and QSL staff as required. Both facilities contain a toilet facility.

The Contractor may make use of the offshore first aid room as an office facility.

The Contractor shall supply and maintain one portable toilet facility on the passing bay that has been provided to the Contractor as part of the Site under their control and management.

3.4 Site inspection

Shortly after the Date of Acceptance of Tender, the Contractor will arrange a time to visit the Site with the Superintendent to conduct a site assessment both on the wharf deck as well as via the Principal's workboat to assist with planning and development of management plans.

3.5 Site Hours

Refer to Annexure Part A Item 42 for the applicable Working Days and Working Hours.

The Site will generally be available 24 hours a day, seven (7) days a week. Standard BST site terminal hours are 7:00am to 4:00pm Monday to Friday. Where night works are included in the Working Hours, the Safety, Environmental and Quality Management Plans will need to include all controls and requirements to ensure WUC carried out at night proceeds safely and in accordance with requirements of the Contract as to quality. Works with high levels of noise will not be permitted at night where that noise impacts residents of the Lucinda township. Commencement of night works is subject to the approval of the Superintendent following submission and implementation of the management plans.

Refer to clause 31 of the General Conditions of Contract. Where the Contractor wishes to work longer hours than the Working Hours identified in Annexure Part A Item 42, the written approval of the prior Superintendent must be obtained in accordance with clause 31 of the General Conditions of Contract. Application for such approval must be accompanied by a risk assessment that describes and provides mitigations for safety, environmental and quality risks associated with the longer work hours. If the longer work hours are approved by the Superintendent, the Contractor shall submit updated management plans for review and approval by the Superintendent prior to commencing the longer work hours. Any additional costs as a result of longer work hours shall be borne by the Contractor.

3.6 Site access

The Contractor must maintain a sign-in register showing their personnel and visitors who are within the LBST site at any time. This register must be able to be given to QSL staff on request and during an emergency. Contractor's personnel will not need to sign in at the Principal's office.

Site access cards will be issued to the Contractor prior to commencement of works. Access is generally through the front gate at the terminal, but during shipping events the entry point is to the rear of the terminal and is monitored by 24-hour security.

If a ship is at berth, it's possible that personnel will require Maritime Safety Identification Cards to access the wharf and this may include the Jetty. If such security is implemented, work may continue in groups where at least one person from each group holds an MSIC and therefore becomes responsible for all others working in the group. Under such arrangements, the MSIC holder must maintain a close proximity and visual contact with personnel until such time as they leave the security area.

3.7 Site and jetty vehicle access

The speed limit on the LBST site is limited to 20km/h. On the jetty the speed is limited to 10km/hr on the tail section and 40km/hr along the general run of the jetty (speed camera enforces speed limits), the Contractor is to ensure that their personnel observe these maximum speed limits.

Persons using the Jetty are to observe the following rules:

- a. Headlights are to be switched on when driving down the Jetty.
- b. There are 6 passing bays down the length of the single lane jetty roadway, in the case of oncoming traffic the closest car to a passing bay is to pull into the passing bay to allow for the other vehicle to pass.

The JMV straddles the Jetty roadway, this vehicle is designed to allow light vehicles to pass under it, however larger vehicles or loads will most likely require that the JMV is removed from the Jetty roadway to either a passing bay or back onshore. Moving the JMV from the jetty roadway is not preferred and is to be avoided where possible, especially for extended period as this is high use machine. The Contractor will need to advise the Superintendent at least two days prior to requiring that the JMV be moved.

The JMV has a clearance underneath of approximately:

2.33m Width

2.15m Height

There is a general 8 Tonne gross load limit on vehicles using the Jetty.

Vehicle parameters:

Maximum Gross Weight = 8 tonne

Maximum Height = 2.1m

Maximum Width = 2.0m

Speed Limit = 40 kph

Where the Principal implements traffic management for other activities the Contractor shall comply with all directions as to Traffic Management on site as directed by the Terminal Manager or their delegate(s).

Personnel must adhere to all marked pedestrian movement walkways where applicable.

3.8 Wharf parking

Unless otherwise agreed with terminal personnel at the daily pre-start, the Contractor must not have more than 2 vehicles parked on the wharf.

3.9 Vehicle hazard to personnel on roadway

The Contractor must enforce a lower speed limit on the jetty roadway in areas where it is working. Where works are underway inside the conveyor gallery the contractor must ensure safety of personnel entering and exiting the gallery. This includes at a minimum:

- Providing warning signage and a lower speed limit in the vicinity of where workers are accessing the gallery.

- Providing barriers at the gallery door exits that serve to ensure personnel exiting the gallery stop and look for vehicles prior to entering roadway. This includes providing a warning area and speed control signs outside of any area where the contractor is working inside the conveyor gallery.

3.10 Use of gantry structures on jetty

The Contractor may employ one or more mobile gantry structures on the Jetty to hold equipment or scaffolding components while allowing vehicles to pass underneath.

The operational procedure for the gantry structure must be submitted for approval by the Superintendent prior to mobilising the structure to site. The following requirements must be met:

- The structure must have a minimum clearance width of 2.4m and a minimum clearance height of 3.2m.
- The design of the structure must be certified by an RPEQ engineer
- Consideration must be made for securing the structure in the event of a cyclone, unless it can be removed from the Jetty in a suitable timeframe.
- An operational and environmental risk assessment must be submitted covering potential hazards associated with the use of the structure.
- The wheel/pad loadings of the structure must not overstress the Jetty roadway girders. The Contractor must provide a loading report for the proposed structure prior to fabrication. QSL will engage independent structural engineers to assess the impacts on the jetty and share this assessment with the Contractor. The Contractor must modify their design based on the results and recommendations of this assessment. The costs of modifying the structure design and meeting the requirements stipulated by the structural engineer shall be borne by the Contractor. Note that there is considerable leeway in the 8t load limit depending on the design, wheel configuration and load distribution of the proposed gantry. For example, the JMV has a total weight of around 18t but has a very wide wheel-base and other features to reduce wheel loads and thus reduce the resulting bending moment induced on the girders/structure, which has previously been identified as the critical design criteria condition. The maximum allowable ultimate limit state moment capacity for the Deck Live Load has previously been assessed to be 964 kNm which is governed by the structure remaining in fair/as designed condition. The moment applied by any proposed gantry must be significantly below this value with allowance for light vehicles passing under.

3.11 Obligation to maintain access along jetty

The Contractor shall facilitate access for vehicles through the Jetty Site (that portion of the Site that includes the jetty roadway) in accordance with the methodology contained at Annexure Part E of the Contract (Methodology).

The Contractor must allow for vehicles to be able to pass through the Jetty Site to access the offshore structure. The following minimum requirements will apply unless otherwise agreed as part of the Contractor's tender submission.

3.11.1 Communications and planning

The Contractor must:

- Have a representative attend the daily site pre-start meeting
- Liaise daily with the nominated BST staff member to advise proposed works for the day and seek information on BST staff or others' movements along the Jetty.
- Provide a procedure for BST staff and others to pass through the Jetty Site. This procedure must be communicated by signage at each end of the Jetty, in the BST office and BST staff crib room. The procedure must not require a site induction process or specialised equipment (except for a mobile phone).

- Ensure a member of the Contractor's personnel is available to co-ordinate and manage the process of vehicle access through the Jetty Site, including being available for communication by radio and mobile phone.

3.11.2 Emergency access

- The Contractor must develop, implement and test a plan to ensure that there is less than two minutes of delay where emergency services access is required through the Jetty Site.

3.11.3 General access

During Bulk Sugar Terminal (BST) site working hours of 7am to 4pm Monday to Friday, and except during shipping activities:

- The Contractor must ensure vehicles seeking to pass the Jetty Site shall experience a wait time of no more than 10 minutes.
- The Contractor must make available 15-minute windows where vehicles may pass the Jetty Site with a wait time of no more than 3 minutes. These windows shall be at regular times each day as co-ordinated with BST staff and include one in the morning, one prior to lunchtime, one after lunchtime, and one at the end of the day.

3.11.4 Closure periods during BST site hours

During working hours of 7am to 4pm Monday to Friday, except for during shipping activities. The Contractor may implement closure periods of up to 3 hours duration. The Contractor will be entitled to enact such closures at least 3 times per week, but may be permitted to enact closures more frequently where the Principal does not require access to the wharf. During these periods the Contractor may, except during an emergency, prevent all vehicles from passing the Jetty Site. Closures must be communicated with a minimum of 5 days' notice. Closures must be clearly communicated by signage at each end of the Jetty, in the BST office and BST staff crib room.

The Superintendent may, by giving a minimum of 5 days' notice, prevent the Contractor from implementing closures on particular days where the Principal requires continuous access for major operational, construction or maintenance activities.

3.11.5 Full closure: Outside of BST site working hours or where Principal does not require access

Outside of the hours of 7am to 4pm Monday to Friday, except during shipping activities, and where there are no other staff or contractors working offshore of the Jetty Site, the Contractor may implement a full closure. During this period, it is not necessary to allow for emergency access, except for emergencies associated with the Contractor's works.

Where the Principal has no planned activities on the wharf or Jetty during working hours and agrees in writing, the Contractor may implement a full closure.

(Note to tenderers: tenderers may describe their requirements for other full closures in their tender where this is required for their chosen methodology. The details may then be negotiated following the tender period and incorporated into the contract. It is expected that the Principal will be able to accommodate reasonable requests by scheduling their own activities to suit).

3.11.6 During shipping activities

Shipping activities include the period where a vessel is berthed on the wharf and any periods prior to berthing and after departure during which the Principal will carry out testing and cleanup activities.

During shipping activities the Contractor must ensure that access through the Jetty Site is continuously maintained with wait times of less than 5 minutes.

3.11.7 Passing bay included in Site

Where the Site includes one or more passing bays for storage of Contractor's materials, equipment and amenities, the Contractor must allow sufficient space for at least one vehicle to pull into the passing bay to allow an oncoming vehicle to pass.

3.12 JMV does not need to pass the jetty site

The JMV will continue to be in operation during the contract period. However, the Contractor need not generally allow for the JMV to be able to pass the site. However, if the Contractor's methodology does not include demobilising over the wet season then it must allow the JMV to pass through the Site at least once per calendar year.

3.13 Safe access

The selection and establishment of a suitable safe access system is entirely the responsibility of the Contractor. Shipping activities must still be able to occur without demobilising the safe access system.

The access system employed for the headstocks must allow for safe inspection by a third party inspector or the Superintendent using appropriate safety equipment. It is acceptable for the Contractor to require that these personnel have a 'working safely at heights' qualification if this is required for the chosen access system. The access system for the piles does not need to allow for access by the Superintendent or nominated inspector. For example, it would be acceptable to use rope access methodologies for the works on the piles.

The Contractor may remove underpans underneath the conveyor gallery to assist with access or services. The Contractor shall be responsible for replacing the underpans and rectifying any damage that has occurred because of their removal and replacement. Underpans must be replaced or the void otherwise secured and covered to the satisfaction of the Superintendent prior to any shipping event.



Figure 6: Removable underpans underneath the conveyor belt

3.14 Shipping activities and conveyor gallery

The WUC area will need to be made safe for shipping activities within a 2-day notice period. This includes making safe of access systems inside the conveyor gallery and replacing any removed underpans underneath the conveyor belt. The Contractor will be permitted to continue works on the Jetty with the following restrictions:

- The Contractor must not access the northern side of the conveyor gallery
- Other access to the conveyor gallery during shipping activities must be minimised and is subject to completion of a risk assessment and appropriate controls being implemented.
- Wrapping of the C7B gallery internal steelwork will not be permitted during shipping activities.

Outside of shipping activities, the Contractor's personnel must use a personal lock on a lock-out box, in accordance with the Principals LOTO procedure, that isolates the conveyor operation when working within the footprint of the lanyard pulls of the belt. This includes any access through the conveyor underpans.

The Principal will be responsible for carrying out the isolation on the belt. The Principal will only carry out isolations during terminal working hours and must generally be given at least one day's notice. The Contractor will be responsible for providing Red Personal Locks, keyed only for an individual, for each of their employees to lock on to the Lock Hasp / Box for the duration of the works. Red Personal Locks must be removed prior to leaving site each day else the responsible personnel may be asked to return to site at their own cost to remove locks after hours if equipment is required to be put back into service prior to the Contractors next working day.

The conveyor belt must not be used as an area for placement of tools and materials. This requirement is due to the risk of personnel leaving tools and equipment on the belt that may cause severe damage and inability to export sugar should they remain on the belt during operation.

The conveyor belt and access systems shall be inspected at the completion of each work day to ensure no items are left on the belt. Prior to shipping activity or completion of an area this inspection must be carried out, documented and such documentary evidence sent to the Superintendent.

3.14.1 Historic shipping data

Previous data for number for shipping events at Lucinda Bulk Sugar Terminal					
	2021	2022	2023	2024	2025
Jan-June	No Data	2	2	4	2 (ongoing)
July-Dec	10	11	9	10	-

The data provided above is only a guide as to future trends and actual shipping in any year can be dependent on the market conditions for that year or other factors. A typical shipping event lasts for 3-4 days depending on ship size. Breakdowns or rain can cause delays. Upcoming and previous shipping events can be viewed at the following link, noting that there is often minimal notice of future shipping events.

[Port of Townsville Shipping Schedule](#)

3.15 Air

There is an air line in the C7B conveyor gallery. The compressor feeding this line is located onshore. The air volume/pressure is generally not sufficient for abrasive blasting works but is generally suitable for air tools or minor spot abrasive blasting etc. The Contractor may use this air on the following conditions:

- Contractor to supply all hoses and fittings as required
- The Principal does not warrant the availability or suitability of the air source. The Principal will, however, arrange to repair the compressor or air supply in a reasonable time commensurate with the nature of the breakdown.
- The Contractor must ensure all air connections and lines are properly secured to totally eliminate the possibility of fouling with the belt during shipping activities or belt operation.
- Where this compressed air is used for spot abrasive blast cleaning, the Contractor shall check that the oil and water traps are empty. Compressed air shall be tested as per ASTM D4285-83 - Standard Test Method for Indicating Oil or Water in Compressed Air. Maintenance and air quality testing shall be conducted at regular intervals and at least once per day.

3.16 Water

There are fire hydrant water points at regular intervals on the Jetty that supply water. The Contractor may make reasonable use of the water. The Contractor is responsible for supplying all necessary fittings. Note the water is not tested and certified as potable for drinking.

3.17 Power

There are no electrical outlets available for use along the jetty. The existing outlets were installed as part of a communication system and do not provide suitable power for other purposes due to voltage drops along the length of the Jetty. These points are not to be used by the Contractor.

3.18 Cyclones

During cyclone season (1 November to 30 April), the Contractor must maintain an appropriate cyclone readiness procedure with tasks at each cyclone level of green, yellow and orange in accordance with the Port of Townsville categories (<https://www.townsville-port.com.au/operations/safety-security/emergency-procedures/cyclones/>). The Contractor must enact their plan throughout the cyclone season and when the risk level increases during a cyclone event. The following minimum requirements apply:

During the cyclone season (condition Green)

- All mobile buildings must be tied down appropriately,
- Where the Contractor leaves site temporarily, all materials and equipment must be secured against cyclonic winds.

During condition yellow/orange

- All loose items must be removed or appropriately restrained,
- All vehicles and mobile equipment must be removed from the Jetty,
- Scaffold encapsulation must be removed, and lower scaffold decks must be removed where possible,
- All personnel to evacuate at condition orange when the direction is given by the Principal, Superintendent, or the Port of Townsville.

The Contractor shall consider the risk of storm surge in their cyclone planning noting the onshore storage area may be subject to flooding due to storm surge.

3.19 Costs for cyclone event preparation

The costs for carrying out preparations in accordance with condition green and managing general cyclone readiness procedures shall be borne by the Contractor.

The reasonable costs of carrying out of preparation activities as part of condition yellow/orange shall be borne by the Principal and claimed as a Variation.

3.20 Site specific safety requirements

The Contractor's attention is drawn to clauses 11A, 12 and 12A of the General Conditions of Contract. Nothing in this clause 3.20 shall be taken to limit or exclude any obligation or liability of the Contractor under the General Conditions of Contract or at law in relation to work, health and safety and the environment

3.20.1 *Fitness for work (including impairment and fatigue management)*

The Contractor is responsible for ensuring that the Contractor's Personnel are fit for work at all times. QSL has a 0% Breath Alcohol Concentration (BrAC) limit for personnel working at Bulk Sugar Terminals. Legal drug use that may impair a person's ability to work, as well as illegal drug use will not be tolerated.

Drug and Alcohol Testing is conducted randomly on QSL sites and must be completed by all personnel in attendance at the time. Failure to comply with this requirement will result in exclusion of the person from Site for the remainder of the Contract.

Fatigue can pose serious risk to safe work. QSL therefore requires that the following minimum controls be implemented:

- Ensure that no employee works greater than 13 hours. This includes 12 hours for the shift with an allowance of 1 hour for shift change over. (Excluding emergency events that require risk assessment and appropriate control measures)
- Provide an adequate period of rest between each rostered day on. A minimum of 10 hours break is required between each work shift.
- Ensure that no more than 7 work periods are worked consecutively.
- Ensure that a minimum of one 24-hour break is taken every 7 work periods.

Any exception to these requirements will require approval from the Superintendent.

3.20.2 *General equipment*

All electrical equipment requires a current test and tag (battery powered equipment using less than 40V is excluded from this requirement).

Due to previous safety issues and investigations into acceptable risk, handheld grinders must meet all of the following minimum safety criteria:

- Anti-vibration handle attached
- Dead-man switch
- Constant speed control
- Kickback protection control

- Overload protection control
- Soft-Start speed control
- 0-Voltage restart protection control
- Disc guard

When conducting initial investigations, the only cordless handheld grinder that QSL could find that met the above criteria was the **Milwaukee M18 Fuel**; there may be other alternatives, but they must comply with the above criteria.

Grinders with discs larger than 125mm are not permitted to be used on Site.

3.20.3 Working on the jetty and wharf (drowning risk)

The Contractor must provide edge protection whenever there are Works within 2 metres of an unprotected edge where there is a risk of drowning. Where personnel are within 2m of an unprotected edge or there is a risk of falling and drowning, they shall wear a personal flotation device and not be working alone.

3.20.4 Materials presenting health risks if disturbed

Dust generated due to work being conducted on Site poses serious health risks. Asbestos, Synthetic Mineral Fibres (SMF), Respirable Crystalline Silica (RCS), and Lead Based Paints are some examples of such materials that may be present on Site that are generally safe if left undisturbed, but have substantial health risks otherwise.

As required by law, QSL maintains Asbestos Registers for all BSTs, and the Contractor is responsible for reviewing these registers if the work they are completing may disturb such areas.

For other such material hazards, the Contractor is responsible for following due diligence protocols, including taking samples and testing to confirm; the results of which will be provided to QSL.

3.20.5 Drone use

The use of drones on QSL sites is generally accepted for inspections. However, the Contractor is responsible for ensuring they meet all current CASA requirements including holding an ARN, Remote Pilots License (RePL), or Remote Operator's Certificates (ReOC) as required, and notifying CASA before flying.

The Contractor must send out an email notification with relevant pilot, drone, and flight details to QSL's Representative, the governing Port Authority, and in the case of Cairns and Townsville, the local Defence Force Coordinators. This email notification must be made at least 48 hours in advance of intended flight.

3.20.6 Storage of material and equipment

As a minimum, all materials and equipment on the Site shall be stored in a manner that prevents mechanical or environmental damage, and storage areas will be kept neat and tidy to minimise potential hazards.

When flammable or combustible materials are to be stored or used on the Site; the Contractor will consult with the Terminal Manager, and shall adhere strictly to any instructions given.

Drums, liquid or gas containers are to be stored away from locations where they could be struck by vehicles and other equipment.

- Drum Storage Sufficient bunds or containment must be provided for drums to contain at least one full drum and 25% of the maximum amount likely to be stored in the location. Drains valves are to be kept closed at all times. Any accumulated stormwater which contains oil or an oil film must be

diverted through an oil separator before discharge. Accumulator type drum storage pallets are permitted for drum storage.

- Gas Cylinders must be stored in a cool, dry and well ventilated place away from combustibles and areas where people are likely to congregate.
- Temporary bunding or dams must have prior approval by Terminal Management before installation.

The Contractor will liaise with the QSL site Supervisors during initial mobilisation to ensure equipment and products are stored appropriately.

The Contractor must ensure they have SDS's for all of the materials they are responsible for while on the Site.

3.20.7 Tank or pit overflows

Persons engaged in refuelling operations or in the filling of tanks or pits must remain at the fill point while this is in progress. Under no circumstances are fuel lines, filling hoses or fuel bowzers to be left running unattended. Fuel dispensing nozzles shall be fitted with automatic cut-offs.

3.20.8 Stormwater

No materials or wastes are permitted to enter drainage systems. Materials should not be stored in locations where it could be released into or otherwise block stormwater drains. Any works must be planned and completed in such a fashion to ensure storm and roof water does not contact sugar or other products or contaminants.

3.20.9 Dry spillages

Bulk products and building materials must be correctly enclosed or covered during transport. Any spillage from tipping or loading operations must be swept up or otherwise collected and placed in the stockpile.

Where dusts may be dispersed in the vicinity of occupied premises measures for dust control such as water sprays or vacuum collection must be used.

3.20.10 Wet spillages

Liquids must be contained so as to prevent entry to drains and gutters. Portable or temporary barriers including cracker dust and sand may be used to contain liquid spillages. Spilled liquids wherever possible should be recovered without contamination and reused. Contaminated materials must be stored correctly and labelled before disposal.

Where the Principal deems it necessary to undertake a clean up or engage a third party to undertake a clean up due to the Contractor's activities, then the cost incurred by the Principal in doing so shall be a debt due and owing to the Principal by the Contractor.

3.20.11 Odour

All noxious or offensive odour must be kept to a minimum. Consideration of the effects of odours which are dispersed by prevailing breezes must be taken into account when storing materials or odorous products.

3.20.12 Noise

Wherever possible, noise minimisation measures should be followed to reduce noise levels and the likelihood of a noise complaint.

The general environmental noise limits levels are shown in the table below.

Time	Environmental Noise Limit
7 am – 6 pm	Background + 10 dB(A)
6 pm – 10 pm	Background + 10 dB(A)
10 pm – 7 am	Background + 8 dB(A)

The Principal may undertake noise monitoring or engage a third party to undertake noise monitoring of the Contractor's activities. Where the monitoring indicates that the Contractor's activities generated noise exceeding the limits identified in the table above, then the cost incurred by the Principal in undertaking the monitoring or engaging a third party to undertake the monitoring shall be a debt due and owing to the Principal by the Contractor.

Contractors should ensure that stationary engines, trucks and other machines are operated in a manner that will not cause noise nuisance or complaints to arise.

Vehicle exhaust brakes should not be used in the vicinity of the terminal.

3.20.13 Oils and greases

Servicing equipment and machines at the sugar terminals is prohibited unless prior approval is given by The Superintendent. No vehicle servicing is permitted unless in designated service bays. Grease and oil residues must be wiped from fillers or grease nipples. All wastes generated by the Contractor during these procedures shall be removed from the Site.

Washing vehicles and equipment is prohibited unless prior approval is given by Terminal Management. No vehicle washing is permitted unless in designated washdown bays.

Spill trays may be placed under parked vehicles however it is expected that prompt attention to repair or remove vehicles that leak fluids will be taken.

3.20.14 Digging

No activities which disturb the surrounds of septic tanks absorption trenches are permitted.

All proposed ground or structure penetrations are to be discussed with the Superintendent and checked for potential buried service interaction prior to undertaking any penetrations.

3.20.15 Contaminated soil and land

Any identified areas of contaminated land (including asbestos or acid sulphates) must remain undisturbed. No traffic or other activities which is likely to disturb the vegetation and soil capping or generate dusts is permitted in these areas.

3.20.16 Incidents and breaches

With the exception of incidents contemplated by clause 11A of the General Conditions, if an incident requires further investigation, and where the Contractor was responsible (either in part or completely), the Contractor and their employees or Sub-Contractors employees will attend any requested discussions and/or investigation at no cost to QSL. Contractors may also be asked to assist in discussions and/or investigations where they may not have been responsible but have knowledge about an incident.

3.20.17 Hot works

The Contractor shall implement a hot work permit system that aligns with QSL's hot work permit requirements. No hot works shall be undertaken without a valid permit. The Contractor shall supply at minimum a 9kg (or larger) dry chemical powder fire extinguisher AND a 9kg (or larger) water fire extinguisher within 4m of the hot work. Particular care must be taken toward protecting the conveyor belt

3.20.18 *Smoking*

Smoking and vaping are not permitted.

3.20.19 *Personal cleanliness*

All persons attending the Lucinda terminal are to maintain a reasonable level of hygiene, no torn or excessively worn clothing or footwear is to be used.

GENERAL NOTES PROTECTIVE COATING

1. TYPICAL PILEBENT SHOWN, PROTECTIVE COATING TREATMENT SAME ON OTHER BENT TYPES
2. SEE ORIGINAL CONSTRUCTION DRAWINGS FOR FURTHER DETAILS OF OTHER PILEBENT TYPES
3. PAINT COATING SYSTEM TO INCLUDE EXTENT OF HEADSTOCK PLUS 100mm BELOW TOP OF DENSO JACKET AS SHOWN HERE. DENSO SYSTEM TO EXTEND TO NO HIGHER THAN 1.9m ABOVE LOWEST ASTRONOMICAL TIDE
4. DETAILS OF PAINT COATING SYSTEM AND PREPARATION AS PER CONTRACT SPECIFICATION AND DULUX SPEC. PCQ-015-0121-4. FOLLOWING NOTES PROVIDE SUMMARY BUT SPECIFICATION TAKES PRECEDENCE:
 - DEGREASE ALL SURFACES. NOTE THERE MAY BE GREASE BENEATH THE BEARING PAD BETWEEN THE HEADSTOCK AND ROADWAY THAT MUST BE PREVENTED FROM CONTAMINATING WORKS
 - ABRASIVE BLAST CLEAN AS1627.4 TO CLASS Sa 2.5 TO VISUAL STANDARD ISO8501-1
 - SEE SPECIFICATION FOR QA SYSTEM REQUIREMENTS AND HOLD POINTS
 - DEEP PITTED AREAS TO BE GROUND SMOOTH WHERE POSSIBLE OR WELD FILLED AND GROUND SMOOTH PER PRINCIPAL'S DIRECTION
 - MASK CONVEYOR SLIDING SURFACE AND EXISITNG DENSO JACKETS.
 - DO NOT APPLY COATING IF RELATIVE HUMIDITY IS GREATER THAN 85% OR STEEL SURFACE IS WITHIN 3 °C OF THE DEW POINT
 - SURFACE SALT MUST BE BELOW 10 MICROGRAMS/CM2
 - APPLY INIITAL COATING WITHIN 4 HOURS OF BLAST CLEANING AND BEFORE DETERIORATION OCCURS. ALL SURFACES TO BE CLEAN AND DRY BEFORE COATING
 - STRIPE COAT AND BRUSH IN ALL EDGES, CORNERS, CORRODED AREAS AND DIFFICULT TO COAT AREAS BOTH COATS
 - 1ST COAT: 500µm DFT DUREMAX GFX
 - 2ND COAT: 500µm DFT DUREMAX GFX
 - MINIMUM RECOAT TIME 14 HOURS, MAXIMUM 14 DAYS
 - INIITAL COATING TO BE WASHED DOWN AND THOROUGHLY DRIED PRIOR TO SUBSEQUENT COATING
 - CONTRASTING COLOURS TO BE USED. FINISH COLOUR MID-GRAY
 - FULL ENCAPSULATION AND COLLECTION OF BLAST MEDIA REQUIRED
5. APPLICATION OF DENSO SEASHIELD 2000FD IN ACCORDANCE WITH SPECIFICATION
6. REMOVE AND COLLECT LOOSE PAINT ON PILES PRIOR TO WATER BLASTING TO PREVENT LOSS OF PAINT TO WATER
7. WHERE REPAIR PLATES ARE INSTALLED ON PILES, ENSURE EDGES OR REPAIR PLATES HAVE DENSO MASTIC APPLIED EXTENDING MINIMUM 100MM FROM EDGE OF PLATE TO SMOOTH OUT PROFILE AND PREVENT GAPS. MASTIC TO BE APPLIED PRIOR TO MARINE PILING TAPE

FOLLOWING APPLICATION OF PAINT COATING SYSTEM, PITTED AREAS DEEPER THAN 2mm ON TOP FACE ARE TO BE FILLED WITH DULUX LUXEPOXY FILLER IN ACCORDANCE WITH MANUFACTURERS INSTRUCTIONS. ROLL OR BRUSH ADDITIONAL COAT DULUX GFX OVER CURED EPOXY FILLER

CONVEYOR GALLERY LEGS SEE SHEET 2

THIS JACKET TO GO ON LAST

DENSO SEASHIELD 2000FD APPLIED TO NOMINATED PILES

MARINE PILING TAPE TO PROTRUDE NO MORE THAN 20MM BELOW BOTTOM OF JACKET

DECK UNIT KEEPER ANGLE REPLACEMENTS SEE SHEET 2

MARK PILEBENT NO. 350mm STENCIL APPLICATION DATE 75mm STENCIL WHITE METALSHIELD EPOXY ENAMEL SPRAY OR OTHER APPROVED

+ 6.7m LAT

1.9m LAT

LAT 0.0

TYPICAL PILEBENT

SECTION B-B

DETAIL C
SEALING AT BEARING PAD
SCALE 1 : 5

STRIPE COAT AT INTERFACE OF RUBBER BEARING PAD AND STEEL, PUSHING UNDER CREVICE. FOLLOWING APPLICATION OF COATING SYSTEM, APPLY SELLEYS PRO-SERIES POLYURETHANE SEALANT TO BASE OF BEARING PAD AND PUSH INTO CREVICE. DO NOT APPLY TO TOP EDGE OF BEARING PAD

PAINT COATING TO EXTEND MINIMUM 100mm UNDERNEATH 2000FD JACKET STOPAQ BANDS, WHERE FITTED, TO BE REMOVED TO MINIMUM 100mm BELOW TOP OF JACKET TO ALLOW PAINT COATING

DETAIL A
TOP OF DENSO SEASHIELD SYSTEM
SCALE 1 : 20

ON PILES WITH EXISITNG JACKETS, TEMPORARILY REMOVE TOP JACKET FOR PAINT COATING APPLICATION. USE NEW PRIMER AND MARINE PILING TAPE DURING REINSTALLATION

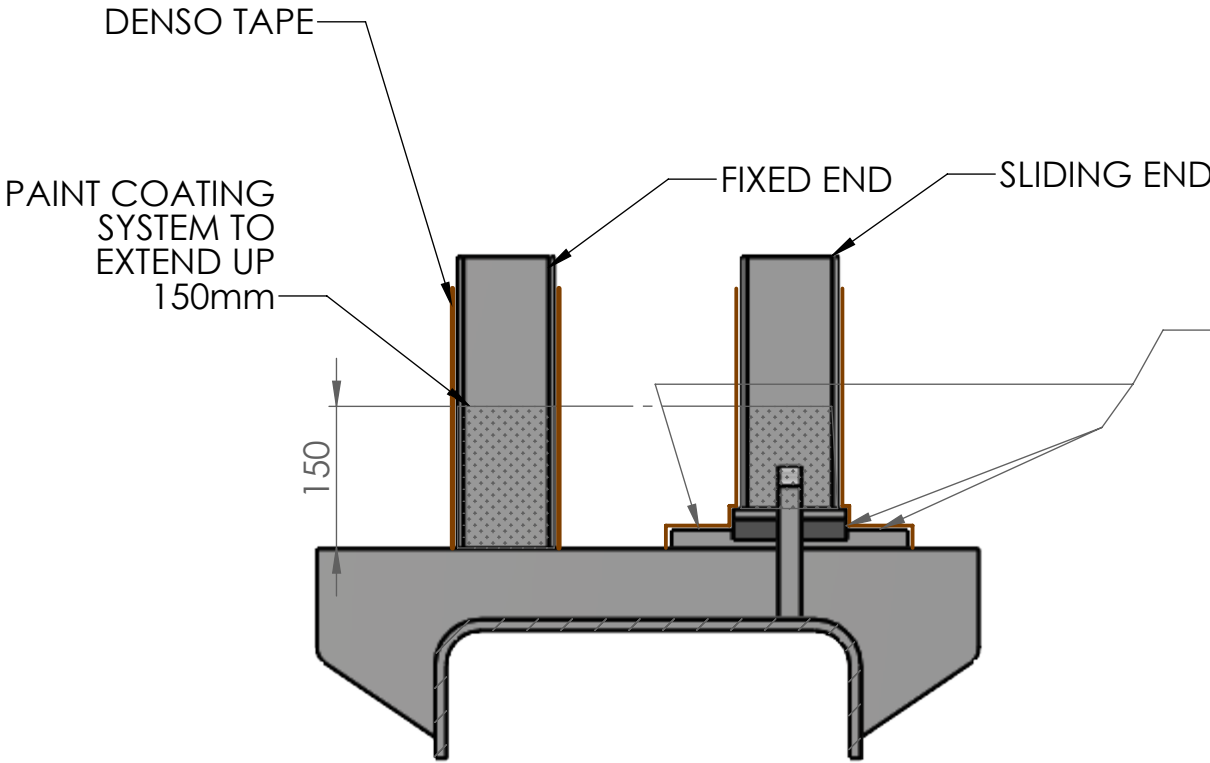


DRAWN	DE	DATE	16/11/2024	REVISION	0
LBST JETTY PILE AND HEADSTOCK REMEDIATION					
TITLE:					
PILEBENT REQUIREMENTS SHEET 1					
DWG NO.				A2	
QSL2501-JPHR-G01-R0					
SCALE:1:40				SHEET 1 OF 2	

TREATMENT AT GALLERY LEGS

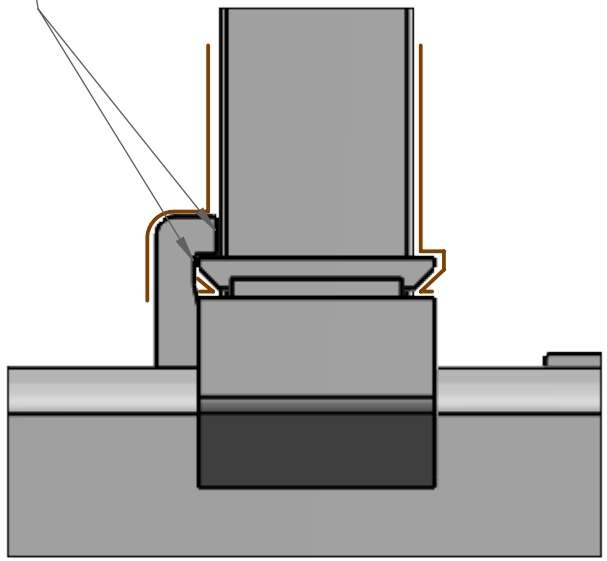
1. REMOVE EXISITNG DENSO TAPE TO MINIMUM 200mm ABOVE FOOT
2. WATER BLAST AND REMOVE ALL CORROSION AROUND SLIDING FOOT
3. ENSURE GAP BETWEEN HOOK AND SLIDING GALLERY FOOT
4. MASK SLIDING SURFACE AND SLIDING FOOT
5. APPLY PAINT COATING SYSTEM TO 150mm ABOVE FOOT
6. REMOVE MASKING AND APPLY THICK LAYER OF DENSO PRIMER TO SLIDING SURFACE AND OTHER SURFACES
7. APPLY 2 LAYERS OF DENSO TAPE TO LEG/FOOT AND HOOK STARTING FROM 400mm ABOVE FOOT DOWN TO BASE OF FOOT. APPLY THICK LAYER OF DENSO PRIMER OVER TAPE SYSTEM.

NOTE: DENSO TAPE TO BE NEATLY APPLIED BY CUTTING CORNERS ETC TO ALLOW TAPE TO CONFORM AROUND EDGES. TO BE COMPLETED TO SATISFACTION OF PRINCIPAL'S REPRESENTATIVE



SECTION E-E SHEET 1
SCALE 1 : 8

ENSURE GAP OF 2mm BETWEEN HOOK AND FOOT
USE HACKSAW BLADE IF REQ'D TO REINSTATE GAP
APPLY DENSO TAPE THROUGH GAP



DETAIL D SHEET 1
SCALE 1 : 8

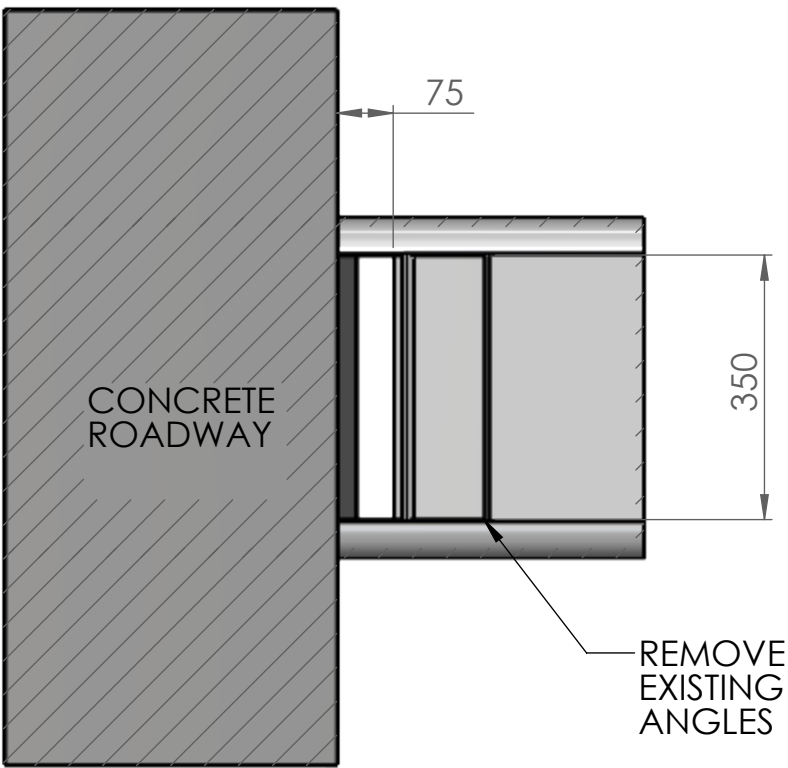
CONVEYOR GALLERY LEGS

DECK UNIT KEEPER ANGLE REPLACEMENT

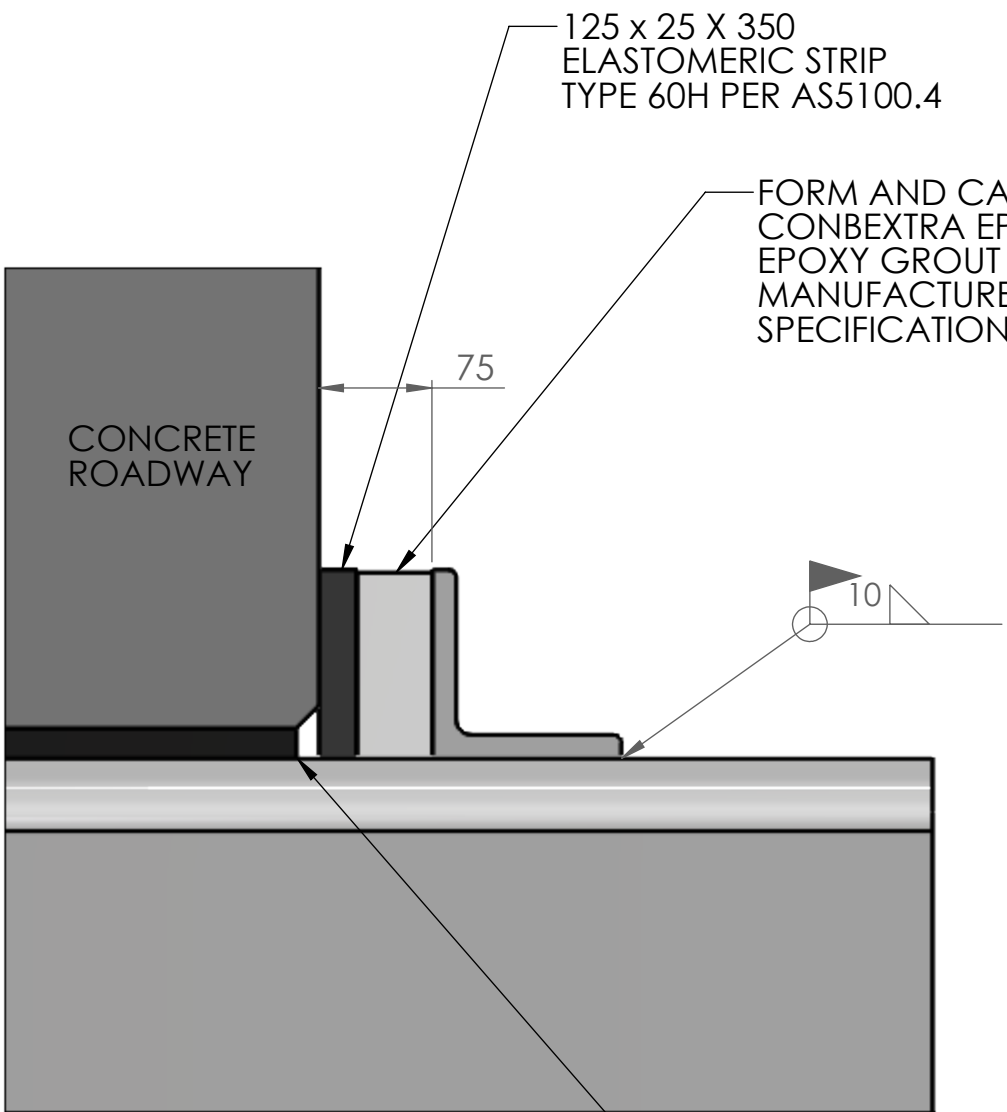
1. REMOVE EXISTING ANGLES BY CUTTING/GRINDING, TAKING CARE NOT TO DAMAGE HEADSTOCK OR CONCRETE ROADWAY
2. WELD NEW ANGLE IN PLACE WITH 75mm GAP TO ROADWAY CONCRETE FACE
3. ABRASIVE BLAST NEW ANGLES TOGETHER WITH REMAINDER OF HEADSTOCK AND APPLY COATING SYSTEM AS PER SPECIFICATION
4. APPLY SIKAFLEX TO BEARING PAD EDGE
5. ADHERE ELASTOMERIC PAD TO ROADWAY
6. FORM AND POUR EPOXY GROUT IN ACCORDANCE WITH MANUFACTURERS REQUIREMENTS

WELDING NOTES:

- BOILERMAKER TO BE SUITABLY QUALIFIED AND EXPERIENCED WITH EVIDENCE OF QUALIFICATION SUBMITTED.
- WELDING TO BE CATEGORY SP TO AS1554 USING E48XX OR W50XX CONSUMABLES
- FINISH PROFILE OF WELDING TO BE SMOOTH AND FREE OF SHARP EDGES OR CREVICES. ALL WELD SPATTER POROSITY AND IRREGULAR WELD TO BE REMOVED AND REPLACED. WELDS TO BE GROUND SMOOTH. UNDERCUTTING NOT PERMITTED. 100% VISUAL INSPECTION



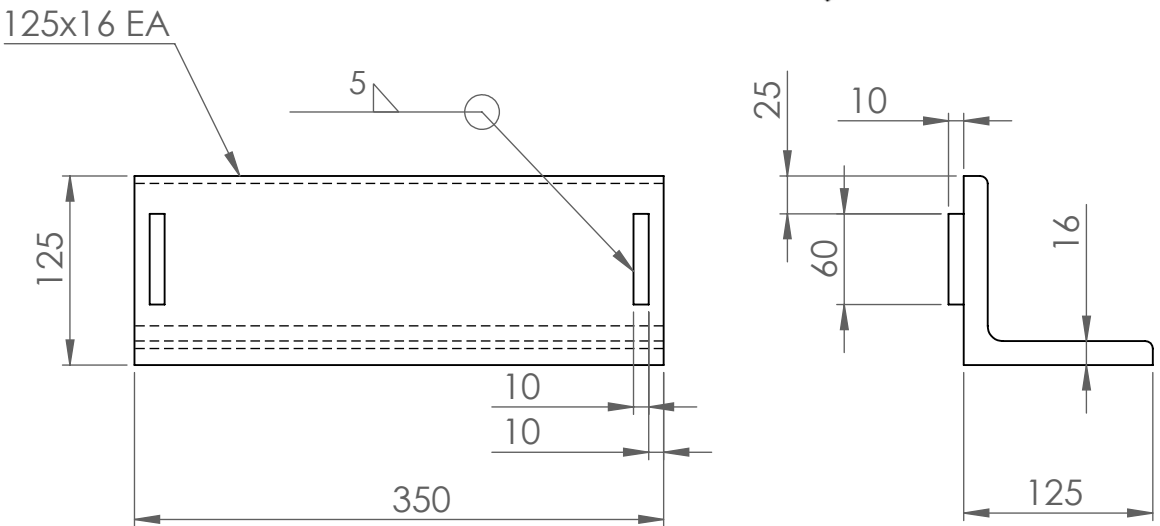
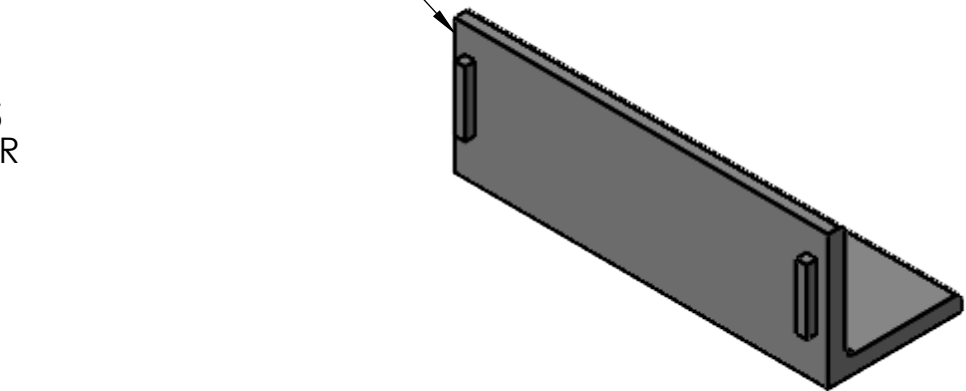
SECTION G-G SHEET 1
SCALE 1 : 10



DETAIL F SHEET 1
SCALE 1 : 5

DECK UNIT KEEPER ANGLE REPLACEMENT

GRIND ALL SHARP EDGES 2mm MINIMUM RADIUS



DECK UNIT KEEPER ANGLE FABRICATION



DRAWN	DE	DATE	16/11/2024	REVISION	0
LBST JETTY PILE AND HEADSTOCK REMEDIATION					
TITLE:					
PILEBENT REQUIREMENTS SHEET 2					
DWG NO.					A2
QSL2501-JPHR-G02-R0					
SCALE:1:50					
SHEET 2 OF 2					

APPENDIX B – UNDERPASS REMEDIATION

QSL2501-JPHR-G03 – Underpass Requirements

UNDERPASS TREATMENT GENERAL

1. REMOVE WALKWAY MESH COMPONENTS, PLACING BOLTS OR STUDS IN THE ATTACHMENT POINTS TO MAINTAIN THEM DURING SUSEQUENT STEPS.
2. REMOVE ALL EXISTING DENSO TAPE
3. CLEAN AND INSPECT FOR SEVERE CORROSION, HOLES ETC. SEEK ADVICE FROM SUPERINTENDENT'S REPRESENTATIVE
4. PAINT PROTECTIVE COATING SYSTEM IN ACCORDANCE WITH DULUX SPECIFICATION PCQ-015-0121-4 AND ANY REQUIREMENTS NOTED IN THE CONTRACT SPECIFICATION
5. CLEAN, ABRASIVE BLAST AND APPLY THE PAINT COATING SYSTEM CONSISTING OF 2 COATS OF 500um DFT DUREMAX GFX TO ALL SURFACES INCLUDING HANDRAILS AND LADDER RUNGS
6. TOPCOAT HANDRAILS AND LADDER RUNGS WITH 125um DFT DULUX WEATHERMAX HBR COLOUR Y14 GOLDEN YELLOW IN ACCORDANCE WITH MANUFACTURERS TECHNICAL DATA SHEETS
7. HOLLOW SECTION MEMBERS FORMING THE FLOOR FRAME AND HANGERS FOR THE UNDERPASS AND UPPER LEVEL PLATFORMS ARE TO BE ADDITIONALLY TREATED WITH THE DENSO SEASHIELD 70 SYSTEM FOLLOWING APPLICATION OF THE PAINT COATING SYSTEM:
 - MINIMUM 20mm DENSO MASTIC TO BE APPLIED AT ALL INTERNAL CORNERS TO BUILD RADIUS
 - APPLY A LAYER OF DENSO SEASHIELD PRIMER
 - APPLY MINIMUM 2 LAYERS OF DENSO MARINE PILING TAPE, CUTTING AS REQUIRED TO NEATLY COVER CORNERSAND EDGES
 - APPLY DENSO GLASS OUTERWRAP UV IN A MINIMUM 55% OVERLAP, ENSURING ALL AIR POCKETS ARE ELIMINATED AND TAPE IS FORMED NEATLY AT EDGES AND CORNERS, WITH NO UPTURNED TAPE EDGES.
 - APPLY TEMPORARY POLYWRAP AND REATTACH WALKWAY MESH TEMPORARILY PRIOR TO GLASS OUTERWRAP CURING SO OUTERWRAP CONFORMS TO MESH DURING CURING.

GENERAL NOTES

8. WUC INCLUDE THE UNDERPASS, THE UPPER LEVEL ACCESS PLATFORM ATTACHED TO THE ROADWAY, BOTH LADDERS, ALL HANDRAILS, AND THE ACCESS PLATFORM ON THE CONVEYOR SIDE EXTENDING UNDERNEATH THE CLADDING WHERE POSSIBLE.
9. FOR THE FIRST UNDERPASS OF EACH TYPE (ORIGINAL AND NEW VARIETIES), A TRIAL WRAPPING SECTION MUST BE UNDERTAKEN TO DETERMINE THE OPTIMUM ARRANGMENT AND ORIENTATION OF TAPE AND OUTERWRAP. THIS SHALL BE COMPLETED TO THE SATISFACTION OF THE SUPERINTENDENT'S REPRESENTATIVE PRIOR TO COMPLETING THE REMAINDER OF THE UNDERPASS. THE AGREED METHOD MUST BE USED ON ALL OTHER UNDERPASSES.
10. REMOVING SECTIONS FOR TREATMENT IN THE WORKSHOP IS PERMISSIBLE.



NEW STYLE UNDERPASS



ORIGINAL UNDERPASS

UPPER LEVEL PLATFORMS INCLUDED

NO TREATMENT REQUIRED ON STAINLESS LADDER CAGES

SEASHIELD 70 SYSTEM TO INCLUDE HANGERS UP TO HEADSTOCK

SEASHIELD 70 SYSTEM TO INCLUDE HANGERS UP TO HEADSTOCK



UPPER LEVEL PLATFORM ON CONVEYOR SIDE

SEASHIELD 70 SYSTEM TO EXTEND SLIGHTLY UNDER SHEETING WHERE POSSIBLE



DRAWN	DE	DATE	16/11/2024	REVISION	0
LBST JETTY PILE AND HEADSTOCK REMEDIATION					
TITLE:					
UNDERPASS REQUIREMENTS					
DWG NO.					A2
QSL25-1-JPHR-G03-R0					
SCALE:1:1					
SHEET 1 OF 1					

APPENDIX C C7B INTERNAL GALLERY WRAPPING REQUIREMENTS

QSL2501-JPHR-G04 – Conveyor Gallery Wrapping Requirements

INTERNAL GALLERY WRAPPING GENERAL NOTES

1. THE CONVEYOR BELT MUST BE LOCKED OUT DURING WORKS AND ALL WORKERS MUST USE A PERSONAL LOCK IN ACCORDANCE WITH THE QSL LOCK-OUT/TAG-OUT PROCEDURE
2. PROTECTION OF THE CONVEYOR BELT IS OF UTMOST IMPORTANCE. A DROP SHEET MUST BE PLACED ON THE BELT WHERE ANY WORK IS OCCURRING ON THE TOP SECTION. THIS DROP SHEET MUST BE REMOVED AT THE COMPLETION OF EACH DAY'S WORK, THE BELT CLEANED AND ANY TOOLS REMOVED FROM ABOVE THE BELT
3. THE AREA TO BE TREATED MUST BE WATER BLASTED ANY LOOSE PAINT AND CORROSION REMOVED. THE CONTRACTOR MUST INSPECT THE AREA FOR SEVERE CORROSION OR HOLING PRIOR TO PROCEEDING. CONCERNS SHOULD BE RAISED WITH THE SUPERINTENDENT'S REPRESENTATIVE FOR FURTHER DIRECTION
4. AN INITIAL SECTION FOR EACH REGION SHALL BE UNDERTAKEN IN ORDER TO CONFIRM THE METHODOLOGY AND WRAPPING SEQUENCE. THE SECTION SHALL BE COMPLETED TO THE SATISFACTION OF THE SUPERINTENDENT'S REPRESENTATIVE AND FURTHER AREAS SHALL BE CONSTRUCTED TO THE SAME STANDARD
5. THE WRAPPING SHALL BE OF A SIMILAR STANDARD TO THE EXISTING GOOD AND NEAT EXAMPLES. A CIGARETTE PAPER STYLE WRAP SHOULD GENERALLY BE USED ON THE TOP CHORD AND SIDE SECTIONS WITH JOINS ON THE TOP OR SIDE OF THE BEAM AND WEATHERBOARDING OF JOINS. ALL AIR POCKETS TO BE REMOVED BY PRESSING TAPE TO SURFACE AND SMOOTHING BY HAND. TERMINATED SECTION SHALL HAVE ONE CIRCUMFERENTIAL WRAP
6. ALL OVERLAPS MUST BE A MINIMUM OF 25mm
7. WRAPPING OF NODES SHALL BE CONSTRUCTED NEATLY BY CUTTING OF TAPE AS REQUIRED TO ENSURE A NEAT FINISH
8. TREATMENT SHALL EXTEND ONTO BOLTED PLATES

TOP SECTION

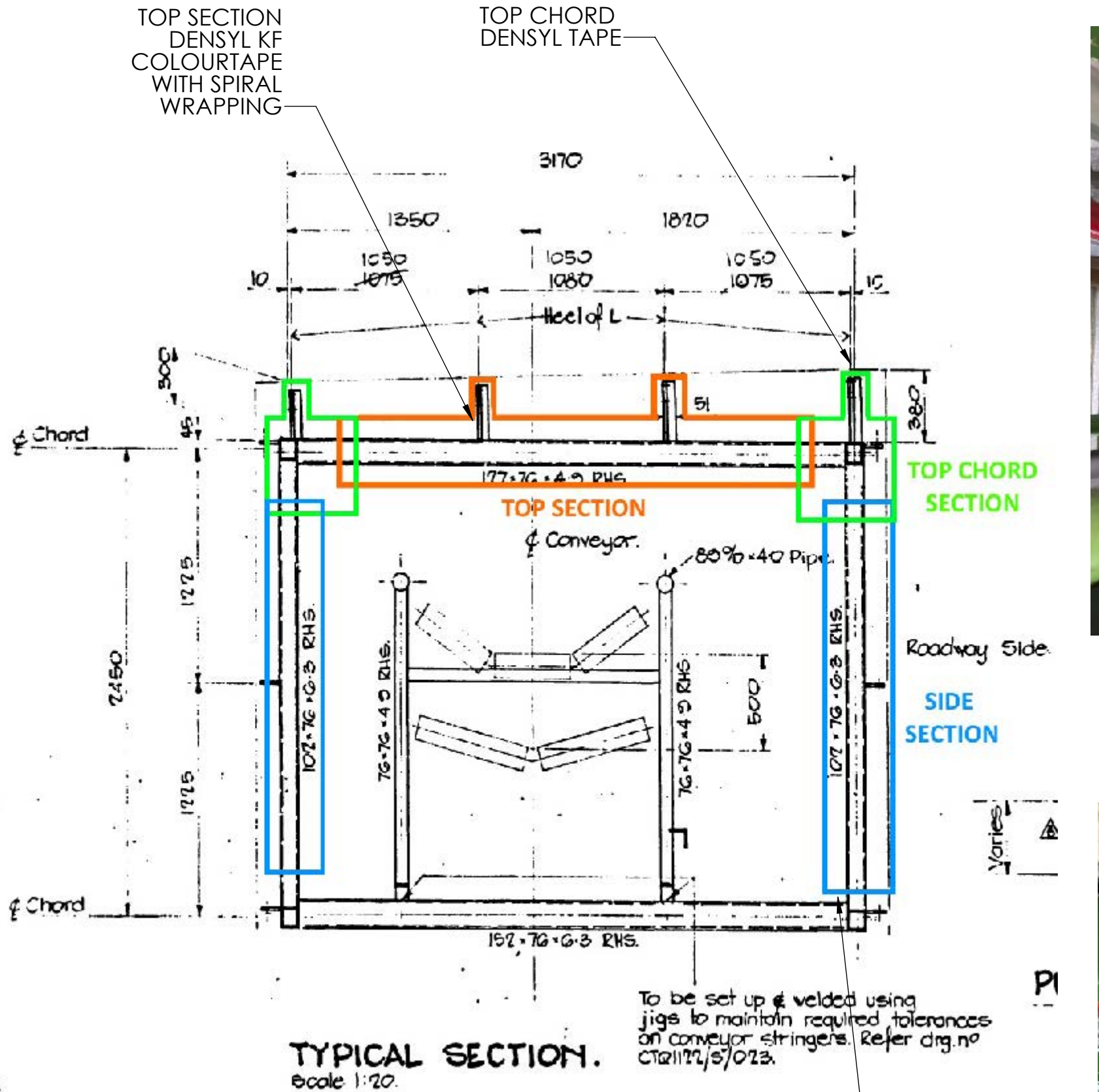
9. THE TOP SECTION IS NOMINALLY INCLUSIVE OF THE AREA 250mm FROM EACH CHORD AND MUST OVERLAP THE TOP CHORD SECTIONS ALREADY COMPLETED
10. THE TOP SECTION SHALL USE THE FOLLOWING SYSTEM:
 - DENSO PRIMER PLACED ON ANY CORRODED AREA PLUS BOLT HEADS
 - A VERY THIN LAYER OF DENSO PRIMER APPLIED TO REMAINING SURFACES
 - DENSYL KF COLOURTAPE SPIRALLY APPLIED WITH MINIMUM 55% OVERLAP
11. THE CABLE TRAY AND FIRE ALARM PLATES SHALL BE LOOSENED WHERE POSSIBLE AND TAPE PUSHED UNDERNEATH
12. TAPE SHALL BE NEATLY FORMED AROUND THE CONDUIT BRACKETS WITH CUT SECTIONS TO FORM CLOSE SEAL. ALTERNATIVELY THESE MAY BE LOOSED WITH TAPE PUSHED UNDERNEATH.

CHORD SECTION

13. THE TOP CHORD SECTION IS INCLUSIVE OF THE AREA 300mm FROM EACH CHORD
14. THE TOP CHORD SECTION SHALL USE THE FOLLOWING SYSTEM:
 - DENSO PRIMER PLACED OVER ALL SURFACES
 - A SINGLE LAYER OF DENSYL TAPE WITH MINIMUM 25mm OVERLAPS

SIDE SECTION

15. THE SIDE SECTION IS INCLUSIVE OF THE AREA 100mm FROM THE BOTTOM CHORD AND 250mm FROM TOP CHORD IT MUST OVERLAP SECTIONS ALREADY COMPLETED
16. THE SIDE SECTION SHALL USE THE FOLLOWING SYSTEM:
 - DENSO PRIMER PLACED OVER ALL SURFACES
 - A NOMINAL SINGLE LAYER OF DENSYL TAPE WITH MINIMUM 25mm OVERLAPS
 - A TOPCOAT OF DENSO SINGLE PACK ELASTOMERIC MEMBRANE APPLIED BY BRUSH OR ROLLER IN A SINGLE THICK COAT OR A SPIRALLY APPLIED LAYER OF DENSO PVC OUTERWRAP WITH 55% OVERLAP



WEATHERBOARDED OVERLAPS

NO OVERLAPS BOTTOM FACE

TYPICAL CIGARETTE STYLE WRAP OF HOLLOW SECTIONS FOR SIDE AND TOP CHORD SECTIONS



TYPICAL VIEW INSIDE GALLERY WITH TOP CHORD SECTIONS COMPLETED



EXAMPLE OF NEAT WORKMANSHIP AT NODE



DRAWN	DE	DATE	16/11/2024	REVISION	0
LBST JETTY PILE AND HEADSTOCK REMEDIATION					
TITLE:					
CONVEYOR GALLERY WRAPPING REQUIREMENTS					
DWG NO.					A2
QSL2501-JPHR-G04-R0					
SCALE:1:1				SHEET 1 OF 1	

APPENDIX D JETTY PILEBENT SCOPE LISTING

APPENDIX D – JettyPilebentScopeofWorks.xlsx

Pile/bent no.	Included in Contract	Pile/bent type	Stopaq bands to remove	Full piles to include	top jacket only to include	Expansion Joint	Headstock Condition (2)	Underpass and type	painted history	SOUTHERN END OF HEADSTOCK -> NORTHERN END OF HEADSTOCKS														Inclusion in Separable Portion (1 to 7)	
										Pile no.	Pile works	Pile no.	Pile works	Pile no.	Pile works	Pile no.	Pile works	Pile no.	Pile works	Pile no.	Pile works	Pile no.	Pile works		
116	Y	Typical					4	OLD							116-2					116-1					
117	Y	Typical					4								117-2					117-1					SP5
118	Y	Typical					4								118-2					118-1					SP5
119	Y	Typical					4								119-2					119-1					SP5
120	Y	Typical	1				4								120-2					120-1					SP5
121	Y	Anchor Bent Type A					4								121-2					121-3					SP5
122	Y	Typical					5								122-2					122-1					SP5
123	Y	Typical					4								123-2					123-1					SP5
124	Y	Typical					4								124-2					124-1					SP5
125	Y	Typical					4								125-2					125-1					SP5
126	Y	Typical					4								126-2					126-1					SP5
127	P	Expansion joint	1				EJ								127-2					127-3					SP5
128	Y	Typical					4								128-2					128-1					SP5
129	Y	Typical					4								129-2					129-1					SP5
130	Y	Typical					4								130-2					130-1					SP5
131	Y	Typical					4								131-2					131-1					SP5
132	Y	Typical					4								132-2					132-1					SP5
133	N						5								133-2					133-3					
134	Y	Typical					4								134-2					134-1					SP5
135	Y	Typical					4								135-2					135-1					SP5
136	Y	Typical	2				5								136-2					136-1					SP5
137	Y	Typical					4								137-2					137-1					SP5
138	Y	Typical	1				5								138-2					138-1					SP5
139	N						EJ								139-2					139-3					
140	Y	Typical	1				4								140-2					140-1					SP5
141	Y	Typical	2				5								141-2					141-1					SP5
142	Y	Typical					4								142-2					142-1					SP5
143	Y	Typical	1				4								143-2					143-1					SP5
144	Y	Typical					5								144-2					144-1					SP5
145	Y	Anchor Bent Type A					4								145-2					145-3					SP5
146	Y	Typical	2				5								146-2					146-1					SP4
147	Y	Typical					4								147-2					147-1					SP4
148	Y	Typical		4	148-2	148-1	SP4																		
149	Y	Typical	1	5	149-2	149-1	SP4																		
150	Y	Typical	2	5	150-2	150-1	SP4																		
151	N																								
152	N																								
153	Y	Typical	2	3	153-2	153-1	SP4																		
154	Y	Typical	1	4	154-2	154-1	SP4																		
155	Y	Typical		3	155-2	155-1	SP4																		
156	Y	Passing Bay	1	3	156-3	156-1	SP4																		
157	Y	Anchor Bent Type A Passing Bay		3	157-2	157-1	SP4																		
158	Y	Typical		4	158-2	158-1	SP4																		
159	Y	Typical		5	159-2	159-1	SP4																		
160	Y	Typical	1	3	160-2	160-1	SP4																		
161	Y	Typical		3	161-2	161-1	SP4																		
162	Y	Typical		3	162-2	162-1	SP4																		
163	P	Expansion joint		EJ	163-2	163-3	SP4																		
164	Y	Typical		5	164-2	164-1	SP4																		
165	Y	Typical	1	3	165-2	165-1	SP4																		
166	Y	Typical		3	166-2	166-1	SP4																		
167	Y	Typical		4	167-2	167-1	SP4																		
168	Y	Anchor Bent Type B		3	168-2	168-3	SP4																		
169	Y	Typical		5	169-2	169-1	SP4																		
170	Y	Typical	1	4	170-2	170-1	SP4																		
171	Y	Typical		5	171-2	171-1	SP4																		
172	Y	Typical		5	172-2	172-1	SP4																		
173	P	Expansion joint	1	EJ	173-2	173-3	SP4																		
174	Y	Typical		5	174-2	174-1	SP3																		
175	Y	Typical		4	175-2	175-1	SP3																		

Pilebent no.	Included in Contract	Pilebent type	Stopaq bands to remove	Full piles to include	top jacket only to include	Expansion Joint	Headstock Condition (2)	Underpass and type	painted history	SOUTHERN END OF HEADSTOCK -> NORTHERN END OF HEADSTOCKS														Inclusion in Separable Portion (1 to 7)							
										Pile no.	Pile works	Pile no.	Pile works	Pile no.	Pile works	Pile no.	Pile works	Pile no.	Pile works	Pile no.	Pile works	Pile no.	Pile works								
176	Y	Typical					5	NEW	painted 2010																SP3						
177	Y	Anchor Bent Type B	2				3																		177-2	177-3	177-4	177-1		SP3	
178	Y	Typical	1				4																		178-2			178-1		SP3	
179	Y	Typical					3																		179-2			179-1		SP3	
180	Y	Typical					3																		180-2			180-1		SP3	
181	P	Expansion joint					EJ																		4	181-2	181-3	181-1	SP4		
182	Y	Typical	3				182-2																		182-1	SP3					
183	Y	Typical	3				183-2																		183-1	SP3					
184	Y	Typical	3				184-2																		184-1	SP3					
185	N						5																		185-2	185-3	185-4	185-1			
186	Y	Typical	3				186-2																		186-1			SP3			
187	Y	Typical	4				187-2																		187-1			SP3			
188	Y	Typical	4				188-2																		188-1			SP3			
189	N						EJ																		5	189-2	189-3	189-1			
190	Y	Typical	3				190-2																		190-1	SP3					
191	Y	Typical	3				191-2																		191-1	SP3					
192	Y	Typical	3				192-2																		192-1	SP3					
193	Y	Anchor Bent Type B					3																		193-2	193-3	193-4	193-1		SP3	
194	Y	Typical	3				194-2																		194-1			SP3			
195	Y	Typical	3	195-2	195-1	SP3																									
196	Y	Typical	2	3	196-2	196-1	SP3																								
197	P	Expansion joint		EJ	3	197-2	197-3	197-1		SP3																					
198	Y	Typical	4	198-2	198-1	SP3																									
199	Y	Typical	1	3	199-2	199-1		SP3																							
200	Y	Passing Bay	2	3	200-2	FULL		200-1		SP3																					
201	Y	Anchor Bent Type B Passing Bay	1	1	3	201-2	201-3	201-4	201-1		SP3																				
202	Y	Typical	1	3	202-2	202-1			FULL		SP3																				
203	Y	Typical	3	203-2	203-1				SP3																						
204	Y	Typical	1	4	204-2	204-1			FULL		SP3																				
205	P	Expansion joint		2	EJ	3	205-2	205-3	TOP	205-1	TOP	SP3																			
206	Y	Typical	1	4	206-2	206-1	FULL			SP3																					
207	Y	Typical	1	4	207-2	207-1				SP3																					
208	Y	Typical	1	5	208-2	208-1	FULL			SP3																					
209	Y	Anchor Bent Type B		1	1	4	209-2	TOP	209-3	209-4	209-1	FULL	SP3																		
210	Y	Typical	1	4	210-2	210-1	FULL	SP3																							
211	Y	Typical	1	4	211-2	211-1	FULL	SP3																							
212	Y	Typical	1	4	212-2	212-1	FULL	SP3																							
213	P	Expansion joint		3		EJ	4	213-2	FULL	213-3	FULL	213-1	FULL	SP3																	
214	Y	Typical	1	1	4	214-2	TOP	214-1	FULL			SP3																			
215	Y	Typical	1	1	4	215-2	TOP	215-1				SP3																			
216	Y	Typical	1	1	4	216-2	TOP	216-1	FULL			SP3																			
217	Y	Anchor Bent Type B	2	1		4	217-2		217-3	217-4	FULL	217-1		SP3																	
218	Y	Typical	1	1	4	218-2						218-1	FULL	SP3																	

Pilebent	Scope statement
1	Pilebent 1 is not included in the contract
2	Pilebent 2 is not included in the contract
3	Pilebent 3 is included in the contract and is of type Typical.
4	Pilebent 4 is included in the contract and is of type Typical.
5	Pilebent 5 is included in the contract and is of type Typical.
6	Pilebent 6 is included in the contract and is of type Typical.
7	Pilebent 7 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint.
8	Pilebent 8 is included in the contract and is of type Typical.
9	Pilebent 9 is included in the contract and is of type Typical.
10	Pilebent 10 is included in the contract and is of type Typical.
11	Pilebent 11 is included in the contract and is of type Typical.
12	Pilebent 12 is included in the contract and is of type Typical.
13	Pilebent 13 is included in the contract and is of type Anchor Bent Type A.
14	Pilebent 14 is included in the contract and is of type Typical.
15	Pilebent 15 is included in the contract and is of type Typical.
16	Pilebent 16 is included in the contract and is of type Typical.
17	Pilebent 17 is included in the contract and is of type Typical. It includes an old style underpass remediation.
18	Pilebent 18 is included in the contract and is of type Typical.
19	Pilebent 19 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint.
20	Pilebent 20 is included in the contract and is of type Typical.
21	Pilebent 21 is included in the contract and is of type Typical.
22	Pilebent 22 is included in the contract and is of type Typical.
23	Pilebent 23 is included in the contract and is of type Typical.
24	Pilebent 24 is included in the contract and is of type Passing Bay. It includes, 1 StopAq band(s) to remove.
25	Pilebent 25 is included in the contract and is of type Anchor Bent Type A Passing Bay.
26	Pilebent 26 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
27	Pilebent 27 is included in the contract and is of type Typical.
28	Pilebent 28 is included in the contract and is of type Typical.
29	Pilebent 29 is included in the contract and is of type Typical.
30	Pilebent 30 is included in the contract and is of type Typical.
31	Pilebent 31 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint.
32	Pilebent 32 is included in the contract and is of type Typical. It includes an old style underpass remediation.
33	Pilebent 33 is included in the contract and is of type Typical.
34	Pilebent 34 is included in the contract and is of type Typical.
35	Pilebent 35 is included in the contract and is of type Typical.
36	Pilebent 36 is included in the contract and is of type Typical.
37	Pilebent 37 is included in the contract and is of type Anchor Bent Type A. It includes, 1 StopAq band(s) to remove.
38	Pilebent 38 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
39	Pilebent 39 is included in the contract and is of type Typical.
40	Pilebent 40 is included in the contract and is of type Typical.
41	Pilebent 41 is included in the contract and is of type Typical.
42	Pilebent 42 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
43	Pilebent 43 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint.
44	Pilebent 44 is included in the contract and is of type Typical.
45	Pilebent 45 is included in the contract and is of type Typical.

Pilebent	Scope statement
46	Pilebent 46 is included in the contract and is of type Typical.
47	Pilebent 47 is included in the contract and is of type Typical. It includes an old style underpass remediation.
48	Pilebent 48 is included in the contract and is of type Typical.
49	Pilebent 49 is not included in the contract
50	Pilebent 50 is included in the contract and is of type Typical.
51	Pilebent 51 is included in the contract and is of type Typical.
52	Pilebent 52 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
53	Pilebent 53 is included in the contract and is of type Typical.
54	Pilebent 54 is included in the contract and is of type Typical.
55	Pilebent 55 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint.
56	Pilebent 56 is included in the contract and is of type Typical.
57	Pilebent 57 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
58	Pilebent 58 is included in the contract and is of type Typical.
59	Pilebent 59 is included in the contract and is of type Typical.
60	Pilebent 60 is not included in the contract
61	Pilebent 61 is not included in the contract
62	Pilebent 62 is included in the contract and is of type Typical. It includes an old style underpass remediation.
63	Pilebent 63 is included in the contract and is of type Typical.
64	Pilebent 64 is included in the contract and is of type Typical.
65	Pilebent 65 is included in the contract and is of type Typical.
66	Pilebent 66 is included in the contract and is of type Typical.
67	Pilebent 67 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint.
68	Pilebent 68 is included in the contract and is of type Typical.
69	Pilebent 69 is included in the contract and is of type Typical.
70	Pilebent 70 is included in the contract and is of type Typical.
71	Pilebent 71 is included in the contract and is of type Typical.
72	Pilebent 72 is included in the contract and is of type Typical.
73	Pilebent 73 is included in the contract and is of type Anchor Bent Type A.
74	Pilebent 74 is included in the contract and is of type Typical.
75	Pilebent 75 is included in the contract and is of type Typical.
76	Pilebent 76 is included in the contract and is of type Typical.
77	Pilebent 77 is included in the contract and is of type Typical. It includes an old style underpass remediation.
78	Pilebent 78 is included in the contract and is of type Typical.
79	Pilebent 79 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint.
80	Pilebent 80 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
81	Pilebent 81 is included in the contract and is of type Typical.
82	Pilebent 82 is included in the contract and is of type Typical.
83	Pilebent 83 is included in the contract and is of type Typical.
84	Pilebent 84 is included in the contract and is of type Typical.
85	Pilebent 85 is included in the contract and is of type Anchor Bent Type A.
86	Pilebent 86 is included in the contract and is of type Typical.
87	Pilebent 87 is included in the contract and is of type Typical.
88	Pilebent 88 is included in the contract and is of type Typical.
89	Pilebent 89 is included in the contract and is of type Typical.
90	Pilebent 90 is included in the contract and is of type Typical.

Pilebent	Scope statement
91	Pilebent 91 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint.
92	Pilebent 92 is included in the contract and is of type Typical. It includes an old style underpass remediation.
93	Pilebent 93 is included in the contract and is of type Typical.
94	Pilebent 94 is included in the contract and is of type Typical.
95	Pilebent 95 is included in the contract and is of type Typical.
96	Pilebent 96 is included in the contract and is of type Typical.
97	Pilebent 97 is included in the contract and is of type Anchor Bent Type A. It includes, 1 StopAq band(s) to remove.
98	Pilebent 98 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
99	Pilebent 99 is included in the contract and is of type Typical.
100	Pilebent 100 is included in the contract and is of type Typical.
101	Pilebent 101 is included in the contract and is of type Typical.
102	Pilebent 102 is included in the contract and is of type Typical.
103	Pilebent 103 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint.
104	Pilebent 104 is included in the contract and is of type Typical.
105	Pilebent 105 is included in the contract and is of type Typical.
106	Pilebent 106 is included in the contract and is of type Typical.
107	Pilebent 107 is included in the contract and is of type Typical. It includes an old style underpass remediation.
108	Pilebent 108 is included in the contract and is of type Passing Bay. It includes, 1 StopAq band(s) to remove.
109	Pilebent 109 is included in the contract and is of type Anchor Bent Type A Passing Bay. It includes, 1 StopAq band(s) to remove.
110	Pilebent 110 is included in the contract and is of type Typical.
111	Pilebent 111 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
112	Pilebent 112 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
113	Pilebent 113 is included in the contract and is of type Typical.
114	Pilebent 114 is included in the contract and is of type Typical.
115	Pilebent 115 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint.
116	Pilebent 116 is included in the contract and is of type Typical.
117	Pilebent 117 is included in the contract and is of type Typical.
118	Pilebent 118 is included in the contract and is of type Typical.
119	Pilebent 119 is included in the contract and is of type Typical.
120	Pilebent 120 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
121	Pilebent 121 is included in the contract and is of type Anchor Bent Type A.
122	Pilebent 122 is included in the contract and is of type Typical. It includes an old style underpass remediation.
123	Pilebent 123 is included in the contract and is of type Typical.
124	Pilebent 124 is included in the contract and is of type Typical.
125	Pilebent 125 is included in the contract and is of type Typical.
126	Pilebent 126 is included in the contract and is of type Typical.
127	Pilebent 127 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint. It includes, 1 StopAq band(s) to remove.
128	Pilebent 128 is included in the contract and is of type Typical.
129	Pilebent 129 is included in the contract and is of type Typical.
130	Pilebent 130 is included in the contract and is of type Typical.
131	Pilebent 131 is included in the contract and is of type Typical.
132	Pilebent 132 is included in the contract and is of type Typical.
133	Pilebent 133 is not included in the contract
134	Pilebent 134 is included in the contract and is of type Typical.
135	Pilebent 135 is included in the contract and is of type Typical.

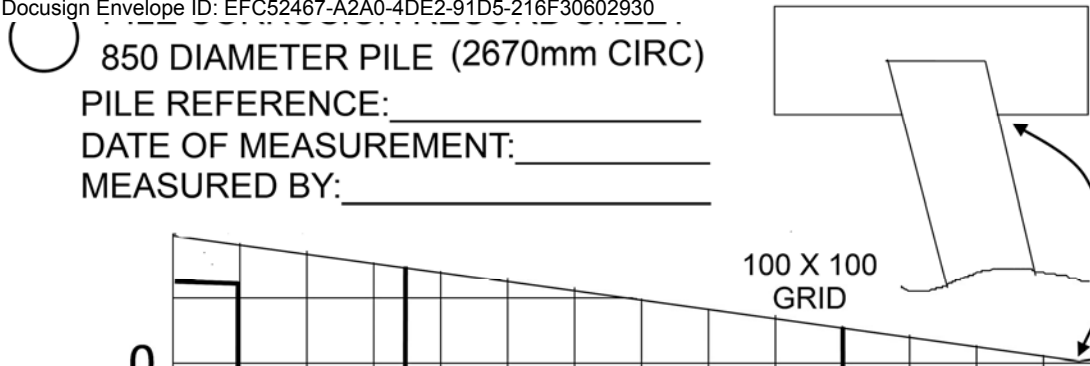
Pilebent	Scope statement
136	Pilebent 136 is included in the contract and is of type Typical. It includes, 2 StopAq band(s) to remove.
137	Pilebent 137 is included in the contract and is of type Typical. It includes an old style underpass remediation.
138	Pilebent 138 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
139	Pilebent 139 is not included in the contract
140	Pilebent 140 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
141	Pilebent 141 is included in the contract and is of type Typical. It includes, 2 StopAq band(s) to remove.
142	Pilebent 142 is included in the contract and is of type Typical.
143	Pilebent 143 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
144	Pilebent 144 is included in the contract and is of type Typical.
145	Pilebent 145 is included in the contract and is of type Anchor Bent Type A.
146	Pilebent 146 is included in the contract and is of type Typical. It includes, 2 StopAq band(s) to remove.
147	Pilebent 147 is included in the contract and is of type Typical.
148	Pilebent 148 is included in the contract and is of type Typical.
149	Pilebent 149 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
150	Pilebent 150 is included in the contract and is of type Typical. It includes, 2 StopAq band(s) to remove.
151	Pilebent 151 is not included in the contract
152	Pilebent 152 is not included in the contract
153	Pilebent 153 is included in the contract and is of type Typical. It includes, 2 StopAq band(s) to remove.
154	Pilebent 154 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
155	Pilebent 155 is included in the contract and is of type Typical.
156	Pilebent 156 is included in the contract and is of type Passing Bay. It includes, 1 StopAq band(s) to remove.
157	Pilebent 157 is included in the contract and is of type Anchor Bent Type A Passing Bay.
158	Pilebent 158 is included in the contract and is of type Typical.
159	Pilebent 159 is included in the contract and is of type Typical.
160	Pilebent 160 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
161	Pilebent 161 is included in the contract and is of type Typical.
162	Pilebent 162 is included in the contract and is of type Typical.
163	Pilebent 163 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint.
164	Pilebent 164 is included in the contract and is of type Typical.
165	Pilebent 165 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
166	Pilebent 166 is included in the contract and is of type Typical.
167	Pilebent 167 is included in the contract and is of type Typical. It includes a new style underpass remediation.
168	Pilebent 168 is included in the contract and is of type Anchor Bent Type B.
169	Pilebent 169 is included in the contract and is of type Typical.
170	Pilebent 170 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
171	Pilebent 171 is included in the contract and is of type Typical.
172	Pilebent 172 is included in the contract and is of type Typical.
173	Pilebent 173 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint. It includes, 1 StopAq band(s) to remove.
174	Pilebent 174 is included in the contract and is of type Typical.
175	Pilebent 175 is included in the contract and is of type Typical.
176	Pilebent 176 is included in the contract and is of type Typical.
177	Pilebent 177 is included in the contract and is of type Anchor Bent Type B. It includes, 2 StopAq band(s) to remove.
178	Pilebent 178 is included in the contract and is of type Typical.
179	Pilebent 179 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
180	Pilebent 180 is included in the contract and is of type Typical.

Pilebent	Scope statement
181	Pilebent 181 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint.
182	Pilebent 182 is included in the contract and is of type Typical. It includes a new style underpass remediation.
183	Pilebent 183 is included in the contract and is of type Typical.
184	Pilebent 184 is included in the contract and is of type Typical.
185	Pilebent 185 is not included in the contract
186	Pilebent 186 is included in the contract and is of type Typical.
187	Pilebent 187 is included in the contract and is of type Typical.
188	Pilebent 188 is included in the contract and is of type Typical.
189	Pilebent 189 is not included in the contract
190	Pilebent 190 is included in the contract and is of type Typical.
191	Pilebent 191 is included in the contract and is of type Typical.
192	Pilebent 192 is included in the contract and is of type Typical.
193	Pilebent 193 is included in the contract and is of type Anchor Bent Type B.
194	Pilebent 194 is included in the contract and is of type Typical. It includes a new style underpass remediation.
195	Pilebent 195 is included in the contract and is of type Typical.
196	Pilebent 196 is included in the contract and is of type Typical. It includes, 2 StopAq band(s) to remove.
197	Pilebent 197 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint.
198	Pilebent 198 is included in the contract and is of type Typical.
199	Pilebent 199 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
200	Pilebent 200 is included in the contract and is of type Passing Bay. It includes, 2 piles requiring scaffold, repair and wrapping.
201	Pilebent 201 is included in the contract and is of type Anchor Bent Type B Passing Bay. It includes, 1 StopAq band(s) to remove, 1 piles requiring scaffold, repair and wrapping of the top jacket only
202	Pilebent 202 is included in the contract and is of type Typical. It includes, 1 piles requiring scaffold, repair and wrapping.
203	Pilebent 203 is included in the contract and is of type Typical.
204	Pilebent 204 is included in the contract and is of type Typical. It includes, 1 piles requiring scaffold, repair and wrapping.
205	Pilebent 205 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint. It includes, 2 piles requiring scaffold, repair and wrapping of the top jacket only
206	Pilebent 206 is included in the contract and is of type Typical. It includes, 1 piles requiring scaffold, repair and wrapping.
207	Pilebent 207 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
208	Pilebent 208 is included in the contract and is of type Typical. It includes, 1 piles requiring scaffold, repair and wrapping.
209	Pilebent 209 is included in the contract and is of type Anchor Bent Type B. It includes, 1 piles requiring scaffold, repair and wrapping, 1 piles requiring scaffold, repair and wrapping of the top jacket only
210	Pilebent 210 is included in the contract and is of type Typical. It includes, 1 piles requiring scaffold, repair and wrapping.
211	Pilebent 211 is included in the contract and is of type Typical. It includes, 1 piles requiring scaffold, repair and wrapping.
212	Pilebent 212 is included in the contract and is of type Typical. It includes, 1 piles requiring scaffold, repair and wrapping. It includes a new style underpass remediation.
213	Pilebent 213 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint. It includes, 3 piles requiring scaffold, repair and wrapping.
214	Pilebent 214 is included in the contract and is of type Typical. It includes, 1 piles requiring scaffold, repair and wrapping, 1 piles requiring scaffold, repair and wrapping of the top jacket only
215	Pilebent 215 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove, 1 piles requiring scaffold, repair and wrapping of the top jacket only
216	Pilebent 216 is included in the contract and is of type Typical. It includes, 1 piles requiring scaffold, repair and wrapping, 1 piles requiring scaffold, repair and wrapping of the top jacket only
217	Pilebent 217 is included in the contract and is of type Anchor Bent Type B. It includes, 2 StopAq band(s) to remove, 1 piles requiring scaffold, repair and wrapping.
218	Pilebent 218 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove, 1 piles requiring scaffold, repair and wrapping.
219	Pilebent 219 is included in the contract and is of type Typical. It includes, 1 piles requiring scaffold, repair and wrapping, 1 piles requiring scaffold, repair and wrapping of the top jacket only
220	Pilebent 220 is included in the contract and is of type Typical.
221	Pilebent 221 is not included in the contract
222	Pilebent 222 is included in the contract and is of type Typical.
223	Pilebent 223 is included in the contract and is of type Typical.
224	Pilebent 224 is included in the contract and is of type Typical.
225	Pilebent 225 is included in the contract and is of type Typical.

Pilebent	Scope statement
226	Pilebent 226 is not included in the contract
227	Pilebent 227 is included in the contract and is of type Typical. It includes, 1 piles requiring scaffold, repair and wrapping. It includes a new style underpass remediation.
228	Pilebent 228 is included in the contract and is of type Typical. It includes, 1 piles requiring scaffold, repair and wrapping, 1 piles requiring scaffold, repair and wrapping of the top jacket only
229	Pilebent 229 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove, 1 piles requiring scaffold, repair and wrapping of the top jacket only
230	Pilebent 230 is included in the contract and is of type Typical.
231	Pilebent 231 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint. It includes, 2 piles requiring scaffold, repair and wrapping, 1 piles requiring scaffold, repair and wrapping of the top jacket only
232	Pilebent 232 is included in the contract and is of type Typical. It includes, 1 piles requiring scaffold, repair and wrapping.
233	Pilebent 233 is included in the contract and is of type Typical.
234	Pilebent 234 is included in the contract and is of type Typical. It includes, 1 piles requiring scaffold, repair and wrapping, 1 piles requiring scaffold, repair and wrapping of the top jacket only
235	Pilebent 235 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove, 1 piles requiring scaffold, repair and wrapping.
236	Pilebent 236 is included in the contract and is of type Typical. It includes, 2 piles requiring scaffold, repair and wrapping.
237	Pilebent 237 is not included in the contract
238	Pilebent 238 is included in the contract and is of type Typical.
239	Pilebent 239 is included in the contract and is of type Typical.
240	Pilebent 240 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove, 1 piles requiring scaffold, repair and wrapping.
241	Pilebent 241 is not included in the contract
242	Pilebent 242 is not included in the contract
243	Pilebent 243 is not included in the contract
244	Pilebent 244 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
245	Pilebent 245 is included in the contract and is of type Typical.
246	Pilebent 246 is included in the contract and is of type Typical. It includes, 2 StopAq band(s) to remove.
247	Pilebent 247 is not included in the contract
248	Pilebent 248 is included in the contract and is of type Passing Bay. It includes, 1 piles requiring scaffold, repair and wrapping.
249	Pilebent 249 is included in the contract and is of type Anchor Bent Type B Passing Bay.
250	Pilebent 250 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove.
251	Pilebent 251 is included in the contract and is of type Typical. It includes, 2 piles requiring scaffold, repair and wrapping.
252	Pilebent 252 is included in the contract and is of type Typical.
253	Pilebent 253 is included in the contract and is of type Typical. It includes, 1 piles requiring scaffold, repair and wrapping.
254	Pilebent 254 is included in the contract and is of type Typical. It includes, 2 piles requiring scaffold, repair and wrapping.
255	Pilebent 255 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint. It includes, 1 StopAq band(s) to remove, 2 piles requiring scaffold, repair and wrapping.
256	Pilebent 256 is included in the contract and is of type Typical. It includes, 2 piles requiring scaffold, repair and wrapping.
257	Pilebent 257 is included in the contract and is of type Typical. It includes, 2 piles requiring scaffold, repair and wrapping. It includes a new style underpass remediation.
258	Pilebent 258 is included in the contract and is of type Typical. It includes, 2 piles requiring scaffold, repair and wrapping.
259	Pilebent 259 is included in the contract and is of type Typical. It includes, 2 piles requiring scaffold, repair and wrapping.
260	Pilebent 260 is included in the contract and is of type Typical. It includes, 2 piles requiring scaffold, repair and wrapping.
261	Pilebent 261 is not included in the contract
262	Pilebent 262 is included in the contract and is of type Typical. It includes, 2 piles requiring scaffold, repair and wrapping.
263	Pilebent 263 is included in the contract and is of type Typical. It includes, 1 StopAq band(s) to remove, 1 piles requiring scaffold, repair and wrapping.
264	Pilebent 264 is included in the contract and is of type Typical. It includes, 2 piles requiring scaffold, repair and wrapping.
265	Pilebent 265 is included in the contract and is of type Typical. It includes, 2 piles requiring scaffold, repair and wrapping.
266	Pilebent 266 is included in the contract and is of type Typical. It includes, 2 piles requiring scaffold, repair and wrapping.
267	Pilebent 267 is included in the contract as 'Provisional, If Ordered' and is of type Expansion joint. It includes, 1 StopAq band(s) to remove.
268	Pilebent 268 is included in the contract and is of type Typical.
269	Pilebent 269 is included in the contract and is of type Typical.
270	Pilebent 270 is included in the contract and is of type Typical.

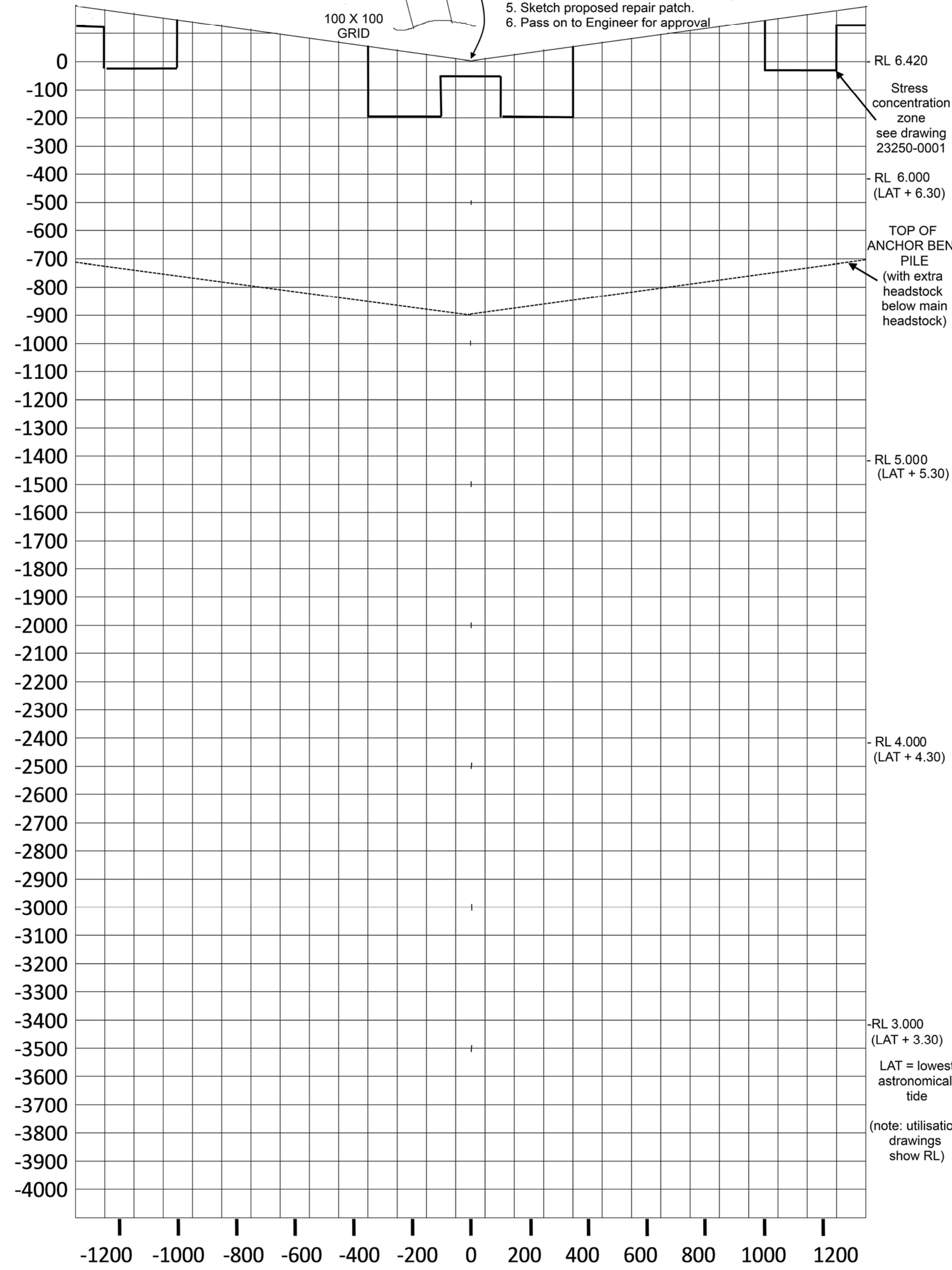
Pilebent	Scope statement
271	Pilebent 271 is included in the contract and is of type Typical.
272	Pilebent 272 is included in the contract and is of type Typical. It includes a new style underpass remediation.
273	Pilebent 273 is included in the contract and is of type Typical.
274	Pilebent 274 is included in the contract and is of type Anchor Bent Type B.
275	Pilebent 275 is included in the contract and is of type 275. It includes, 1 StopAq band(s) to remove, 2 piles requiring scaffold, repair and wrapping.
276	Pilebent 276 is not included in the contract
277	Pilebent 277 is not included in the contract
278	Pilebent 278 is not included in the contract
279	Pilebent 279 is not included in the contract
280	Pilebent 280 is not included in the contract
281	Pilebent 281 is included in the contract and is of type 281. It includes, 6 piles requiring scaffold, repair and wrapping.
282	Pilebent 282 is not included in the contract
283	Pilebent 283 is not included in the contract
284	Pilebent 284 is not included in the contract

APPENDIX I PILE INSPECTION RECORD SHEET



1. Refer to utilisation diagram for particular pile
2. Locate areas larger than 100mm across, where remaining thickness is less than allowable over the area (repair areas)
3. Sketch repair areas on grid below together with any measurements
4. For each repair area, mark thicknesses and uncorroded areas around circumference of pile, 500mm above and below the area. Uncorroded sections may be coloured green.
5. Sketch proposed repair patch.
6. Pass on to Engineer for approval

850 DIAMETER PILE (2670mm CIRC)
PILE REFERENCE: _____
DATE OF MEASUREMENT: _____
MEASURED BY: _____



APPENDIX J LUCINDA JETTY PILE UTILISATION AND REPAIR DIAGRAMS

Note these are likely to be modified during the contract or tender period

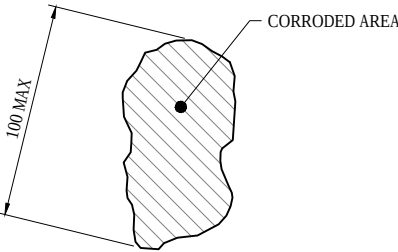
REPAIR PROCEDURES

READ UTILISATION AT OBSERVED DAMAGE LOCATION FOR BENT AND RL FROM DRAWINGS 232350-0002 AND 0003. USE 100% UTILISATION AND NO ALLOWANCE FOR CONCENTRATED CORROSION "Y" IF DAMAGE IS WITHIN STRESS CONCENTRATION ZONE.

1. IS CORRODED AREA HOLED?
YES: PLATE REPAIR IN ACCORDANCE WITH REPAIRS SHOWN ON THIS DRAWING. REFER TO THE ENGINEER FOR FURTHER INVESTIGATION
NO: GO TO STEP 2
2. IS THE OVERALL CORRODED AREA LARGER THAN "Y"?
NO: NO REPAIR REQUIRED
YES: GO TO STEP 3
3. IS THE MINIMUM RESIDUAL THICKNESS GREATER THAN "Z"?
YES: NO REPAIR REQUIRED
NO: IS THE SPECIFIC AREA UNDER THICKNESS LESS THAN "Y"?
A). NO REPAIR REQUIRED
OR
B). PAD WELD TO THICKNESS GREATER THAN "Z"

STEELWORK NOTES

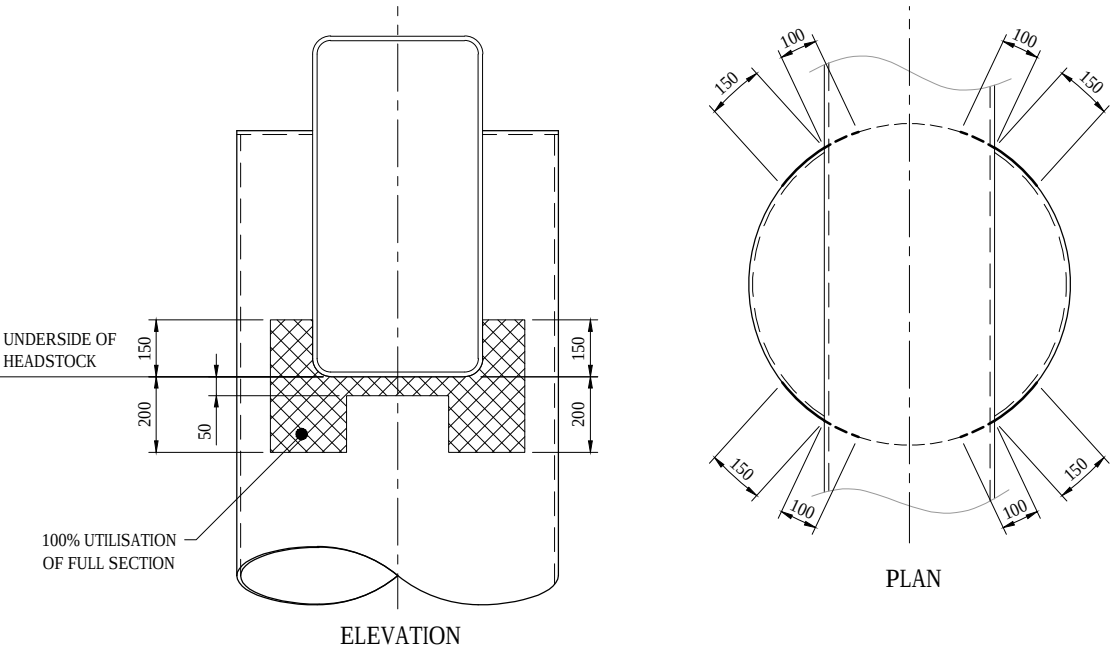
1. ALL STEEL PLATES TO AS3678 AND 3679.1 U.N.O.
2. ALL PLATES TO BE GR250 U.N.O.
3. FABRICATION TO BE IN ACCORDANCE WITH AS 4100
4. ALL WELDING IN ACCORDANCE WITH AS 1554.1 CATEGORY S.P.
5. ALL WELDS TO BE 8mm CONTINUOUS FILET WELD E41XX U.N.O.
6. REMOVE ALL SHARP EDGES, BURRS, CORNERS, WELD SLAG ETC
7. GRIND ALL CORNERS AND EDGES OF REPAIR PLATE TO MINIMUM 2mm RADIUS
8. NO PIN HOLES PERMITTED IN WELD



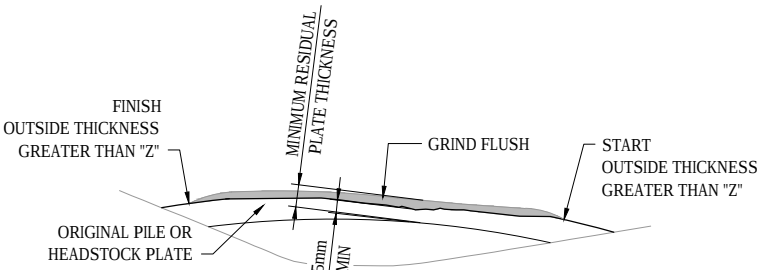
MAXIMUM ALLOWABLE AREA OF CONCENTRATED CORROSION (N.T.S.)

PERCENTAGE OF FULL SECTION UTILISATION	DIMENSION "Z"	
	RESIDUAL PLATE THICKNESS IN NOMINAL 10mm PLATE	RESIDUAL HEADSTOCK PLATE THICKNESS IN NOMINAL 12mm PLATE
90 - 100%	9.5mm*	11.4mm*
80 - 90%	9.3mm	11.3mm
70 - 80%	8.6mm	10.6mm
60 - 70%	7.9mm	10.0mm
50 - 60%	7.1mm	9.2mm
40 - 50%	6.3mm	8.4mm
0 - 40%	6.3mm	7.6mm

UTILISATION RESIDUAL PLATE THICKNESS TABLE FOR PILE AND HEADSTOCK MEMBERS

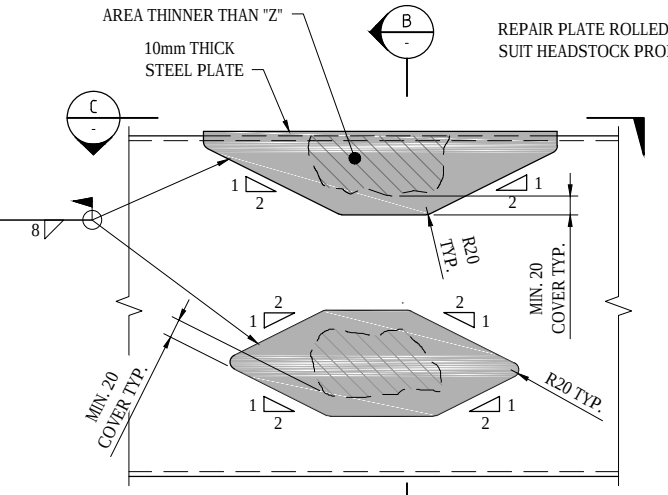


DEFINITION OF STRESS CONCENTRATION ZONE

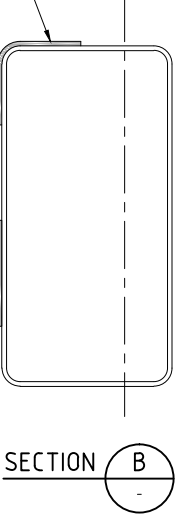


PAD WELD WITH ONE RUN MAX 3mm THICK. GRIND FLUSH.

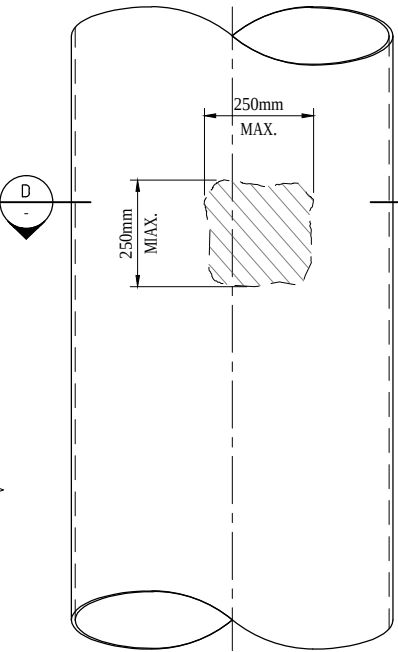
TYPICAL PAD WELD REPAIR



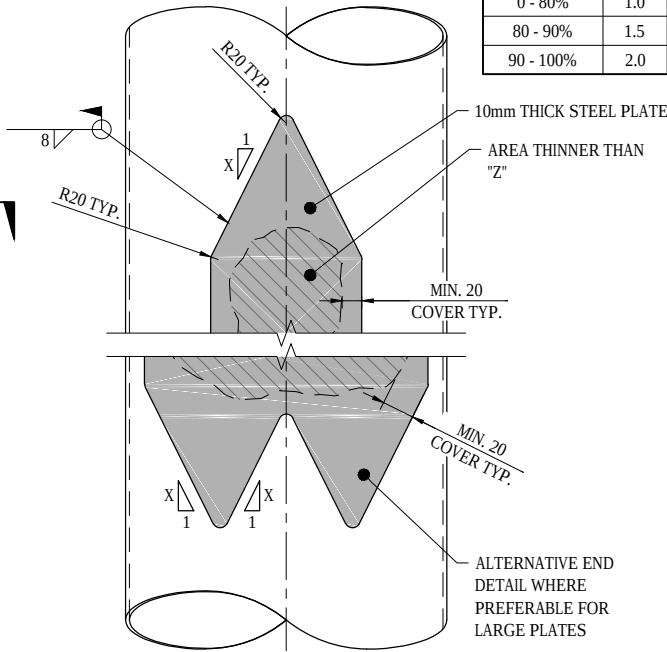
STRUCTURAL REPAIR FOR CORROSION TO HEADSTOCK



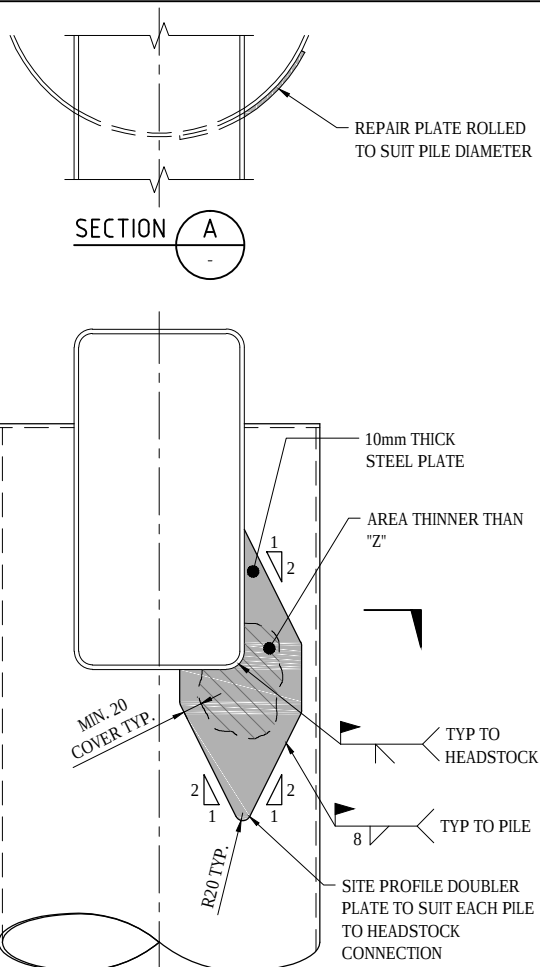
PAD WELD REPAIR TO HEADSTOCK



TYPICAL PAD WELD REPAIR



STRUCTURAL REPAIR FOR CORROSION TO PILE



STRUCTURAL REPAIR FOR CORROSION IN VICINITY OF PILE / HEADSTOCK CONNECTION

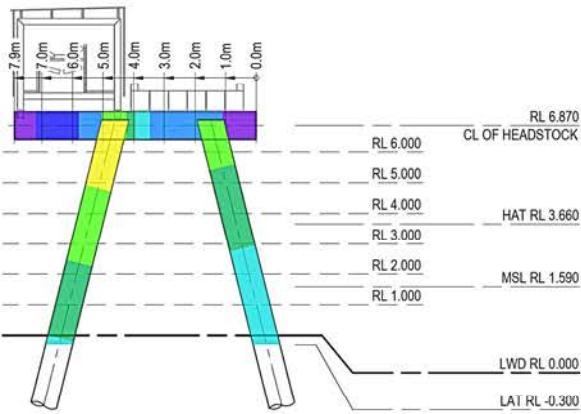
NOTE: GEOMETRY DETAILED IS FOR DEVELOPED PLATE

UTILISATION	X
0 - 80%	1.0
80 - 90%	1.5
90 - 100%	2.0

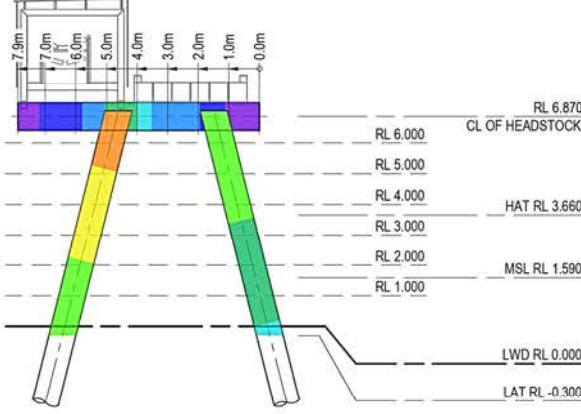
REV	DATE	REVISION DETAILS	APP
A	6.02.13	ISSUED FOR CONSTRUCTION	AON

DRAWN R. HOLMAN	DESIGNED A. FLEER
CHECKED A. NILSEN	
APPROVED	DATE 6/02/13
A. NILSEN	

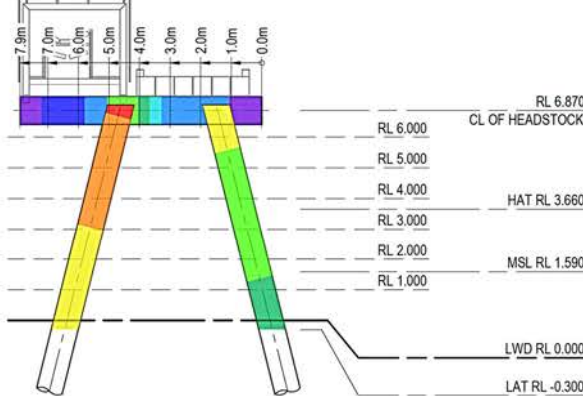
PROJECT No. 232350	SIZE A1
SCALE 1:10	REV A
DRAWING No. 0001	



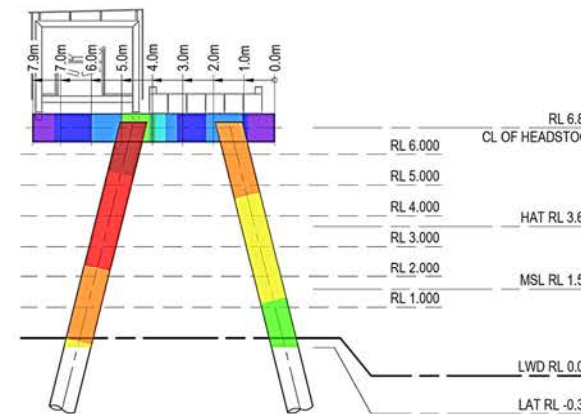
TYPICAL JETTY BENT 8500 PILES
REPRESENTS BENT 3 UP TO 46°



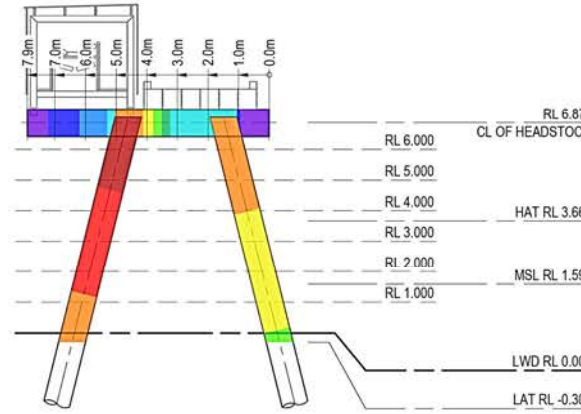
TYPICAL JETTY BENT 8500 PILES
REPRESENTS BENT 47 UP TO 187°



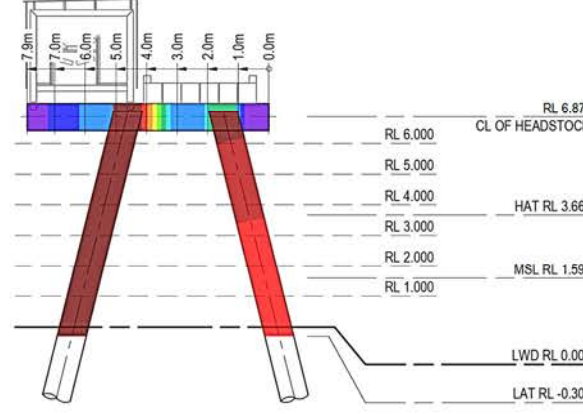
TYPICAL JETTY BENT 8500 PILES
REPRESENTS BENT 188 UP TO 212°



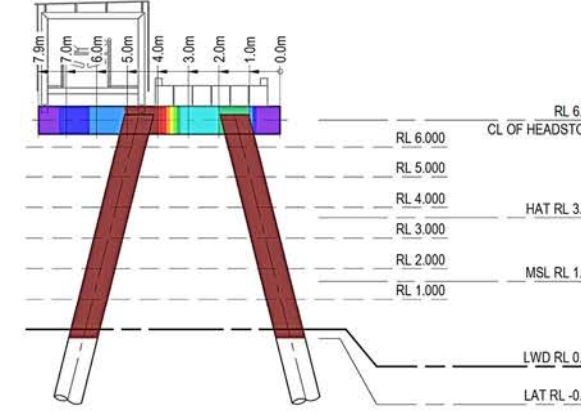
TYPICAL JETTY BENT 8500 PILES
REPRESENTS BENT 213 UP TO 236°



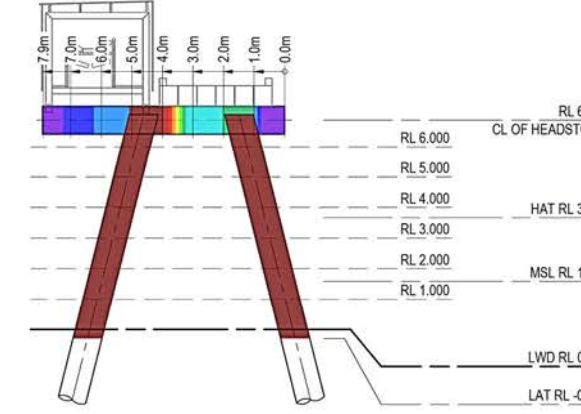
TYPICAL JETTY BENT 8500 PILES
REPRESENTS BENT 237 UP TO 257°



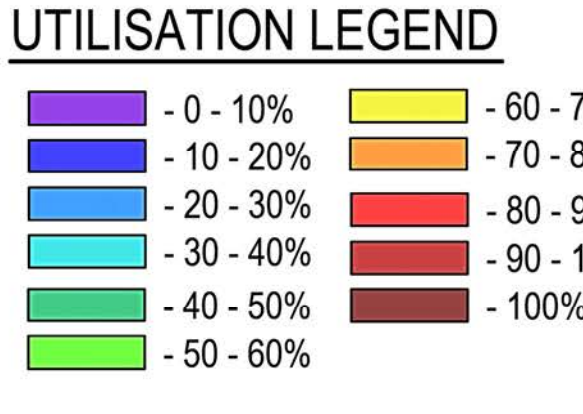
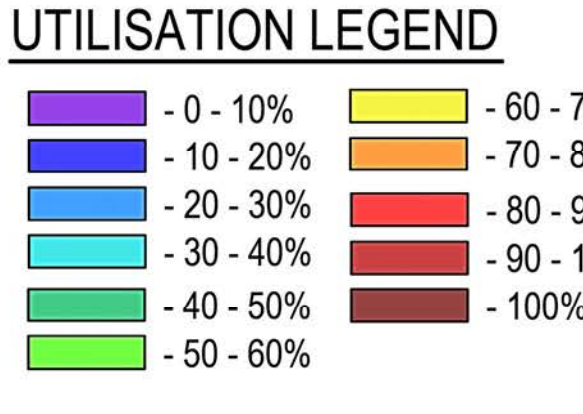
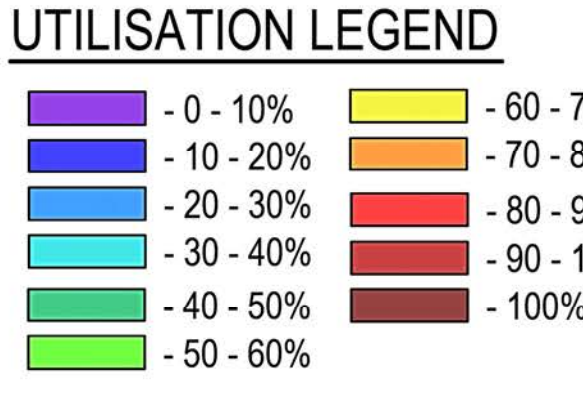
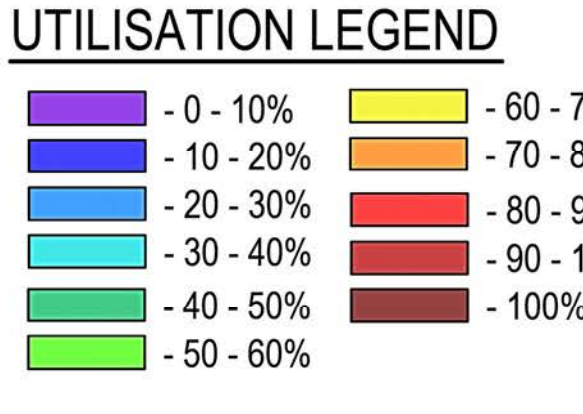
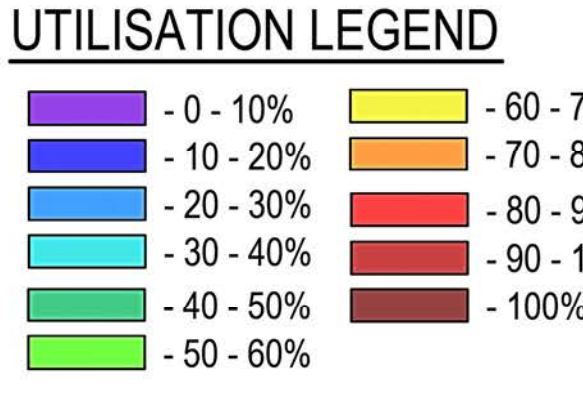
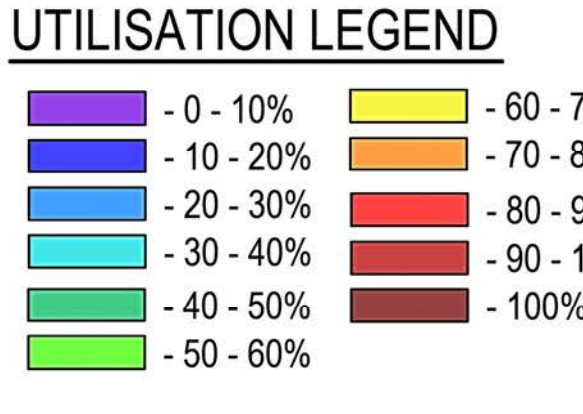
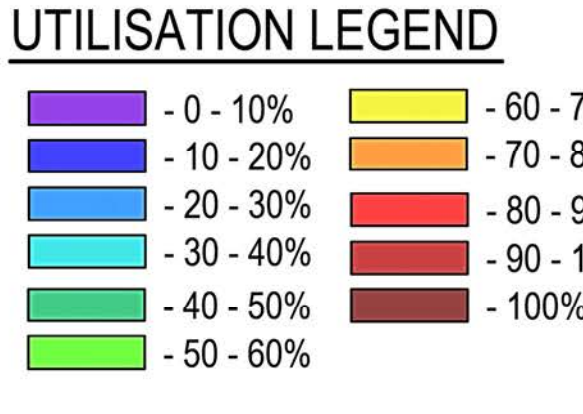
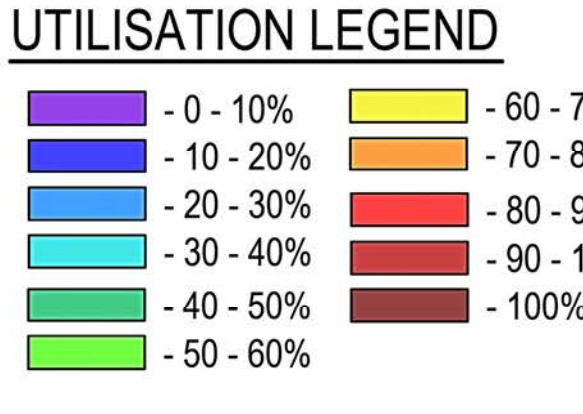
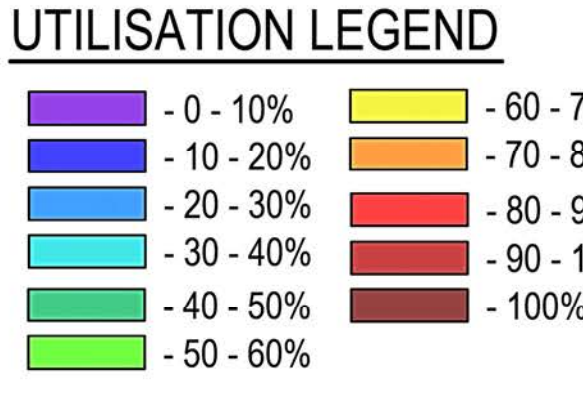
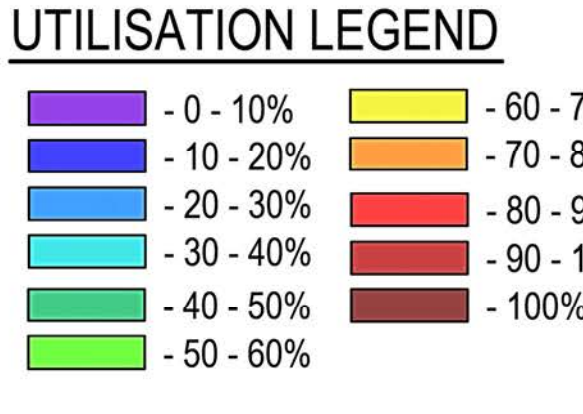
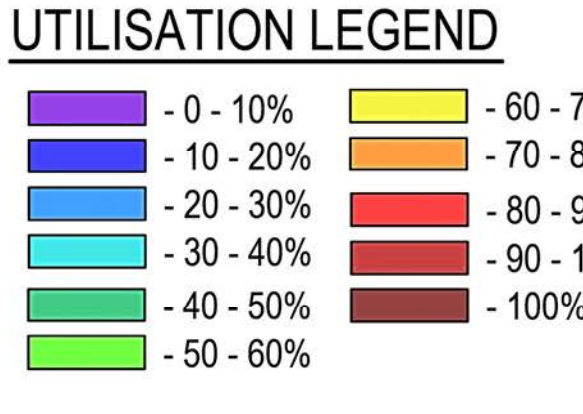
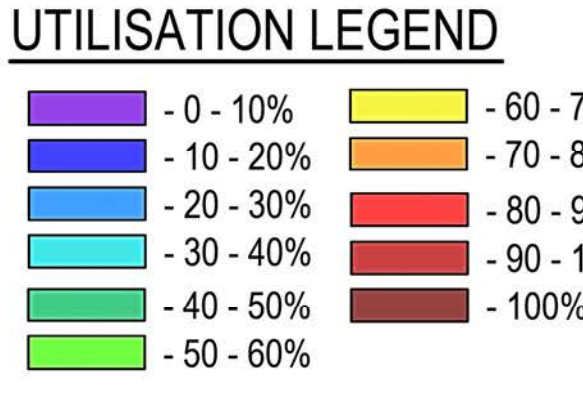
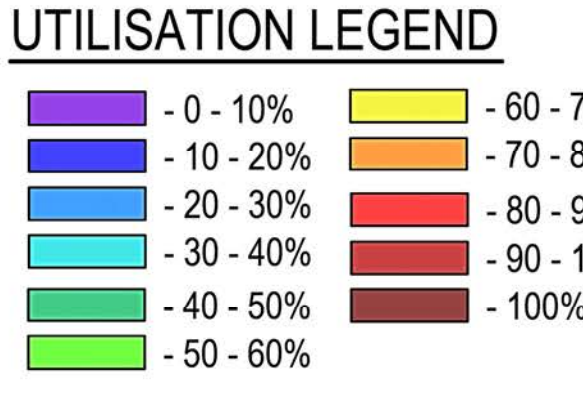
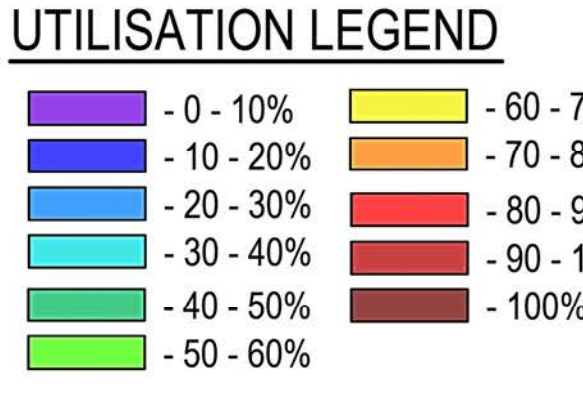
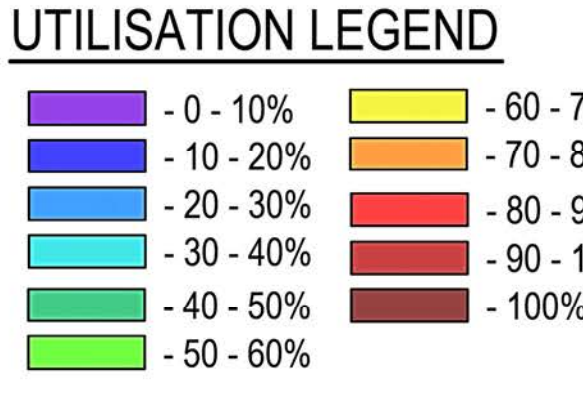
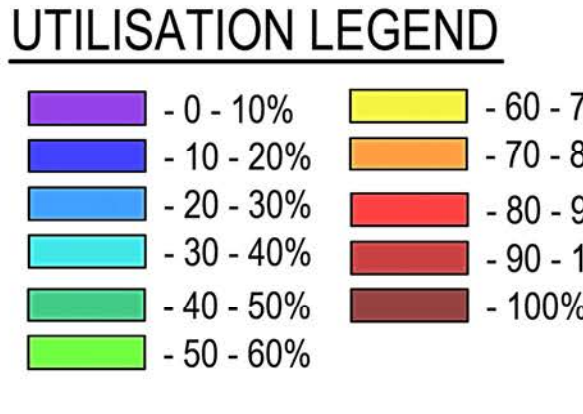
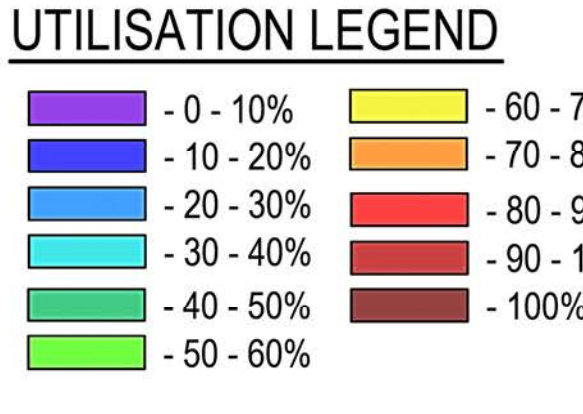
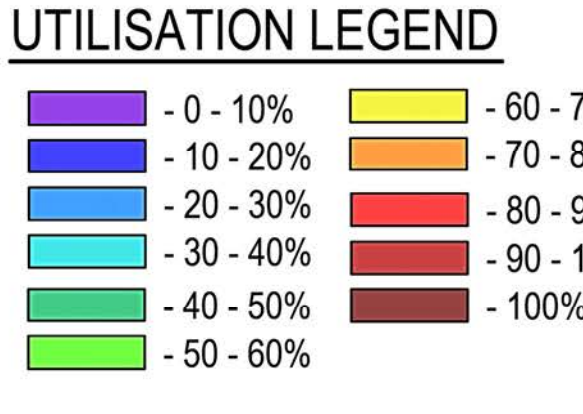
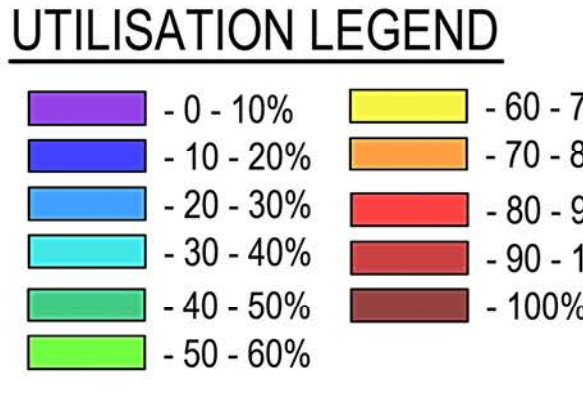
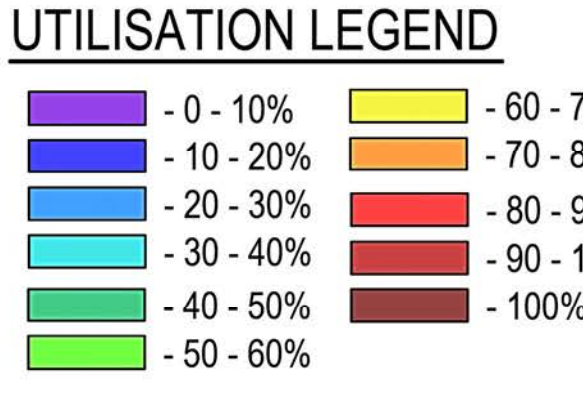
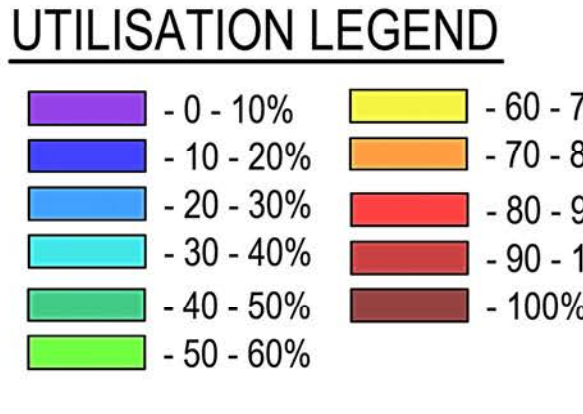
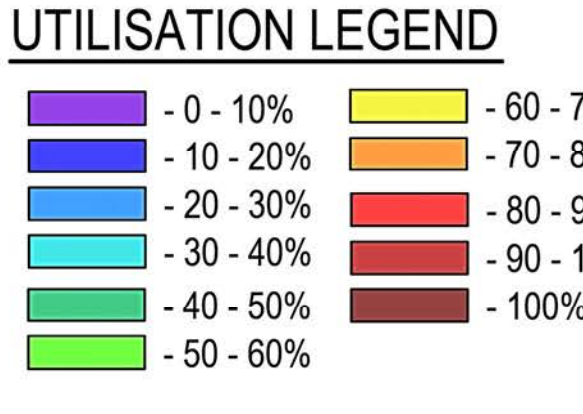
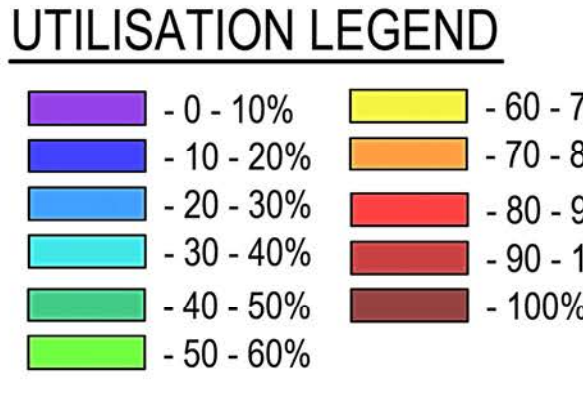
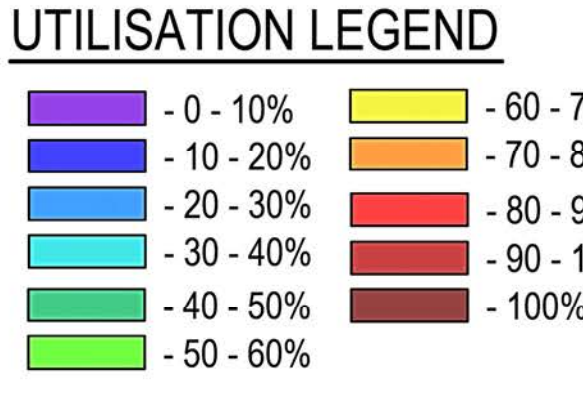
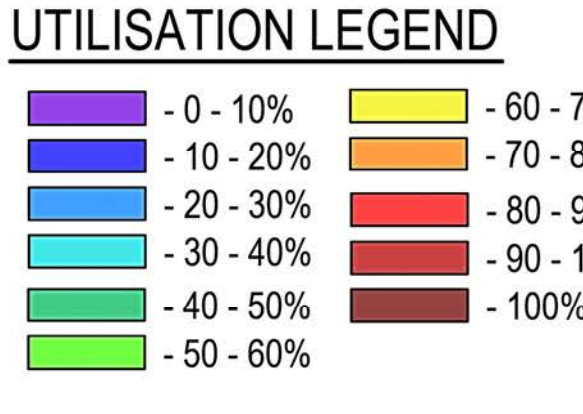
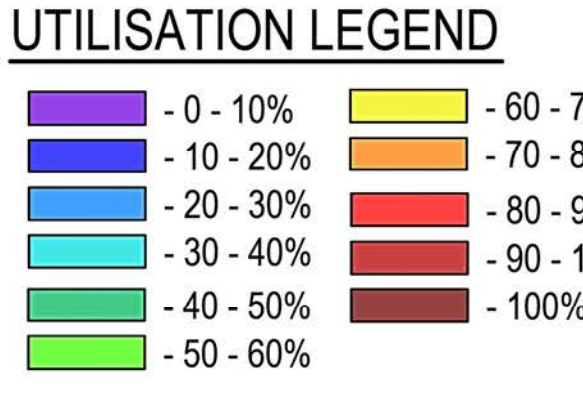
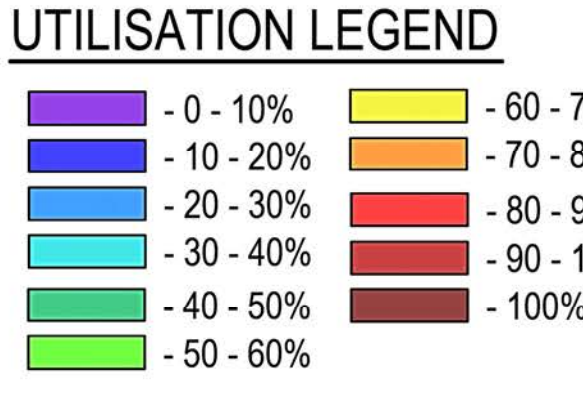
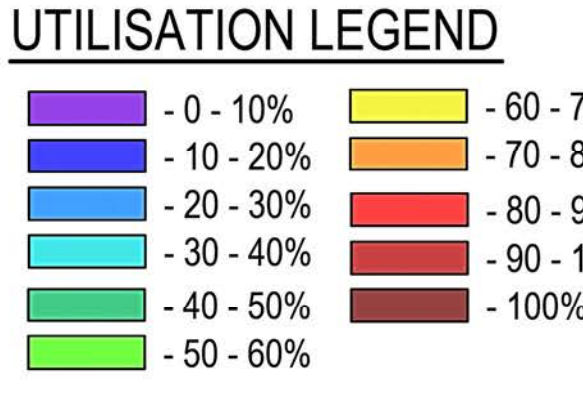
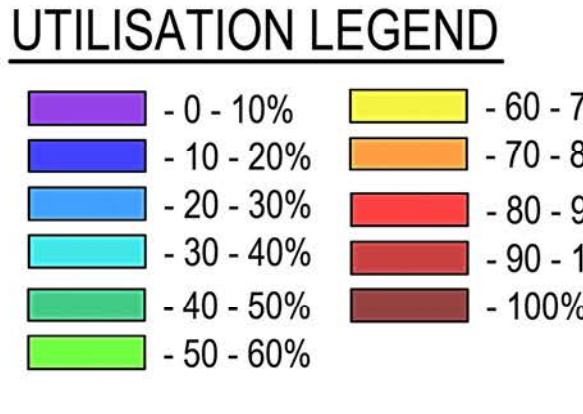
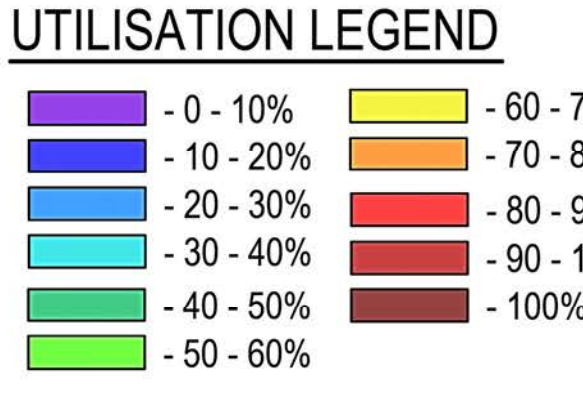
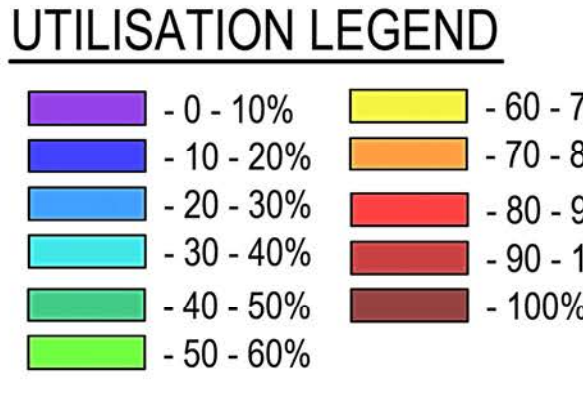
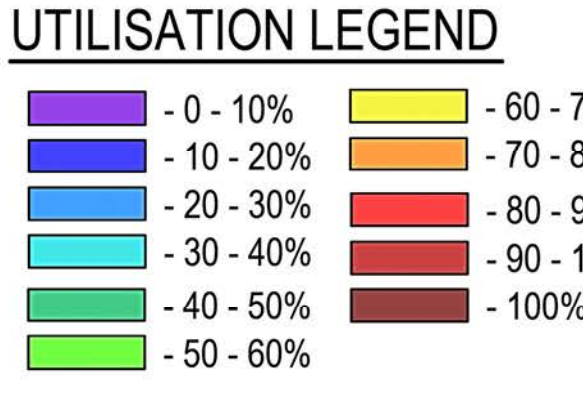
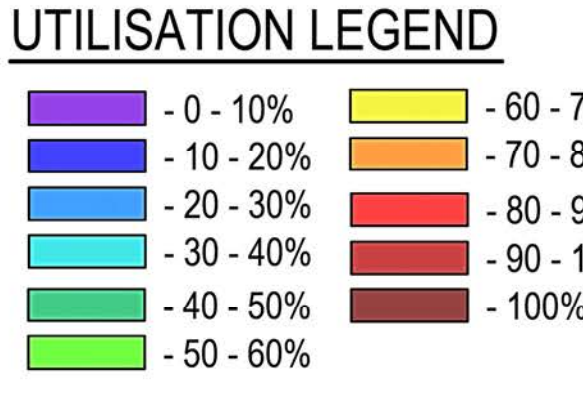
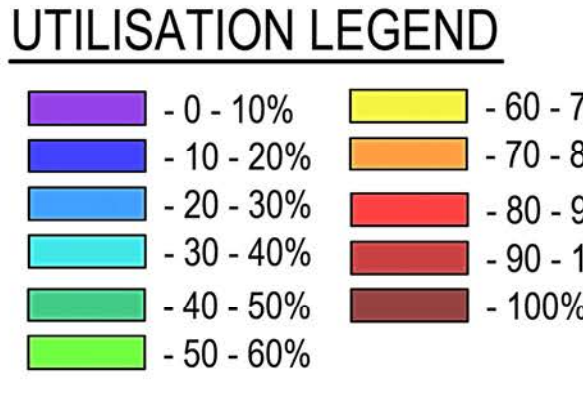
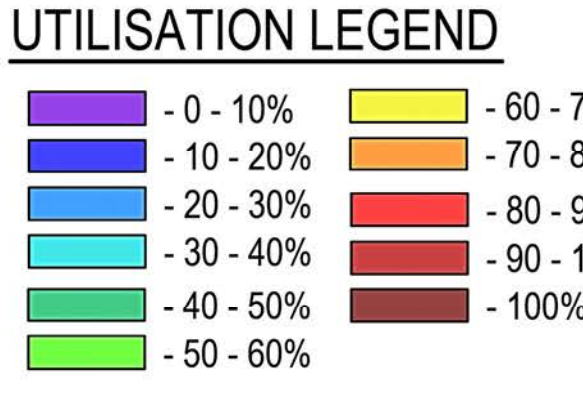
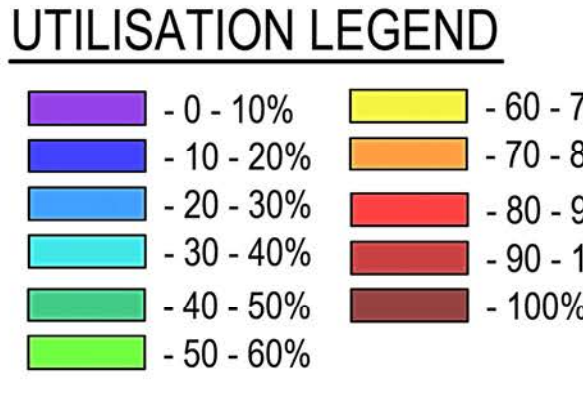
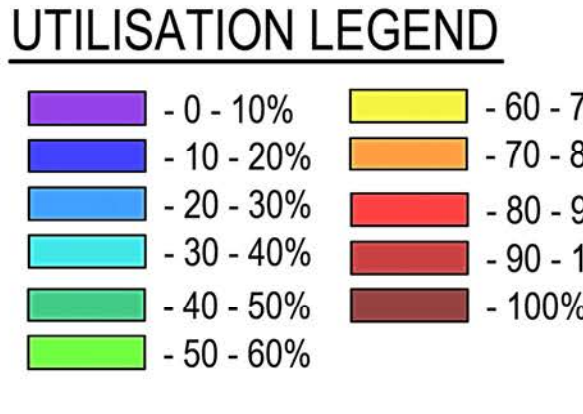
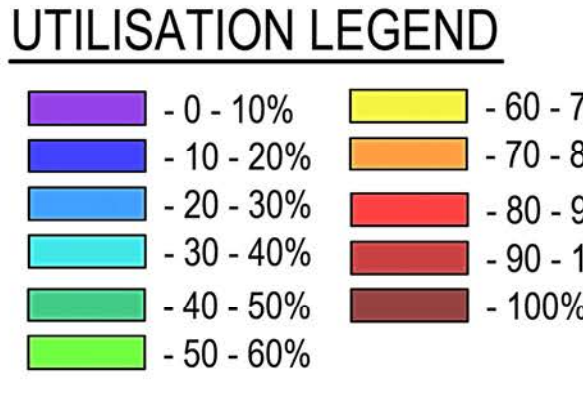
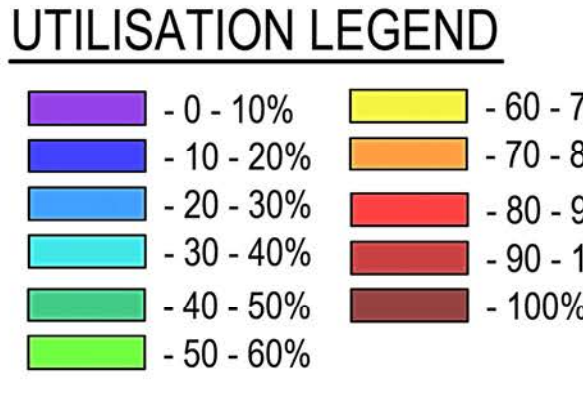
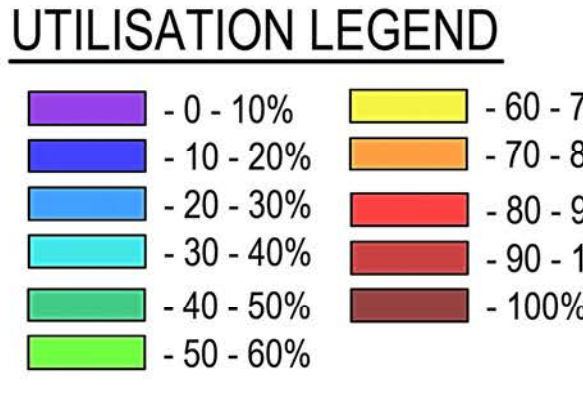
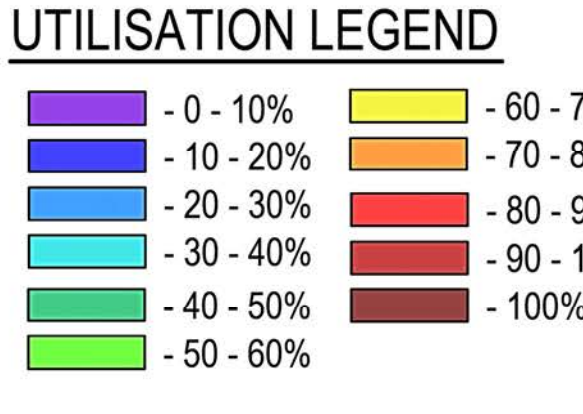
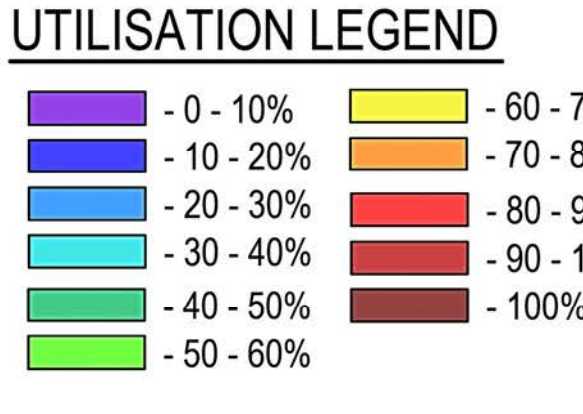
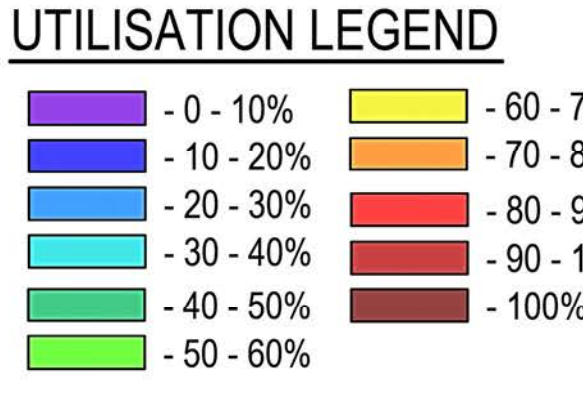
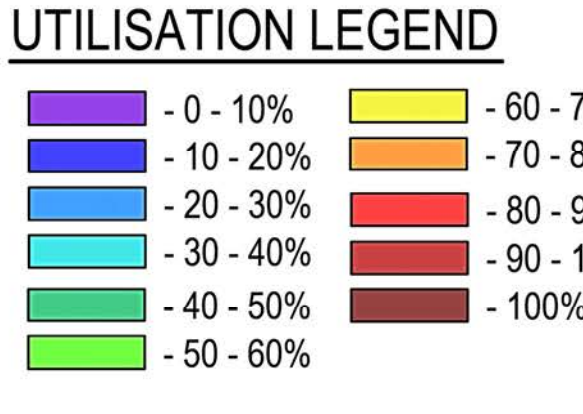
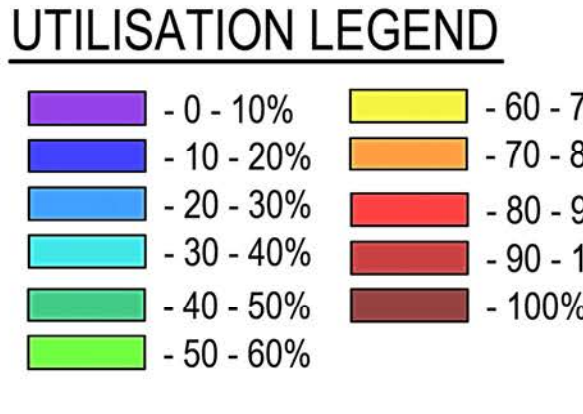
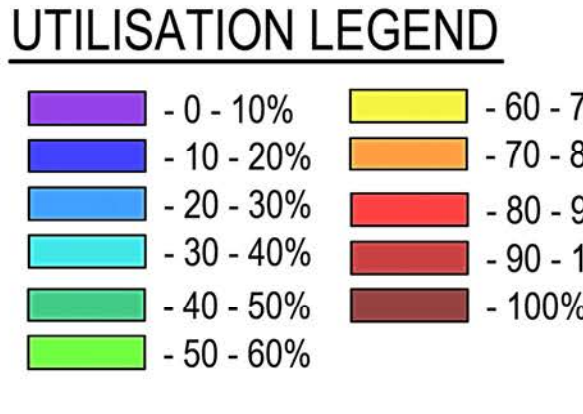
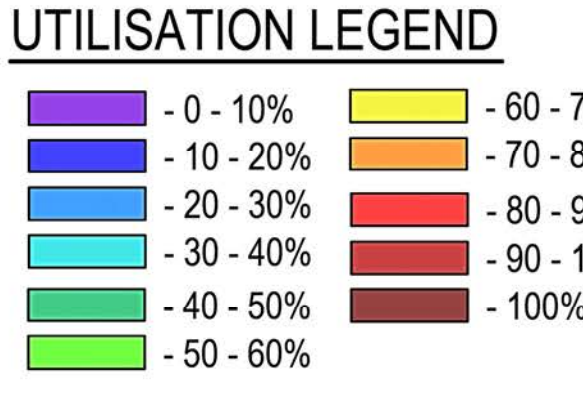
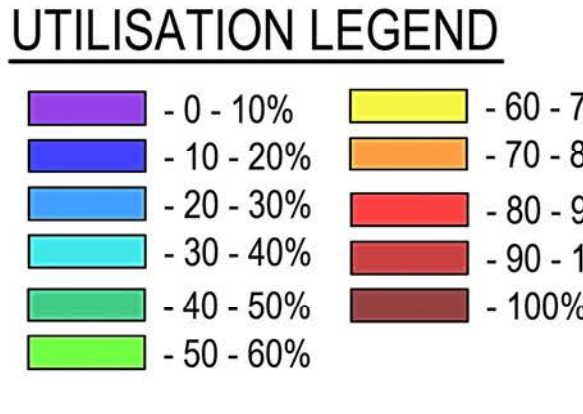
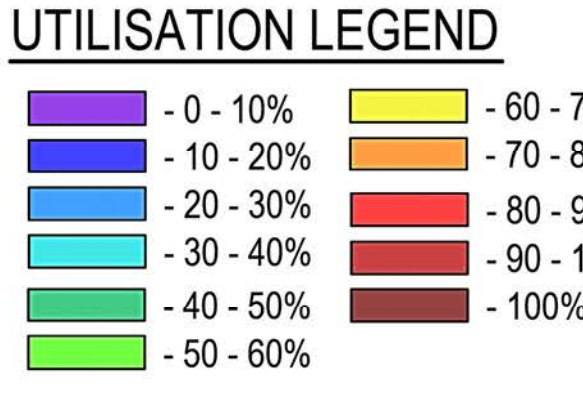
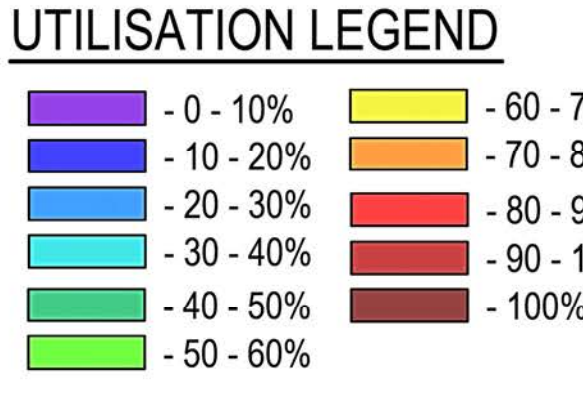
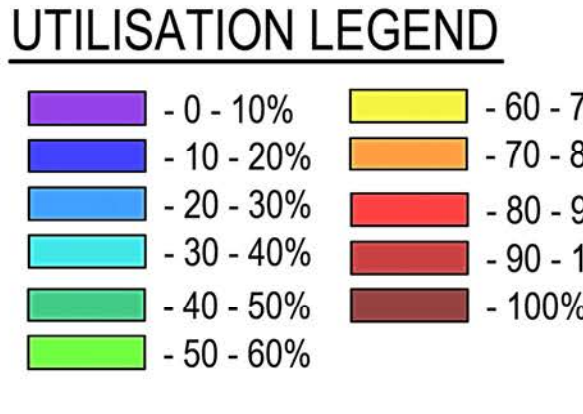
TYPICAL JETTY BENT 9200 PILES
REPRESENTS BENT 253 UP TO 261°



TYPICAL JETTY BENT 9200 PILES
REPRESENTS BENT 262 UP TO 268°

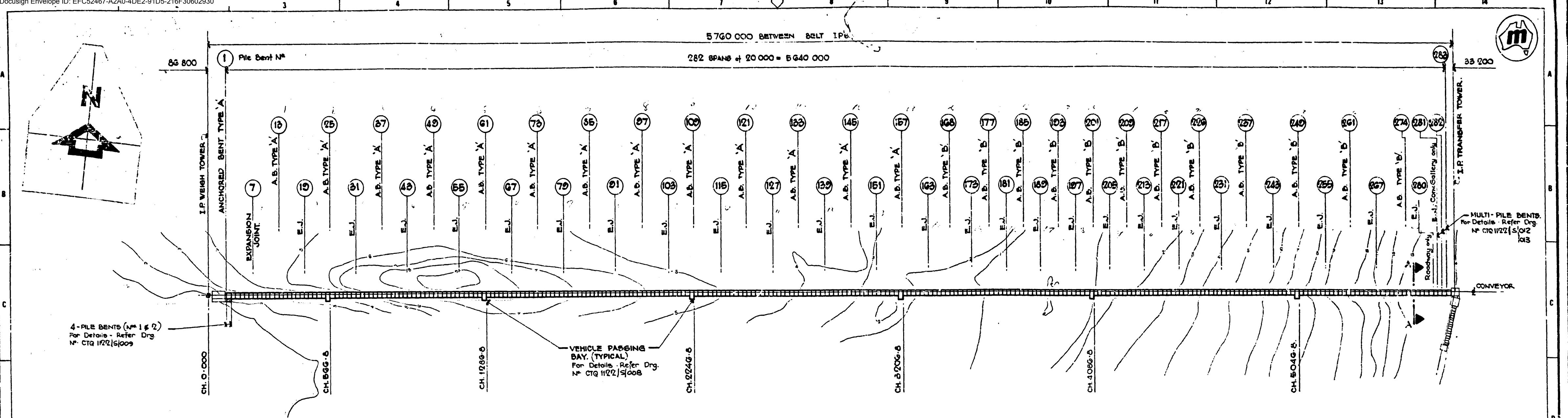


TYPICAL JETTY BENT 9200 PILES
REPRESENTS BENT 269 UP TO 273°



APPENDIX K JETTY DRAWINGS

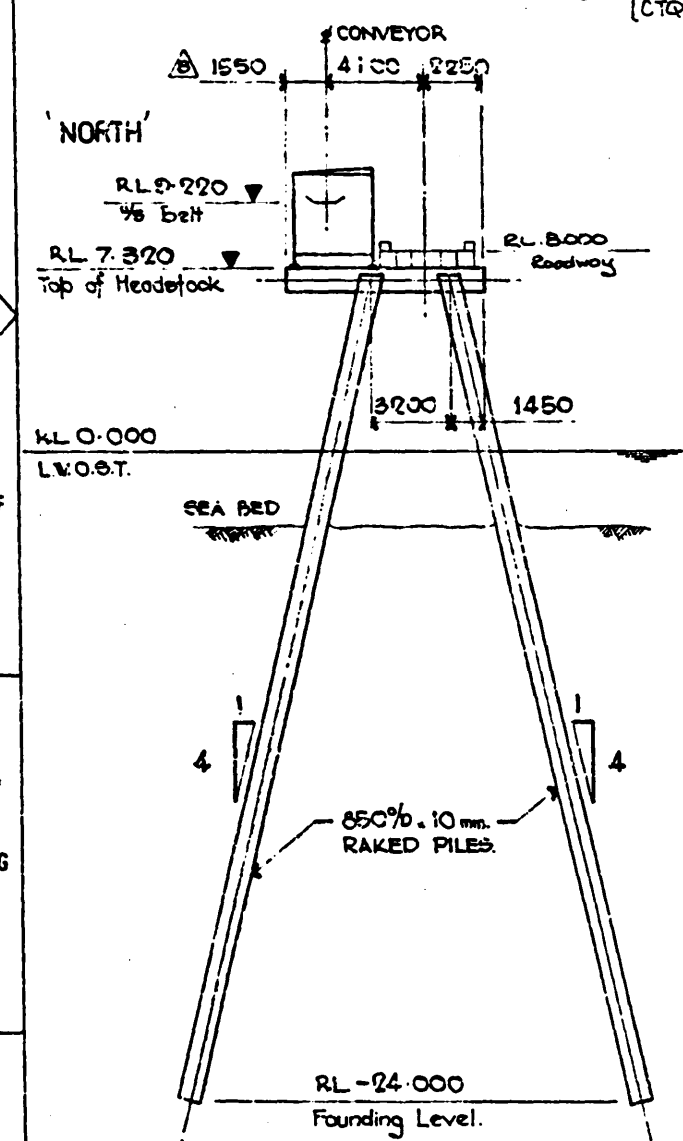
Drawing number	Description
CTQ-1122-S-001	Approach jetty - General plan and sections
CTQ-1122-S-002	Approach jetty - Pile schedule
CTQ-1122-S-003	Approach jetty - Sheet 1 superstructure layout
CTQ-1122-S-004	Approach jetty - Sheet 2 superstructure layout
CTQ-1122-S-005	Approach jetty - Typical cross section
CTQ-1122-S-006	Approach jetty - Type 'A' anchor bent
CTQ-1122-S-007	Approach jetty - Type 'B' anchor bent
CTQ-1122-S-008	Approach jetty - Passing bay details
CTQ-1122-S-010	Approach jetty - Pilebents no.s 275-279
CTQ-1122-S-012	Approach jetty - Pilebent no. 281
CTQ-1122-S-015	Approach jetty - Miscellaneous details
CTQ-1122-S-018	Approach jetty - Standard conveyor gallery
CTQ-1122-S-020	Approach jetty - Standard span cladding
CTQ-1122-S-021	Approach jetty - Gallery flashing details
CTQ-1122-S-022	Approach jetty - Conveyor stringer details
CTQ-1122-S-023	Approach jetty - Gallery tolerances
CTQ-1122-S-024	Approach jetty - Variations to standard gallery
CTQ-1122-S-025	Approach jetty - Variations to cladding
CTQ-1122-S-026	Approach jetty - Cross-under details
CTQ-1122-S-030	Approach jetty - Pre-cast roadway GA
CTQ-1122-S-031	Approach jetty - Standard roadway girder details
CTQ-1122-S-032	Approach jetty - Expansion joint details
CTQ-1122-S-033	Approach jetty - Pre-cast roadway girder arrangement and details
CTQ-1122-S-034	Approach jetty - Pre-cast roadway girder details sht 2
CTQ-1122-S-038	Approach jetty - As-constructed piles
CTQ-1122-S-039	Approach jetty - As-constructed piles



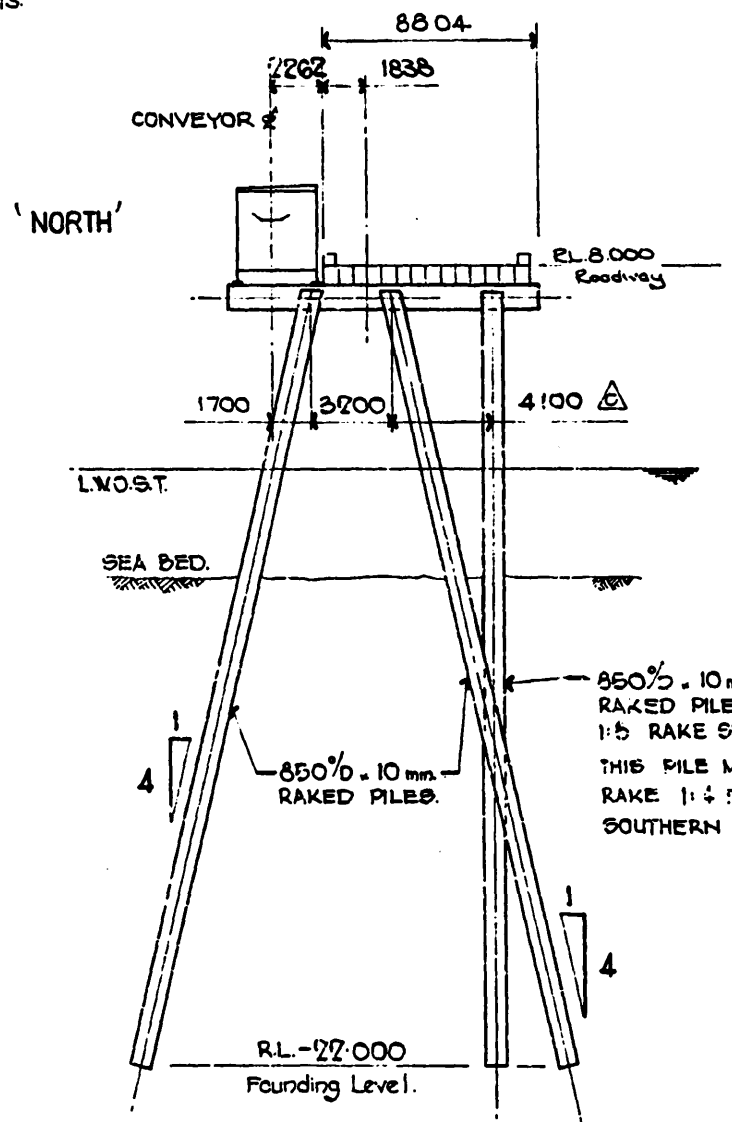
NOTE:
 A.B. = ANCHORED BENT.
 ANCHORED BENTS TYPE 'A' = 1 ANCHOR PILE - Refer Drg. N° CTQ 1122/S/006 for Details.
 ANCHORED BENTS TYPE 'B' = 2 ANCHOR PILES - Refer Drg. N° CTQ 1122/S/007 for Details.
 E.J. = EXPANSION JOINT. - Refer Drgs N° CTQ 1122/S/003 for Details.
 [CTQ 1122/S/032]

PLAN - APPROACH JETTY.

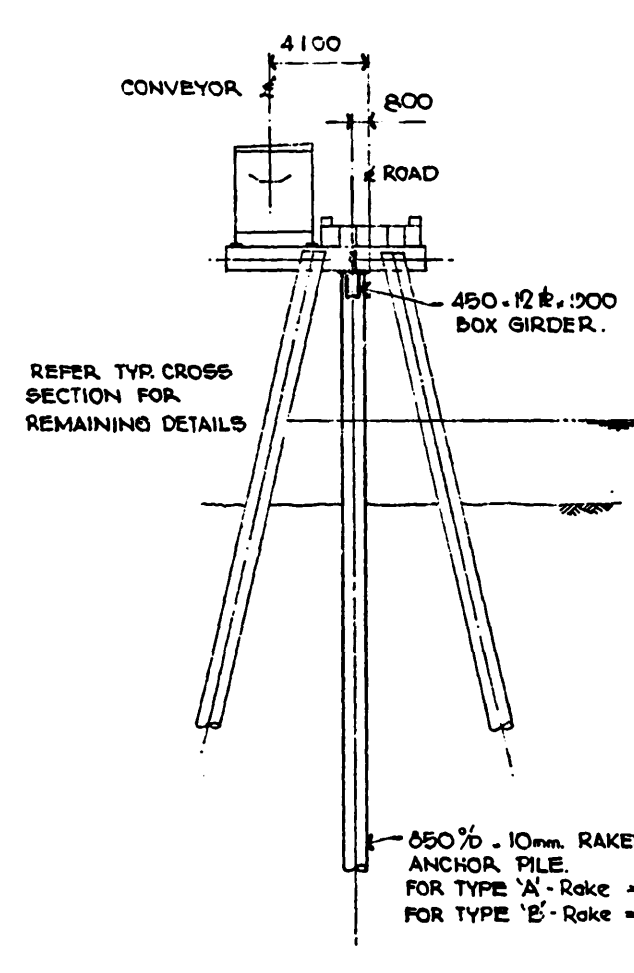
SCALE: 1:7,500.



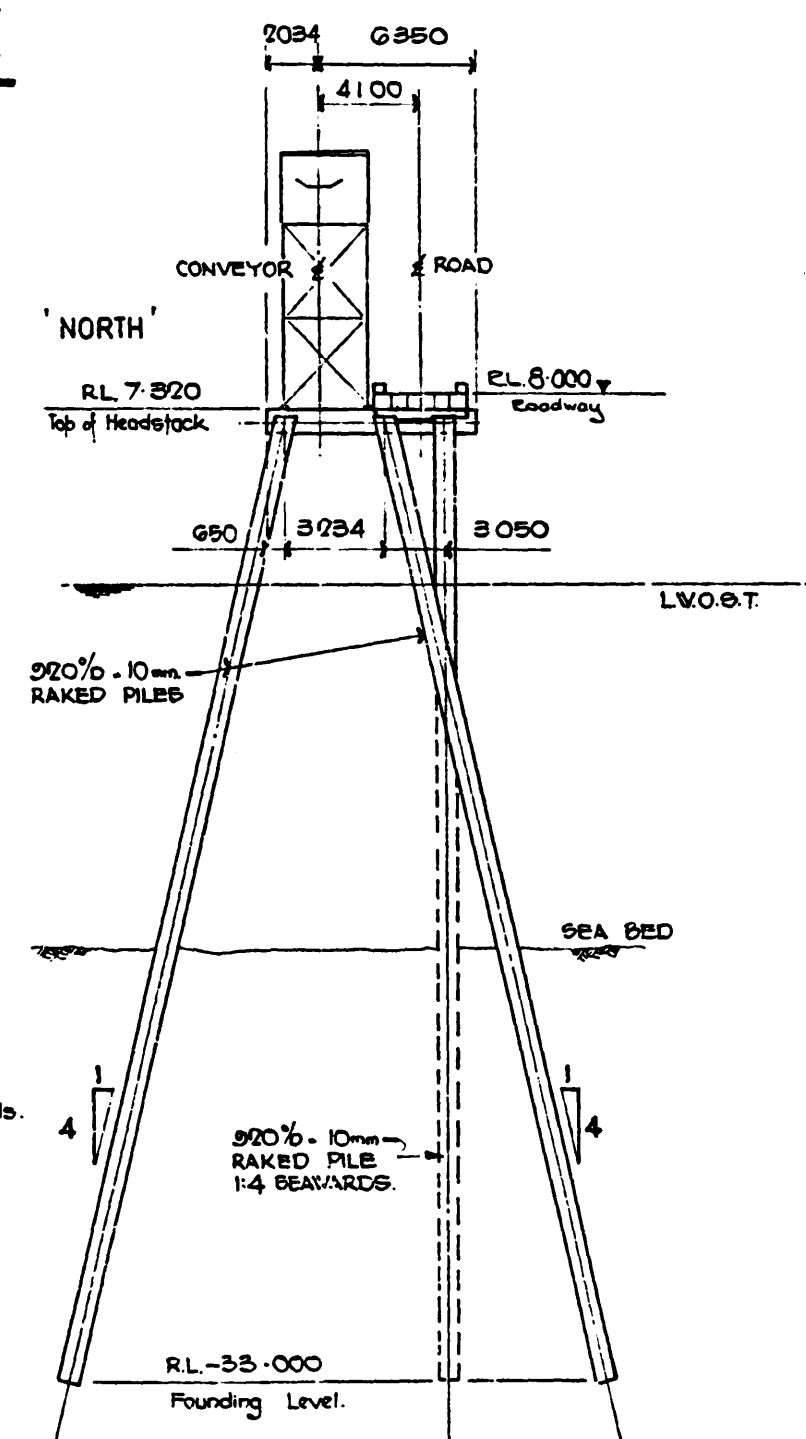
TYP CROSS-SECTION.
 PILE BENT N° 10 SHOWN.



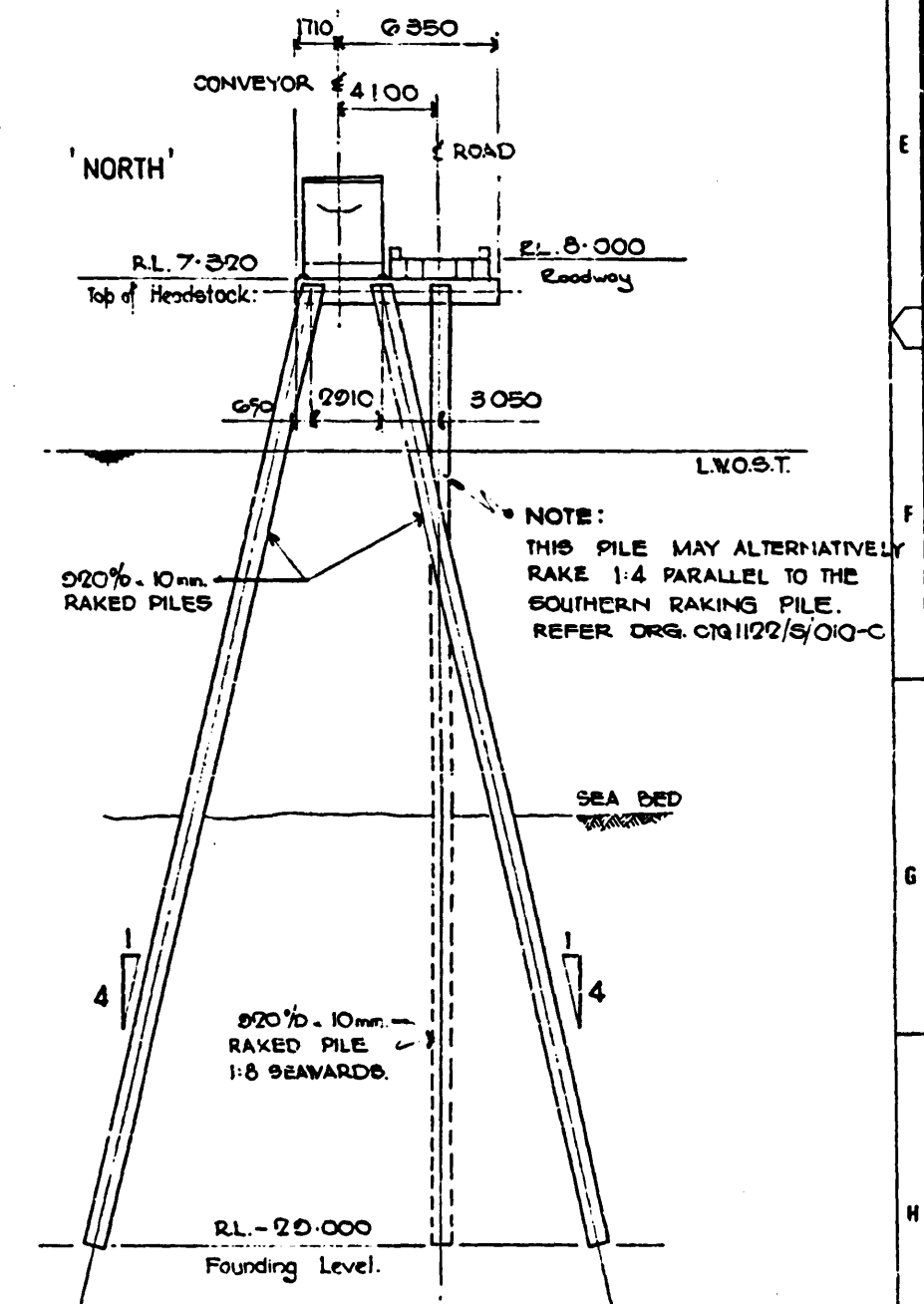
TYP PASSING-BAY SECTION.
 BAY AT CHS. 500-5 SHOWN.



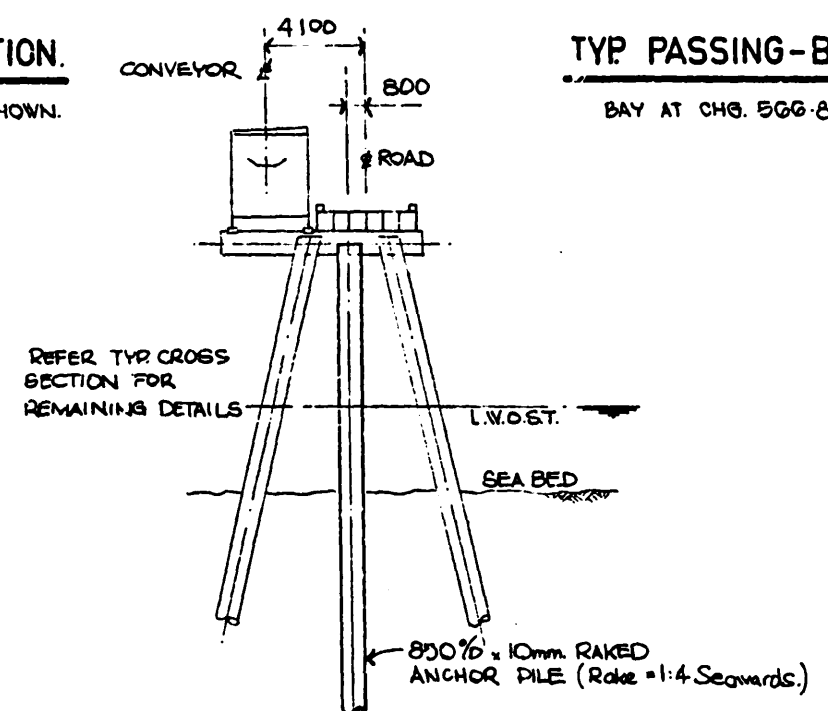
TYP ANCHORED BENT SECTION.



PILE BENT N° 280.



SECTION A-A
 TYPICAL FOR PILE BENT N°s
 275, 276, 277, 278, 279.



TYP EXPANSION JOINT BENT SECTION

NOTES:

- FOR PILE SIZES, FOUNDING LEVELS & SEA BED LEVELS - REFER DRS. N° CTQ 1122/S/002.
- ROADWAY LEVEL FROM PILE BENT N° 1 TO PILE BENT N° 281 TO BE R.L. 8.000. FROM PILE BENT N° 282 SEAVARDS, ROADWAY LEVEL TO BE R.L. 7.700.
- PILE BENT N° 284 (Not drawn for clarity) IS LOCATED 18 METRES SEAVARD (EAST) OF PILE BENT N° 283.

THE CORPORATION OF THE TREASURER OF QUEENSLAND			
THE SUGAR BOARD BULK SUGAR TERMINALS			
B.S.T. LUCINDA - STAGE III APPROACH JETTY.			
GENERAL PLAN & SECTIONS.			
MACDONALD WAGNER & FRIDDLE PTY. LTD.			
CSR LIMITED			
DRAWING NO. CTQ 1122/S/001			
DATE 2-7-77			
DRAWN BY			
CHECKED BY			
APPROVED BY			
DATE 2-7-77			
FOR PILE BENT N° 280			
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FOR PILE BENT N° 456			

PILE BENT NO.	1 & 2	1 & 2	3, 4, 5, 6 & 7	8-10	20-31	32-43	44-55	56-67	68-70	80-91	92-103	104-115	116-127
PILE NO.	1N, 2N, 1S, 2S.	1S1, 1S2, 2S1, 2S2.	3N, 4N, 5N, 6N, 7N.	8N, 9N, 10N, 11N, 12N, 10N, 13N, 14N, 15N, 16N, 17N, 18N.	20N, 21N, 22N, 23N, 24N, 31N, 25N, 26N, 27N, 28N, 29N, 30N.	32N, 33N, 34N, 35N, 36N, 37N, 38N, 39N, 40N, 41N, 42N, 43N.	44N, 45N, 46N, 47N, 48N, 49N, 50N, 51N, 52N, 53N, 54N, 55N.	56N, 57N, 58N, 59N, 60N, 61N, 62N, 63N, 64N, 65N, 66N, 67N.	68N, 69N, 70N, 71N, 72N, 73N, 74N, 75N, 76N, 77N, 78N, 79N.	80N, 81N, 82N, 83N, 84N, 85N, 86N, 87N, 88N, 89N, 90N, 91N.	92N, 93N, 94N, 95N, 96N, 97N, 98N, 99N, 100N, 101N, 102N, 103N, 104N, 105N, 106N, 107N, 108N, 109N, 110N, 111N, 112N, 113N, 114N, 115N.	104N, 105N, 106N, 107N, 108N, 109N, 110N, 111N, 112N, 113N, 114N, 115N.	116N, 117N, 118N, 119N, 120N, 121N, 122N, 123N, 124N, 125N, 126N, 127N.
CHAINAGE (Metres)	868 to 106.8	868 to 106.8	126.8 to 206.8.	226.8 to 446.6	466.8 to 686.8	706.8 to 926.8	946.8 to 1166.8	1186.8 to 1406.8	1426.8 to 1646.8	1666.8 to 1886.8	1906.8 to 2126.8	2146.8 to 2366.8	2386.8 to 2606.8
PROVISIONAL TOE R.L.	-28.000	-28.000	-24.000	-24.500	-22.000	-24.500	-22.000	-22.000	-22.500	-24.000	-27.000	-24.500	-22.000
PROVISIONAL LENGTH.	36.140	35.240 35.060	32.020	22.530	29.060	32.530	29.060	29.060	30.470	32.020	35.110	32.530	29.060
BED LEVEL. (APPROX)	-2.000	-2.000	-2.750	-3.500	-5.500	-7.500	-8.500	-8.000	-7.750	-7.000	-6.000	-5.000	-4.000

PILE BENT NO	128 - 132	140 - 151	162 - 163	164 - 173	174 - 181	182 - 189	190 - 197	198 - 205	206 - 213	214 - 221	222 - 231	232 - 243	244 - 252
PILE NO.	128N, 129N, 130N, 131N, 132N, 133N, 134N, 135N, 136N, 137N, 138N, 139N, 140N, 141N, 142N, 143N, 144N, 145N, 146N, 147N, 148N, 149N, 150N, 151N, 152N, 153N, 154N, 155N, 156N, 157N, 158N, 159N, 160N, 161N, 162N, 163N, 164N, 165N, 166N, 167N, 168N, 169N, 170N, 171N, 172N, 173N, 174N, 175N, 176N, 177N, 178N, 179N, 180N, 181N, 182N, 183N, 184N, 185N, 186N, 187N, 188N, 189N, 190N, 191N, 192N, 193N, 194N, 195N, 196N, 197N, 198N, 199N, 200N, 201N, 202N, 203N, 204N, 205N, 206N, 207N, 208N, 209N, 210N, 211N, 212N, 213N, 214N, 215N, 216N, 217N, 218N, 219N, 220N, 221N, 222N, 223N, 224N, 225N, 226N, 227N, 228N, 229N, 230N, 231N, 232N, 233N, 234N, 235N, 236N, 237N, 238N, 239N, 240N, 241N, 242N, 243N, 244N, 245N, 246N, 247N, 248N, 249N, 250N, 251N, 252N, 253N, 254N, 255N, 256N, 257N, 258N, 259N, 260N, 261N, 262N, 263N, 264N, 265N, 266N, 267N, 268N, 269N, 270N, 271N, 272N, 273N, 274N, 275N, 276N, 277N, 278N, 279N, 280N, 281N, 282N, 283N, 284N, 285N, 286N, 287N, 288N, 289N, 290N, 291N, 292N, 293N, 294N, 295N, 296N, 297N, 298N, 299N, 300N, 301N, 302N, 303N, 304N, 305N, 306N, 307N, 308N, 309N, 310N, 311N, 312N, 313N, 314N, 315N, 316N, 317N, 318N, 319N, 320N, 321N, 322N, 323N, 324N, 325N, 326N, 327N, 328N, 329N, 330N, 331N, 332N, 333N, 334N, 335N, 336N, 337N, 338N, 339N, 340N, 341N, 342N, 343N, 344N, 345N, 346N, 347N, 348N, 349N, 350N, 351N, 352N, 353N, 354N, 355N, 356N, 357N, 358N, 359N, 360N, 361N, 362N, 363N, 364N, 365N, 366N, 367N, 368N, 369N, 370N, 371N, 372N, 373N, 374N, 375N, 376N, 377N, 378N, 379N, 380N, 381N, 382N, 383N, 384N, 385N, 386N, 387N, 388N, 389N, 390N, 391N, 392N, 393N, 394N, 395N, 396N, 397N, 398N, 399N, 400N, 401N, 402N, 403N, 404N, 405N, 406N, 407N, 408N, 409N, 410N, 411N, 412N, 413N, 414N, 415N, 416N, 417N, 418N, 419N, 420N, 421N, 422N, 423N, 424N, 425N, 426N, 427N, 428N, 429N, 430N, 431N, 432N, 433N, 434N, 435N, 436N, 437N, 438N, 439N, 440N, 441N, 442N, 443N, 444N, 445N, 446N, 447N, 448N, 449N, 450N, 451N, 452N, 453N, 454N, 455N, 456N, 457N, 458N, 459N, 460N, 461N, 462N, 463N, 464N, 465N, 466N, 467N, 468N, 469N, 470N, 471N, 472N, 473N, 474N, 475N, 476N, 477N, 478N, 479N, 480N, 481N, 482N, 483N, 484N, 485N, 486N, 487N, 488N, 489N, 490N, 491N, 492N, 493N, 494N, 495N, 496N, 497N, 498N, 499N, 500N, 501N, 502N, 503N, 504N, 505N, 506N, 507N, 508N, 509N, 510N, 511N, 512N, 513N, 514N, 515N, 516N, 517N, 518N, 519N, 520N, 521N, 522N, 523N, 524N, 525N, 526N, 527N, 528N, 529N, 530N, 531N, 532N, 533N, 534N, 535N, 536N, 537N, 538N, 539N, 540N, 541N, 542N, 543N, 544N, 545N, 546N, 547N, 548N, 549N, 550N, 551N, 552N, 553N, 554N, 555N, 556N, 557N, 558N, 559N, 560N, 561N, 562N, 563N, 564N, 565N, 566N, 567N, 568N, 569N, 570N, 571N, 572N, 573N, 574N, 575N, 576N, 577N, 578N, 579N, 580N, 581N, 582N, 583N, 584N, 585N, 586N, 587N, 588N, 589N, 590N, 591N, 592N, 593N, 594N, 595N, 596N, 597N, 598N, 599N, 600N, 601N, 602N, 603N, 604N, 605N, 606N, 607N, 608N, 609N, 610N, 611N, 612N, 613N, 614N, 615N, 616N, 617N, 618N, 619N, 620N, 621N, 622N, 623N, 624N, 625N, 626N, 627N, 628N, 629N, 630N, 631N, 632N, 633N, 634N, 635N, 636N, 637N, 638N, 639N, 640N, 641N, 642N, 643N, 644N, 645N, 646N, 647N, 648N, 649N, 650N, 651N, 652N, 653N, 654N, 655N, 656N, 657N, 658N, 659N, 660N, 661N, 662N, 663N, 664N, 665N, 666N, 667N, 668N, 669N, 670N, 671N, 672N, 673N, 674N, 675N, 676N, 677N, 678N, 679N, 680N, 681N, 682N, 683N, 684N, 685N, 686N, 687N, 688N, 689N, 690N, 691N, 692N, 693N, 694N, 695N, 696N, 697N, 698N, 699N, 700N, 701N, 702N, 703N, 704N, 705N, 706N, 707N, 708N, 709N, 710N, 711N, 712N, 713N, 714N, 715N, 716N, 717N, 718N, 719N, 720N, 721N, 722N, 723N, 724N, 725N, 726N, 727N, 728N, 729N, 730N, 731N, 732N, 733N, 734N, 735N, 736N, 737N, 738N, 739N, 740N, 741N, 742N, 743N, 744N, 745N, 746N, 747N, 748N, 749N, 750N, 751N, 752N, 753N, 754N, 755N, 756N, 757N, 758N, 759N, 760N, 761N, 762N, 763N, 764N, 765N, 766N, 767N, 768N, 769N, 770N, 771N, 772N, 773N, 774N, 77												

PILE BENT NO	253 - 255.	256 - 267.	268 - 274	275 - 280	275 - 279	281	281	282, 283.	282, 283	281, 282, 283	280	284	284
PILE NO.	253 N, 254 N, 255 N.	256 N, 257 N, 258 N, 259 N, 260 N, 261 N, 262 N - 267 N.	268 N, 269 N, 270 N, 271 N, 272 N, 273 N, 274 N.	275 N1 - 280 N1.	275 S, 276 S, 277 S.	281 N1 281 N2	281 N3 281 N4.	282 N1, 282 N2, 282 N3, 282 N4.	282 N5, 283 N5	281 S1, 282 S1, 283 S1		284 N	
	255 S, 254 S, 255 S.	256 S, 257 S, 258 S, 259 S, 260 S, 261 S, 262 S, 263 S, 264 S, 265 S, 266 S, 267 S.	268 S, 269 S, 270 S, 271 S, 272 S, 273 S, 274 S.	275 N2 - 280 N2.	276 S, 275 S.			283 N1, 283 N2, 283 N3, 283 N4.	282 N6, 283 N6	281 S2, 282 S2, 283 S2	280 S		284 S
CHAINAGE (Metres)	5126.8 to 5166.8	5185.8 to 5406.8	5426.8 to 5546.8	5566.8 to 5606.8	5566.8 to 5646.8	5686.8	5686.8	5706.8 to 5726.8	5706.8 to 5726.8	5686.8 to 5726.8	5666.8	5744.8	5744.8
PROVISIONAL TOE R.L.	-24.700	-26.100	-30.500	-31.000	-31.000	-33.000	-33.000	-33.000	-33.000	-33.000	-31.000	-33.000	-33.000
PROVISIONAL LENGTH	32.740	34.180	38.720	39.030	38.160	41.100	40.470	41.020	40.650	40.040 40.650	39.030	41.020	39.860
BED LEVEL (APPROX)	- 9.500	-11.500	-12.500	-13.500	-13.500	-13.500	-13.500	-13.500	-13.500	-13.500	-13.500	-13.500	-13.500
HEAD R.L.	7.058	7.058	7.058	6.858	6.858	6.858	6.258	6.858	6.858	6.258 6.858	6.858	6.858	6.858

PILE BENT No	1	13	25	37	49	61	73	85	97	109	121	133	145	157	168	177	185	193	201	209	217	226	237	248	261	274
PILE No	1A	13A	25A	37A	49A	61A	73A	85A	97A	109A	121A	133A	145A	157A	168A, 168A1	177A1, 177A1	185A, 185A1	193A, 193A1	201A, 201A1	209A, 209A1	217A, 217A1	226A, 226A1	237A, 237A1	248A, 248A1	261A, 261A1	274A, 274A1
CHAINAGE (Metres)	87.4	327.4	567.4	807.4	1047.4	1287.4	1527.4	1767.4	2007.4	2247.4	2487.4	2727.4	2967.4	3207.4	3426.2, 3427.4	3606.2, 3607.4	3766.2, 3767.4	3926.2, 3927.4	4086.2, 4087.4	4246.2, 4247.4	4406.2, 4407.4	4586.2, 4587.4	4806.2, 4807.4	5046.2, 5047.4	5286.2, 5287.4	5546.2, 5547.4
PROVISIONAL TOE R.L.	-28.000	-24.500	-22.000	-24.500	-22.000	-22.000	-22.500	-24.000	-27.000	-24.500	-22.000	-25.000	-26.000	-24.000	-24.000	-23.000	-25.000	-25.000	-25.500	-29.000	-24.500	-25.500	-25.500	-24.500	-26.100	-30.500
PROVISIONAL LENGTH	35.320	31.710	29.130	31.710	29.130	29.130	29.650	31.190	34.282	31.710	29.130	32.220	33.260	31.190	30.260, 31.830	29.260, 30.180	31.260, 32.220	31.260, 32.220	31.760, 32.740	33.760, 34.750	35.760, 36.750	37.760, 38.750	39.760, 40.750	41.760, 42.750	43.760, 44.750	45.760, 46.750
BED LEVEL (APPROX)	-2.000	-3.500	-5.500	-7.500	-8.500	-8.000	-7.750	-7.000	-6.000	-5.000	-4.000	-3.500	-2.500	-2.000	-1.500	-1.000	-0.500	-0.500	-1.500	-2.500	-3.000	-4.250	-5.500	-7.250	-11.500	-12.500

PILE BENT N ^o .	24, 25	60, 61	108, 109	153, 157	200, 201	248, 249
PILE N ^o	2452, 2552	6052, 6152	10652, 10952	15652, 15752	20052, 20152	24852, 24952
CHAINAGE (Metres)	546.8 to 560.8	1207.8 to 1280.8	2226.8 to 2240.8	3186.8 to 3200.8	4066.8 to 4080.8	5026.8 to 5040.8
PROVISIONAL TOE R.L.	-22.000	-21.000	-24.500	-24.000	-25.500	-24.500
PROVISIONAL LENGTH	20.000	20.000	22.530	22.020	23.570	27.580
BED LEVEL (APPROX)	-5.500	-6.800	-5.000	-2.000	-1.500	-7.250

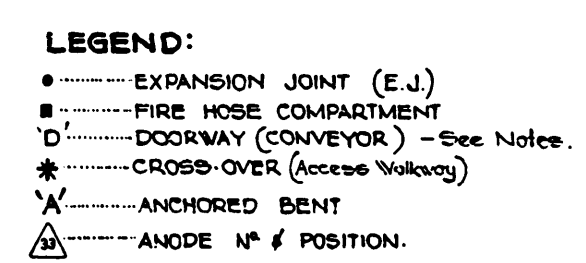
PILE BENT NO.	7	19	31	43	55	67	79	91	103	115	127	139	151	163	173	181	189	197	205	213	221	231	243	255	267
PROVISIONAL TOE R.L.	-24.000	-24.500	-25.000	-24.500	-25.000	-25.000	-22.500	-24.000	-25.000	-24.500	-22.000	-25.000	-26.000	-24.000	-24.600	-23.000	-25.000	-25.000	-25.500	-29.000	-24.500	-29.500	-25.500	-24.700	-26.100
PROVISIONAL LENGTH	31.970	32.480	29.910	25.480	73.010	29.910	30.420	31.370	35.080	32.480	29.910	33.000	34.030	31.970	32.590	30.040	33.000	33.000	33.510	37.120	32.480	37.640	33.510	32.600	34.130

GROUP	TOTAL LENGTH.
TABLE 1	16 550 metres.
TABLE 2	3 235 metres.
TABLE 3	1 227 metres.
TABLE 4	381 metres.
TABLE 5	816 metres.
TOTAL LENGTH 860 TO PILES.	18 800 metres.
TOTAL LENGTH 920 TO PILES.	3 385 metres.

1. Pile Head R.L. for all piles to be R.L. 7-058 unless noted.
2. Anchor Pile Head R.L. to be R.L. 7-258.
3. For Layout of Pile Bents & Anchored Bents & Passing
Boys - Refer Dwg. No. CTQ112215/001
4. Pile Head R.L. for Expansion Joint Anchor piles to be R.L. 7-008.

NO.		DATE	AMENDMENTS	THE SUGAR BOARD BULK SUGAR TERMINALS		DRAWING NO. CTQ 1122/S/002	
A	16-2-77		HEADSTOCK LEVEL RAISED FOR CONC. ROAD DESIGN.	SCALES 1:200 DRAWN J.T. PASSED <i>[Signature]</i> DATE 22-7-79	B.S.T. LUCINDA STAGE III APPROACH JETTY PILE SCHEDULE		B
B	20-10-77		PASSING-BAY PILES AMENDED.				
C.S.R. REFERENCE NO.				MACDONALD WAGNER & PRIDDLE PTY. LTD. CONSULTING ENGINEERS 313 ST PAULS TERRACE PORTHURST VALLEY OLD 4004 PH 52 7152 IN ASSOCIATION WITH CSR LIMITED			

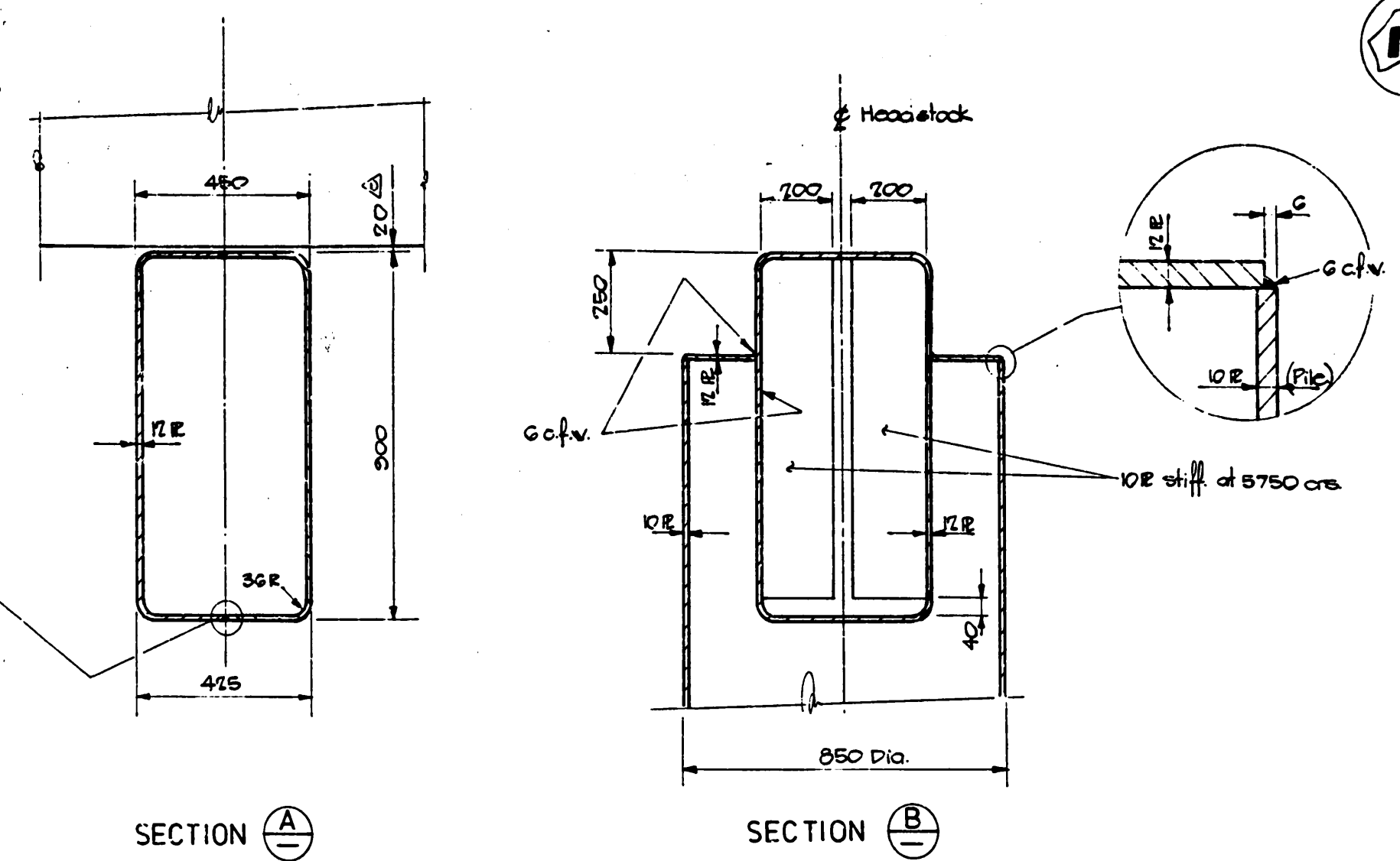




NOTES:

- DOORWAYS IN CONV. GALLERY TO BE ON THE ROAD-SIDE ONLY EXCEPT AT CROSS-OVER POSITIONS WHERE DOORWAYS ARE BOTH SIDES OF CONV. GALLERY.
- FOR CONTINUATION OF APPROACH STRUCTURE - REFER SHEET 1.
- FOR DETAILS OF CROSS-OVER WALKWAYS - REFER DRG NO. CTQ1122/S/026 (ALL CROSS-OVER POSITIONS ARE SEAWARD OF THE MARKED PILE BENT.)
- FOR DETAILS OF EXPANSION JOINTS REFER DRGS NO. CTQ1122/S/006 & CTQ1122/S/017.

THE CORPORATION OF THE TREASURER OF QUEENSLAND									
NO.	DATE	AMENDMENT	THE SUGAR BOARD BULK SUGAR TERMINALS						
A	29.11.76	REE HOSE COMPARTMENT ADDED AT BENT 185.	SCALES 1:500 DRAWN J.T. PASSED <u>J.W. GARY</u> DATE 28.8.76				B.S.T. LUCINDA STAGE III APPROACH JETTY - SHEET 2. SUPERSTRUCTURE LAYOUT.		
							DRAWING NO.		
							CTQ1122/s/0004		
							A		
C.S.R. REFERENCE NO.			MACDONALD WAGNER & PRIDDLE PTY. LTD. CONSULTING ENGINEERS 212 ST MARKS TERRACE PORTFOLIO VALLEY QLD 4066 PH 32 7172 IN ASSOCIATION WITH CSR LIMITED						



Refer Dwg. No. C/10112/S/015
for details of allowable
tolerances for headstocks
& piles.

SECTION C—

φ Pile Bent.

7 min.
17 RE
7-5' 6"
48 min.
10 RE

Incomplete penetration
single-bevel butt weld.

A technical diagram showing a side view of a suspended conveyor idler assembly. A horizontal pipe, labeled "80 ϕ $\times$ 4.0 wall conveyor pipe stringer", is supported by a "Conveyor idler support bracket (By others)". A "Suspended conveyor idlers (By others)" is shown below the pipe, with a dimension of 150 indicated between the pipe and the idler. The idler is a cylindrical roller with a central shaft. The entire assembly is shown in a perspective view, with the idler and support bracket angled downwards relative to the horizontal pipe stringer.

THE CORPORATION OF THE TREASURER OF QUEENSLAND		
NO	DATE	AMENDMENTS
A	10-12-76	Roadway construction altered.
B	4-1-77	Dimension changed.
C	18-1-77	Roadway & to Conveyor & altered. Neoprene bearing thk. amended. Dimensions added.
D	18-1-77	Dimensions changed.
E	1-2-77	Headstock length altered to 7900
C S R REFERENCE NO.		

THE SUGAR BOARD BULK SUGAR TERMINALS	
SCALES 1:20:110 DRAWN R. Holmes PASSED DATE DEC 76.	B.S.T. LUCINDA STAGE III APPROACH JETTY TYPICAL CROSS-SECTION.
DRAIVING NO. CT01122/5/005	
E	

MACDONALD WAGNER & PRIDDLE PTY.LTD.
 CONSULTING ENGINEERS 212 ST MARKS TERRACE, NORTHQUEEN HILL QLD 4066 PH 31 7112
 IN ASSOCIATION WITH
CSR LIMITED

φ Pile Bent & Headstock.

φ Conveyor

φ Roadway

φ Headstock.

G cont. fillet weld.

12" Beel 12" (cut to suit)

RL. G-420 Δ

Incomplete penetration single bevel butt weld.

G cont. fillet weld.

SEAWARD. →

φ Raked Pile 850% Hovoll.

ELEVATION

4- 200. 12 E STIFFENERS
C.F.V. TO INSIDE H'STOCK.
Δ Refer Drg CTQ1122/S/005
for SHIT* details.

φ Headstock.

φ Pile Bent

φ Pile.

PLAN

ANCHOR PILE AT EXPANSION JOINT.

ELEVATION

SECTION A

315 φ's of 10R stiff in Headstock.

340 (Side R)

327 (Stiff)

315 (Cover)

10 Cover R

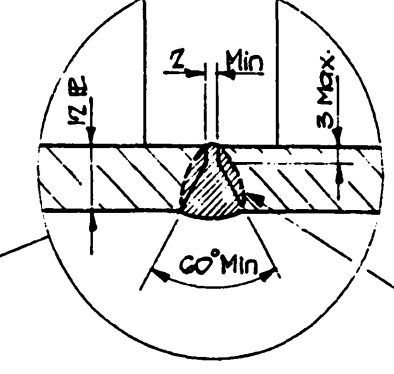
12 R stiff

12 R stiff

12 Side R

12 Base R

10 R stiffeners
G cont. fillet weld
all round both sides



Qualified incomplete penetration single V butt weld.

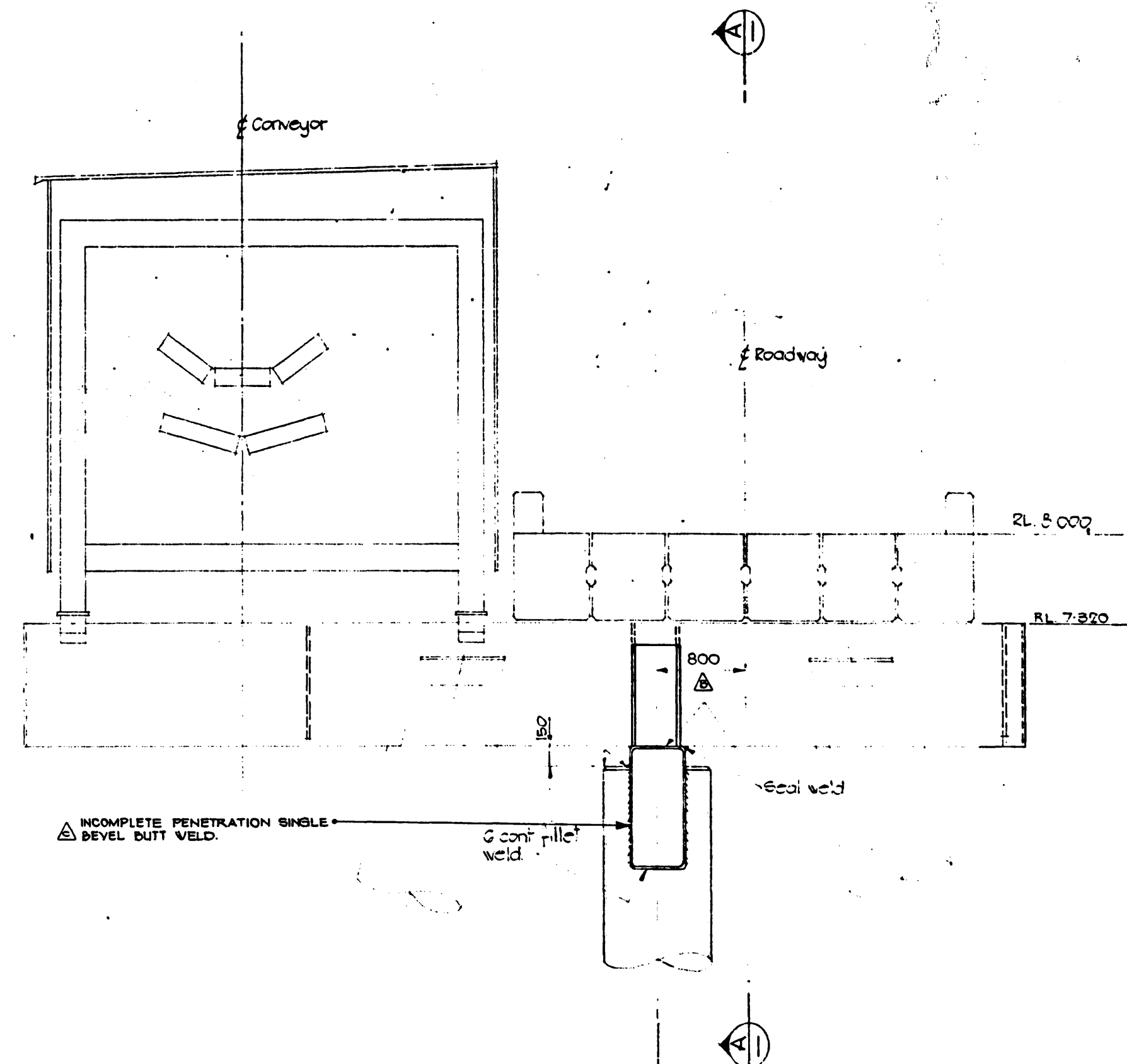
SECTION B

850 %

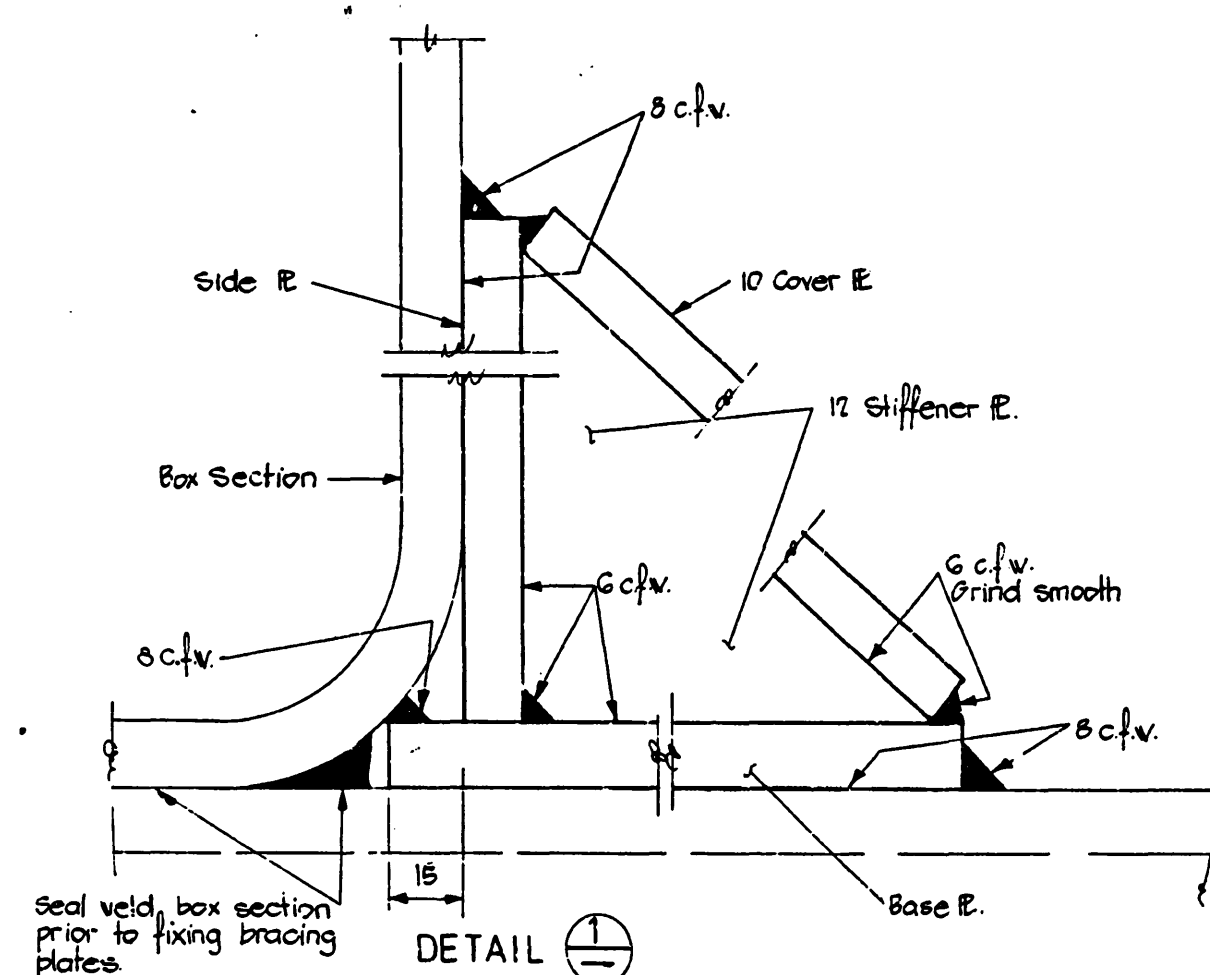
NOTE

For Type B' Anchored Pile Bent - Refer Drg No CTQ1122/S/007
For Typical Cross-Section of Approach Jetty - refer Drg No CTQ1122/S/005.

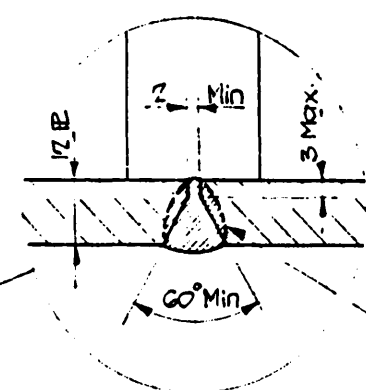
THE CORPORATION OF THE TREASURER OF QUEENSLAND									
NO		DATE		AMENDMENTS		THE SUGAR BOARD BULK SUGAR TERMINALS			
A		20-12-76		Roadway construction altered.					
B		1-2-77		DIM. TO FILE FROM ROAD OF AMEND. RL HEADSTOCK 4"6 AMENDED.		B.S.T. LUCINDA STAGE III APPROACH JETTY TYPE 'A' TYPICAL ONE-PILE ANCHOR BENT.			
C		23-8-77		PLATE SIZES AMENDED.					
						SCALES 1/32 1/10 DRAWN R. Holmes PASSED DATE DEC-76			
						DRAWING NO CTQ1122/S/006 C			
						MACDONALD WAGNER & PRIDDLE PTY. LTD. CONSULTING ENGINEERS 213 ST PAULS TERRACE PORTSIDE VALLEY, QLD 4006 PH. 52 7123 IN ASSOCIATION WITH			
C S R REFERENCE NO.						CSR LIMITED			



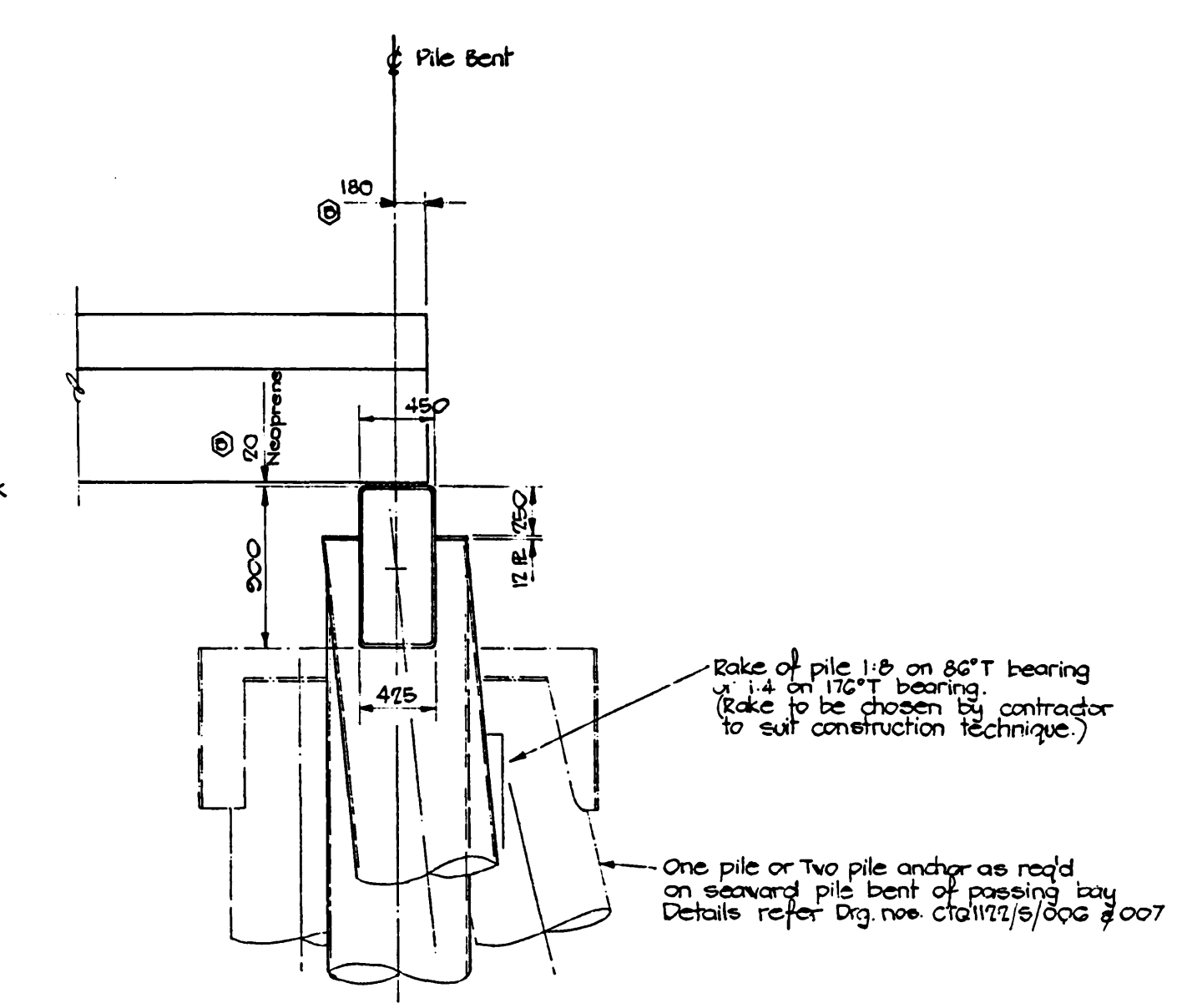
Raked Pile
 650% $\times$ 10 vall.
 N.B. 920% $\times$ 10 vall for pile bent
 No 261 & 274.



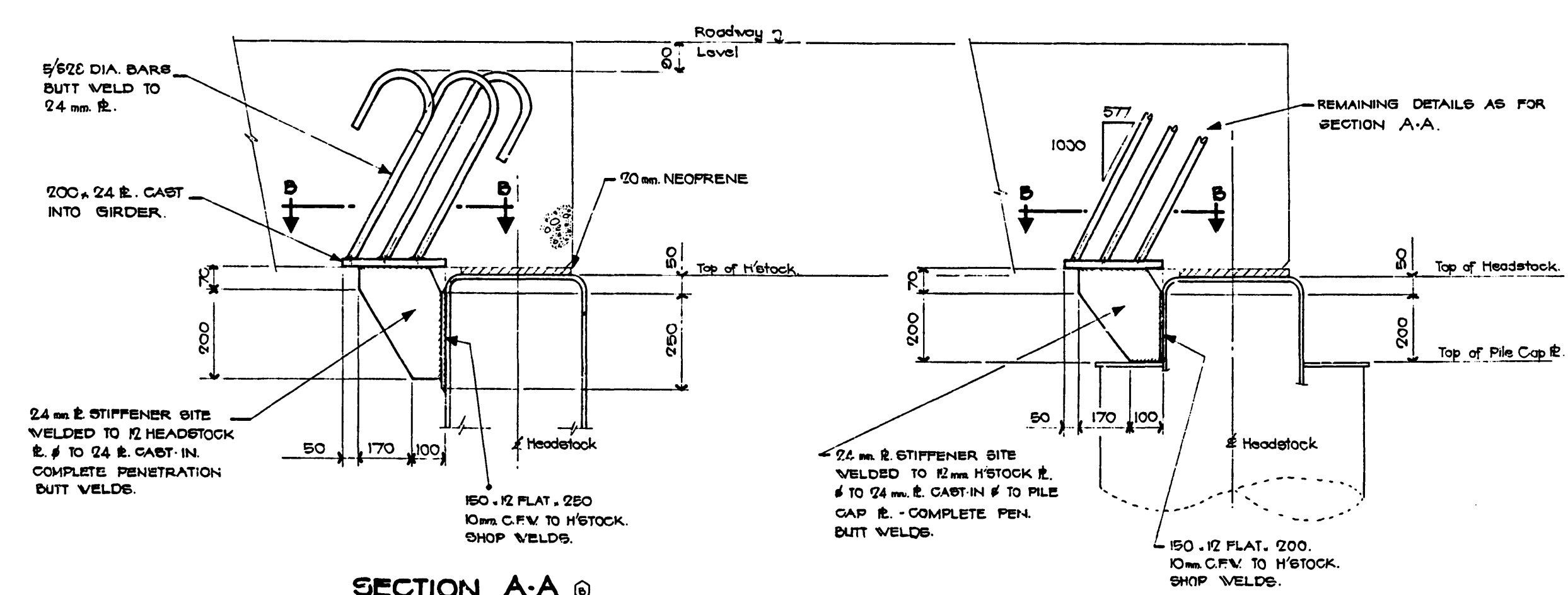
For 'Type A' Anchor File Bent - Refer Drg. No CTR/172/S/OOS .
For Typical Cross Section of Approach - (85% pile) Refer Drg. No CTR/172/S/OOS
 - (20%)

SECTION 

THE CORPORATION OF THE TREASURER OF QUEENSLAND			
NO	DATE	AMENDMENTS	
A	20-12-76	Roadway construction altered.	THE SUGAR BOARD BULK SUGAR TERMINALS B.S.T. LUCINDA STAGE III APPROACH JETTY TYPE 'B' TYPICAL TWO-PILE ANCHOR BENT. DRAWING NO CTO1122/S/007 C
B	1-2-77	DIM. TO PILE & FROM ROAD & AMENDED. RL AMEND.	
C	20-5-77	PLATE SIZES AMENDED.	
			SCALES 1/25 1/20 DRAWN R. Holmes PASSED DATE DEC. 76.
C.S.R. REFERENCE NO.			MACDONALD WAGNER & PRIDDLE PTY. LTD. CONSULTING ENGINEERS 215 B PINE STREET MOUNTAIN VIEW QLD 4064 PH 32 7131 AN ASSOCIATION WITH CSR LIMITED



END VIEW



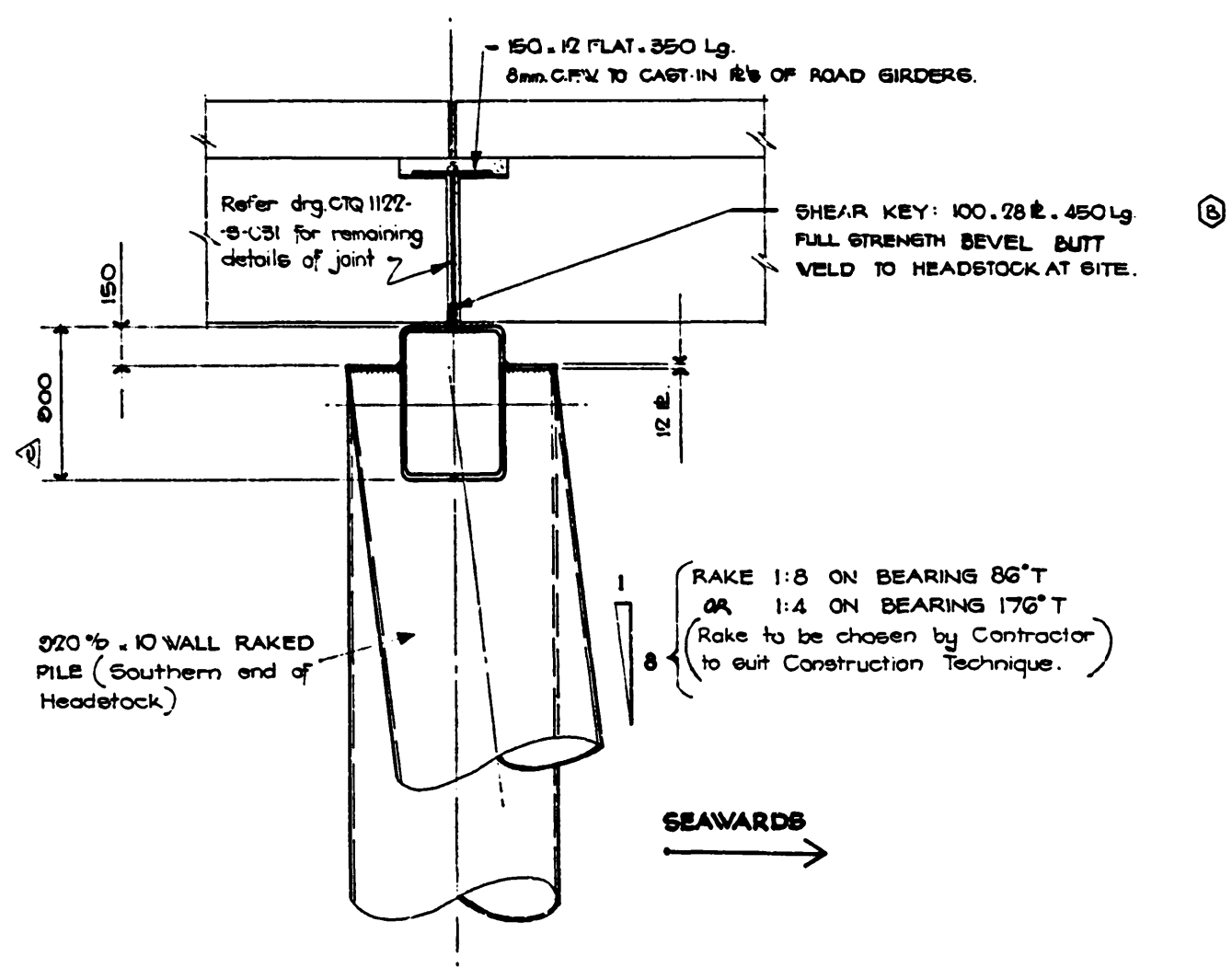
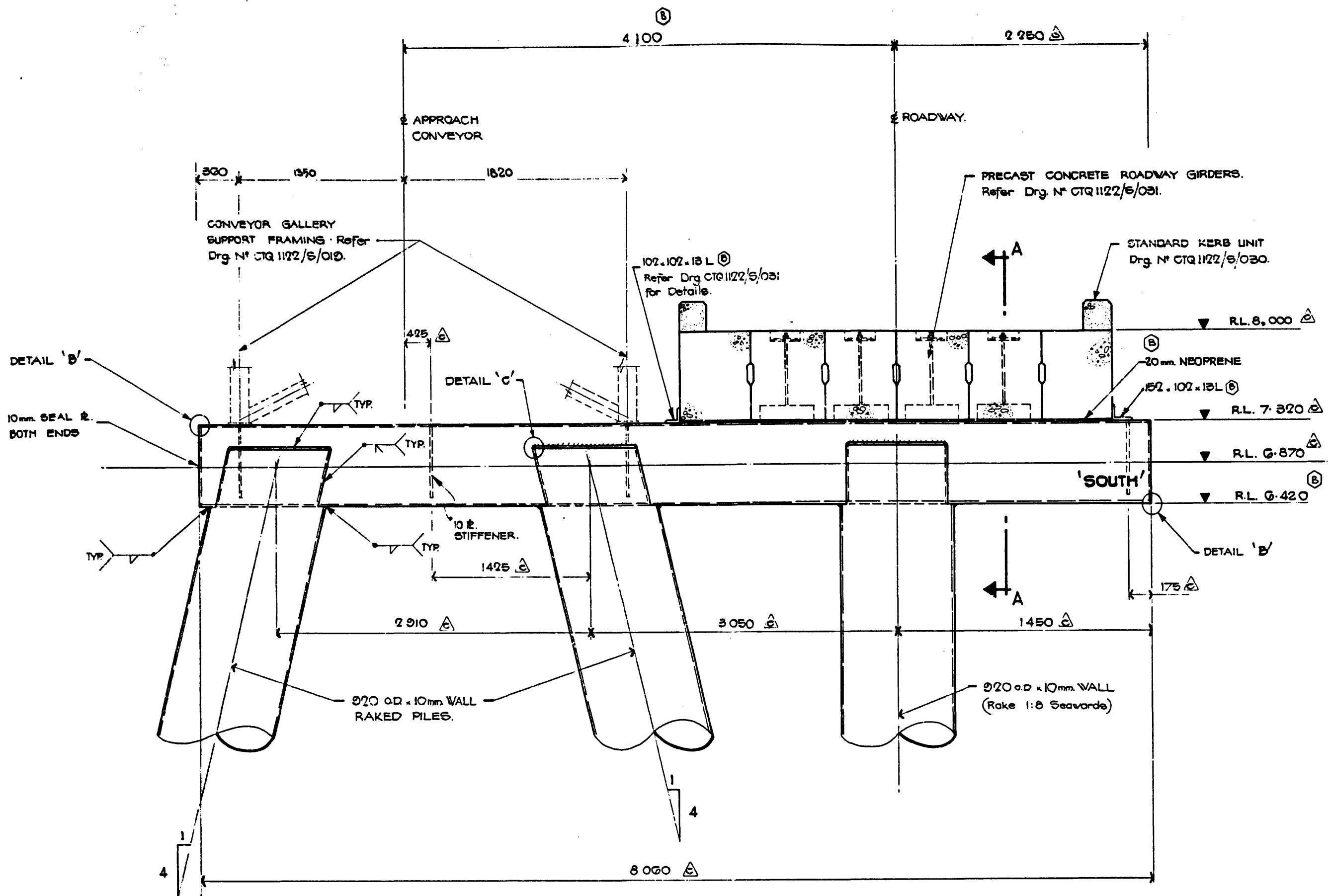
SECTION C-C

SECTION B-B

NOTES:

- THE CONTRACTOR MAY MODIFY BOX GIRDER DETAILS TO SUIT POST-TENSIONING TECHNIQUE, SUBJECT TO ENGINEERS APPROVAL.
- FOR STANDARD CONC. ROADWAY GIRDERS # STANDARD JOINT DETAILS - REFER DRG. CTQ 1122/5/031.
- FOR REMAINING HEADSTOCK # PILE DETAILS - REFER DRG. N° CTQ 1122/5/005.

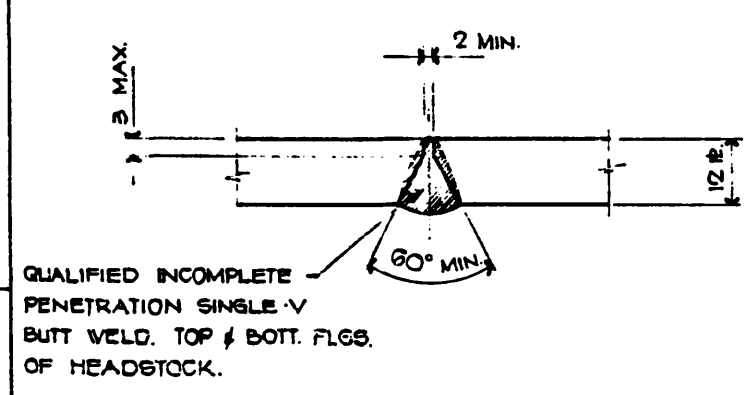
THE CORPORATION OF QUEENSLAND			THE SUGAR BOARD		BULK SUGAR TERMINALS	
NO.	DATE	AMENDMENTS	SCALES 1:125 1:100 DRAWN R. Holmes PASSED DATE DEC-76		B.S.T. LUCINDA STAGE III APPROACH JETTY PASSING BAY DETAILS.	
A	20-12-76	Roadway construction altered.	DRAWING NO. CTON22/5/008		E	
B	JAN 77	Conv. of dim. to Road if amend. Deck unit hold down details added. Girder Length alt.				
C	1-2-77	HEADSTOCK LENGTH ALTERED.				
D	16-5-77	File Spacing amended to suit Contractor Construction. Methods.				
E	19-8-77	Shop weld edge angles to suit Contractor	MACDONALD WAGNER & PRIDDLE PTY.LTD. CONSULTING ENGINEERS 313 ST PAUL'S TERRACE PORTHURST VALLEY, OLD 4004 PH 52 7132 AN ASSOCIATION WITH CSR LIMITED		C.S.R. REFERENCE NO.	



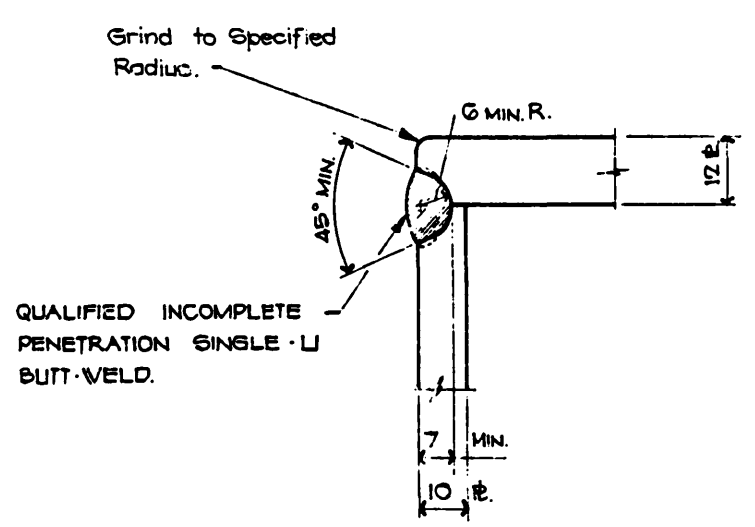
SECTION A-A.
1:20.

PILE BENTS 275, 276, 277, 278, 279.

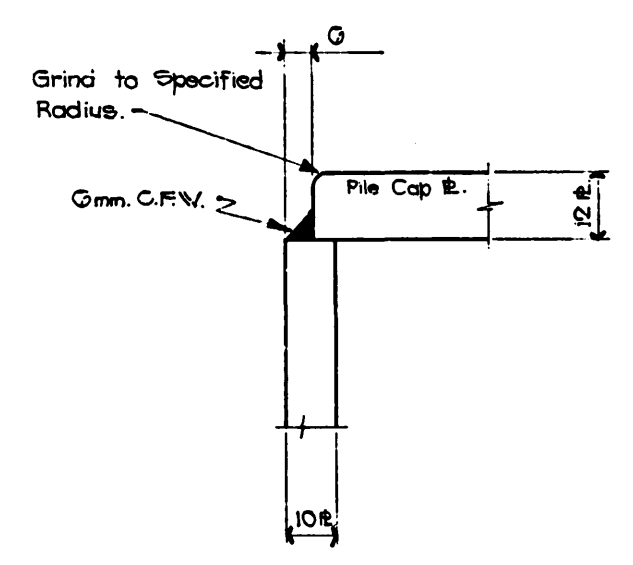
Scale: 1:20



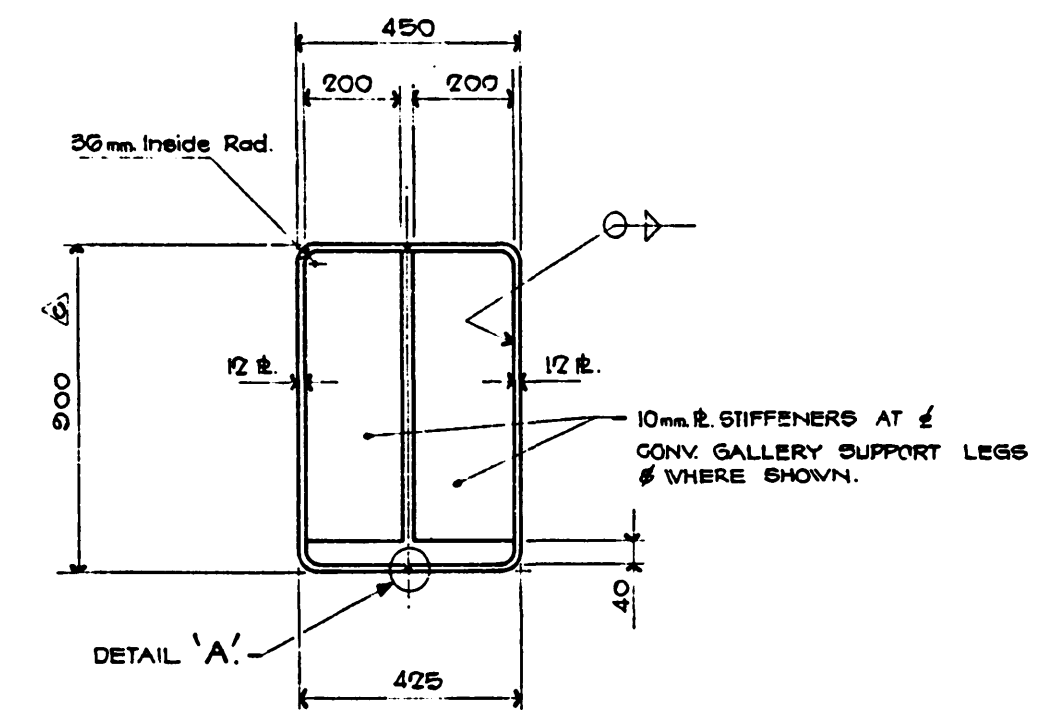
DETAIL 'A'



DETAIL 'B'



DETAIL 'C'

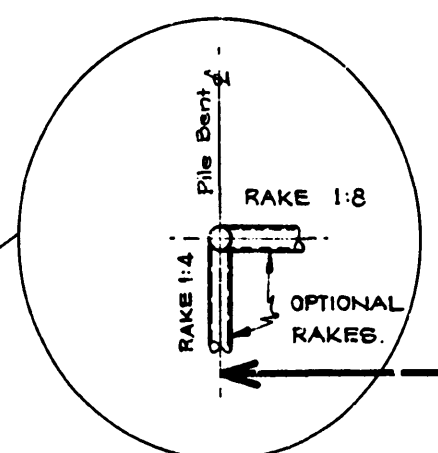


HEADSTOCK SECTION.

NOTES:

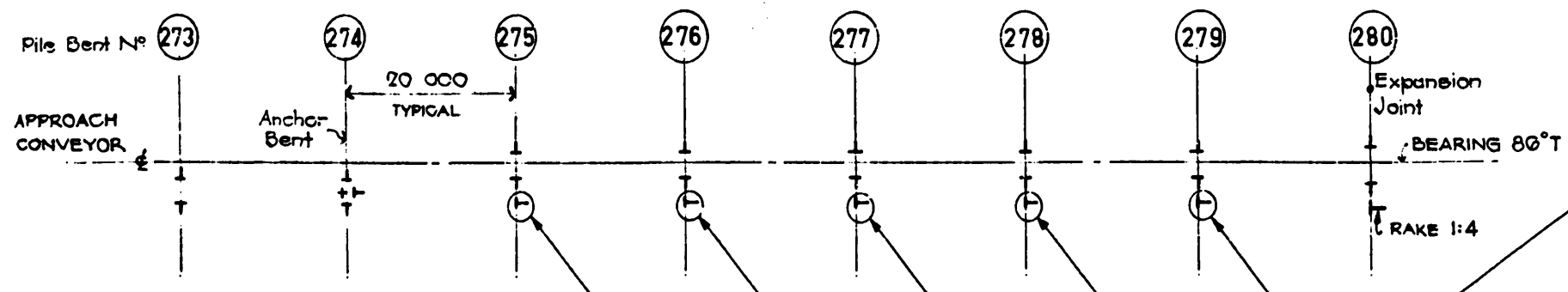
1. FOR LOCATION OF PILE BENTS - REFER GEN. PLAN DRG. N° CTQ1122/S/001
2. FOR A TYPICAL PILE BENT CROSS-SECTION REFER DRG. N° CTQ1122/S/005
3. ALL WELDS - 6mm. C.F.V. UNLESS NOTED.
4. FOR STANDARD ROADWAY JOINT DETAILS - REFER DRG. N° CTQ1122/S/031.

NOTE: WITH THE PILES IN THE AMENDED POSITIONS RE. REVISION C - ONLY THE 1:4 RAKE IN THE PLANE OF THE BENT IS ALLOWABLE



PILE BENT LOCATION PLAN.

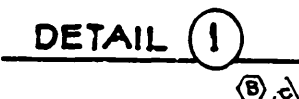
NOTE: All Piles rake 1:4 unless noted otherwise.



THE CORPORATION OF THE TREASURER OF QUEENSLAND					
Nº	DATE	AMENDMENTS	THE SUGAR BOARD BULK SUGAR TERMINALS		
A	20-12-76	Roadway construction altered.	SCALE 1:10 U.N. DRAWN J.T. PASSED B.S.T. LUCINDA STAGE III APPROACH JETTY. PILE BENTS Nº 275-279 inc.		
B	JAN-77	£ ROAD TO CONV. CL'S AMENDED.			
C	15-11-77	PILE LOCATIONS & HEADSTOCK LENGTH AMENDED TO SUIT CONTRACTOR'S CONSTRUCT	DATE DEC. '76		
			DRAWING Nº CTQ1122/S/010		
			C		
CSR REFERENCE Nº			MACDONALD WAGNER & PRIDDLE PTY.LTD. CONSULTING ENGINEERS 213 J. RAILS TERRACE, FORTITUDE VALLEY, Q.L.D. 4006 PH 52 7122 IN ASSOCIATION WITH CSR LIMITED		



PILE BENT 28!

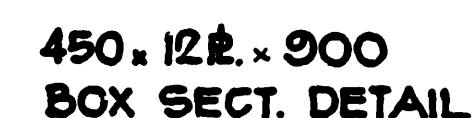


SECTION B-B

NOTES :

- ALL WELDS 6mm. C.F.V. UNLESS NOTE
- FOR DETAILS OF STD. DECK UNITS - refer drg. nr. CTG 1122/6/031.

Scale 1:12.



Scale 1:10

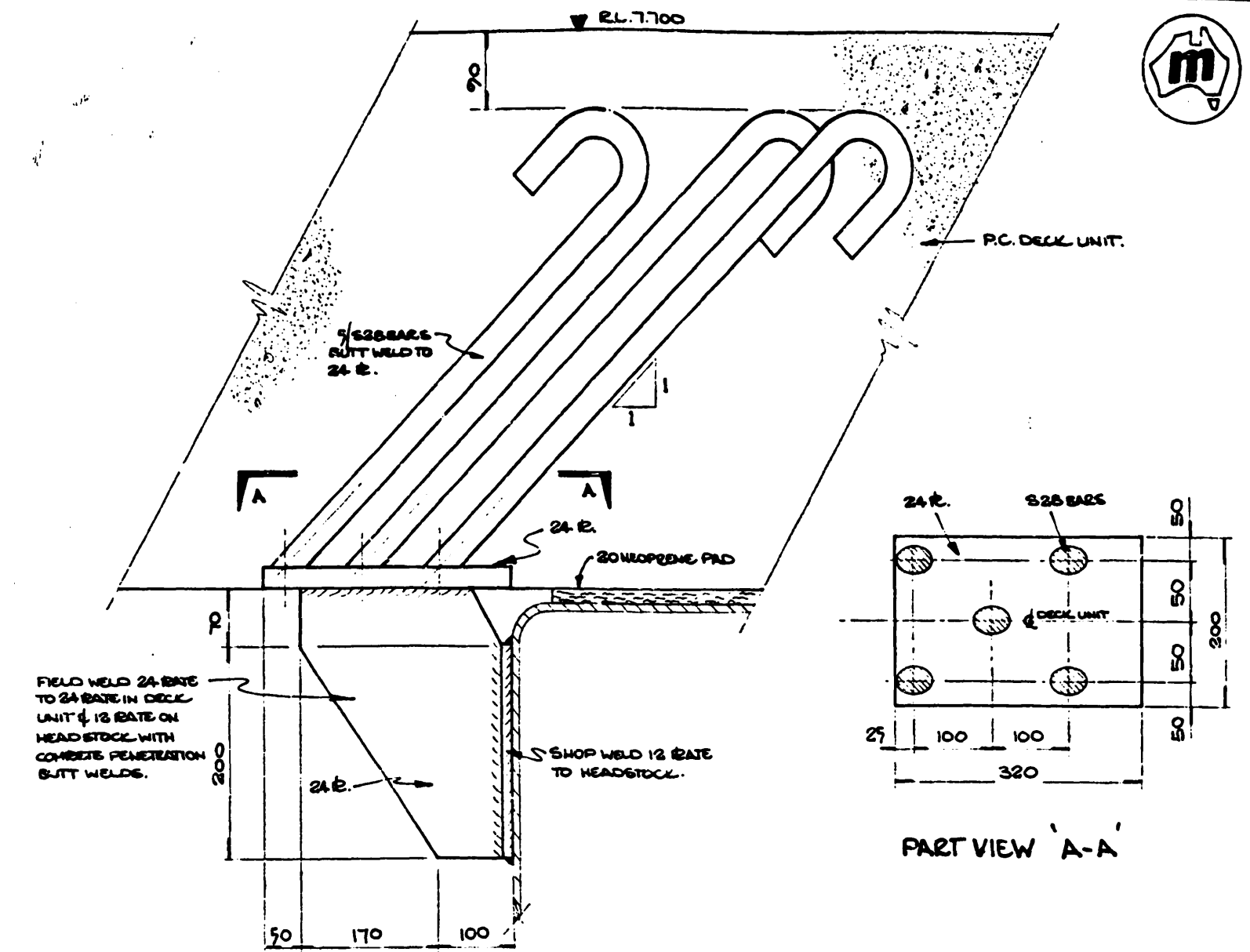
THE SUGAR BOARD BULK SUGAR TERMINALS

NO.	DATE	AMENDMENTS	THE SUGAR BOARD BULK SUGAR TERMINALS		
A	20-12-78	Roadway construction altered.	SCALES 1:20 DRAWN J.T. PASSED DATE DEC. '78	B.S.T. LUCINDA STAGE III APPROACH JETTY. PILE BENT NO. 281.	DRAWING NO. CTO 122/S/012 E
B	JAN. 77	DIMENSION 4100 WAS 4000. NEOPRENE AMEND			
C	10-5-77	DETAIL ① AMENDED			
D	12-5-77	SITE WELDING OF 152 L AMENDED TO SHOR WELDING TO SUIT CONTRACTOR.			
E	18-11-77	PILE BENT AMENDED TO SUIT CONTRACTORS CONSTRUCTION METHOD.			
C.S.R. REFERENCE NO.			MACDONALD WAGNER & PRIDDLE PTY.LTD. CONSULTING ENGINEERS 713 ST PAULS TERRACE, PORTHURST VALLEY, QLD 4006 PH. 51 7122 IN ASSOCIATION WITH CSR LIMITED		

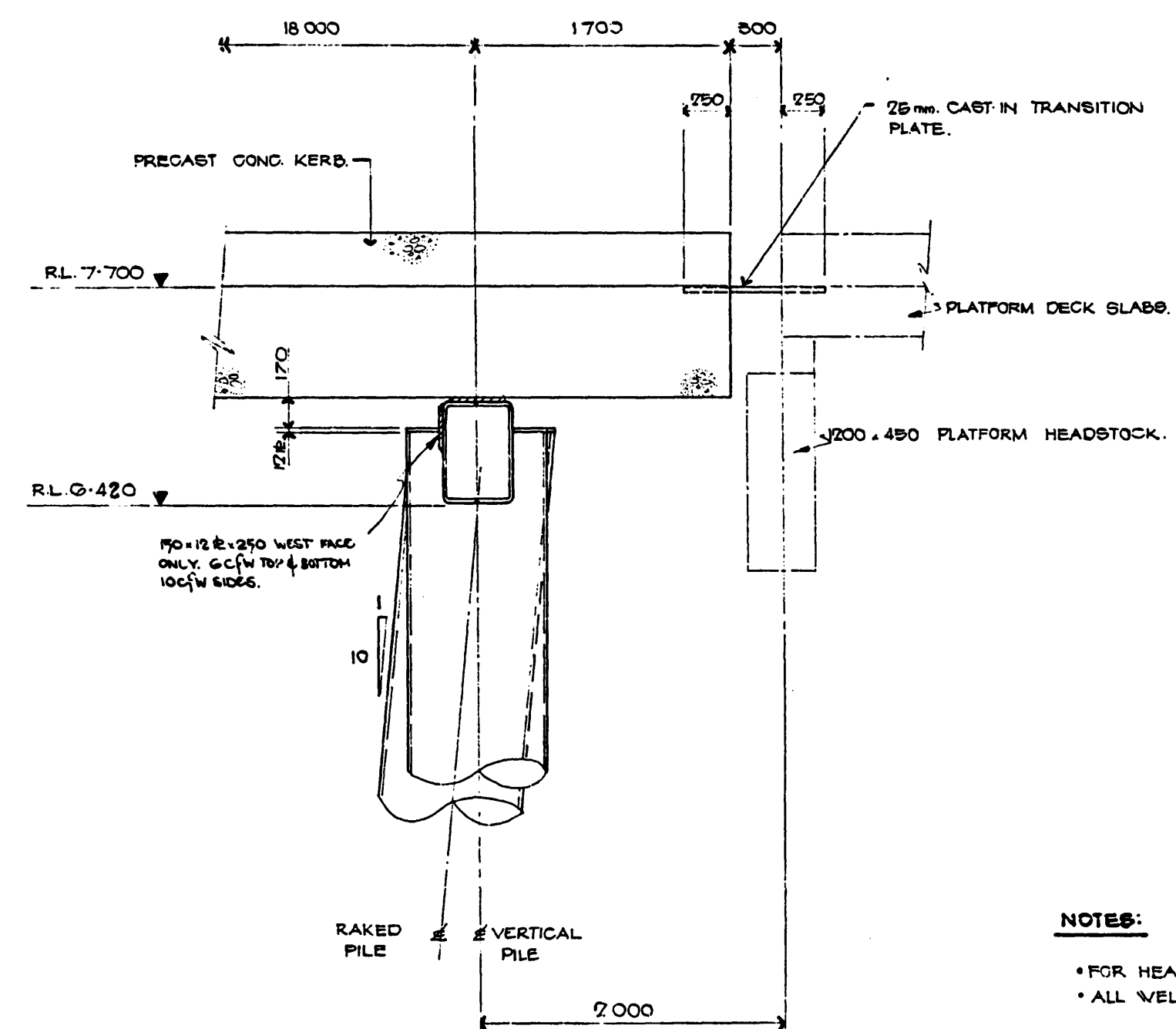
MACDONALD WAGNER & PRIDDLE PTY. LTD

IN ASSOCIATION WITH
CSR LIMITED





DECK UNIT RETAINING BRACKET
TYPICAL FOR FILE BENT NOS 282, 283 & 284.



END VIEW

NOTES:

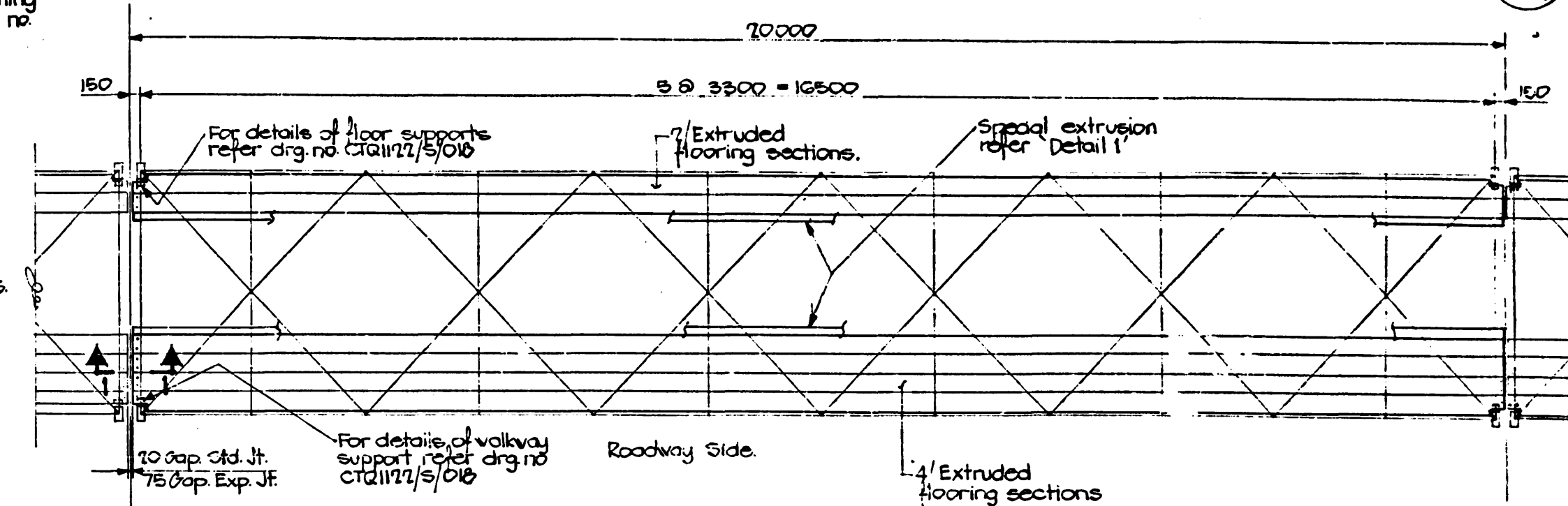
- FOR HEADSTOCK DETAILS - REFER DRG. CIG1122/S/OIO
- ALL WELDS $\varnothing$ mm. C.F.W. UNLESS NOTED.

THE CORPORATION OF THE TREASURER OF QUEENSLAND

NO.	DATE	AMENDMENTS	THE SUGAR BOARD BULK SUGAR TERMINALS	
A	20-12-76	Roadway construction altered.	SCALES 1:100 & 1:25 DRAWN K.K.L. PASSEN DATE DEC/ 70	B.S.T LUCINDA STAGE III APPROACH JETTY PILE BENT N ^a 284
B	JAN '77	PILE CENTERS AMENDED DECK UNIT NO BRACKET DETAILS ADDED		DRAWING NO CTQ1122/S/014
C	SEPT '77	Dimension altered for Bent Position.		C
			MACDONALD WAGNER & PRIDDLE PTY.LTD. CORPORATE ENGINEERS 313 ST PAULS TERRACE PORTPHILL BRISBANE, QLD 4006 PH. 52 7172. IN ASSOCIATION WITH CSR LIMITED	
C.S.R. REFERENCE NO.				







V-Rib (all sheeting local)

30meco V-Rib VS-900 Roof sheeting or equiv.

Roadway Side

3/ Fibreglass panels to match VS-360 Roof sheeting or equiv.

450

Extra aluminium girt req'd at every 3rd span where fire hydrant occurs. Refer Dwg no CTQ1122/3/045 for details.

Aluminium girts as detailed.

Comalco V-Rib VS-300 wall sheeting or equiv.

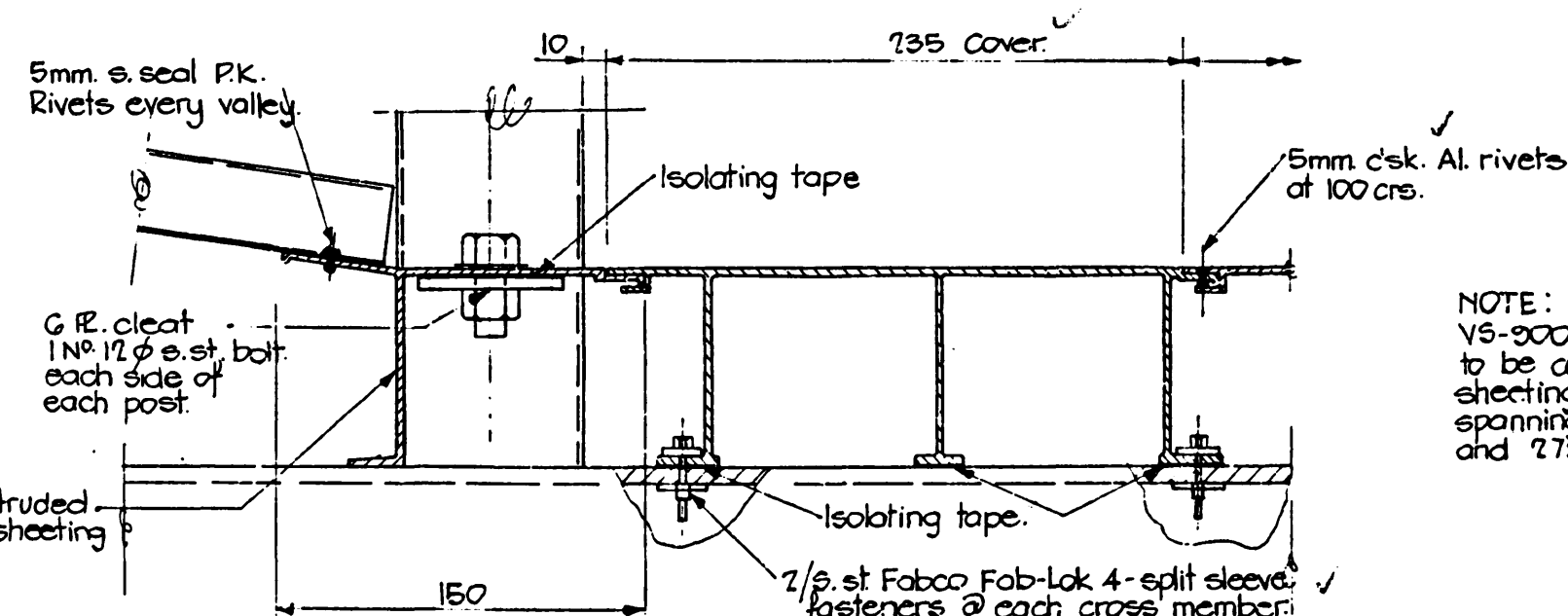
SIDE ELEVATION. (ROADWAY SIDE)

Technical drawing of an aluminium extruded floor section. The drawing shows a cross-section of the floor assembly. On the left, a vertical dimension of 75 is indicated. A horizontal dimension of 735 is labeled "Coverage". The floor surface is shown with "Knurled non-skid longit. ribs". On the right, a vertical dimension of 70 is indicated. A horizontal dimension of 10 is indicated. A vertical dimension of 3 is indicated. The text "3mm Al. Pl. Top surface to be blasted and anti-skid epoxy treated." is written. Below this, the text "Hard fix disk 5mm rivets every 100mm." is written. The drawing is labeled "ALUMINIUM EXTRUDED FLOOR SECTION." at the bottom.

Hand-drawn cross-section diagram of a composite floor system. The diagram shows a concrete slab on top of a metal deck. Key details include:

- Reinforcement:** 3mm Al. RE. treated as noted above. (3mm diameter reinforcement bars).
- Dimensions:**
 - 65 (distance from knurled rib to first bar)
 - 70 (distance between first and second bars)
 - 75 Nom (nominal distance between bars)
 - 50 (distance from last bar toarris corner)
 - 3 (thickness of metal deck)
- Annotations:**
 - Knurled rib ground back.
 - C/sk screws 100mm lock nut at 100mm c/c.
 - 50x8 Slot Hole.
 - Arreis corners to floor level (typ).
 - Hard fix c/sk 6mm rivets every 100mm.

Special Al. extruded -
section for sheeting
support.



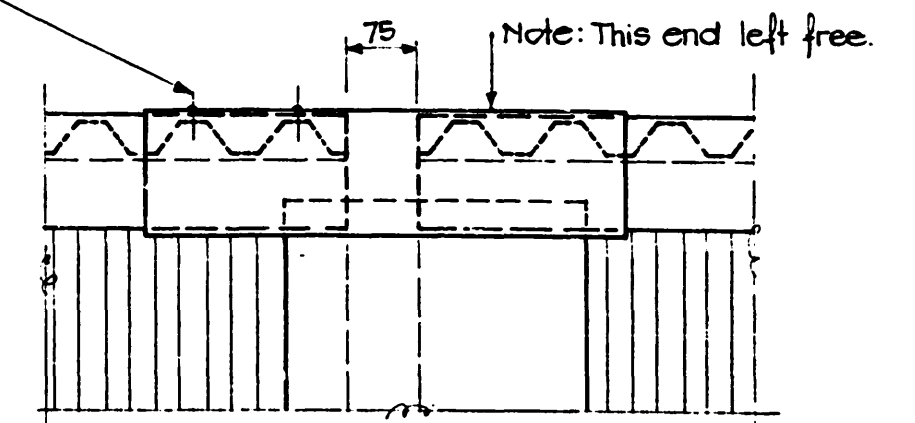
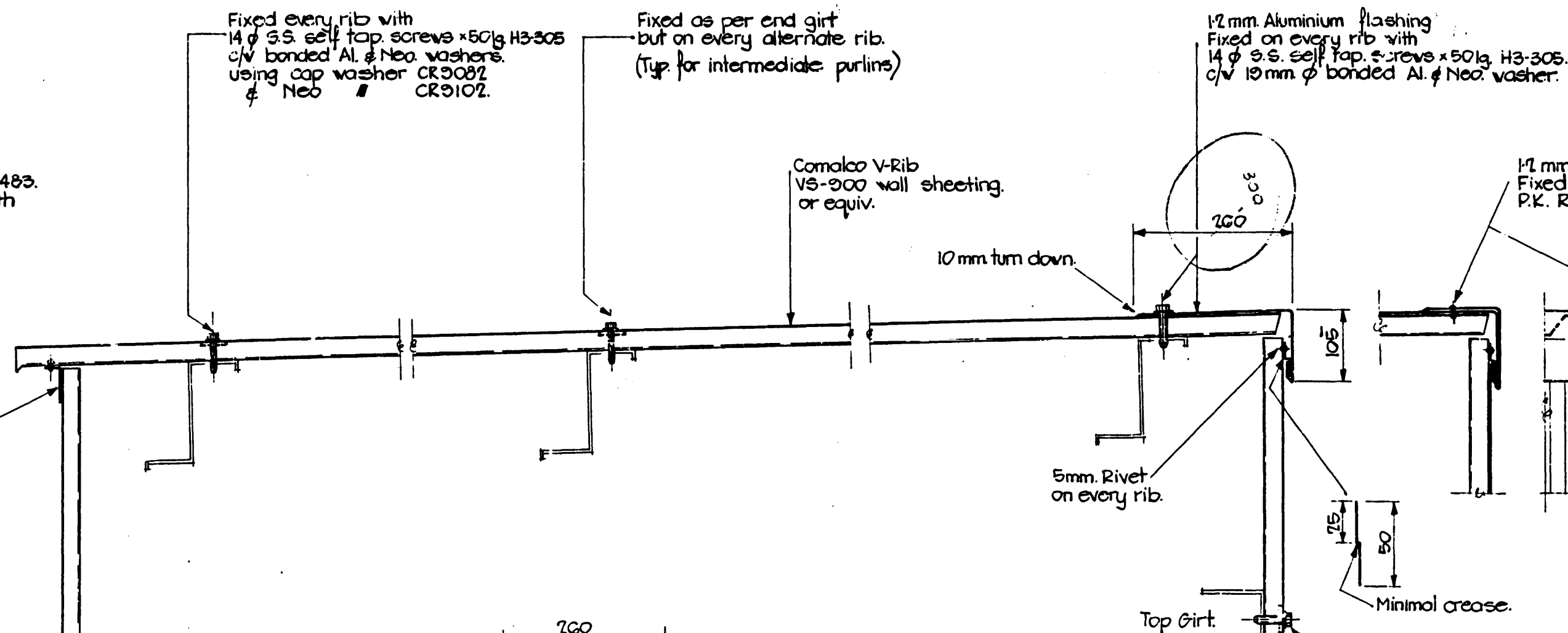
NOTE:
V5-900 sheeting under conveyor belt is to be covered with 1mm. thk. flat aluminium sheeting riveted to V-Rib for Galleries spanning between pile bents no 1 to 3 and 779-781 only.

THE CORPORATION OF THE TREASURER OF QUEENSLAND

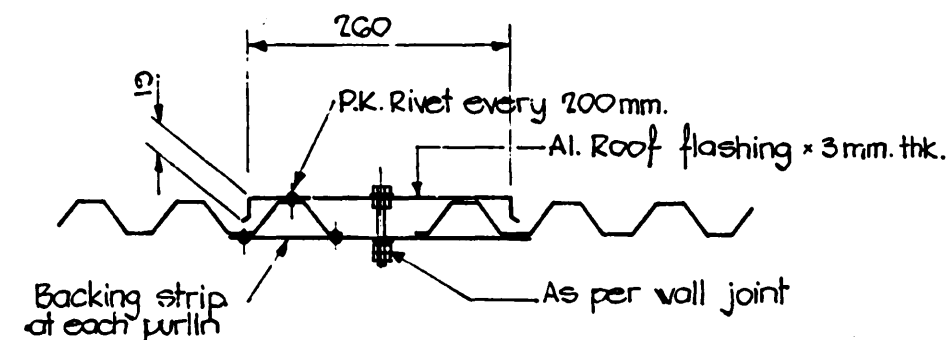
Nº	DATE	AMENDMENTS	THE SUGAR BOARD BULK SUGAR TERMINALS		DRAWING Nº
A	24-2-77	DIMENSION ALTERED. FULFILL CLERT ALTERED. GIRT ADDED	SCALES 1"=50' DRAWN E.C. HOLMES PASSED <i>[Signature]</i> DATE <i>24/2/77</i>	B.S.T. LUCINDA STAGE III APPROACH JETTY. STANDARD SPAN CLADDING	CTQ1122/S020 A
CSR REFERENCE Nº.			MACDONALD WAGNER & PRIDDLE PTY.LTD. CONSULTING ENGINEERS 312 ST PAULS TERRACE PORTPHU VALLEY QLD 4008 PH 32 7122 IN ASSOCIATION WITH CSR LIMITED		

WALL SHEET. FLASHING JOINT

Comalco V-Rib VS-900
wall sheeting or equiv.



TYP. SKILLION FLASHING JOINT.



ROOF SHEET. DETAIL AT TRUSS JOINT

NOTE: Provide similar cover flashing over VS-200 sheeting beneath conveyor belt. (3 backing strips reqd.)

FIXING NOTES:

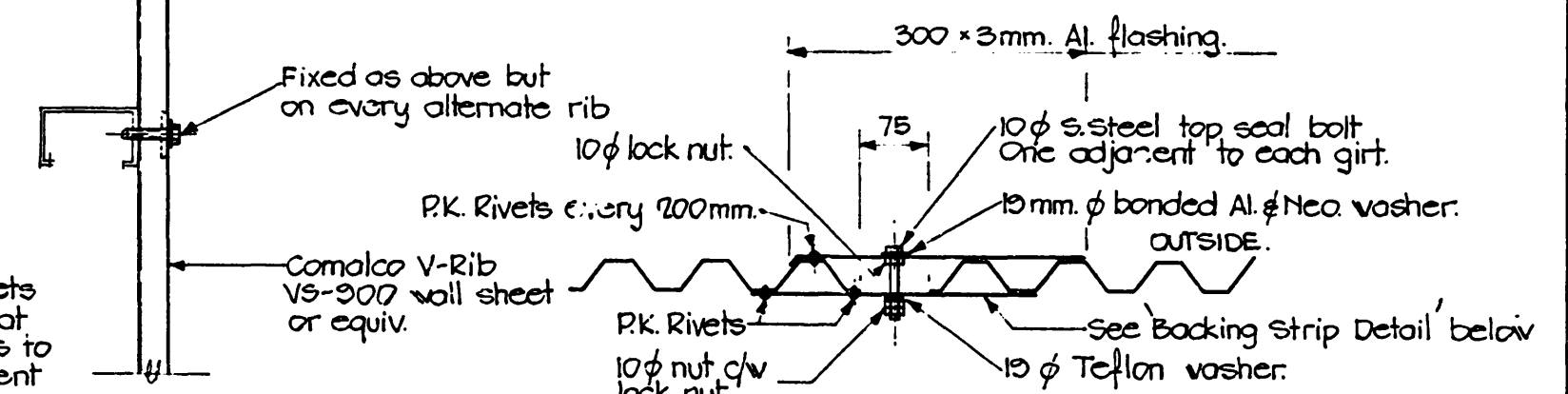
STITCHING:
Fix all Aluminium roof & wall sheeting at
200 mm. c/s. using 5mm. dia. P.K. Al. self sealing rivets.

FIBRE GLASS STITCHING:
Fix at 100mm. crs. using 5mm. ϕ P.K. rivets with washers front & back of sheeting.

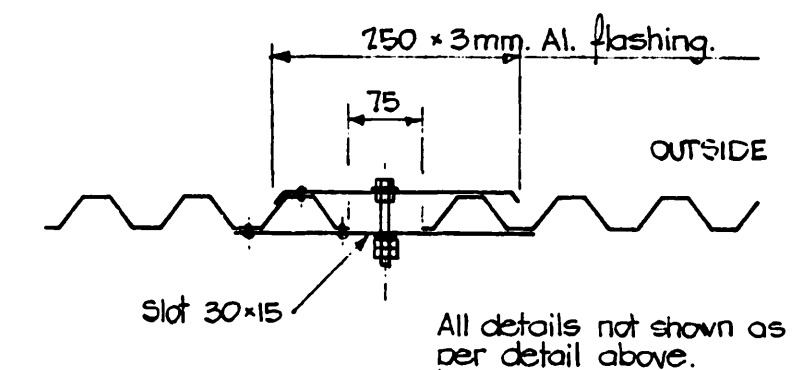
CORNERS:
Fix of every rib where noted above & adjacent
to head & tail end structures.

P.K. RIVETS :
Denotes:-Aluminium self sealing 5mm. ϕ rivets
used in conjunction with 12mm. ϕ Al. & Neo. bonded
washers. Sizes -ADG4, G6, & G8 to suit.

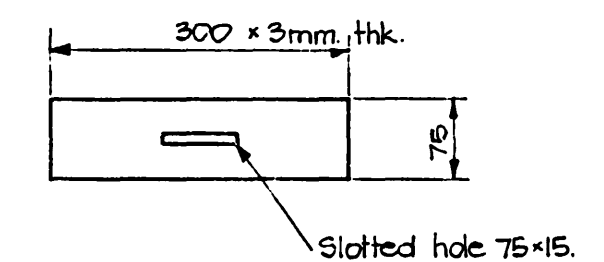
NOTE! Fibre Glass sheets to be fixed on all girts at every rib. This applies to 300 mm. on each adjacent V-Rib sheeting.



WALL SHEET. DETAIL AT EXPANSION JOINT.

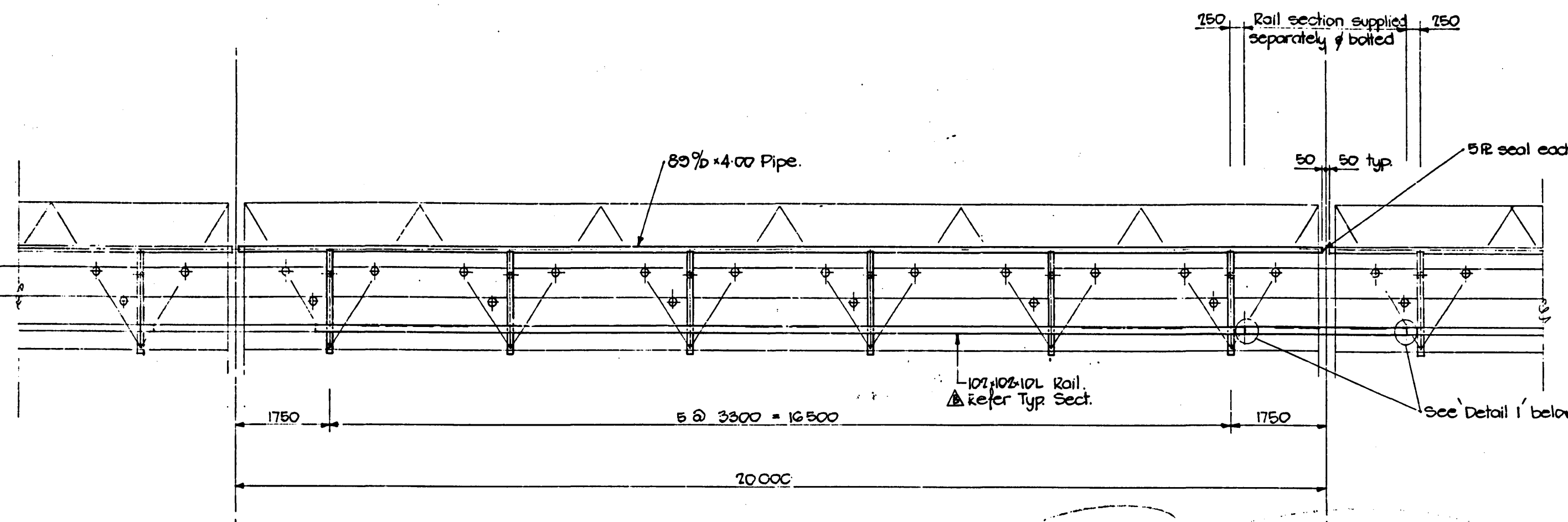


WALL SHEET. DETAIL AT STANDARD JOINT.

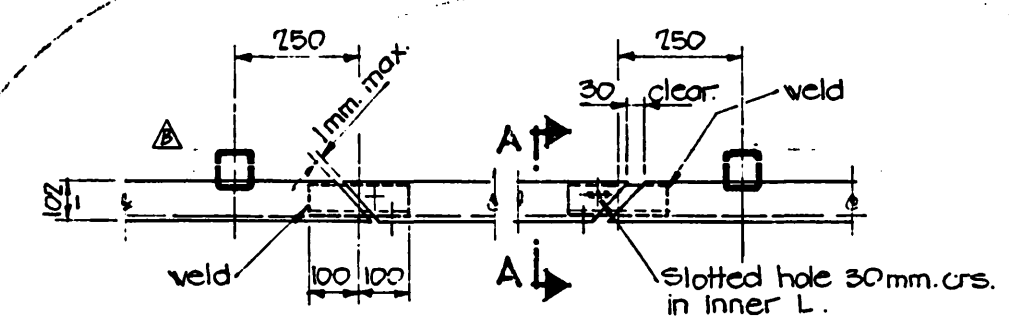


BACKING STRIP DETAIL.
Provide Backing Strip adjacent to each girt.

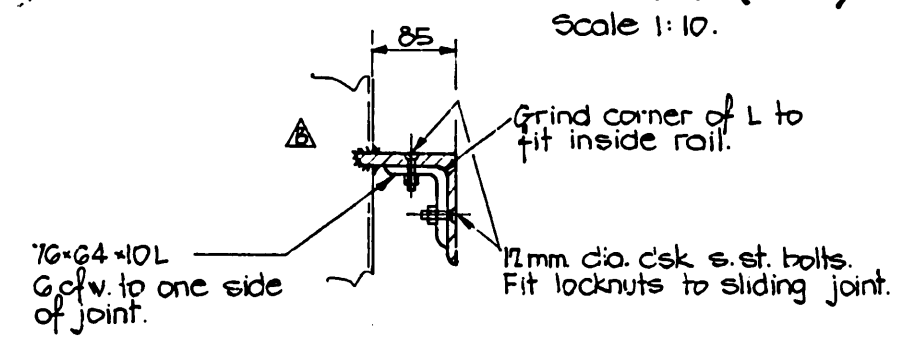
THE CORPORATION OF THE TREASURER OF QUEENSLAND									
NO.	DATE	AMENDMENTS		THE SUGAR BOARD BULK SUGAR TERMINALS					
				SCALES 1:5 DRAWN R.G. HOLMES PASSED <i>MB</i> DATE <i>AUG 76</i>	B.S.T. LUCINDA STAGE III APPROACH JETTY GALLERY FLASHING DETAILS				DRAWING NO. CTQ1122/S/021
				MACDONALD WAGNER & PRIDDLE PTY.LTD. CONSULTING ENGINEERS 312 ST. PAULS TERRACE, PORTHURST VALLEY, QLD. 4006 PH 51 7122					
C.S.R. REFERENCE NO.			IN ASSOCIATION WITH C.S.R. LIMITED						



ELEVATION
Scale: 1:50

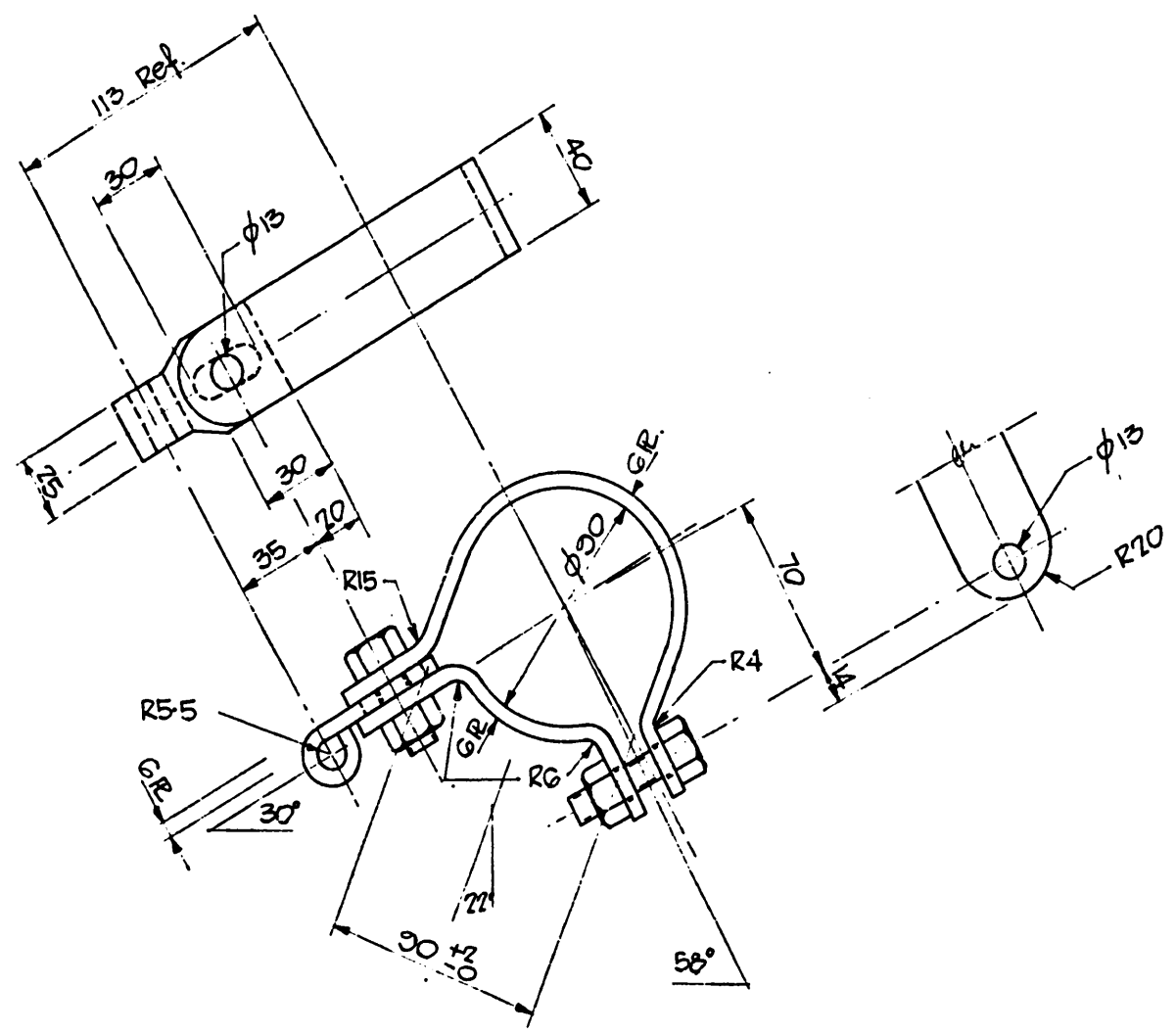


DETAIL 1 (PLAN)
Scale: 1:10.



SECTION A-A
Scale: 1:5.

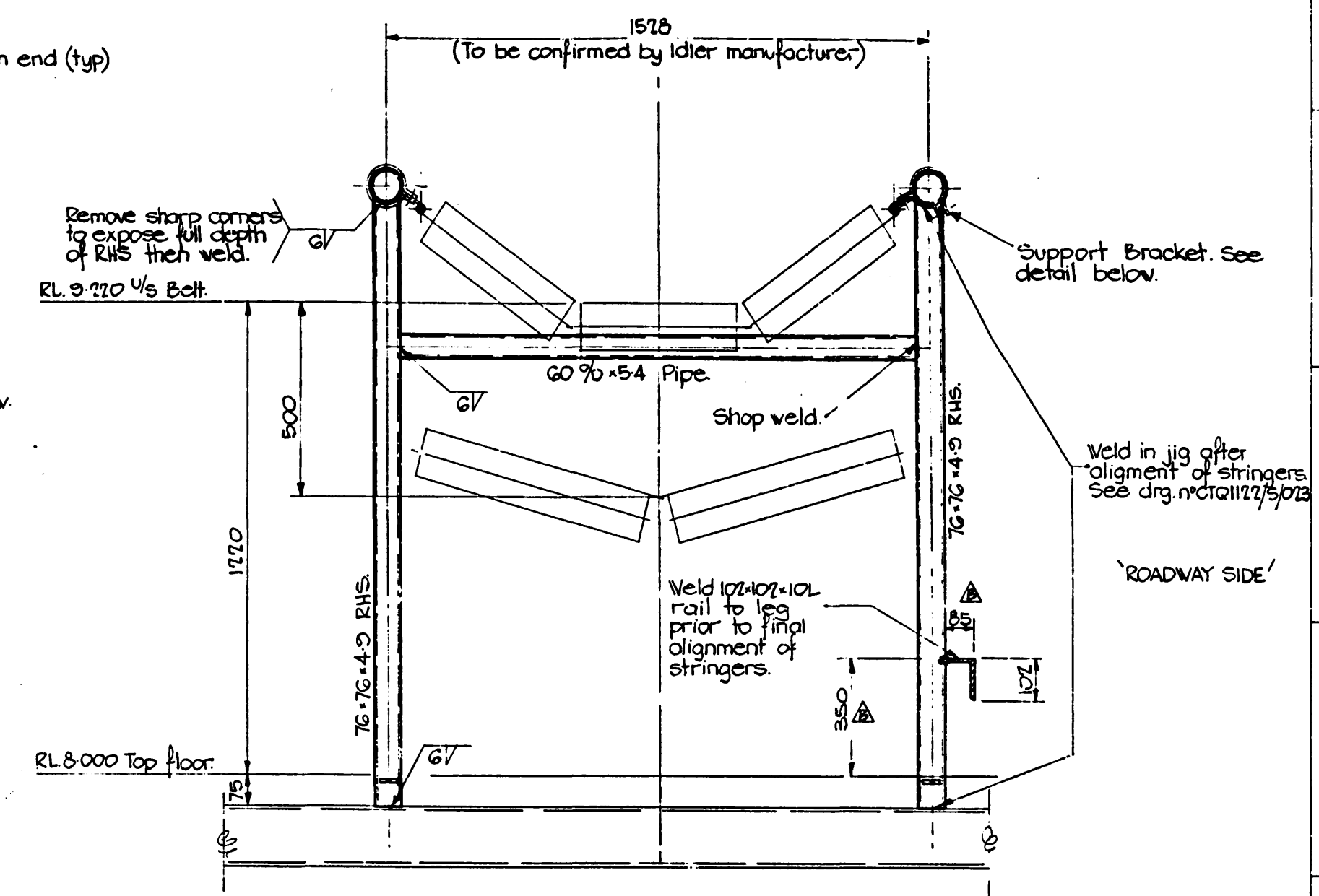
HOLD V



CARRYING IDLER SUPPORT BRACKET. SUPPLIED & FITTED UNDER NOMINATED CONTRACT NO 14C.
Scale: 1:2

Note: Dimensions to be checked for conformity with idlers.

Return idler Brackets to be supplied & welded to stringer support leg under sub-contract No 14C.



TYP. SECTION.
Scale: 1:10.

Stringer and Angle Rail by Mechanical sub-contractor.

Details as above.

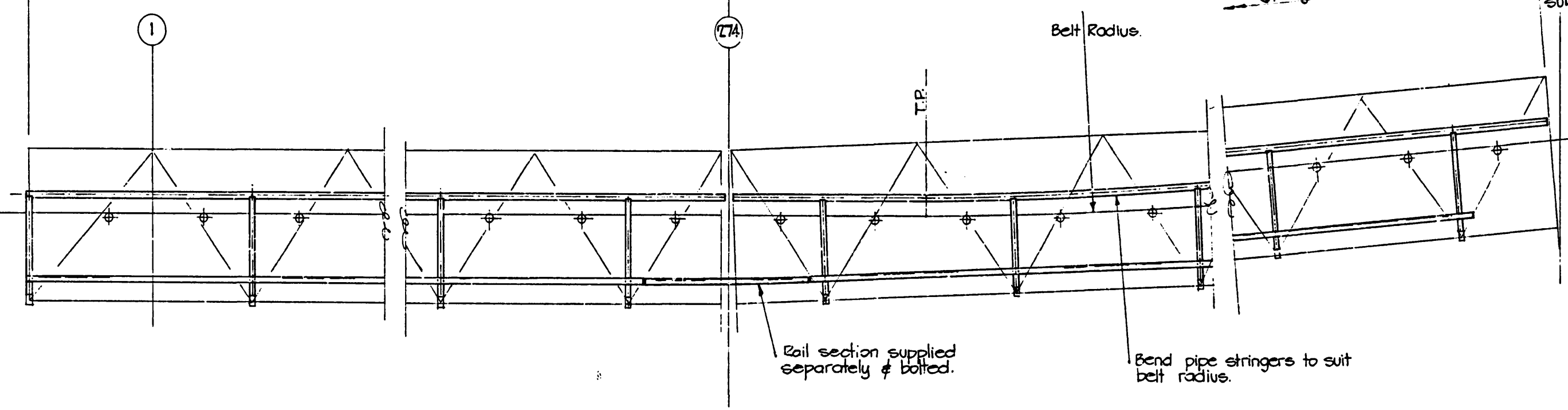
Standard spans.

Belt Radius.

Stringer details as above.

(281)

Stringers by Mechanical sub-contractor.

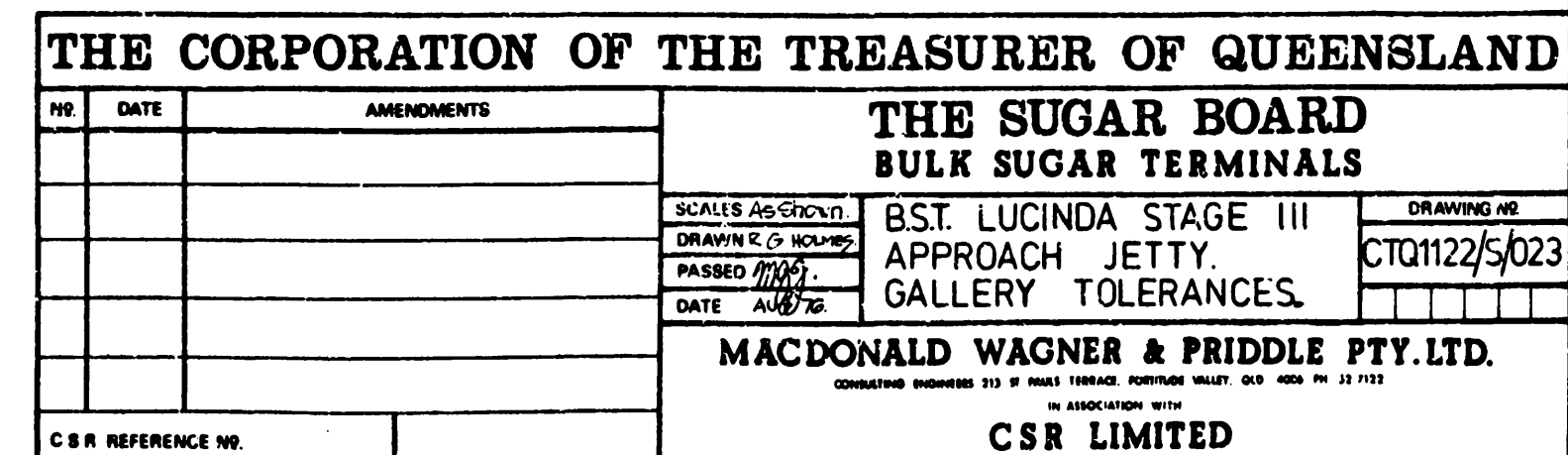


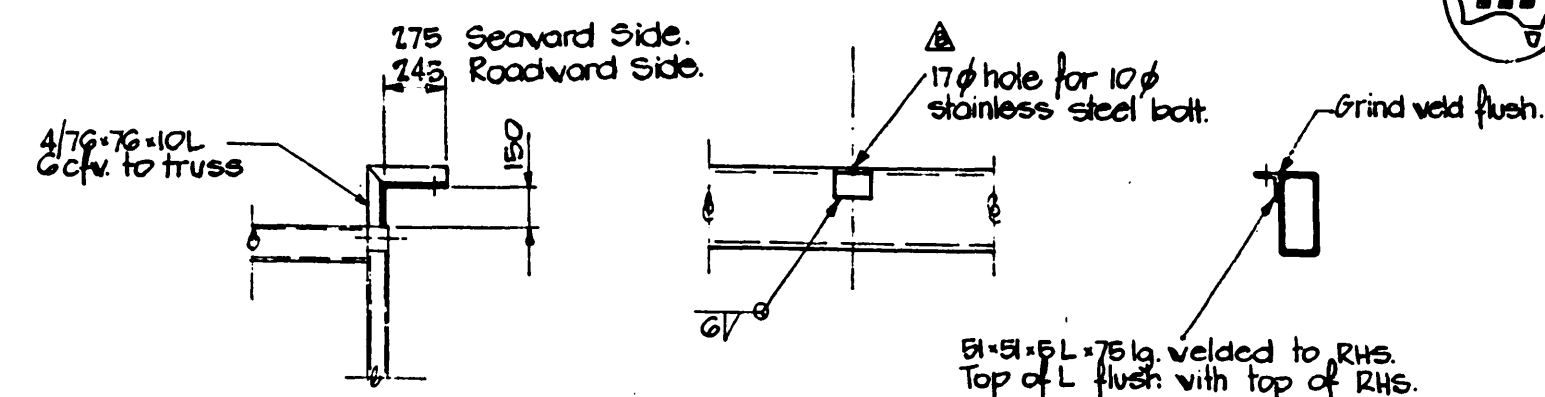
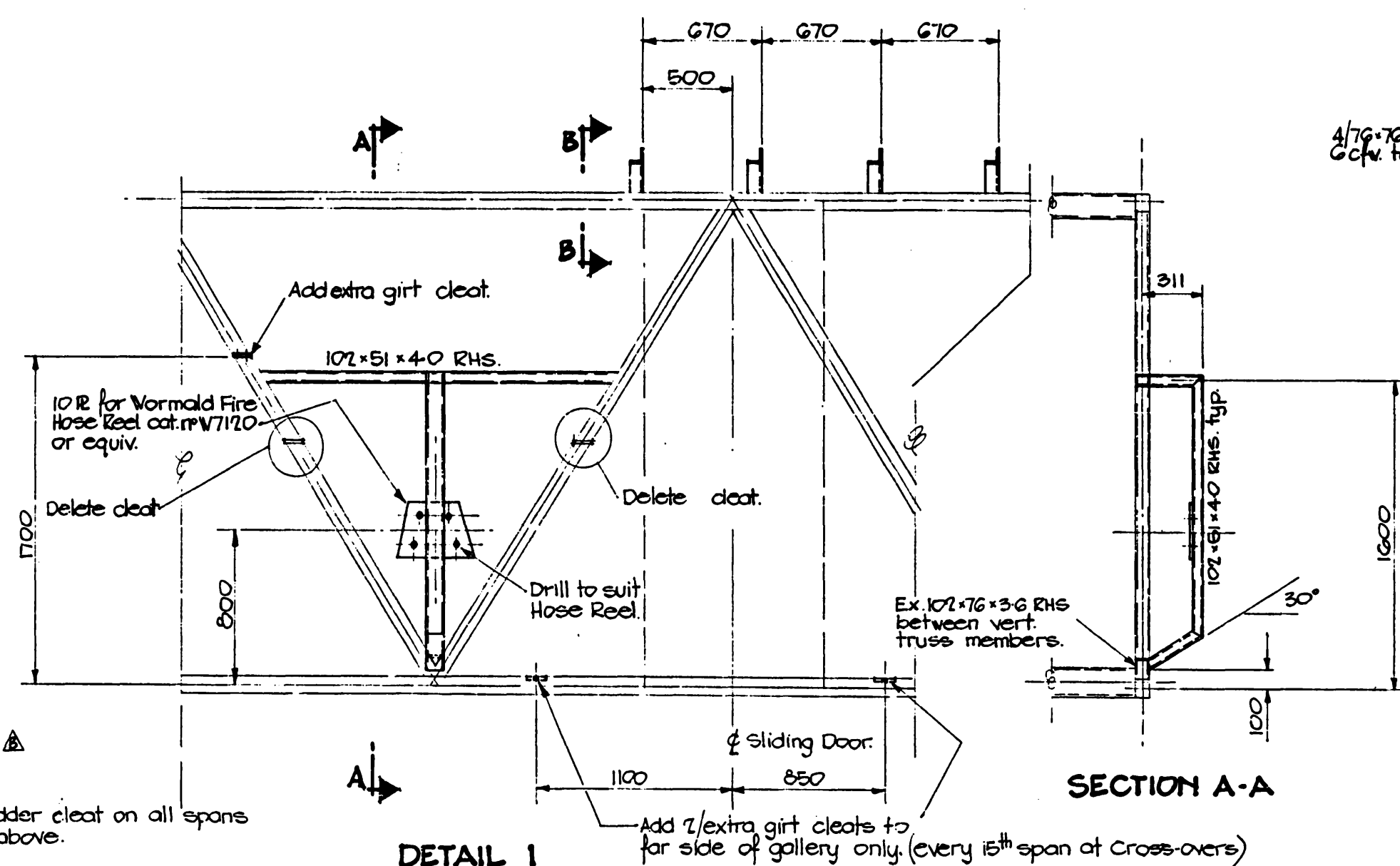
VARIATIONS TO STRINGERS
TAIL END. PILE BENT 1

VARIATIONS TO STRINGERS. PILE BENTS 174 - 281

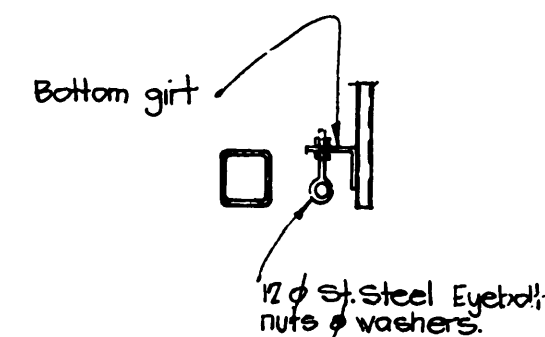
THE CORPORATION OF THE TREASURER OF QUEENSLAND									
NO.	DATE	AMENDMENTS		THE SUGAR BOARD BULK SUGAR TERMINALS					
A	29-11-76	CONTRACT NO 11C NOW READS NO 14C.		SCALES AS SHOWN.		BST. LUCINDA STAGE III APPROACH JETTY.		DRAWING NO.	
B	24-2-77	RAIL ALTERED AND EXTENDED UP INCUR.		DRAWN BY R. HOLMES.		CONV. STRINGER DETAILS.		CT01122/5/022	
				PASSED <u>M.B.</u>				X/B	
				DATE <u>AUG 86</u>					
				MACDONALD WAGNER & PRIDDLE PTY. LTD.					
				CONSULTING ENGINEERS 213 ST PAULS TERRACE PORTLAND VILLAGE QLD 4008 PH 32 7172					
				IN ASSOCIATION WITH					
CSR REFERENCE NO.				CSR LIMITED					







CABLE LADDER FIXING CLEAT.



SECTION C-C. 
ANODE SUPPORT.

DETAIL 1

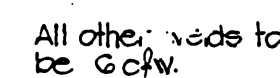
Expansion Joint

Anchored Bent

Expansion Joint

• Fixed End
○ Free End

For location of Anchored bents & expansion joints - Refer drg.no CTQ1172/S/003 & 004.

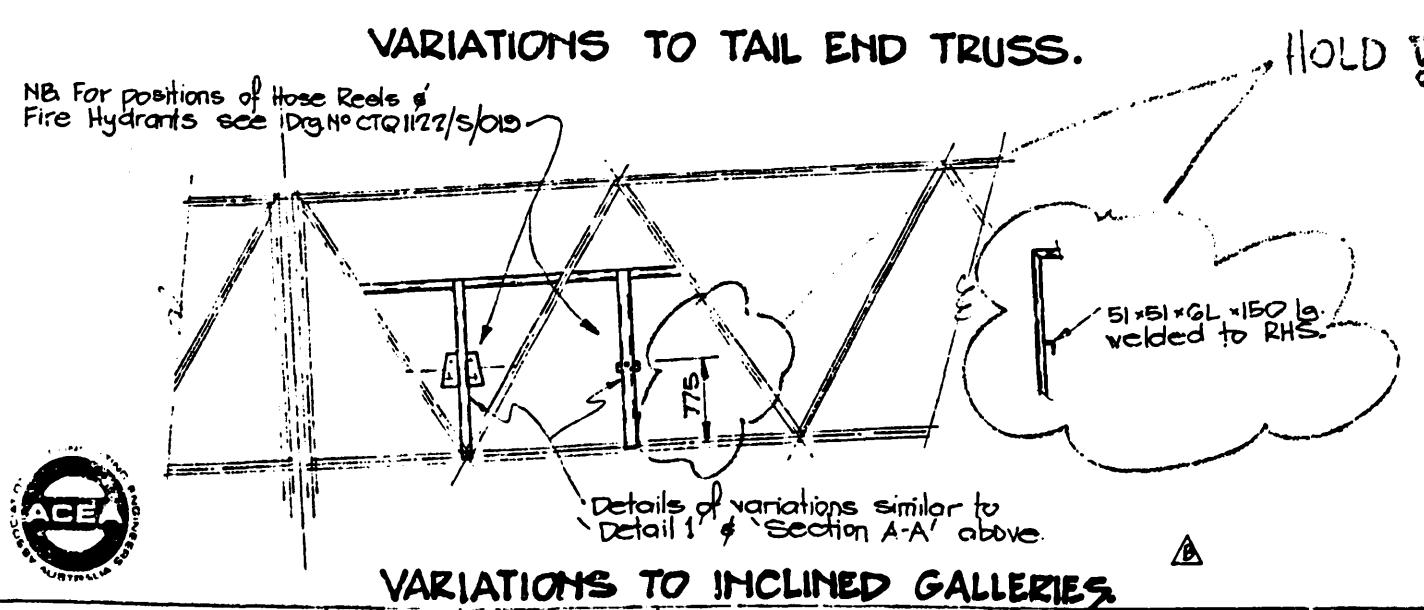


TOP CHORD

SIDE ELEVATION

BOTTOM CHORD.

NA For positions of Hose Reels &
Fire Hydrants see Drg No CTQ 1122/S/012



NOTE: All other details as per
standard truss Drg. no 0121122/5/018

1. All exposed corners to be ground smooth to a min. radius of 1mm.
2. Welding, grinding & alignment of the entire gallery section & conv. stringers to be completed prior to commencement of surface preparations & painting.
3. Final painting of structure to be completed prior to the installation of aluminium girts, purlins, floors, cladding, cable trays etc.

CHORDS.
Butt welds in all 107-107 RHS to be qualified complete penetration butt welds made over an internal backing ring. Butt welds to be located not more than 500mm from the intersection of 107-76 web diagonals & will be permitted only outside the middle third of the truss spans.

SIDE WEB MEMBERS.
Fillet welds to face of chords - Gmm. cont.
Welds to edge of chords - Detail 1 drg. no CTQ1122/5/016
Welds at intersection of web members - Detail 1 drg. no CTQ1122/5/016
Grind flush where necessary to permit assembly of top & bottom chord cross members.

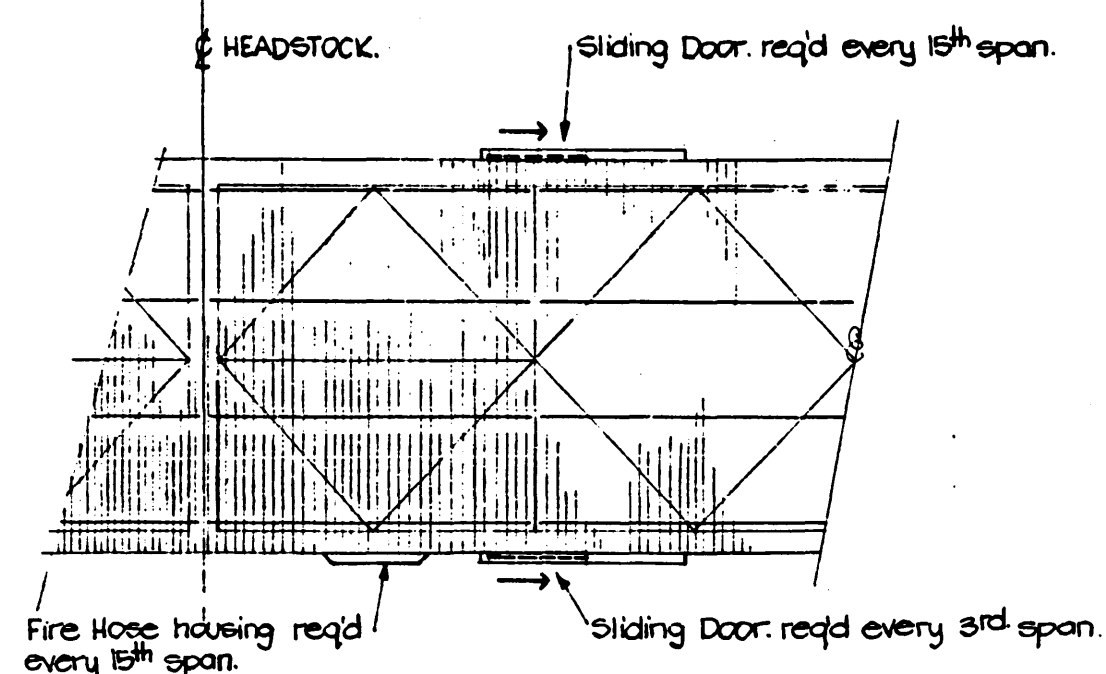
Weld to faces of side trusses - Gmm. cont. fillet.
Weld to edges of top & bottom chord - Detail 1 drg. no CTQ1177/5/013

Cont: 5mm fillet to all connections except where welds are made to the exposed thickness of members, weld size shall be limited to the member thickness.

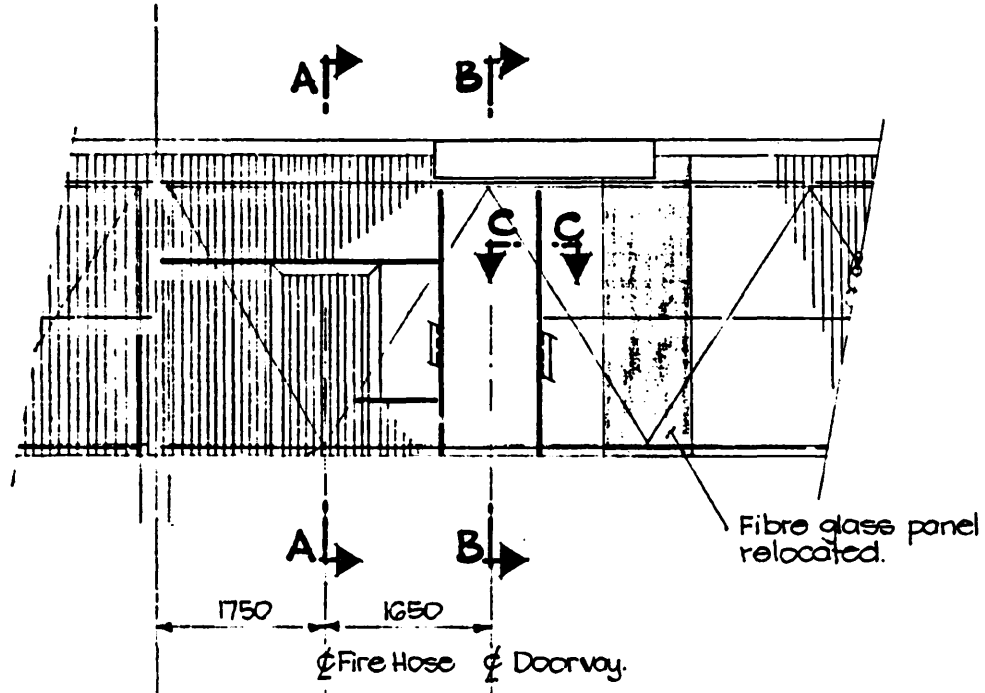
For welding of corners refer Detail 2 drg. no CTQ1122/S/018

All RHS shall conform to AS1163, Gd. 250 & shall be free from all oil or paint during fabrication. Ends of all members shall be saw cut. Incorrect fit-up of joints will subject the member to rejection by the Engineer. Alternatively the Engineer may require fillet weld sizes to be increased to compensate for incorrect fill-up at joints. Refer Specification.

THE CORPORATION OF THE TREASURER OF QUEENSLAND			THE SUGAR BOARD BULK SUGAR TERMINALS	
NO	DATE	AMENDMENTS	SCALES 1:100 1/20 DRAWN E.G. HOLMES PASSED <i>mgk</i> DATE <i>24/6/72</i>	B.S.T. LUCINDA STAGE III APPROACH JETTY. VARIATIONS TO STD. GALLERY
A	4.1.77	Diagonal brace omitted.		DRAWING NO.
B	24.2.77	HOLE DIA. ALTERED & TRA BRACKETS ADDED. VARIATIONS TO INCLUDE GALLERIES ADDED.		CTQ1122/5/02
				CTQ1122/5/02 1/8
C.S.R. REFERENCE NO.			MACDONALD WAGNER & PRIDDLE PTY.LTD. CONSULTING ENGINEERS 215 ST. PETERS TERRACE, PORTPHORO VALLEY, QLD. 4006 PH. 32 7122 IN ASSOCIATION WITH C.S.R. LIMITED	

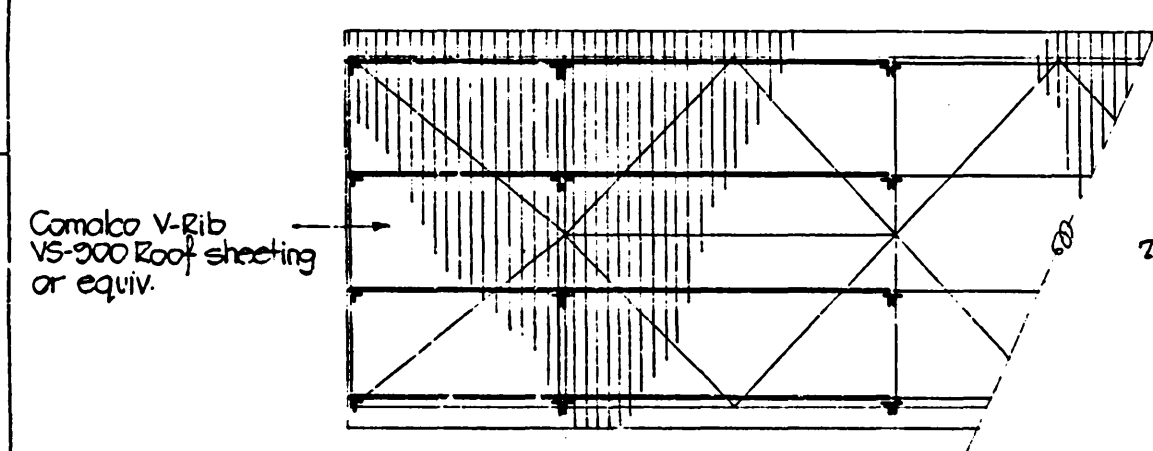


ROOF PLAN

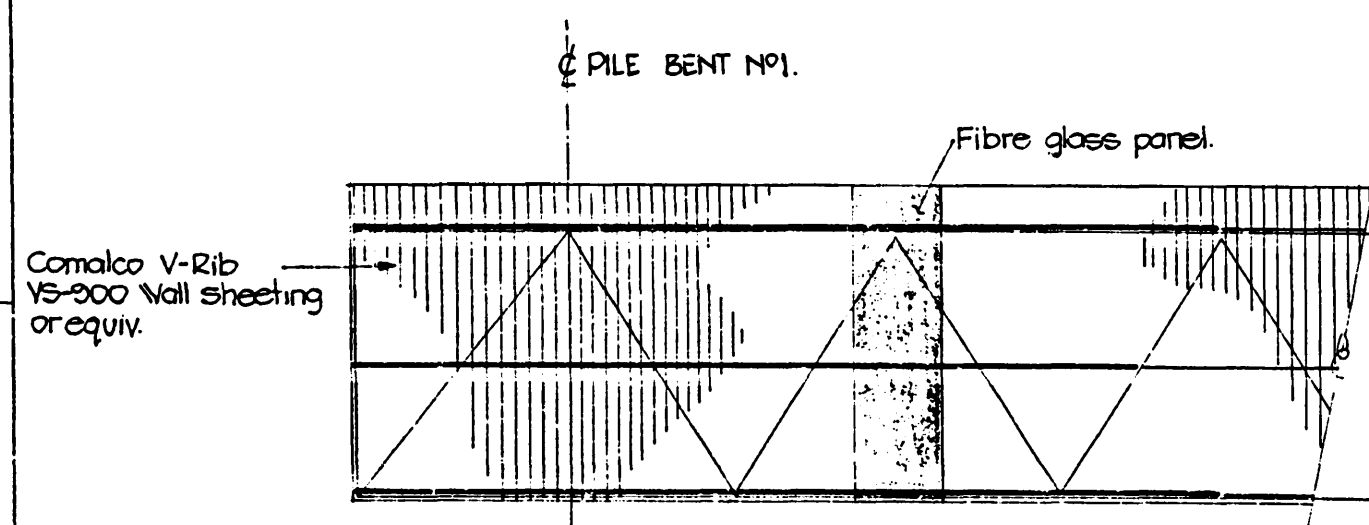


SIDE ELEVATION. (ROADWAY SIDE)

VARIATIONS TO STANDARD TRUSS.

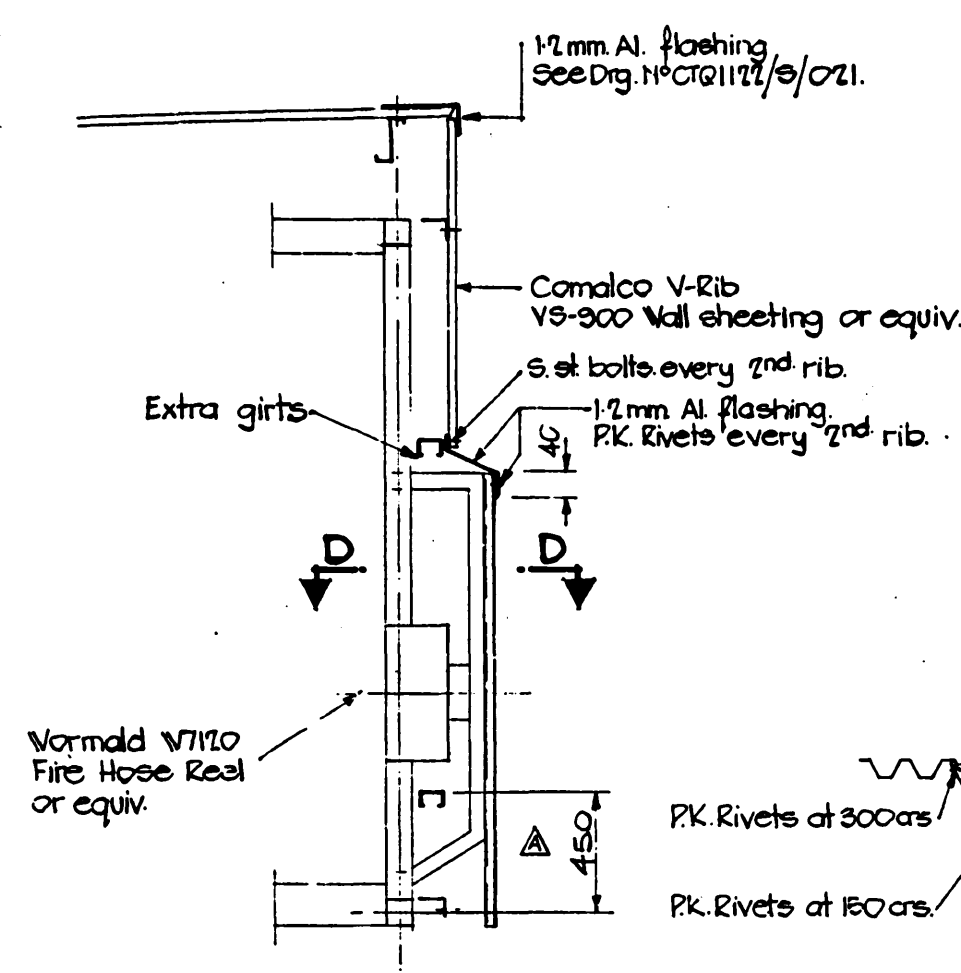


ROOF PLAN.

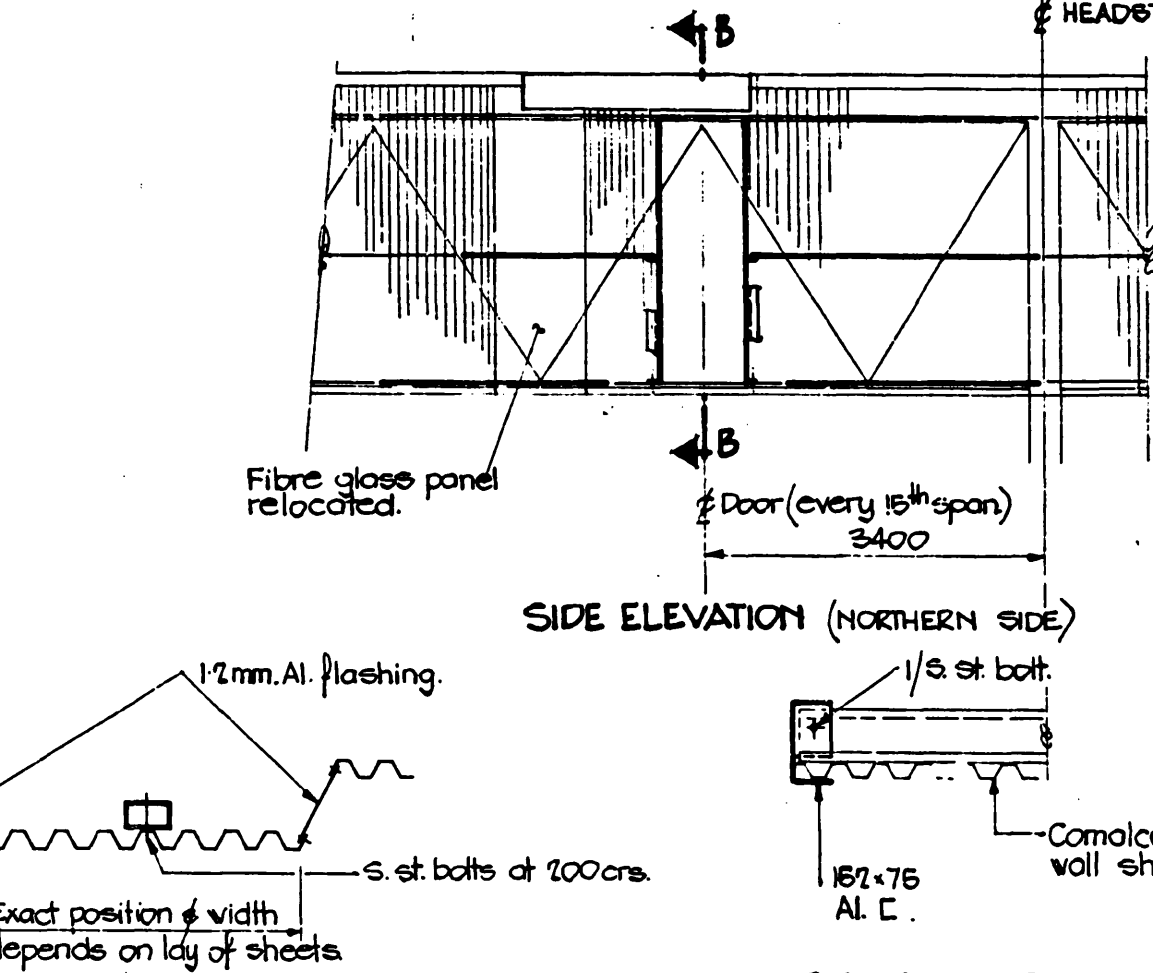


SIDE ELEVATION.

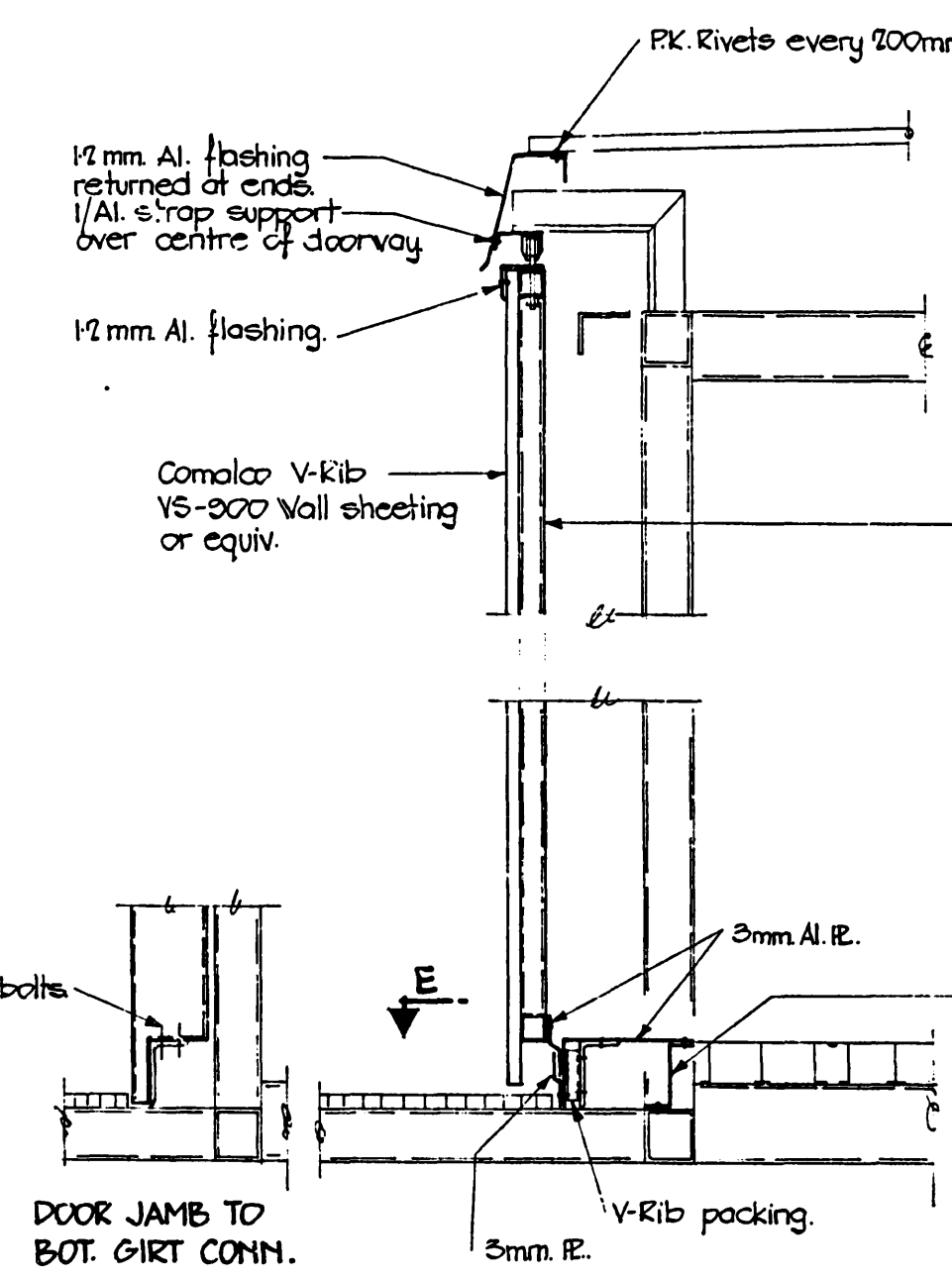
VARIATIONS TO TAIL END TRUSS.



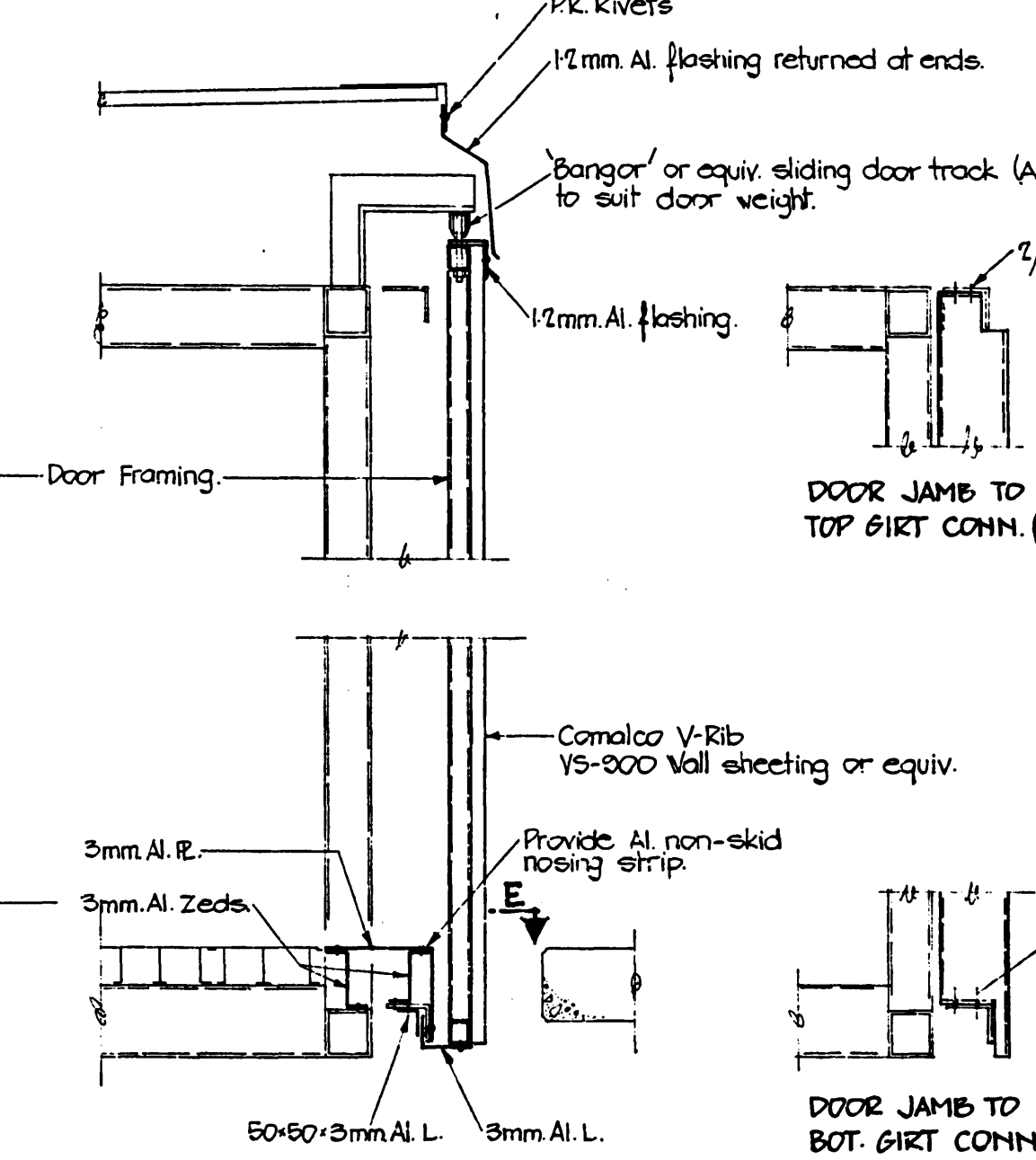
SECTION A-A



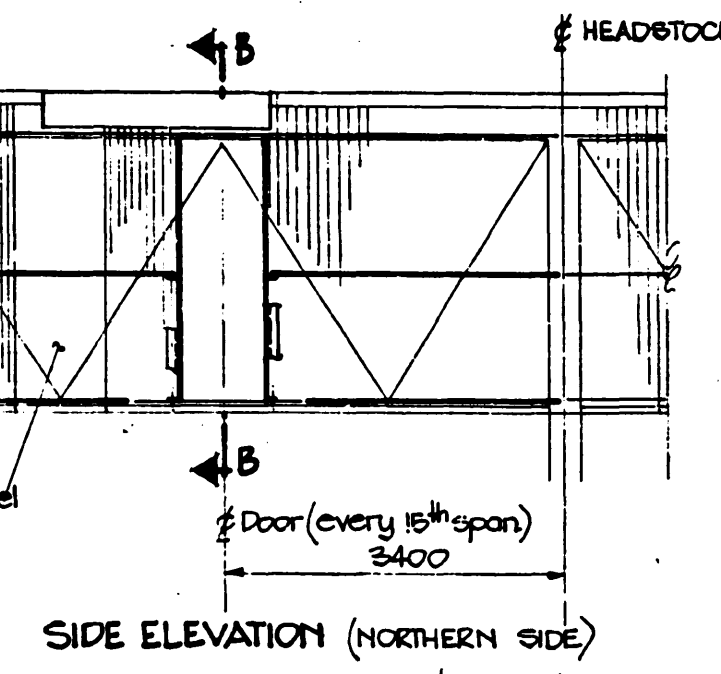
SECTION D-D.



SECTION B-B

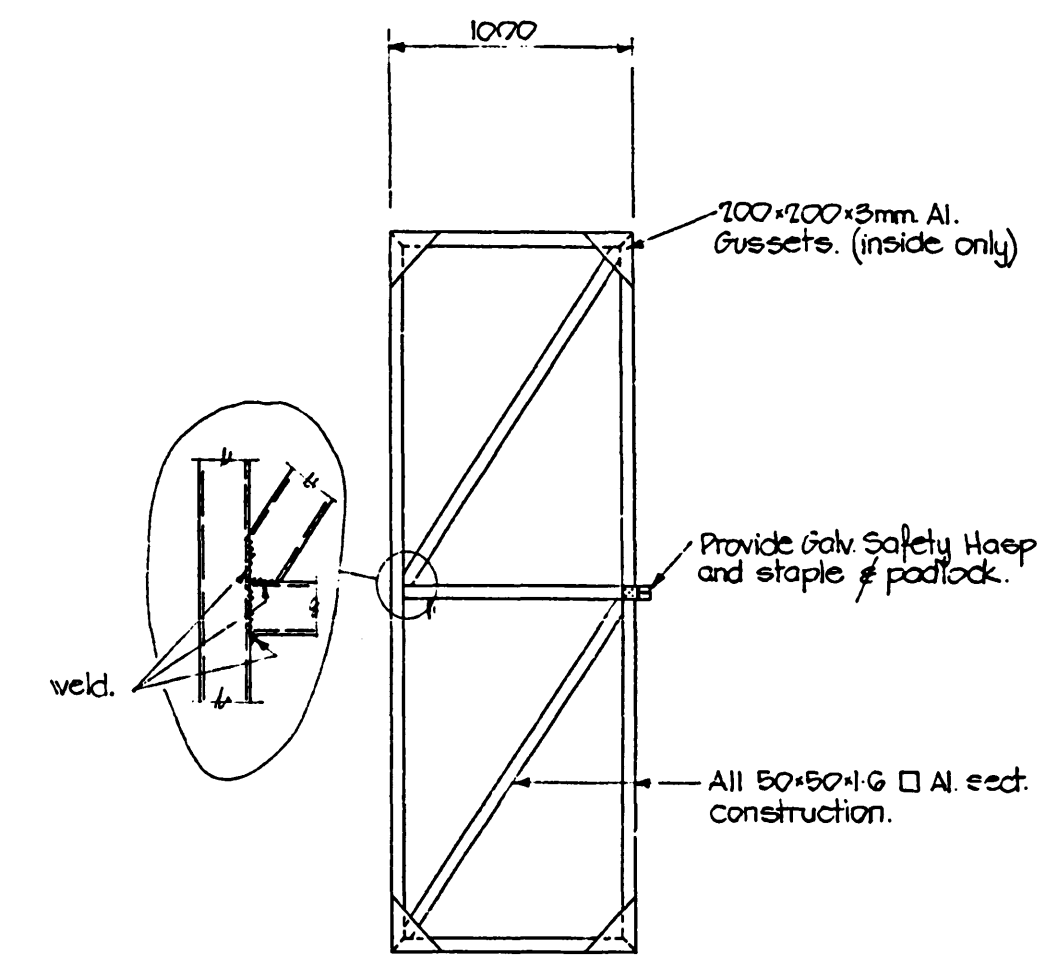


PLAN SECT. E-E.



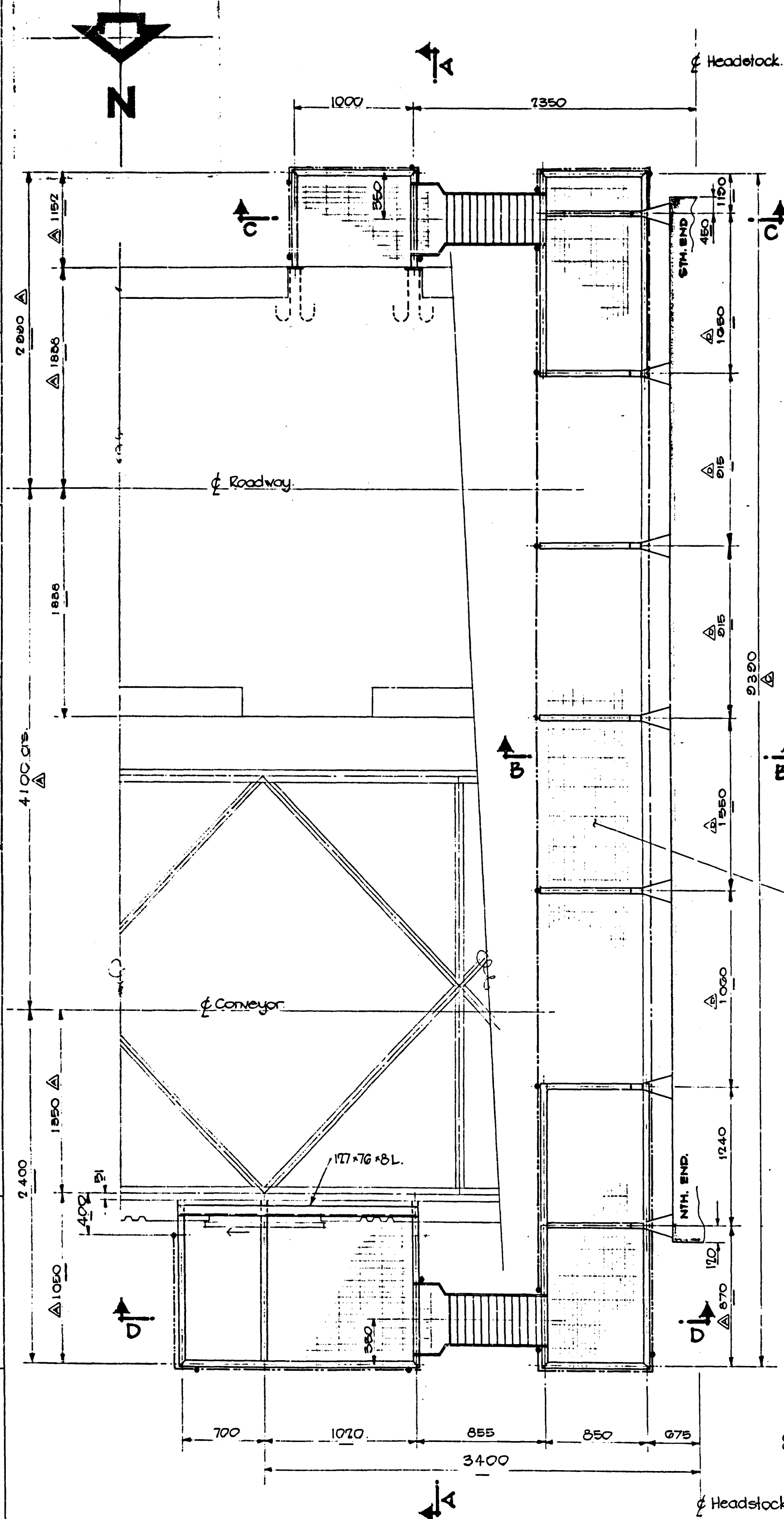
SIDE ELEVATION (NORTHERN SIDE)

SECTION C-C.



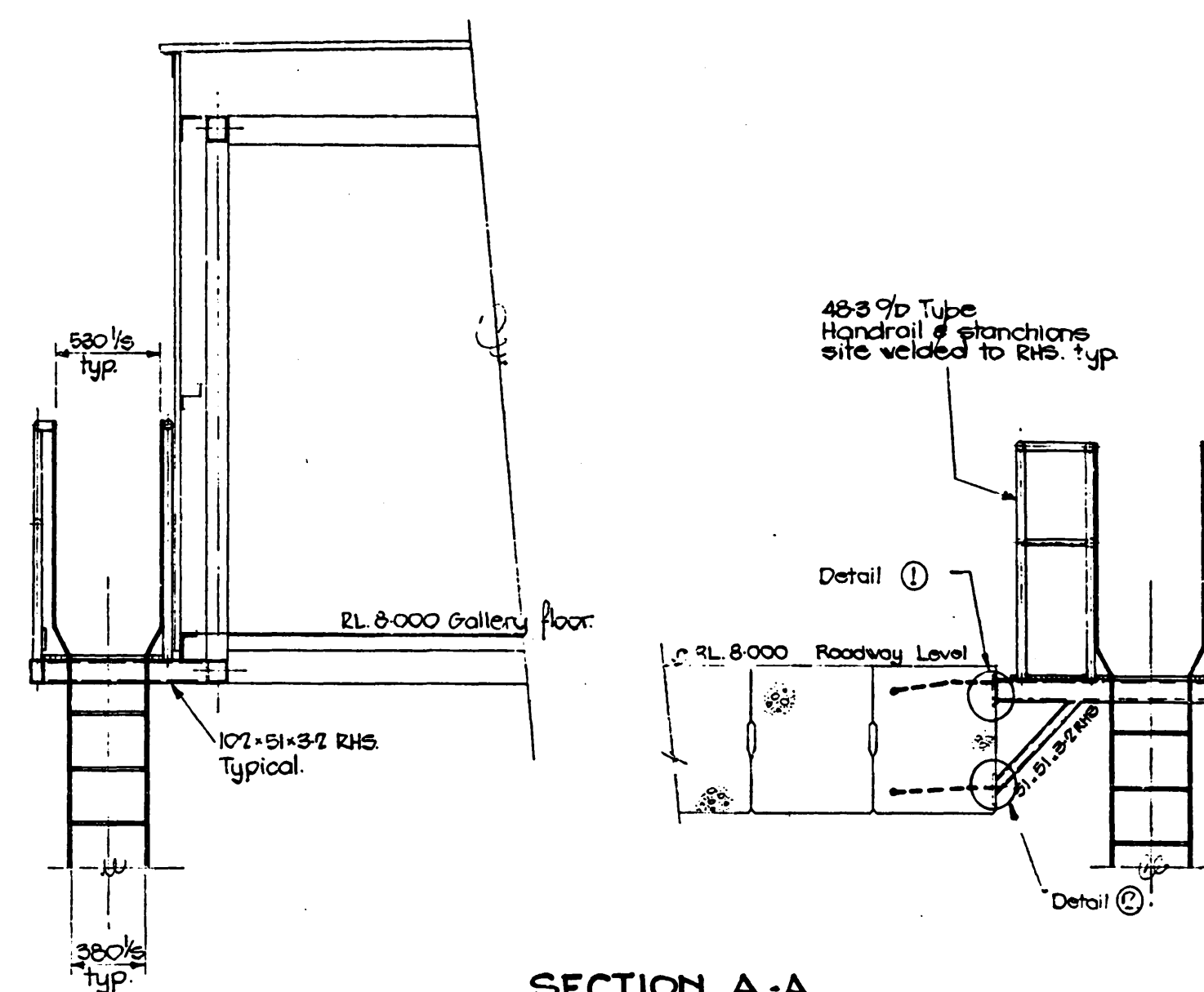
AL. SLIDING DOOR (INSIDE VIEW)

THE CORPORATION OF THE TREASURER OF QUEENSLAND									
THE SUGAR BOARD					BULK SUGAR TERMINALS				
B.S.T. LUCINDA STAGE III					APPROACH JETTY				
VARIATIONS TO CLADDING.					DRAWING NO.				
MACDONALD WAGNER & PRIDDLE PTY. LTD.					CTQ1122/5025				
CONSULTING ENGINEERS 212 ST. PETERS TERRACE PORTLAND VALLEY, QLV. 4008 PH. 32 7132					IN ASSOCIATION WITH				
C.S.R. LIMITED					C.S.R. LIMITED				

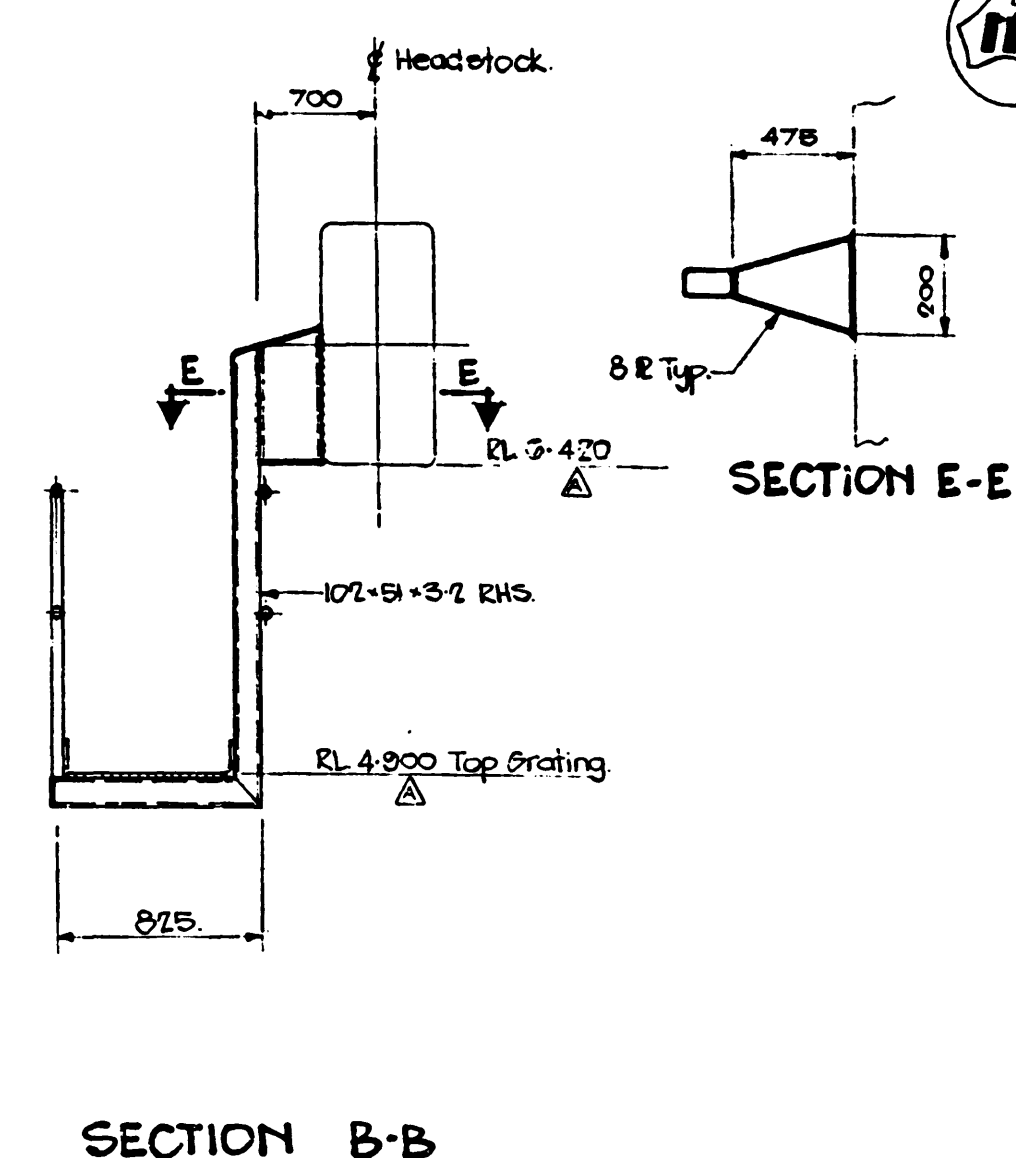


PLAN ON CROSS-OVER.

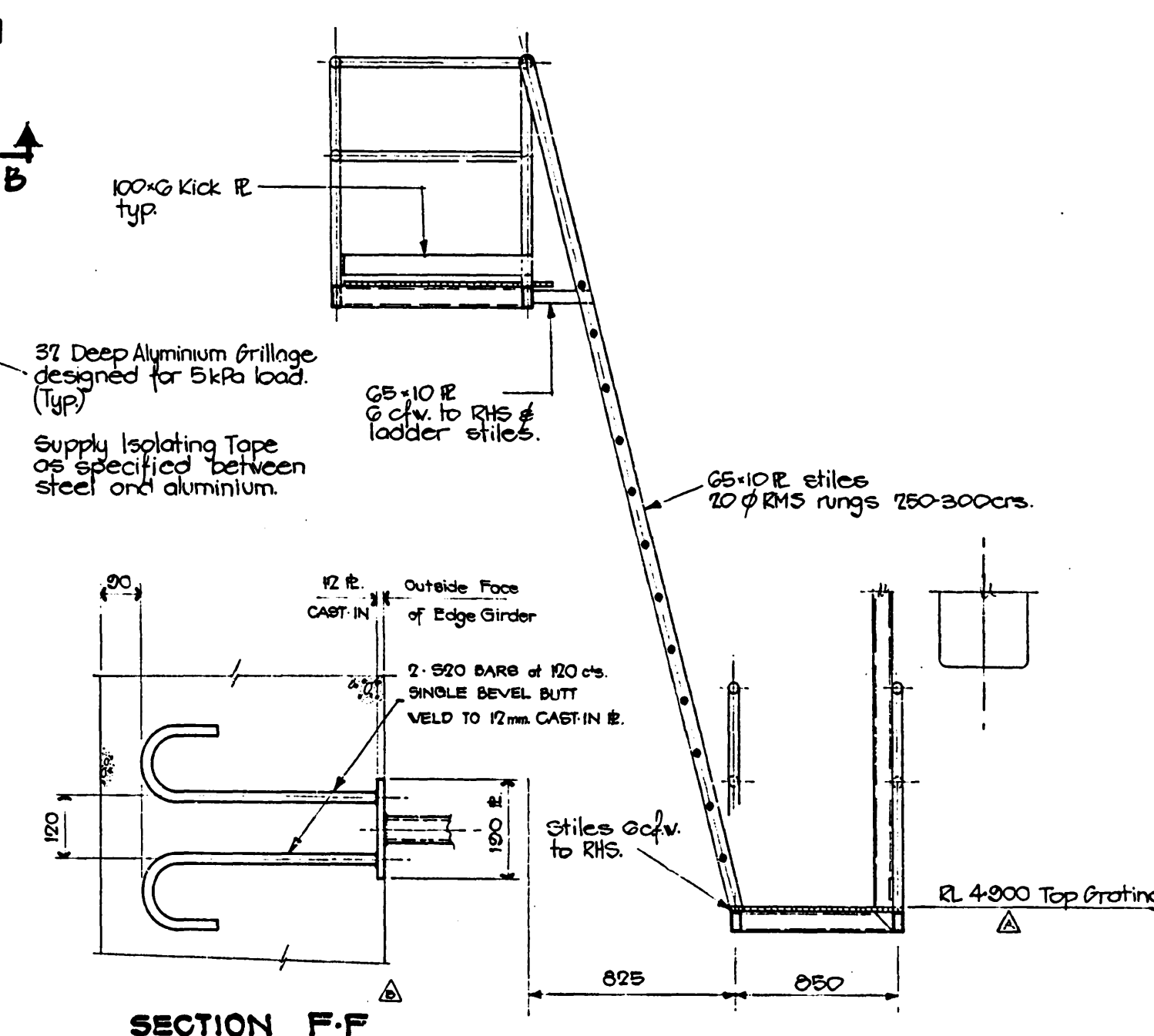
For locations refer 'Super Structure Layout Sh. i #2'.



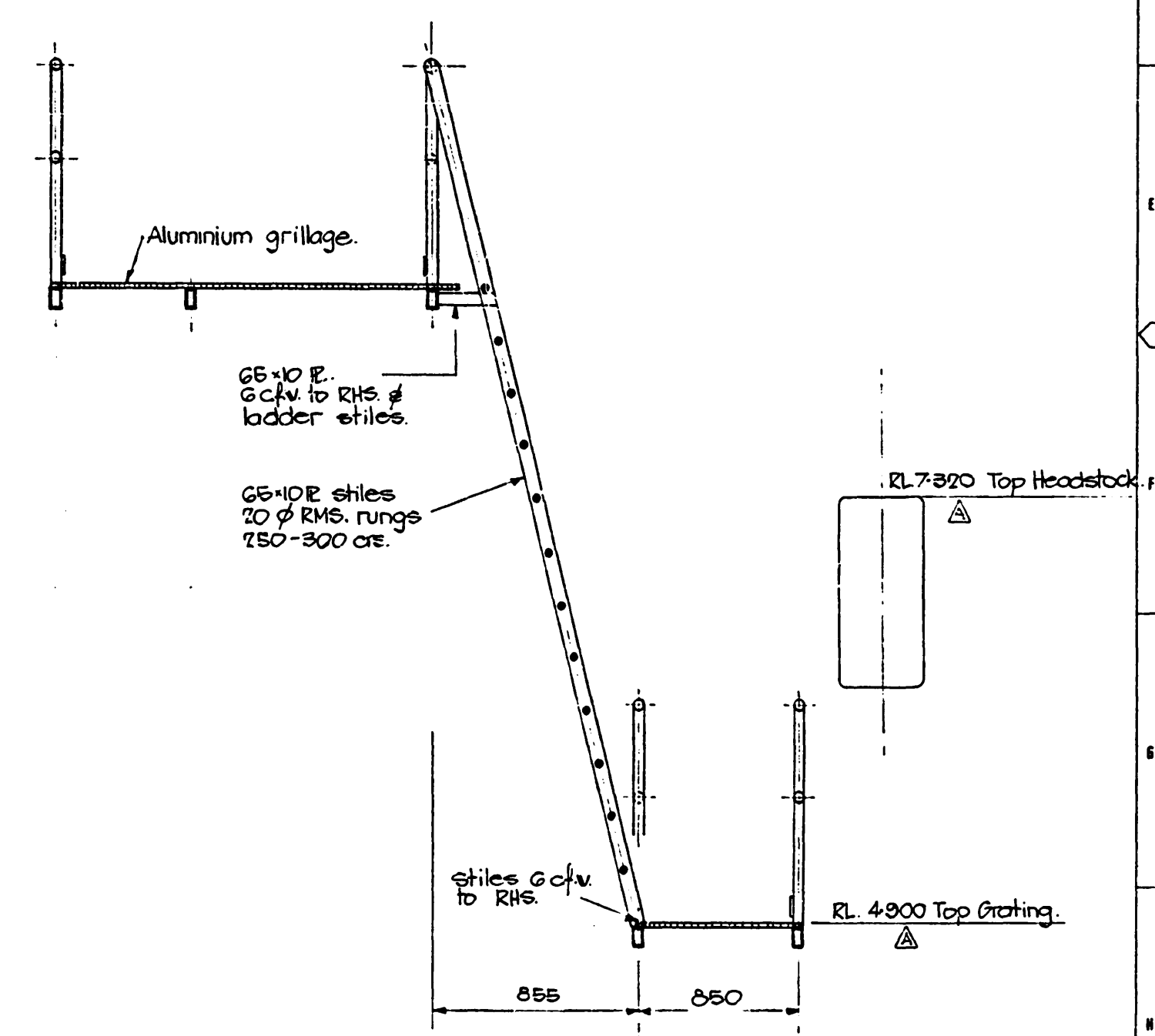
SECTION A-A



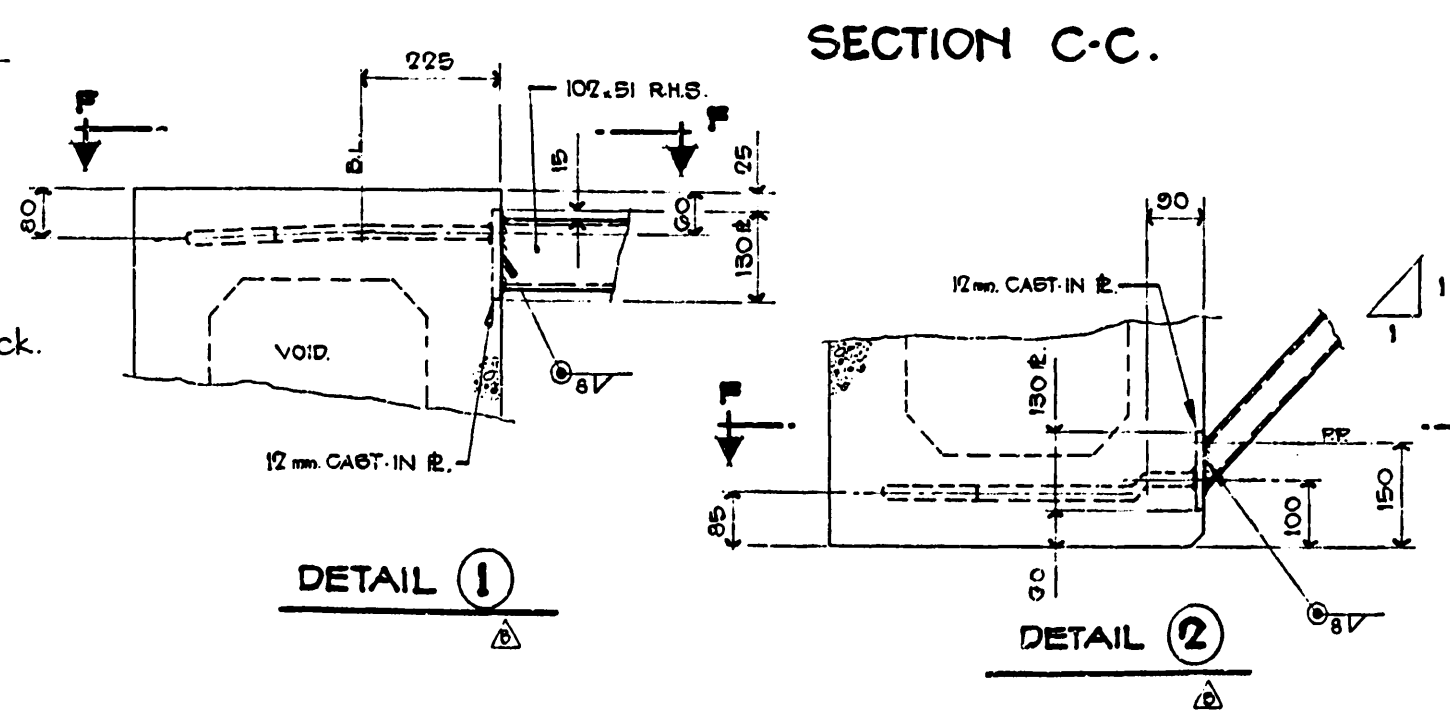
SECTION B-B



SECTION F.F



SECTION D-D

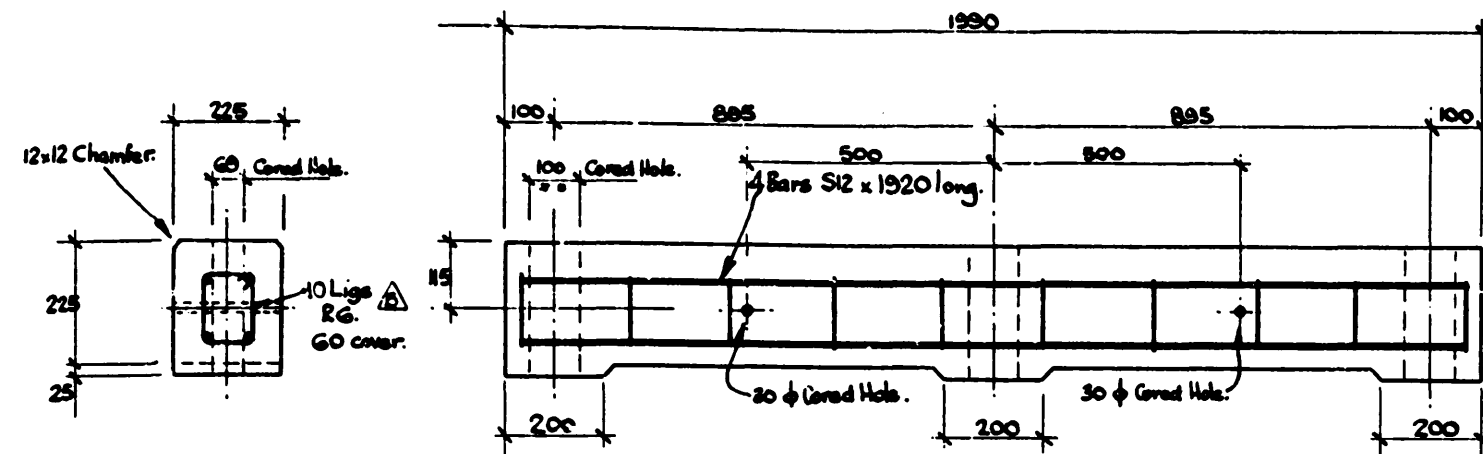


SECTION C-C.

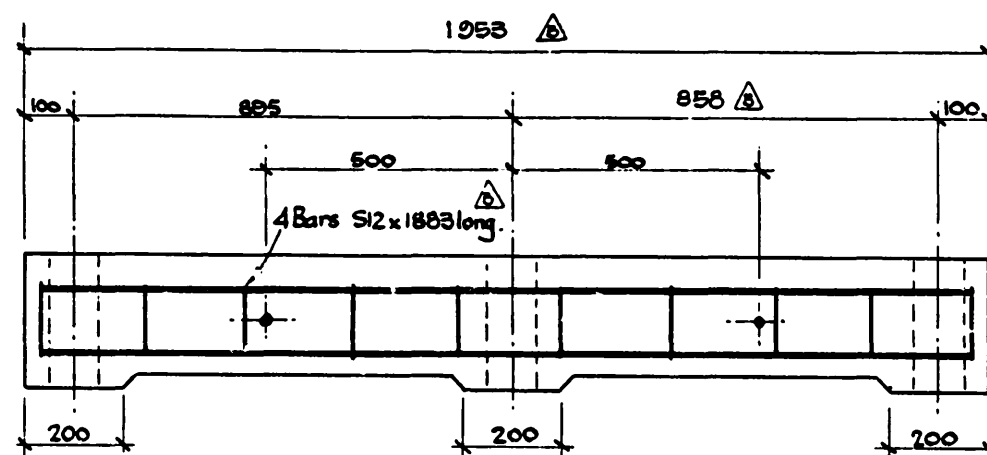
DETAIL ①

DETAIL (2)

THE CORPORATION OF THE TREASURER OF QUEENSLAND			THE SUGAR BOARD BULK SUGAR TERMINALS	
NO	DATE	AMENDMENTS	SCALES 1:70	DRAWING NO
A	20-1-77	Org. revised for conc. roadway girder design.	DRAWN R.C. HOLMES	CTM22/5026
B	2-2-77	Details 1 & 2 amended for Cast in E. with Tendons. Section FF added.	PASSED MAY	
C	26-5-77	9/A DIMENSION ALTERED.	DATE AUG 76	APPROACH JETTY CROSS-OVER DETAILS.
D	25-10-77	Walkway Hangers Relocated.	MACDONALD WAGNER & PRIDDLE PTY.LTD. CONSULTING ENGINEERS 215 ST PHILAS TERRACE, PORTLAND VICT. 3207 AUSA PH 52 7121 C.S.R. LIMITED	
E	27-10-77	Walkway Hangers Extended to clear Piles at extreme toleranced position.		
C.S.R. REFERENCE NO.				

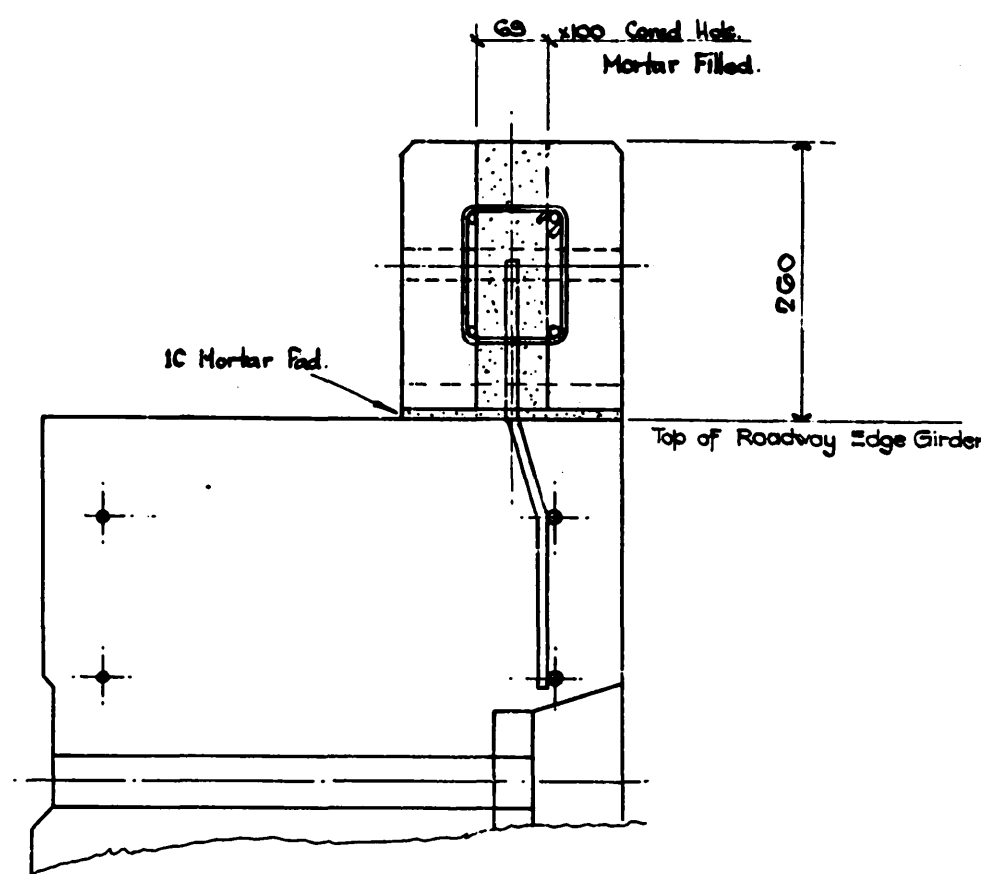


1:10.

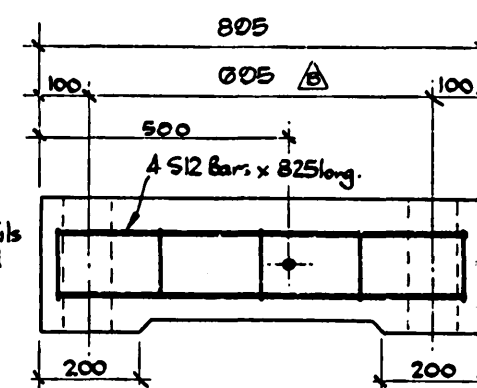


1:10.

Mark this end
'TO EXPANSION JOINT.'

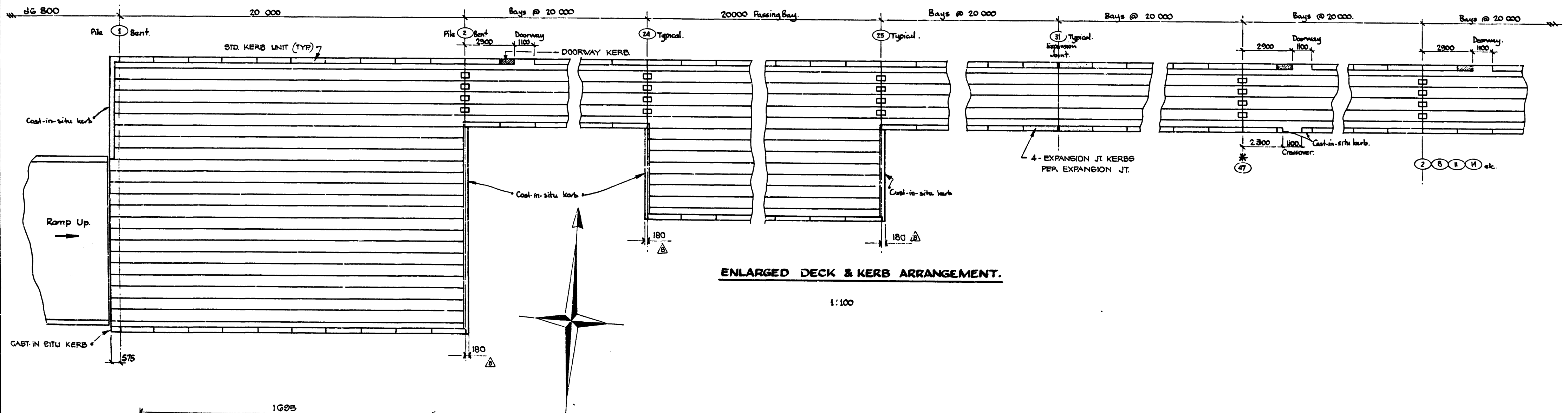


1:5.

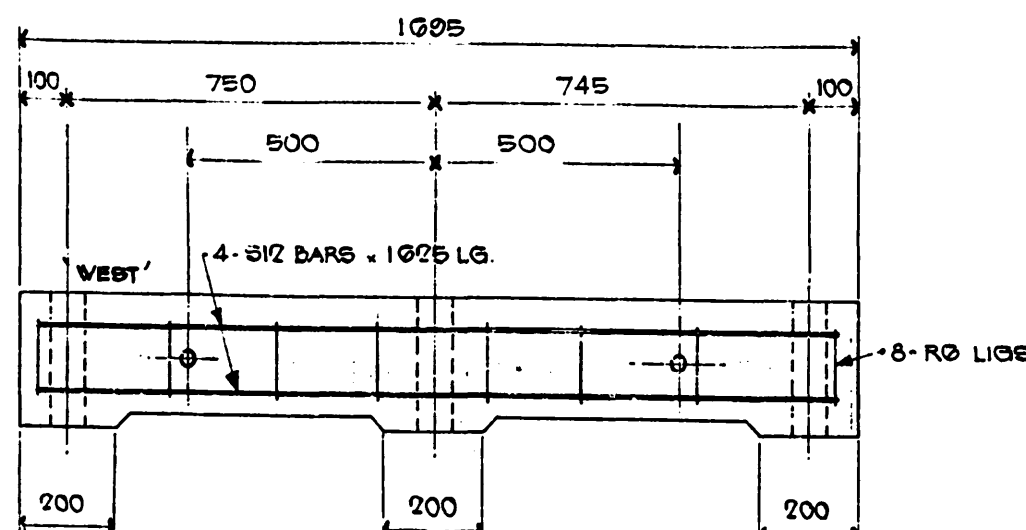


1:10

Remaining details
as for standard
kerb unit.



1:100



FOR FILE BENT Nº 284 ONLY.

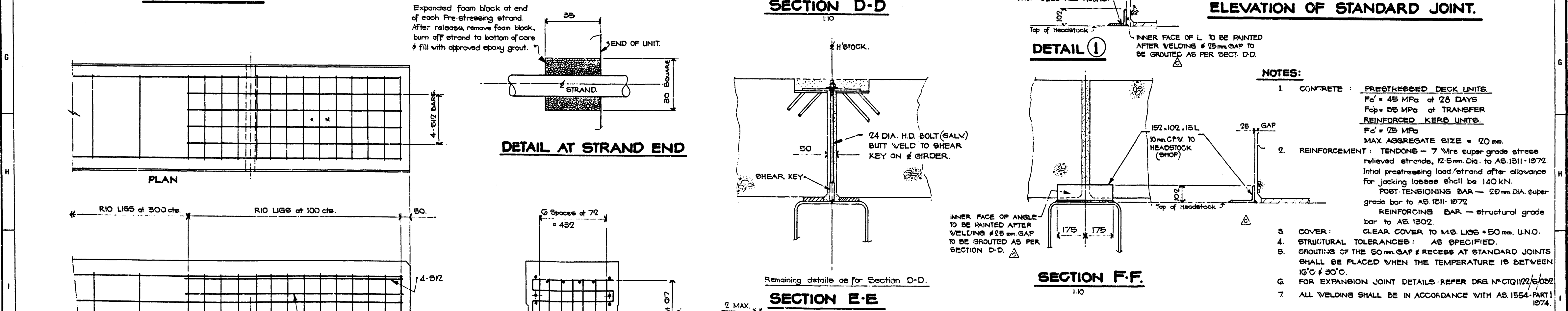
1:10

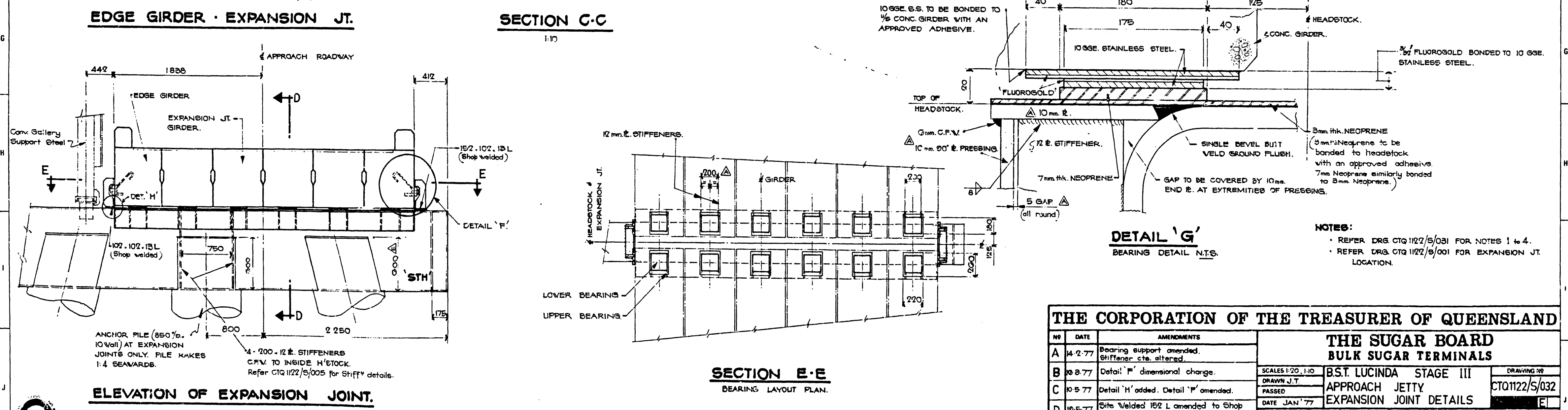
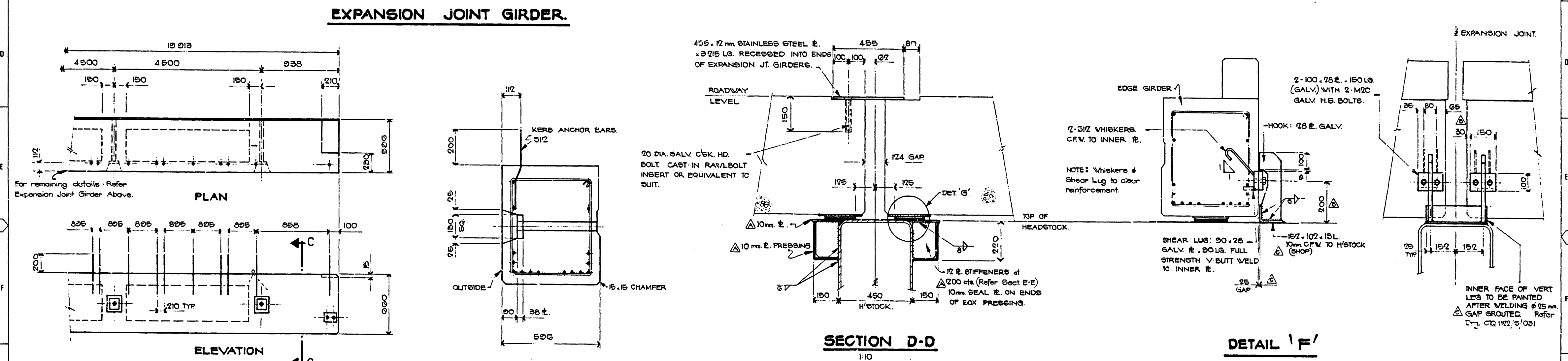
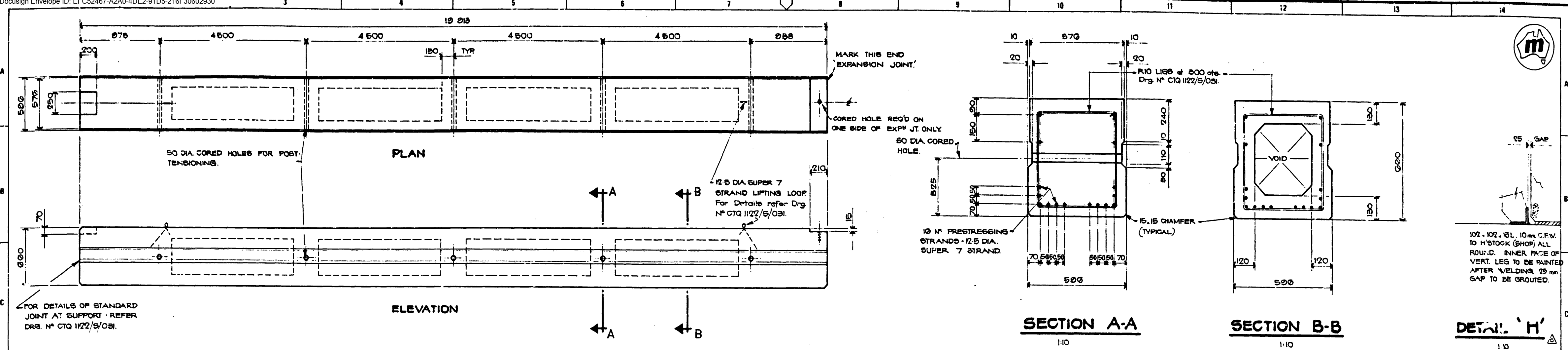
Remaining details as
for standard kerb unit.

1. CONCRETE: Reinforced kerb Units.
 $F_c' = 25 \text{ MPa}$. Max. agg. size = 20 mm.
2. FOR LAYOUT OF APPROACH JETTY - REFER
DRGS. CTG/1172/3/003 & 004.

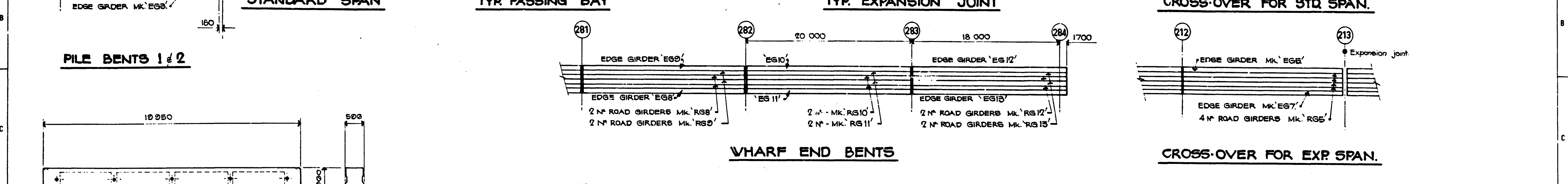
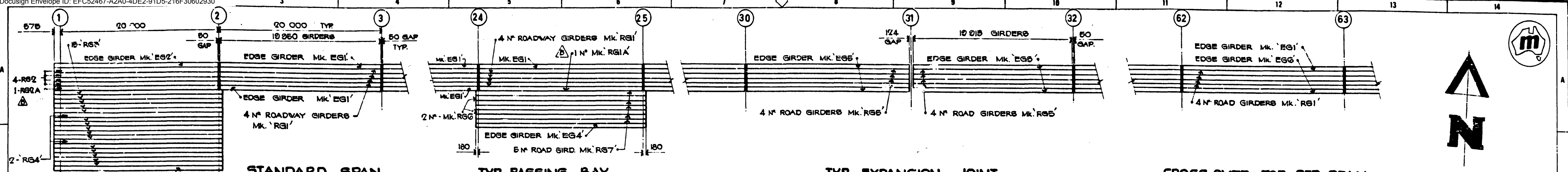
THE CORPORATION OF THE TREASURER OF QUEENSLAND			
NR.	DATE	AMENDMENTS	
A	27-11-76	SAY NOS 18 & 19 NOW EXPANSION TYPE.	THE SUGAR BOARD BULK SUGAR TERMINALS SCALES <i>As Shown</i> B.S.T. LUCINDA - STAGE III DRAWN <i>Whealey</i> OFFSHORE LOADING FACILITY. PASSED <i>Whealey</i> ALTERNATE PRECAST ROADWAY. DATE <i>28-11</i> GENERAL ARRANGEMENT. MACDONALD WAGNER & PRIDDLE PTY. LTD. CONSULTING ENGINEERS 215 W. PALES TERRACE, PORTLAND VALLEY, QLD 4006 PH 52 7121 IN ASSOCIATION WITH CSR LIMITED
B	16-12-77	Drawing amended.	
CSR REFERENCE NO.			DRAWING I/R CTA1122/5/030 B



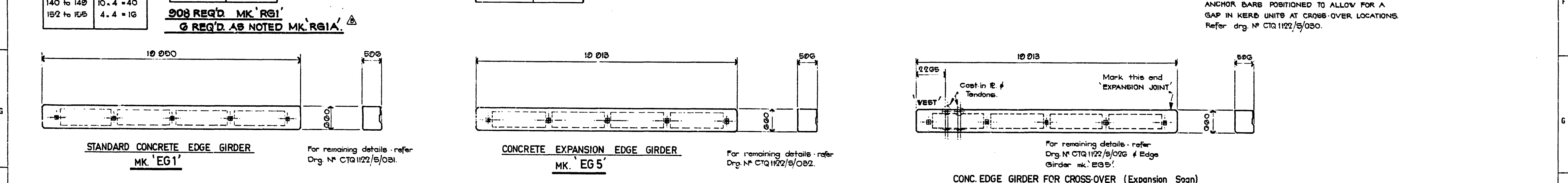




THE CORPORATION OF THE TREASURER OF QUEENSLAND					THE SUGAR BOARD		BULK SUGAR TERMINALS	
NO	DATE	AMENDMENTS			B.S.T. LUCINDA STAGE III		DRAWING NO.	
A	14-2-77	Bearing support amended. Stiffener cts. altered.			APPROACH JETTY		CTQ1122/5/032	
B	10-3-77	Detail 'F' dimensional change.			EXPANSION JOINT DETAILS		E	
C	10-5-77	Detail 'H' added. Detail 'F' amended.			MACDONALD WAGNER & PRIDDLE PTY. LTD.		CONSULTING ENGINEERS 213 ST PAULS TERRACE, PORTHURST VALLEY QLD 4006 PH 33 7122	
D	10-5-77	Site Welded 152 L amended to Shop welded to suit Contractor.			IN ASSOCIATION WITH		CSR LIMITED	
E	10-8-77	Pile Height amended on headstock.			CSR REFERENCE NO.			



File Bent No.	No. Req'd.	File Bent No.	No. Req'd.	File Bent No.	No. Req'd.	File Bent No.	No. Req'd.
2 to 5	4.4 = 16	156 (PB)	1.5 = 5	168, 180	2.4 = 8	17	1
8 to 17	10.4 = 40	157 to 161	5.4 = 20	186, 187	2.4 = 8	32	1
20 to 23	4.4 = 16	164 to 171	8.4 = 32	204, 205	2.4 = 8	47	1
24 (PB)	1.5 = 5	174 to 179	6.4 = 24	212, 213	2.4 = 8	62	1
25 to 29	5.4 = 20	182 to 187	6.4 = 24	220, 221	2.4 = 8	77	1
32 to 41	10.4 = 40	190 to 195	6.4 = 24	230, 231	2.4 = 8	92	1
44 to 53	10.4 = 40	196 & 199	2.4 = 8	242, 243	2.4 = 8	107	1
56 to 59	4.4 = 16	200 (PB)	1.5 = 5	254, 255	2.4 = 8	122	1
60 (PB)	1.5 = 5	201 to 203	3.4 = 12	266, 267	2.4 = 8	137	1
61 to 65	5.4 = 20	206 to 211	6.4 = 24	278, 280	2.4 = 8	152	1
68 to 77	10.4 = 40	214 to 219	6.4 = 24	150, 151	2.4 = 8		
80 to 89	10.4 = 40	222 to 229	8.4 = 32				
92 to 101	10.4 = 40	232 to 241	10.4 = 40				
104 to 107	4.4 = 16	244 to 247	4.4 = 16				
108 (PB)	1.5 = 5	248 (PB)	1.5 = 5				
109 to 113	5.4 = 20	249 to 253	5.4 = 20				
116 to 125	10.4 = 40	256 to 258	10.4 = 40				
128 to 137	10.4 = 40	268 to 278	11.4 = 44				
140 to 149	10.4 = 40						
152 to 155	4.4 = 16						



File Bent No.	No. Req'd.	File Bent No.	No. Req'd.	File Bent No.	No. Req'd.	File Bent No.	No. Req'd.
2 to 5	4.2 = 8	116 to 121	6.2 = 12	227 (co)	1.1 = 1	186, 187	2.2 = 4
8 to 10	3.2 = 12	122 (co)	1.1 = 1	228 & 229	2.2 = 4	196, 197	2.2 = 4
17 (co)	1.1 = 1	123 to 125	3.2 = 6	232 to 241	10.2 = 20	204, 205	2.2 = 4
20 to 23	4.2 = 8	128 to 136	3.2 = 12	244 to 247	4.2 = 8	212	1.1 = 1
24 (PB)	1.1 = 1	137 (co)	1.1 = 1	248 (PB)	1.1 = 1	220, 221	2.2 = 4
25 to 29	5.2 = 10	140 to 149	10.2 = 20	249 to 253	5.2 = 10	230, 231	2.2 = 4
32 (co)	1.1 = 1	152 (co)	1.1 = 1	256	1.2 = 2	242	1.1 = 1
33 to 41	10.2 = 18	153 to 155	3.2 = 6	257 (co)	1.1 = 1	254, 255	2.2 = 4
44 to 46	3.2 = 6	156 (PB)	1.1 = 1	258 to 265	8.2 = 16	266, 267	2.2 = 4
47 (co)	1.1 = 1	157 to 161	5.2 = 10	268 to 271	4.2 = 8	278, 280	2.2 = 4
48 to 53	5.2 = 12	164 to 166	3.2 = 6	272 (co)	1.1 = 1	213	1.2 = 2
56 to 59	4.2 = 8	167 (co)	1.1 = 1	273 to 275	6.2 = 12	243	1.2 = 2
60 (PB)	1.1 = 1	168 to 171	4.2 = 8			150, 151	2.2 = 4
61	1.2 = 2	174 to 179	6.2 = 12				
62 (co)	1.1 = 1	182 (co)	1.1 = 1				
63 to 65	3.2 = 6	183 to 187	5.2 = 10				
68 to 70	2.2 = 18	190 to 195	4.2 = 8				
77 (co)	1.1 = 1	196	1.1 = 1				
80 to 89	10.2 = 20	197	1.2 = 2				
92 (co)	1.1 = 1	198 to 199	2.2 = 4				
93 to 101	9.2 = 18	200 (PB)	1.1 = 1				
104 to 106	3.2 = 6	201 to 203	3.2 = 6				
107 (co)	1.1 = 1	206 to 211	6.2 = 12				
108 (PB)	1.1 = 1	214 to 219	6.2 = 12				
109 to 113	5.2 = 10	222 to 226	5.2 = 10				

NOTES:

- FOR ALL REINFORCEMENT DETAILS - REFER DRG. N° CTQ 1122/5/031
- FOR GENERAL LAYOUT OF APPROACH JETTY - REFER DRG. N° CTQ 1122/5/003 & 004.
- FOR REMAINING GIRDER DETAILS - REFER SHEET 2 - DRG. N° CTQ 1122/5/034.

THE CORPORATION OF THE TREASURER OF QUEENSLAND

THE SUGAR BOARD

BULK SUGAR TERMINALS

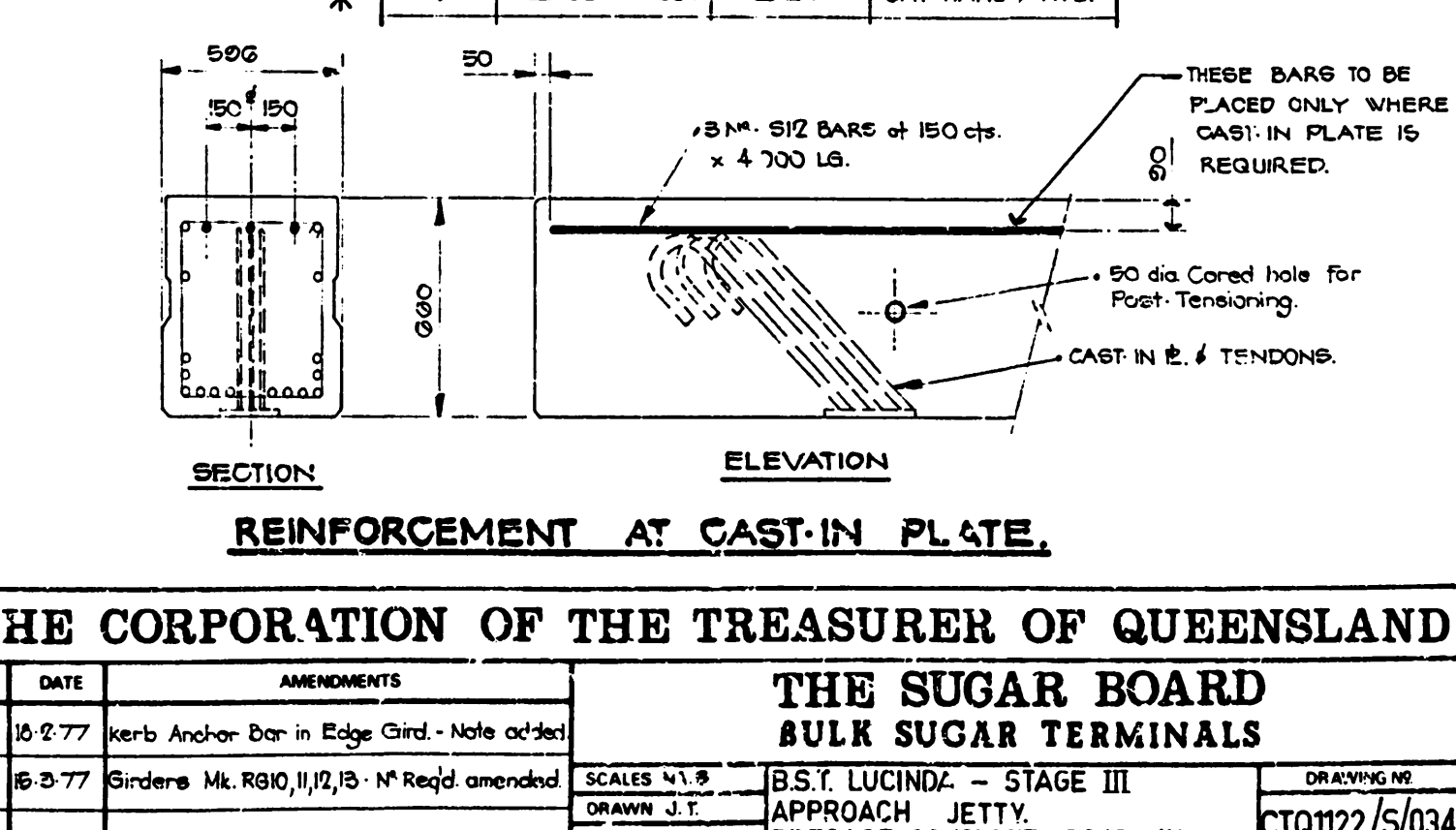
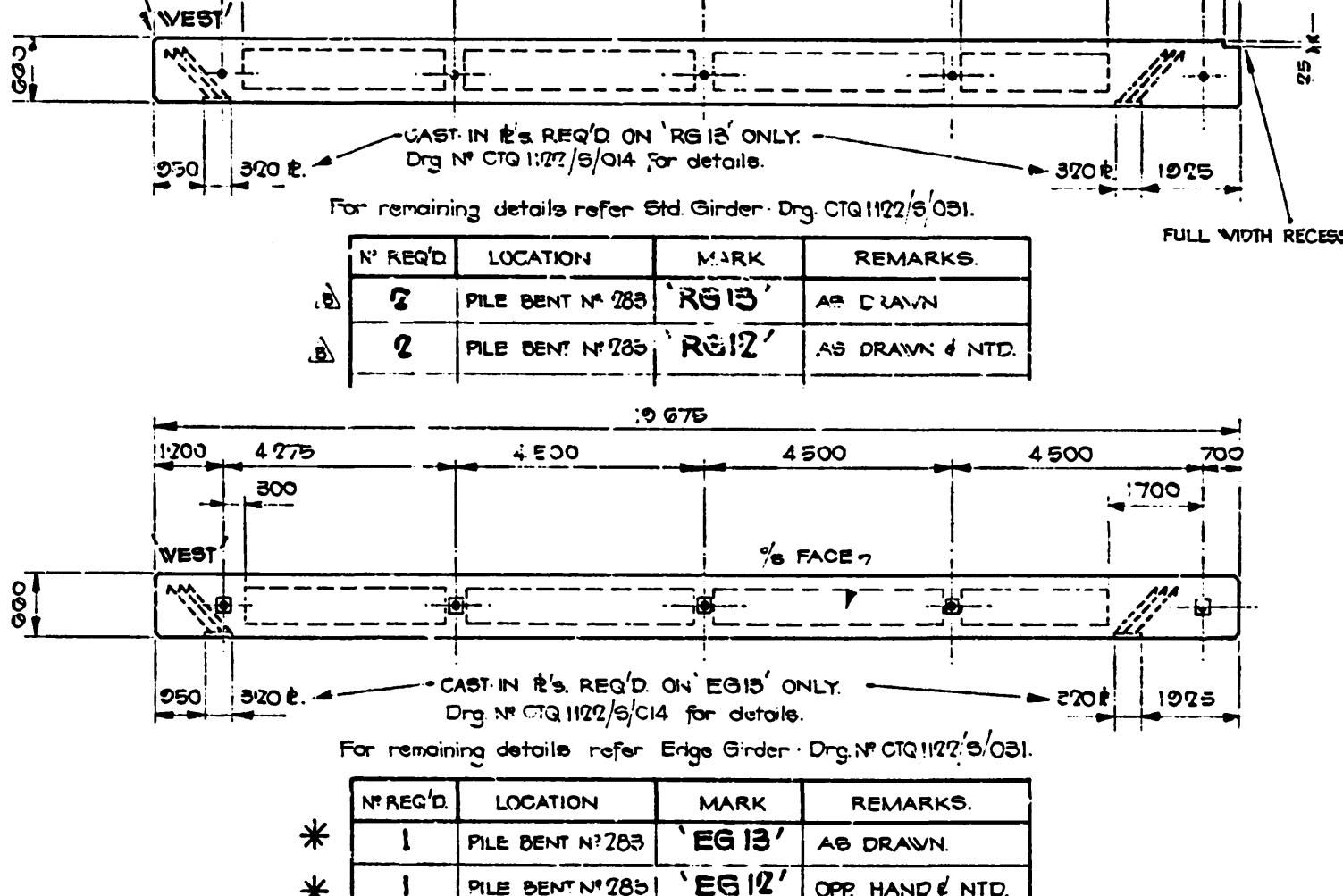
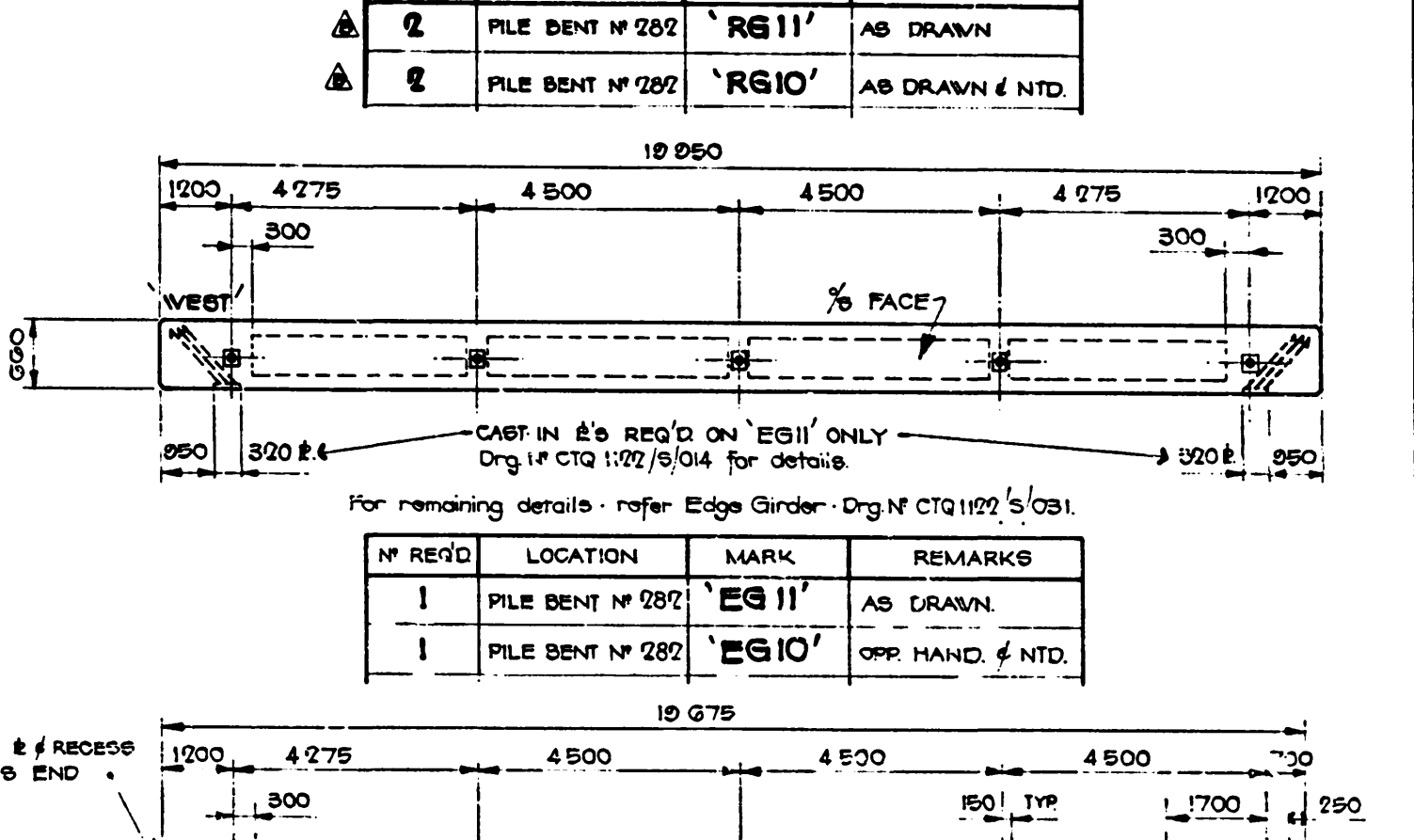
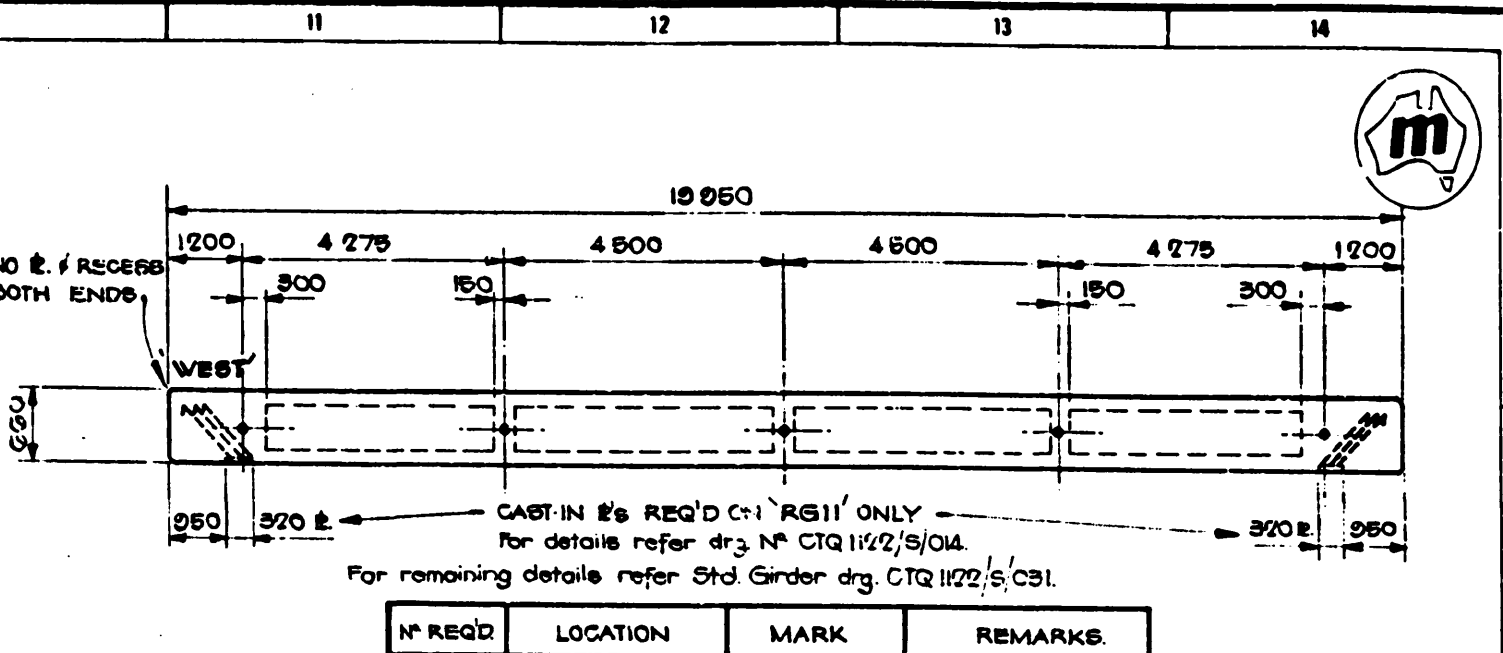
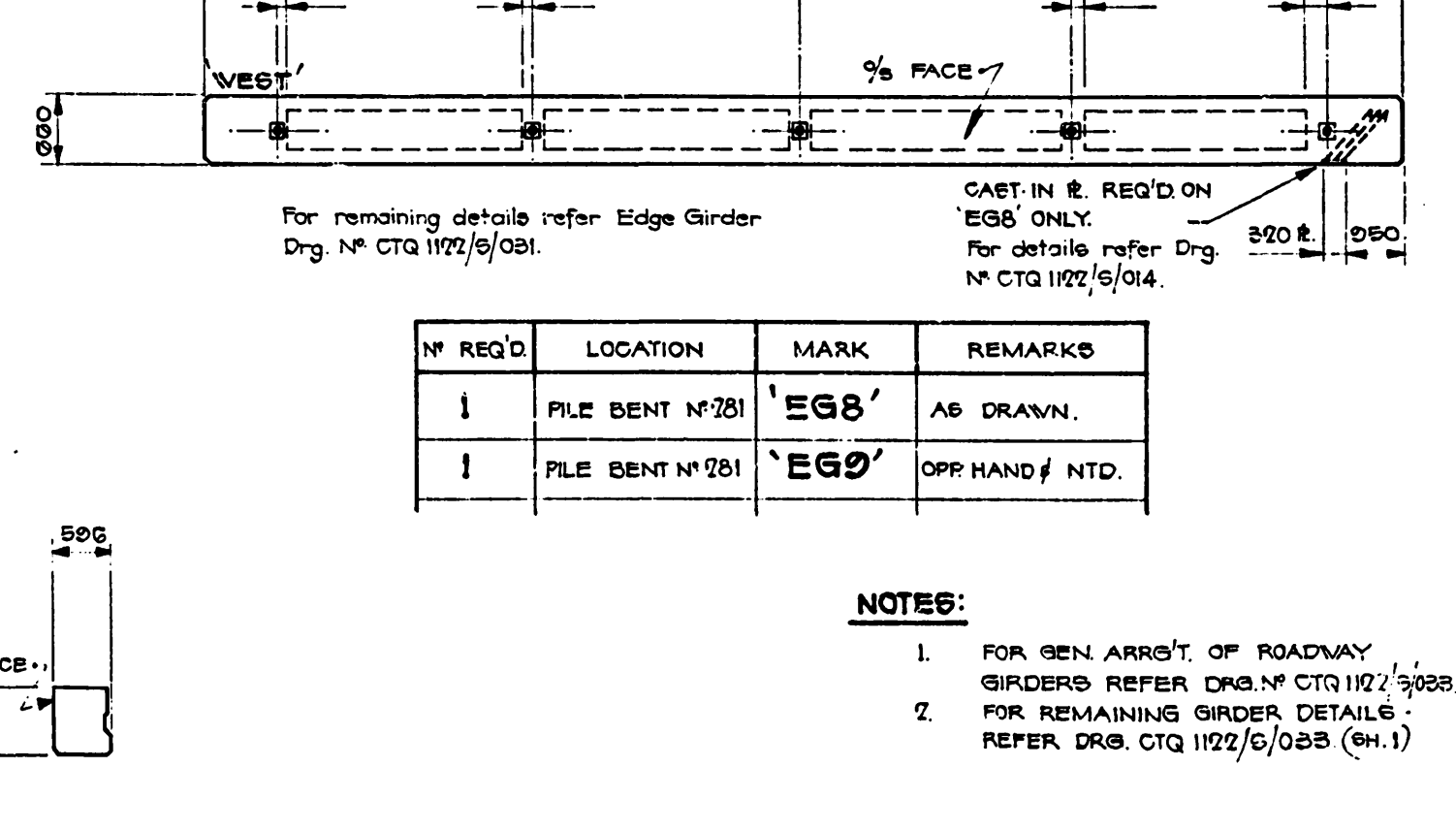
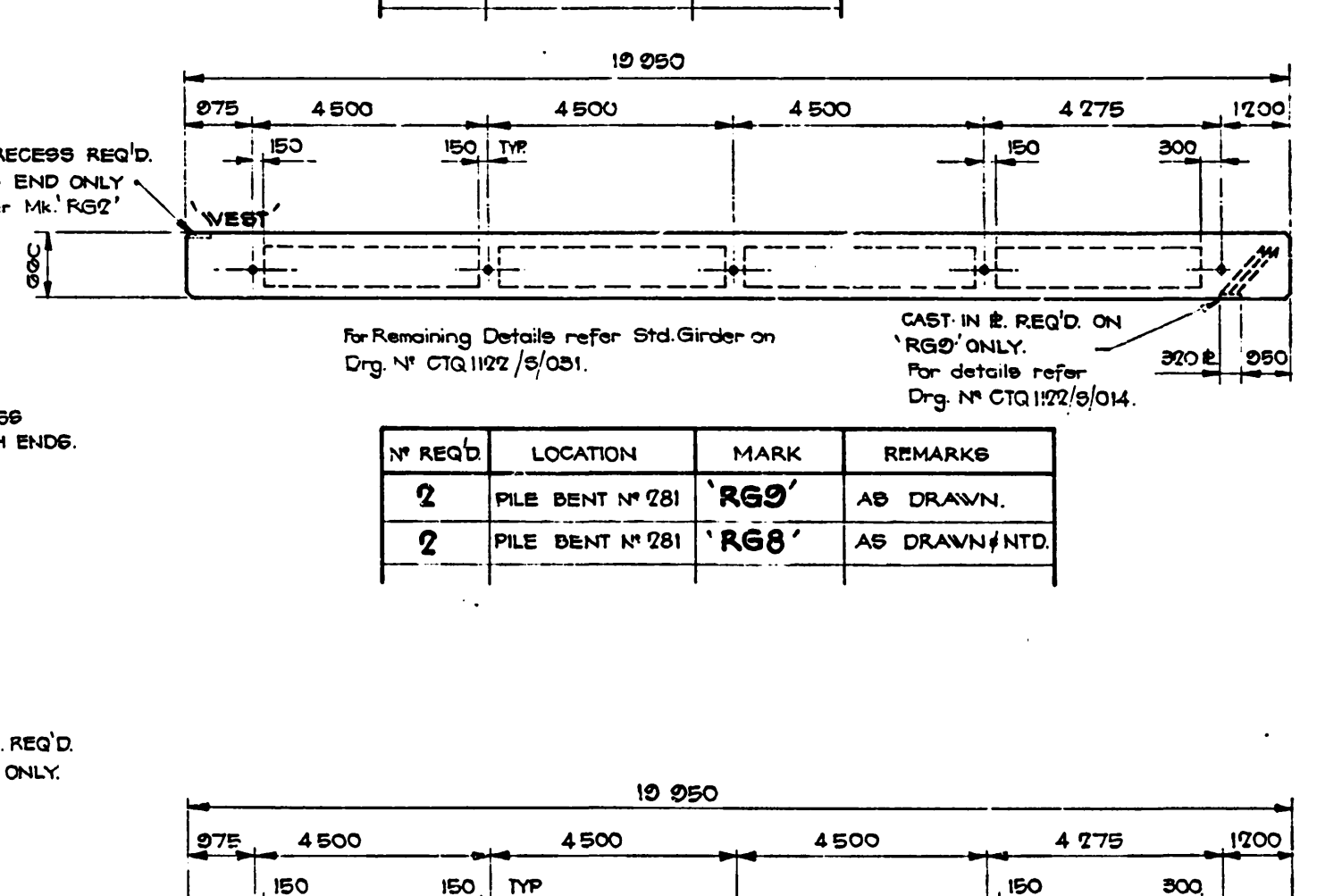
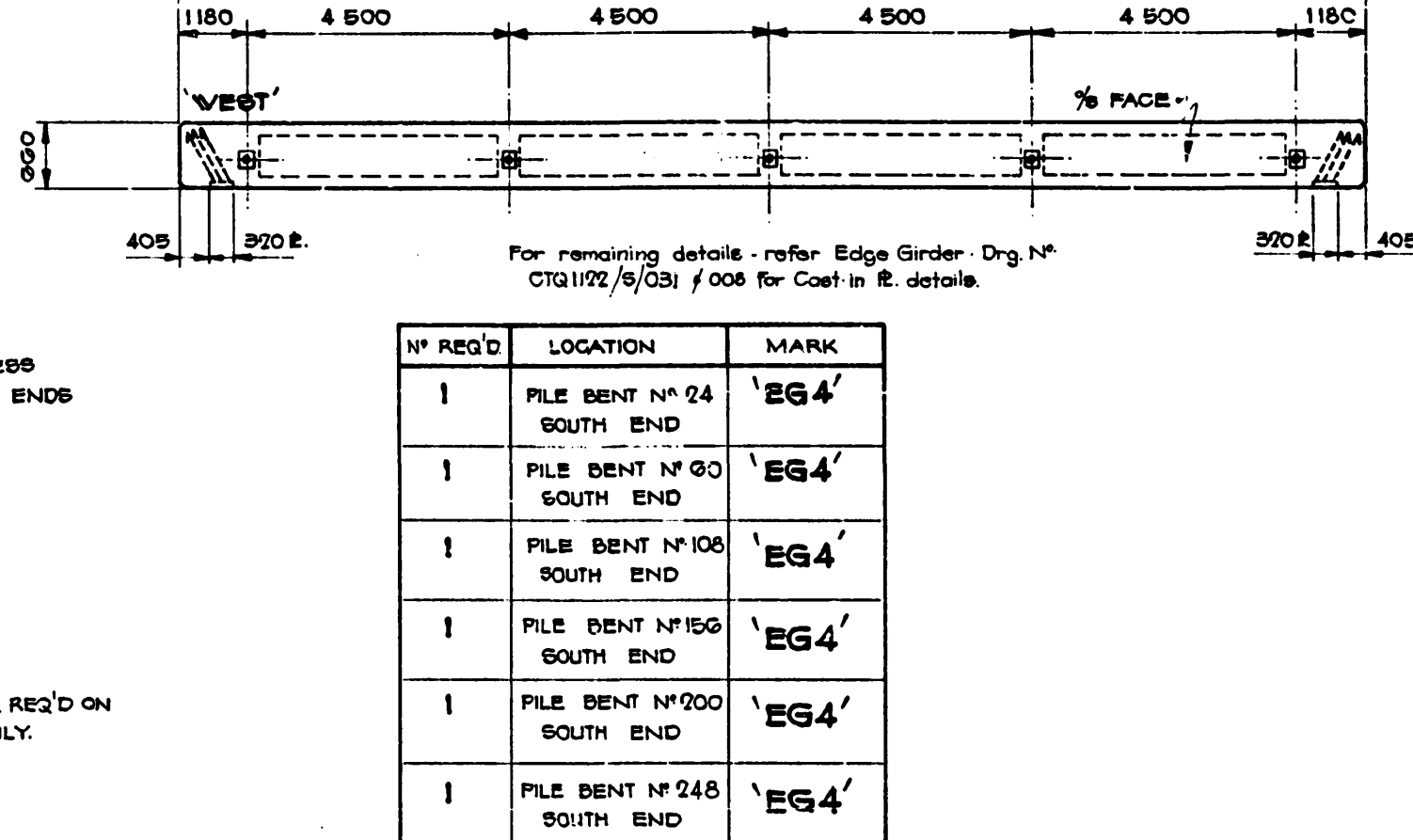
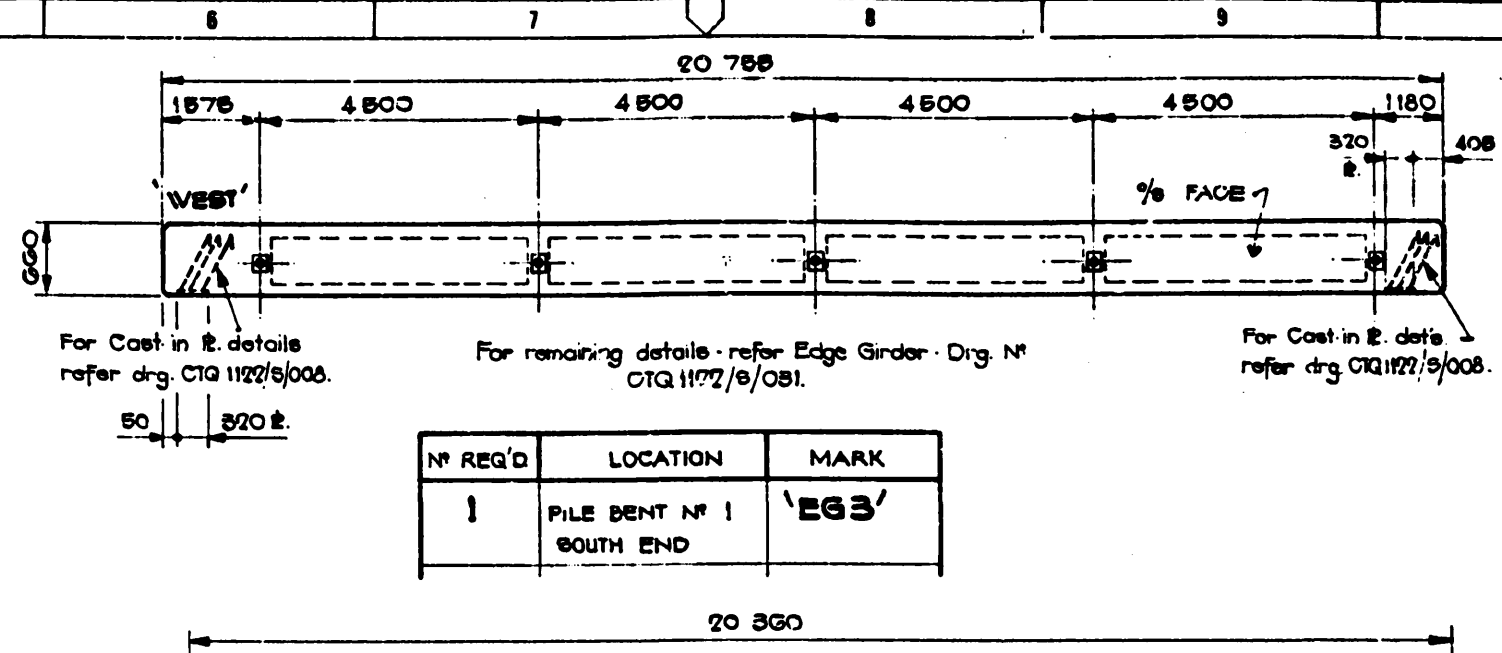
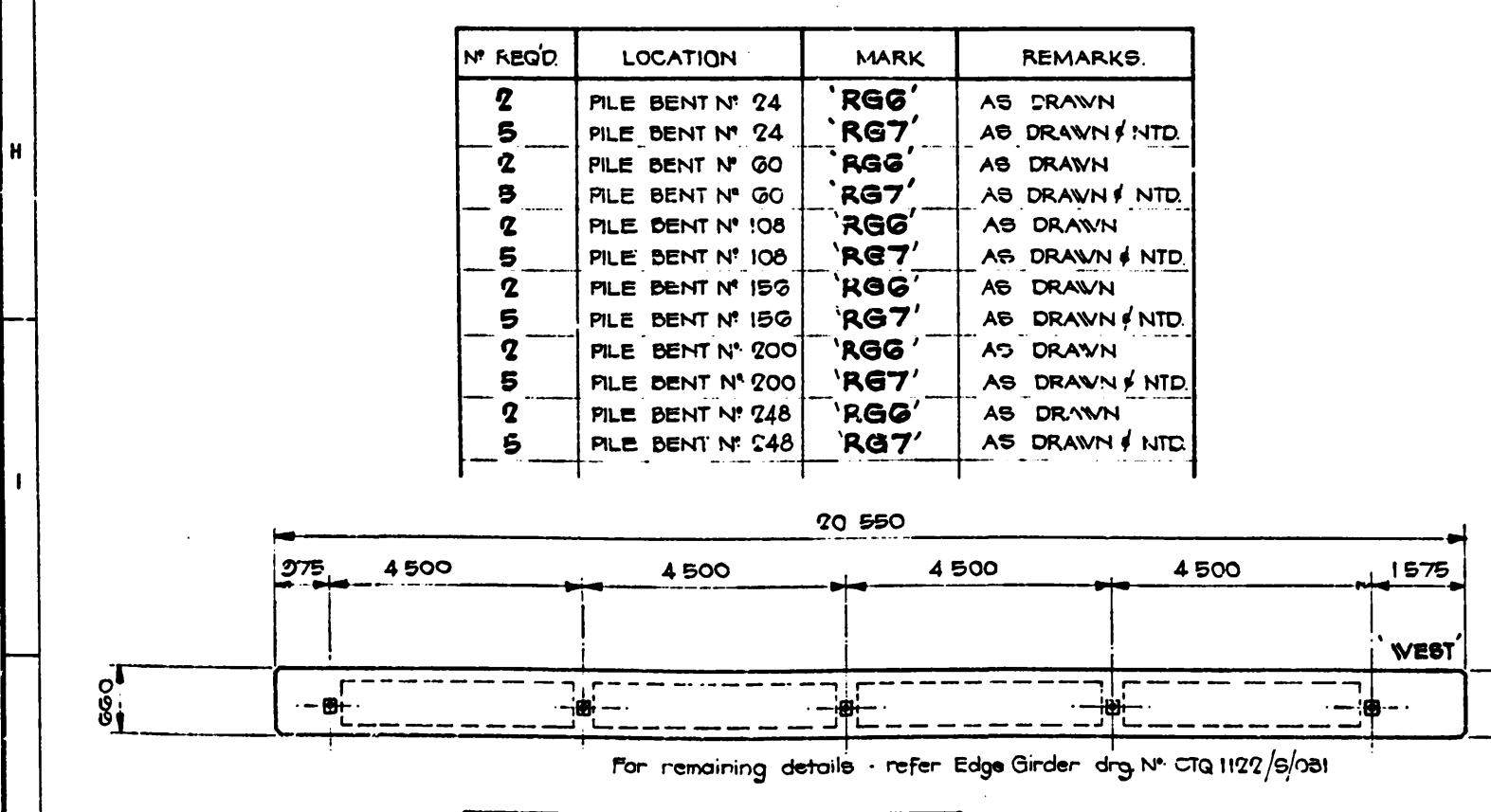
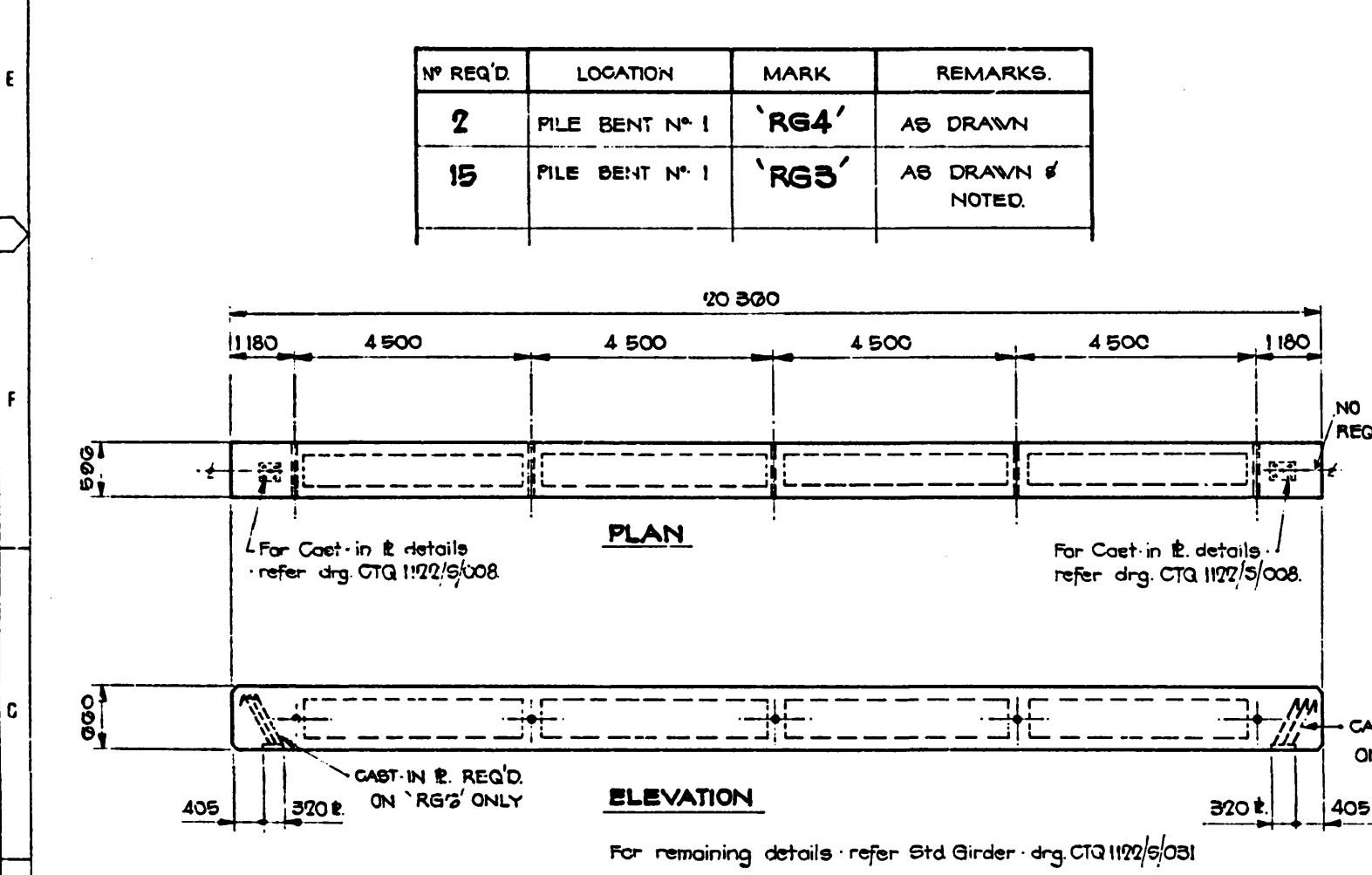
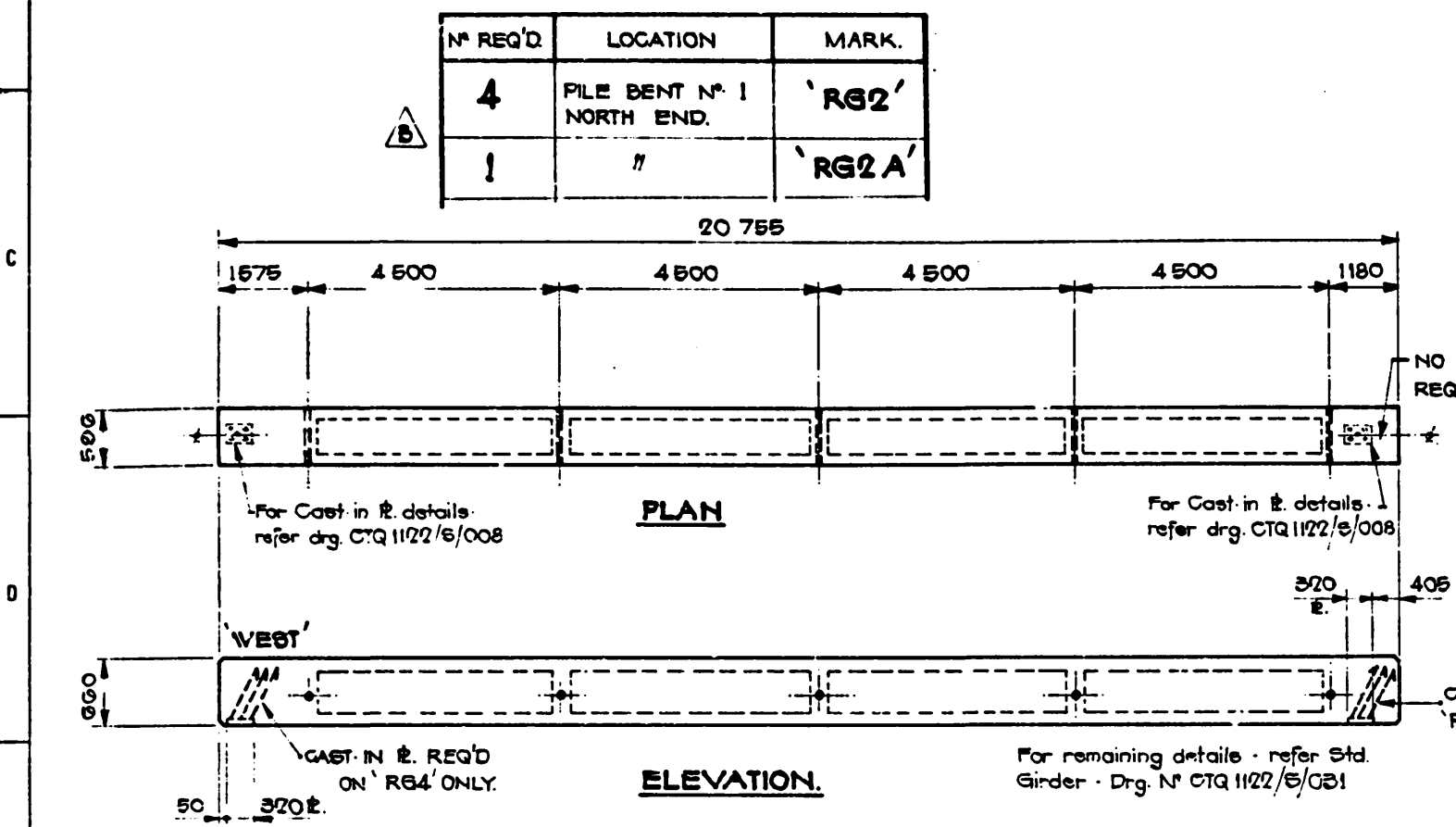
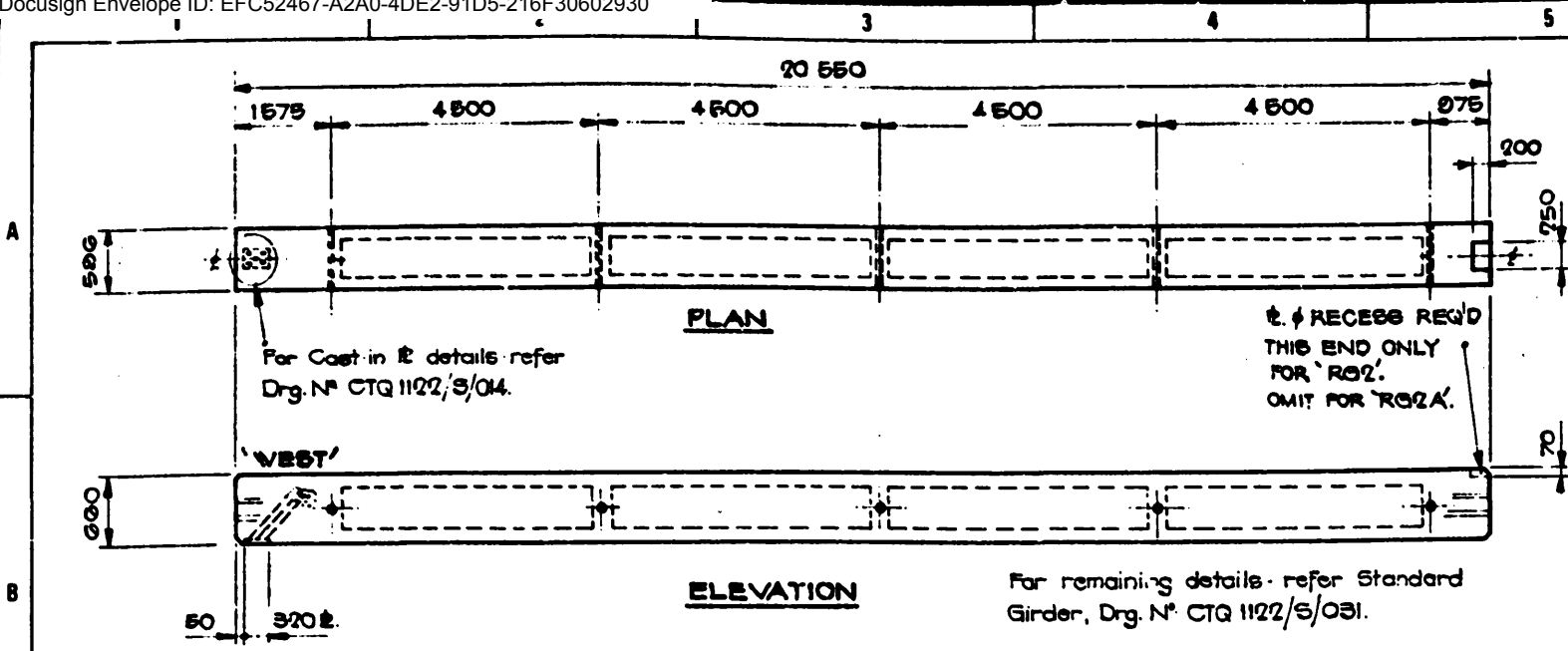
STAGE III

APPROACH JETTY

PRECAST CONCRETE ROADWAY GIRDER ARRGT. & DETAILS (SHEET 1)

MACDONALD WAGNER & PRIDDLE PTY. LTD.

CSR LIMITED



NOTES:

- FOR GEN ARR'GT. OF ROADWAY GIRDERS REFER DRG. N° CTQ 1122/S/038.
- FOR REMAINING GIRDER DETAILS REFER DRG. CTQ 1122/S/038 (SH.1)

* EDGE GIRDERS 'EG12' & 'EG15' TO HAVE KERB ANCHOR BARS ON EAST END POSITIONED TO ALLOW FOR SPECIAL KERB UNIT. - DRG. N° CTQ 1122/S/030.

THE CORPORATION OF THE TREASURER OF QUEENSLAND

THE SUGAR BOARD

BULK SUGAR TERMINALS

SCALE: 1/4" = 1'-0"

DRAWN: J.T.

PASS'D:

DATE: FEB '77

MACDONALD WAGNER & PRIDDLE PTY. LTD.

IN ASSOCIATION WITH

CSR LIMITED

ST. LUCINDA - STAGE III

APPROACH JETTY

PRECAST CONCRETE ROADWAY

GIRDER DETAILS - SHEET 2

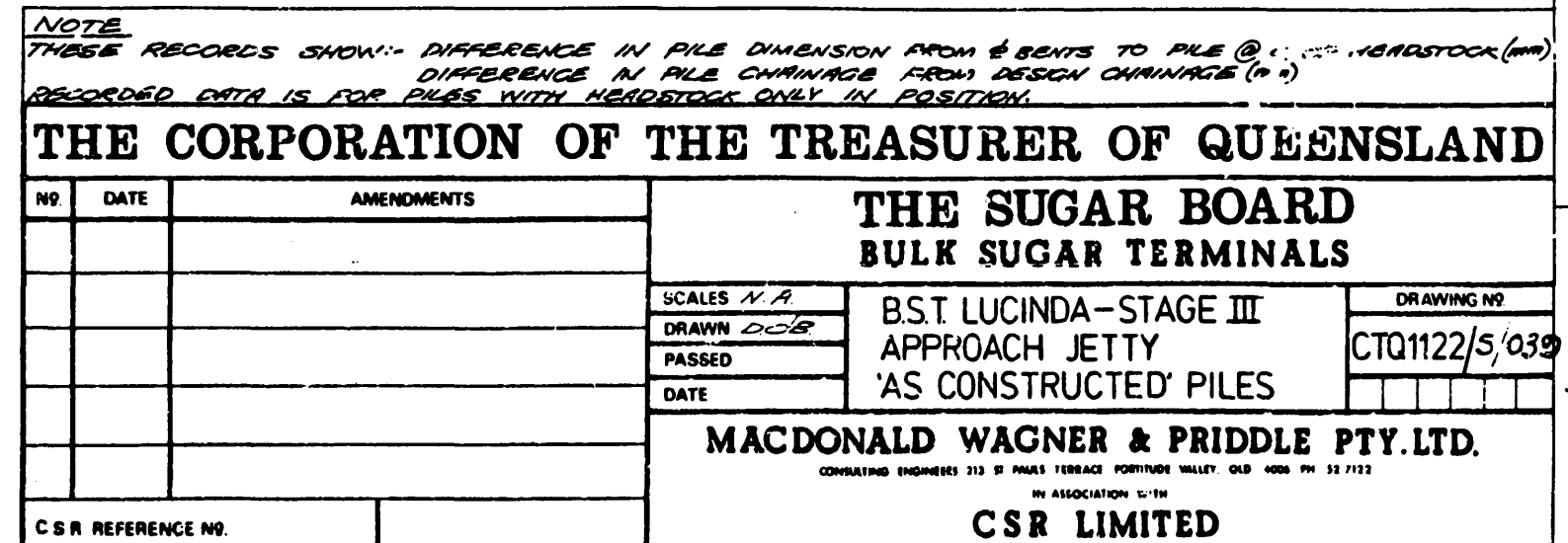
DRAWING NO. CTQ1122/S/034



DETAIL 3.

10		11		12		13		14	
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Handover of one jetty 'spider' unit at completion of contract

The Contractor shall hand over one of the Jetty 'spiders' to the Principal at the completion of the contract. The unit handed over shall be of the Contractor's choice. Any additions made to the unit since the initial design and approval may be removed by the Contractor or left in place.

The unit handed over shall be in an as-is condition. Upon handover, the unit does not form part of 'The Works' as per Clause 1 in the General Conditions of Contract. Upon handover, no warranty, guarantee or defects liability period applies. The Principal assumes all risk with respect to further use of the unit for any purpose.

Annexure Part E – Methodology

RESPONSE SCHEDULES SCHEDULE E – METHODOLOGY

Schedule E1 – Methodology

Provide a statement of the Tenderer's proposed arrangements, procedures and methodologies for carrying out and completing WUC. This should include:

1. *An overview of the methodology proposed by the Tenderer for the carrying out and completion of WUC;*

GENERAL:

Advanced Aqua Blasting (AAB) will have on site depending on work schedule

- 1 x AAB supervisory staff
- 1 x APP Group Quality Assurance person
- 2 x Ingham Agencies (IA) welding teams consisting of 2 men each with the relevant qualifications
- 1 x AAB abrasive blasting and painting team consisting of 3 men
- 1 x AAB Denso Seashield installation team consisting of 3 men
- 1 x Benchmark Scaffolding (BM) scaffold team consisting of 1 supervisor and 4 scaffolders

All AAB team members will be interchangeable depending on skills.

SCAFFOLD

AAB will seek permission to **work on Sundays** during scaffold erection / dismantling to minimize traffic interruptions. Scaffolders will work their 7 shifts including weekends. Their RDO will be weekdays.

Start the erection of 6 scaffolds from PRIORITY A PILEBENT 257.

It is anticipated that 5 pilebents will be fully scaffolded at any one time with work proceeding on them, and with 1 being dismantled and 1 being erected. All scaffolds build plans will be engineered.

The scaffolding will then proceed in numerical order, priority A and B to the last pilebent 281.

AAB will then relocate to pilebent 256 and proceed inwards to complete pilebents priority A and B in numerical order.

The scaffold will be erected so that access to the headstock and piles is on the southern side of the roadway and not impede roadway spider movement.

Initial access will also be from removing the gallery underpans. Replaced during shipping.

The scaffold will be erected so that the roadway spiders can travel along the roadway.

The headstock scaffolding will be erected to facilitate encapsulation and to allow the removal of the petroleum tape and masking around the gallery feet and the removal of any tape / jacket system down to > 600mm on the piles to allow for abrasive blasting and painting. The scaffold will also be built to allow removal of any part of the structure in contact with the headstock to be removed for abrasive blasting and painting.

****See attached scaffold design****

HEADSTOCKS / PILES – similar methodology for all PRIORITY A and B pile bent (Appendix D)

1) Welding repairs

On the completion of the scaffold on pile bent 257. AAB will

- a) Position spider 1W (1W is the welding equipped spider).
- b) High pressure water wash.
- c) Start steel thickness inspections and recordings as per **Contract Technical Specification 4.1.2 and the HOLD and WITNESS Points** in the Inspection and Test Plan (ITP).
- d) Steel repairs on both headstock and piles. (Specification and drawing APPENDIX J).
- e) Replacement of deck unit keeper (QSL 2501-JPHR-G01 APPENDIX A).
- f) After all welding repairs are completed, repeat the above step for the next scaffolded pile bent

2) Denso / Blast and Paint

On the completion of all welding repairs.

- a) Encapsulation and removal / protection of Denso wrapping around gallery feet and removal any tape/wrapping to >600mm down the piles
- b) Position spider 2B (2B is the abrasive blasting and painting equipped spider)
- c) Abrasive blast and paint
- d) Collect waste for transportation offshore and remove encapsulation
- e) Set up the spider 2B on the next scaffolded pilebent.
- f) Denso Seashield jacket fitting **Contract Technical Specification 2.2.3 and the HOLD and WITNESS Points** in the Inspection and Test Plan (ITP).

NOTE: The second coat of protective coating may be applied using QSL gallery air.

Scaffold to remain for the jacket fitting if required

*****See attached Drawings for Jetty Spider*****

2. ***Approach with regards to mobilisation for the works over a multi-year program;***

All scaffold and the 2 roadway spiders will remain at the QSL contractors yard during the demobilisation.

The 40' smoko and office donga's will remain at the QSL contractors yard during the demobilisation.

AAB staff on demobilisation will be re employed on other works.

Ingham Agencies (IA) staff on demobilisation will be re employed on other works.

AAB and IA'S staff are local, so they will be available for mobilisation each year.

Benchmark Scaffolding (BM) staff on demobilisation will return home.

All plant will be returned to AAB's Ingham workshop for servicing and under cover storage.

**** See attached drawings - proposed allocated space at the QSL contractors' yard****

3. ***Methodology for managing periods of inclement weather with respect to avoiding delays and maintaining productivity.***

AAB will present to the Principal a "*Stand down agreement*" for consideration. This has worked well in the past tenders and has the following main points:

During periods of Strong wind warning from the BOM, work will cease on the pilebents and staff will work on the C7B Conveyor gallery as per Contract Technical Specification 2.3

It is envisaged that some staff will return to work at the AAB and IA's workshop if the standdown is of a prolonged time frame.

CYCLONES

It is anticipated that no work will be ongoing during the normal cyclone season December through to April. If the BOM issues a cyclone alert all outside of this time frame all efforts will be made to secure the site.

During cyclone season (1 November to 30 April), AAB will maintain an appropriate cyclone readiness procedure with tasks at each cyclone level of green, yellow and orange in accordance with the Port of Townsville categories.

4. ***Methodology for managing jetty roadway access and details of how it proposes to co-ordinate and work with the Principal's staff and with other contractors:***

- (a) ***during the works;***
- (b) ***during shipping activities; and***
- (c) ***during major maintenance or construction activities on the wharf.***

(a). ROADWAY

AAB will attend or convey to QSL local staff the daily scope of works particularly road closures and the likely impact on QSL activities. This will convey the passing bay being used and the times of the 3hr closures if any.

AAB anticipates that scaffold erection and dismantling will have the most impact on roadway access.

AAB will use weekends for this activity whenever possible. Built scaffold will have no impact on vehicle roadway travel.

The spider design allows unimpeded normal vehicle passing on the roadway.

AAB will have a ***Traffic Management Plan*** which includes signage at the start and end of the work with speed limits, mobile contact numbers and names if access is required outside of the prearranged times. 2 way communication will also be used.

(b). SHIPPING

All removed gallery underpans will be replaced for a shipping event.

An inspection before shipping as well as normal end of day inspections will be carried out to ensure nothing impedes the operation of the belt and no material is left on the belt during gallery work.

QSL and other contractors staff have priority during shipping. AAB will have a Maritime Safety Identification Card (MSIC) holder on site during shipping.

(c). OTHER ACTIVITIES

AAB sees no issues with QSL engaged contractors or visitors such as CSIRO staff travelling through the designated work site. The worksite will be designated by signage and kept as compact as possible. This will be aided by the work methodology of working on consecutive headstocks.

5. ***defect rectification management proposed;***

AAB will submit a detailed Quality Plan as well Inspection and Test Plans (ITP's). that met its requirements as outlined in the Contract General Specification S9.2. This will include defect identification and repair.

This is outlined in detail in SCHEDULE H3 Quality Management.

6. ***provide a summary of any cost saving initiatives or opportunities that it has identified or recommends;***

AAB will employ local people and trades whenever possible to eliminate accommodation costs.

Advanced Aqua Blasting and Ingham Agencies are local companies with a high profile in the community.

AAB will purchase plant and equipment new where ever possible to minimize the possibility of *break downs over the contract life span.*

7. ***provide a summary of innovative work procedures or any other innovation that it recommends or offers during the course of carrying out and completion of WUC; and***

The use of 2 roadway spiders to maximise clearance for vehicles designed by Doug Lee from NQ Engineering, RPEQ certified and built locally by Ingham Agencies,. These spiders will carry equipment for welding, abrasive blasting and painting. They are steerable and towable.

AAB proposed to have on site 2 x 6 seater Polaris vehicles. These will transport all staff engaged on the contract from the staff donga at the contractor yard after sign on, Daily pre start to the work site where they will remain on the assigned passing bay. The use of 2 vehicles allows flexibility in lunch time travel to the off shore crib room, bringing people in and out etc.

AAB will mark the roadway in the work area with yellow paint to identify the 2m from the gallery wall where Personal Flotation Devices (PFD) are not required to be worn. The remaining area is considered an exposed edge and PFD wearing is mandatory. The identification line will be extended as the work site moves.

8. ***details of how it proposes to co-ordinate and work with the Principal's staff and with other contractors which might be retained by the Principal on the same Site***

(a) for the duration of WUC; and

AAB will hold a formal induction for all AAB and contractors staff with QSL, STL invited to attend.

At this induction all staff will also do the QSL Lucinda BST site specific induction.

This induction will be only for staff working on the contract and will safety oriented. QSL and other contractors staff need not be inducted unless working in AAB's designated work site. People needing the full AAB site specific induction will normally be inducted on a Monday at the AAB site office in the contractors yard.

AAB will provide an escort within reason for small tasks within the AAB worksite.

AAB will attend or convey to QSL local staff the daily scope of works particularly road closures and the likely impact on QSL activities.

(b) in respect to any shut downs of the Principal's facilities which may occur during the course of carrying out and completing WUC."

AAB will be using main gate passes on a limited basis. Entry to QSL will be through the gate in the contractors yard so main gate issues are not a problem.

AAB will be using QSL gallery water and air as needed. If interruption to these services occur AAB will make alternate arrangements.

AAB will use the off shore crib room while the work site is closest to the wharf. AAB will require electrical power for fridges, micro waves etc. AAB will provide a generator for any prolonged outage.

Please nominate proposed working hours to be inserted in Item 42 of the Contract. Where night works are nominated, please describe the proposed works and include a methodology for managing risks associated with night works.

AAB will seek permission to **work on Sundays** during scaffold erection / dismantling to minimize traffic interruptions. Scaffolders will work their 6 shifts including weekends. Their RDO will be a week day. Normal work hours 0600 – 1700.

For AAB and IA staff – Normal work hours 0600 – 1700 Monday / Saturday with Sundays as the RDO. This may alter to accommodate favorable tide cycles for pile jacket fitting.

AAB will also make provision for seeking approval for a night shift during favorable tide cycles for welding repairs and or Seashield jacket fitting.

The following form part of the Methodology

Source	Topic	Description
RFI01+Interview responses	QA/QC Representative role implementation	Ben Eames from APP Group will be utilised to provide and set up the Dashpivot system for Quality Assurance record management and other management and reporting functions. Refer to further description of the proposed system in slides from tender interview in annexure Part I - Key Personnel. Rod Cockle will commence in a QA/QC officer role until a QA/QC person can be permanently added to AAB Staff. The permanent person is likely to be Wade Mitchell, who will also assist in scheduling, safety and other project management tasks.
RFI01 + Interview responses	Online quality reporting tool	AAB will utilise Dashpivot software system and upload data daily during weekdays. Refer to further description of the proposed Dashpivot system in slides from tender interview in annexure Part I - Key Personnel.
RFI01	Use of Polaris vehicles or similar	Principal RFI: Schedule E methodology notes the use of Polaris vehicles. QSL has a speed limit of 20km/h for these vehicles in accordance with state regulations. Please confirm that AAB has reviewed the applicable state regulations and agrees to implement any/all safety measures, as required, in the use of Polaris vehicles along the jetty/wharf. Contractor Response: AAB understand the state regulations and have taken that into considerations. ATV to be governed to 20K/hour. All doors to be solid hinged. AAB choose this as a transportation method to minimize vehicle footprint onsite and allow more flowing worksite for breaks, etc.
RFI01	Conveyor gallery access	Principal RFI: Please indicate how workers will access the conveyor gallery wrapping above the belt. Contractor Response: AAB to have engineered and approved platforms with access ladders that straddle the conveyor system. Design to be approved by QSL to prior to fabrication. See attched proposal design. Platform to be made from aluminium. Bolt on ladders for ease of handling and dismantling in a shipping event. 3 x to be made so multiple work fronts can be accessed. Platform to be 1200mm wide.
RFI02	Expansion joint scaffold access	Principal RFI: Could you please re-examine and explain pricing for item 2.9.1 (access for expansion joints). As the price is much higher than access for similar pilebents, we would like to understand the additional effort, modified design and inclusions that drive this price increase. Contractor Response: The variance in pricing is due to a design change required to support the scaffolding. Benchmark proposes to weld a steel member to the end of the headstock to provide support for the southern end of the scaffold. The quoted price includes: 1) Labour hours for a boilermaker to complete the welding works; 2) Temporary scaffolding to enable safe access for the boilermaker to carry out the welding; and 3) Subsequent installation of the scaffolding to access the headstock.
RFI05	Embedding of learnings from safety incidents	Principal RFI: please detail how AAB plan to embed past incident learnings from all projects into their daily WHS practices to prevent recurrence of similar injuries, such as manual handling strains, eye injuries from foreign bodies, and lacerations. Contractor Response: All AAB Incident investigation and resolution is around the "Hierarchy of Controls" . During this process, AAB will make what ever change is warranted regardless of the inconvenience and the cost as AAB puts worker safety first. To have Incident reporting, you must first have an Incident reporting culture and forums to encourage this. AAB holds weekly toolbox meetings at AAB workshop with items on the agenda covering incidents or nearmisses, safety shares and any recommended changes to SWMS. These are also shared between site works to allow all staff to be informed of any issues or changes. Through this we have implemented changed like Foam Backed Glasses to be worn through out identified scopes of work. For the LBST Jetty Remediation Project, AAB will hold a monthly safety toolbox BBQ (additional to weekly toolbox) with workers being awarded for Hazard Identification, safety culture and take 5 inputs. On the agenda AAB will find incidents and challenges in industries similar to ours to find learnings we could use to implement changes to our WHS Practices. AAB values Third Party Audits to review our processes and will make suggested changes as soon as practicable and ensure compliance and reporting feedback .
RFI08	Recovery of loose and flaking paint during preparation	Principal RFI: Please provide brief details of how AAB will minimise loose and flaking paint that is removed during waterblasting of headstocks and piles from entering the environment Contractor Response: The headstock scaffold will be fully encapsulated on all sides with fire resistant powerclad before water washing, with geofab on the floor to allow water to be strained before release. After water washing the geofab, removed loose coating and rust will be removed, the powerclad on the sides will remain. After welding repairs, powerclad or similiar will be placed on the scaffold floor to complete encapsulation. It will not be a negative pressure encapsulation so a small amount of nuisance dust will escape during abrasive blasting. This is permitted under AAB'S IEMS. The removed coating and spent garnet will be collected and transported inshore for collection by MAMS Environmental services on an as need basis. The majority of loose coatings and rust on the piles will be removed manually during the steel inspection process before water washing and will be collected in a funnel type arrangement. This funnel will stay inplace during water washing of the piles allowing strained water to pass through a geofab filter material. The removed loose coating and rust will also be collected and transported inshore for collection by MAMS Environmental services. This collection funnel will be installed above the barnacle line on the pile.

CONCEPT NOTES:

1. This is a proof of concept drawing only. RPEQ licenced engineer Doug Lee who is familiar with the JMV concept and requirements at Lucinda sugar terminal has attended site with Advanced Aqua Blasting and confirms that development of this concept design is feasible and if progressed will be able to be detail designed, certified and manufactured for use in accordance with all required QLD regulation, AS standards, and QSL (Lucinda site) standards/requirements.
2. This is a concept for a load rated platform trailer that can be towed into position along the jetty, it is intended to be loaded with painting equipment by Advanced Aqua Blasting for the purpose of jetty refurbishment.

MATERIAL NOTES:

1. Structural beams 250UB
2. Drawbar 75x6 SHS
3. Platform, stairs, handrails all Webforge Monowills system typical
4. Solid tyres 8.25-15, no suspension, no service brake, park brake only
5. Linkages / hinges / hydraulics detail design to be developed

FABRICATION NOTES

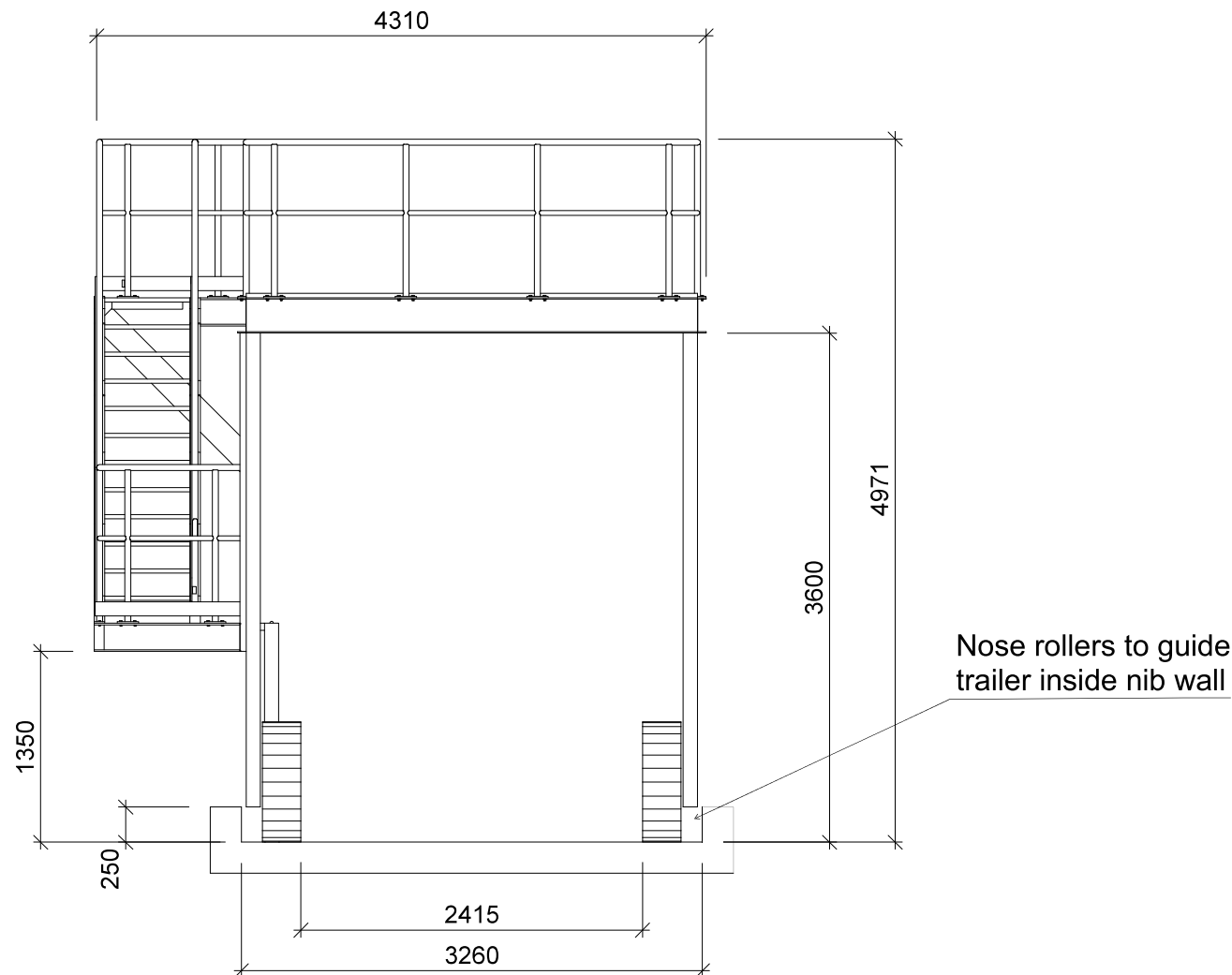
1. The main portals are fully welded
2. Ancillary equipment can be bolted or be removable as required
3. The drawbar is fully welded mounted on special swivelling/locking hinges
4. Painting to owners specification

DIMENSIONAL NOTES:

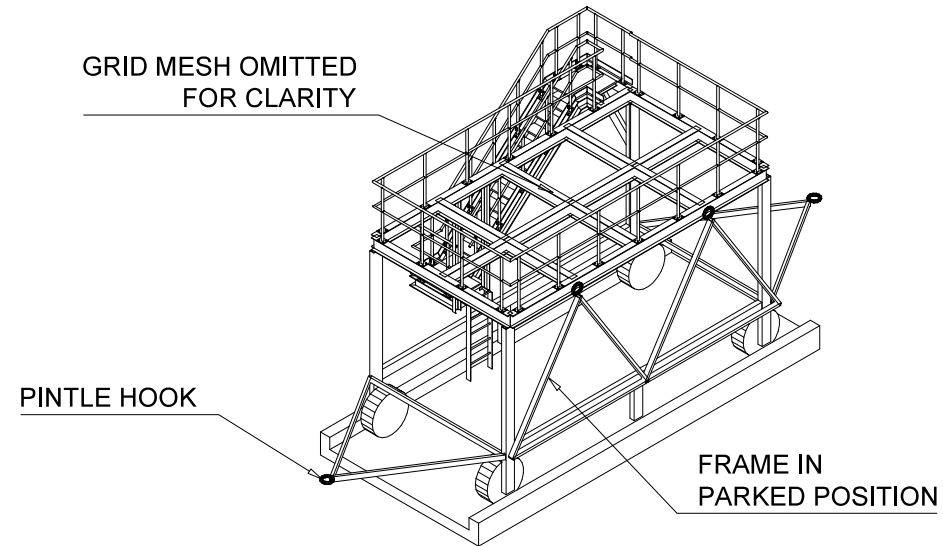
1. Trailer platform length between 4000 & 6000 mm
2. Under trailer clearances to be confirmed by emergency service vehicle requirements.
3. Exact location of point loads (welders, blast pots, garnet storage etc) to be specified

CAPACITY NOTES:

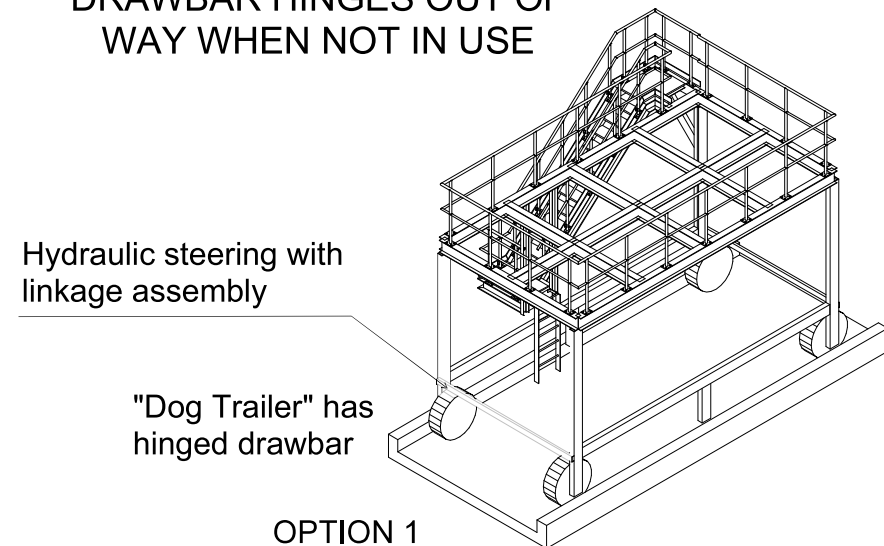
1. Aggregate Trailer Mass = 3500 kg
2. Tare mass estimated = 1800 kg
3. The load platform is rated to hold payload up to 3500kg
4. Tow vehicle must be rated to pull trailer, an unbraked, loaded trailer may not be suitable for towing with light vehicle ute. If towing with ute electric brakes to be fitted at higher laden weights.



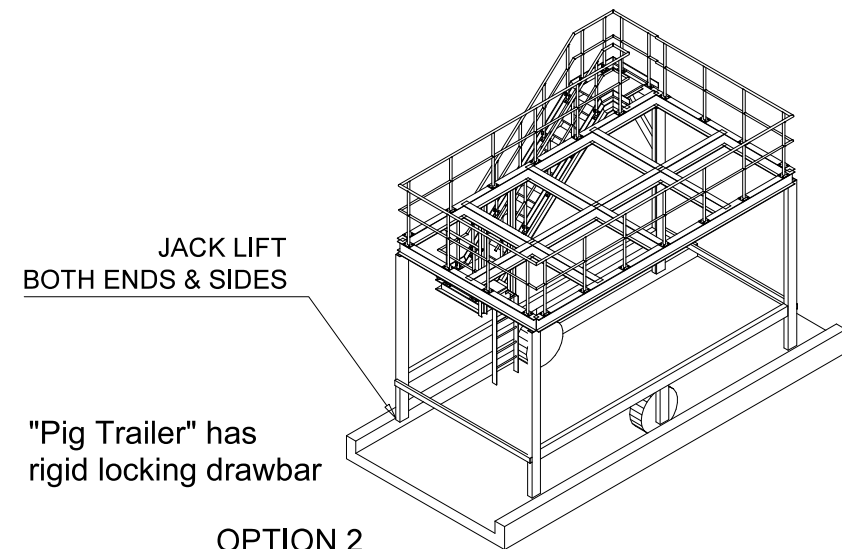
FRONT VIEW



DRAWBAR TYPICAL
DRAWBAR HINGES OUT OF WAY WHEN NOT IN USE

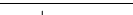


OPTION 1
4 WHEELED STEERING TRAILER



OPTION 2
2 WHEELED TRAILER WITH JACKS

CONCEPT REVIEW

DRAWN TO AS1100 TECHNICAL DRAWING STANDARDS ALL DIMENSIONS IN	 North Queensland Engineering	Job No.: 240723 Client : ADVANCED AQUA BLASTING Project : JETTY SPIDER	Mark No :		 3RD ANGLE PROJECTION	
SCALE: As Shown			Revision Date : 14.11.2024		Scale : 1:125,1:50	
			Revision : A		Drawn : G.RICHTER	
COPYRIGHT © This drawing remains the property of North Queensland Engineering Pty Ltd and shall not be passed on in any electronic format to any third party without prior written approval from North Queensland Engineering Pty Ltd.			DRG No. : 240723 -A- 001			
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LUCINDA BULK SUGAR TERMINAL TYPICALS BENCHMARK

GENERAL NOTES

- IF IN DOUBT, PLEASE ASK,
- DO NOT OBTAIN DIMENSIONS BY SCALING FROM THESE DRAWINGS.
- VERIFY ON SITE ALL SET OUT DIMENSIONS SHOWN ON THESE DRAWINGS.
- ALL DIMENSIONS ARE IN MILLIMETERS UNLESS NOTED OTHERWISE ON THESE DRAWINGS.
- ALL PERMANENT STRUCTURE DIMENSIONS ARE TO BE CHECKED & VERIFIED PRIOR TO ERECTION OF SCAFFOLD. THE SCAFFOLD DESIGN IS BASED ON THE DIMENSIONS SHOWN ON THESE DRAWINGS. RE-DESIGN MAY BE REQUIRED WHERE ACTUAL DIMENSIONS DIFFER.
- NO ALTERATIONS TO THE DESIGN SHOWN ON THESE DRAWINGS ARE PERMITTED WITHOUT THE PRIOR APPROVAL OF THE DESIGNER. SUPPORT LAYOUT MAYBE RE-ARRANGED TO SUIT SITE CONDITIONS, PROVIDING THAT THE MAXIMUM BAYS SIZES AND LEG LOADS DO NOT EXCEED THE VALUES INDICATED IN THE NOTES.
- THE DESIGN DEFECTED ON THESE DRAWINGS IS VALID FOR THE NOMINATED PROJECT & STRUCTURE ONLY.
- THE SCAFFOLD SHOWN IN THESE DRAWINGS HAS GENERALLY BEEN DESIGNED IN ACCORDANCE WITH THE RELEVANT SECTIONS OF THE FOLLOWING AS AUSTRALIAN STANDARDS & DOCUMENTS:
 - AS/NZS 1170.0:2002 STRUCTURAL DESIGN ACTIONS PART 0: GENERAL PRINCIPALS
 - AS/NZS 1170.2:2021 STRUCTURAL DESIGN ACTIONS PART 2: WIND ACTIONS
 - AS/NZS 1576,1:2019 SCAFFOLDING PART 1: GENERAL REQUIREMENTS
 - AS 4100:2020 STEEL STRUCTURES
 - AS 4576:2020 GUIDELINES FOR SCAFFOLDING
 - RELEVANT SCAFFOLD CODE OF PRACTICE
- THE DESIGN HAS BEEN BASED ON THE USE OF ACROW FORMWORK & SCAFFOLDING EQUIPMENT ONLY. OTHER EQUIPMENT MAY HAVE A DIFFERENT SAFE WORKING LOAD AND THUS MAY NOT BE ADEQUATE IN THE APPLICATION SHOWN ON THIS DRAWING.
- IRRESPECTIVE OF ANY RECOMMENDATIONS SHOWN ON THE DRAWINGS, THE USER IS RESPONSIBLE FOR THE STRUCTURAL ADEQUACY OF MATERIAL OF THEIR OWN SUPPLY AND FOR ENSURING THAT THE APPROVED WORKING LOAD LIMITS OF ACROW FORMWORK & SCAFFOLDING EQUIPMENT ARE NOT EXCEEDED.
- THE CLIENT MUST ENSURE THAT THE PERMANENT STRUCTURE HAS ADEQUATE STRENGTH TO WITHSTAND THE IMPOSED LOADS NOTED. ALL THE LOADS MUST BE TAKEN BY THE PERMANENT STRUCTURE.
- THE CLIENT IS TO ENSURE THAT THE SUPPORTING GROUND (WHERE APPLICABLE) HAS ADEQUATE CAPACITY TO SUPPORT THE LOADS FROM THE SCAFFOLD WITHOUT EXCESSIVE SETTLEMENT INCLUDING DIFFERENTIAL SETTLEMENT.
- ALL TYPICAL DETAILS ARE TO BE USED AS EXAMPLES ONLY AND IF SPECIAL OR DIFFERENT DETAILS ARE NOTED IN THE CONSTRUCTION DRAWINGS VARY. TYPICAL DETAILS ARE TO BE DISREGARDED.
- ALL SCAFFOLD SHALL BE INSPECTED & STRUCTURE MUST BE CHECKED BY CLIENT PERIODICALLY AND AFTER INCLEMENT WEATHER OR MAIN LOADING EVENT TO ENSURE THAT ALL COMPONENTS AND CONNECTIONS ARE TIGHT AND THE INTEGRITY OF THE STRUCTURE IS INTACT.
- LOADS SHOWN ARE UNFACTORED WORKING LOADS.

SCAFFOLD NOTES

- THE STABILITY OF THE SCAFFOLD MUST BE MAINTAINED AT ALL TIMES, AND IF NECESSARY, ADDITIONAL BRACING AND ANCHORING MUST BE PROVIDED.
- IMPACT LOADS HAVE NOT BEEN ALLOWED FOR IN THIS DESIGN. IT SHALL REMAIN THE CLIENTS RESPONSIBILITY TO ENSURE THAT SUFFICIENT PRECAUTIONS ARE TAKEN TO AVOID IMPACT LOADS ON THE SCAFFOLD DURING USE AND CRANAGE & VEHICULAR OPERATIONS.
- THIS DESIGN HAS BEEN PREPARED FROM THE INFORMATION SUPPLIED TO THE DESIGNER BY, OR ON BEHALF OF THE SITE TEAM. THE SITE TEAM MUST CHECK THAT WE HAVE CORRECTLY INTERPRETED THEIR REQUIREMENTS AND THAT WHERE APPLICABLE, ALL LOADINGS, DIMENSIONS, DETAILS, ERECTION AND SEQUENCES ETC, ARE AS REQUIRED AND PRACTICABLE.
- ALL ERECTION WORK IS TO BE CARRIED OUT IN A SAFE TRADESMANLIKE MANNER AND BE IN ACCORDANCE WITH ACROW FORMWORK & SCAFFOLDING GUIDELINES, AS1576, AS4576, LOCAL AUTHORITY REQUIREMENTS AND OCCUPATIONAL HEALTH AND SAFETY REGULATIONS.
- TIES OR OTHER COMPONENTS ARE NOT TO BE MOVED OR TAMPERED WITH IN ANY WAY EXCEPT BY ACROW FORMWORK & SCAFFOLDING OR OUR NOMINATED SUB-CONTRACTORS.
- UNLESS NOTED OTHERWISE ON DRAWING THE DISTANCE BETWEEN BRACED BAYS IS NOT TO EXCEED FIVE BAYS.
- UNLESS NOTED OTHERWISE ON THE DRAWING, ALL RETURNS OR BREAKS IN SCAFFOLD ARE TO BE TUBE TIE, COMPLETE WITH PLANKING, GUARDRAILS OR MESH INFILL AS REQUIRED ON ADJACENT SCAFFOLD.
- UNLESS NOTED OTHERWISE ON THE DRAWING, SCAFFOLD MUST BE A MAXIMUM OF 225mm FROM THE WORKING FACE UNLESS DIRECTED OTHERWISE BY LOCAL SAFETY AUTHORITIES. GAPS TO WORKING FACE GREATER THAN 225mm TO BE FILLED AS REQUIRED ON SITE.
- INDEPENDENT SCAFFOLD BAYS TO BE LINKED USING SCAFFOLD TUBE AND FITTING AT LEDGER POINTS NO GREATER THAN 2.00m VERTICALLY.
- UNLESS NOTED OTHERWISE ON DRAWING PLATFORM BRACKETS ("HOP UPS") ARE NOT TO BE LOADED WITH MATERIALS. THEY ARE TO BE USED FOR ACCESS BY PERSONNEL ONLY.
- BELOW SYMBOLS SHOWN ON THE DRAWINGS INDICATE LAPS AND RAKERS/TIES.

LAP DETAIL MARKER INDICATES SCAFFOLD TUBES, FITTINGS & LAPPING BOARDS TO JOIN ADJACENT SCAFFOLDS. TUBE TIE AT 2000mm CENTERS VERTICALLY (INTERNAL & EXTERNAL STD)



TIE DETAIL MARKER INDICATES SCAFFOLD TIES FIXED TO SLAB EDGE, THROUGH WINDOW OPENINGS AND TO THE SIDE OF EDGE BEAMS. HORIZONTAL DISTANCE BETWEEN TIES AS SHOWN AND VERTICAL DISTANCE BETWEEN TIES MUST NOT EXCEED 4000mm UNDO. TIES TO BE TIED TO BOTH STANDARDS UNDO.



TIES AND BRACING

THE PRINCIPAL CONTRACTOR IS RESPONSIBLE FOR ENSURING ALL SCAFFOLD STRUCTURES REMAIN ADEQUATELY TIED AND BRACED TO CARRY THE TIE FORCES AND ENSURE STABILITY. TIES OR HOP-UPS ARE NOT PERMITTED TO BE MOVED BY OTHER TRADES REGARDLESS IF THEY HOLD A VALID SCAFFOLD TICKET WITHOUT CONSULTING AFS.

- CONSULT ACROW FORMWORK & SCAFFOLDING FOR SCAFFOLD TIE AND BRACING REQUIREMENTS FOR SCAFFOLD EXCEEDING 50m TO THE TWP.
- REFER DRAWING FOR REQUIREMENTS REGARDING DOUBLED LEGS OR 1m LIFTS IF REQUIRED.
- NO JOINTS IN STANDARDS ABOVE TOP TIE IS ALLOWED UNLESS PINNED OR SPLICED AS PER THE TYPICAL DETAIL.
- PRESCRIBED CONTAINMENT/LINING NOT TO EXTEND MORE THAN 2m PAST LAST TIE. UNO.
- TIES ARE TO BE INSTALLED AS PER STANDARD TIE DETAILS SHOWN IN THESE DRAWINGS. ALL TIE LOCATIONS MAY UTILISE ANY OF THESE DETAILS TO SUIT THE PARTICULAR TIE POINT IF APPLICABLE.
- TIES MUST BE INSTALLED WITHIN 300mm OF A CONNECTED NODE OR SCAFFOLD LEG TO BE DOUBLED AS PER TYPICAL SINGLE LEG TIE DETAIL IN MIDDLE OF STANDARD.
- TUBE IN SCAFFOLD COUPLERS MUST PROTRUDE BY AT LEAST 10mm OUT OF THE COUPLER. IT MAY ALSO BE PRUDENT TO WRAP END OF TIES IF THE SCAFFOLD TUBE EXTENDS INTO THE WALKWAY AREA.
- IF WIND SPEEDS ARE LIKELY TO REACH OR EXCEED 100km/hr, PRESCRIBED CONTAINMENT SHEETING TO BE REMOVED.
- SCAFFOLD BOARDS SHALL BE STRAPPED DOWN OR REMOVED IF WIND SPEEDS ARE FORECAST TO EXCEED 60km/hr.
- REFER TO UNIMESH FR2 INSTALLATION GUIDE FOR INSTRUCTIONS ON HOW TO ATTACH CONTAINMENT TO PERIMETER SCAFFOLD.
- WHERE ANCHORS INTO CONCRETE ARE REQUIRED, THEY SHALL BE SET AS PER MANUFACTURERS SPECIFICATION CENTRALLY INTO MIN. 200mm THICK SLAB @ 25MPa MIN.

DESIGN LOADS

- THE SCAFFOLD HAS BEEN DESIGN ED USING THE FOLLOWING PARAMETERS:
SCAFFOLD INTENDED USE - ACCESS AND EDGE PROTECTION
TYPE OF SCAFFOLD - RINGLOCK (AT-PAC)

MAX. SCAFFOLD HEIGHT TO TOP DECK = xx m
NO. OF DECK LEVELS = x
NO. OF WORKING LEVELS AT ANY ONE TIME = x
MAX. ADJUSTABLE BASE JACK EXTENSION = 450 mm
MAX. LEG LOAD (EXCL. STAIRS) = xx kN
MAX. WORKING TIE LOAD = 6 kN
- THE SCAFFOLD DUTY RATING, MAXIMUM LOAD APPLIED UNIFORMLY OVER THE ENTIRE BAY & MAXIMUM NUMBER OF DECKS TO BE USED AT ANY ONE TIME SHALL BE IN ACCORDANCE WITH TABLE 1.0.

TABLE 1.0: DUTY RATING		
RATING	MAX BAY LOAD	MAX NO LEVELS
LIGHT DUTY	225 KG	0
MEDIUM DUTY	450 KG	0
HEAVY DUTY	675 KG	0

- LOADING BAYS UNIFORMLY DISTRIBUTED LOAD = 5 kPa
- STAIRS UNIFORMLY DISTRIBUTED LOAD = 2.5 kPa
- WIND LOADS CALCULATED IN ACCORDANCE WITH AS/NZS 1170.2 USING THE FOLLOWING PARAMETERS:
 - IMPORTANCE LEVEL 2 (CONSTRUCTION)
 - REGION (A / B / C)
 - TERRAIN CATEGORY 2
 - V_{ref} (ULTIMATE) = [41 / 48 / 56] m/s (WITHOUT HOARDING), 27.8 m/s (WITH HOARDING)

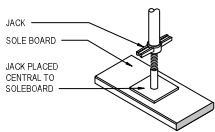
CERTIFICATION

- IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO OBTAIN DESIGN CERTIFICATION IN ACCORDANCE WITH THE RELEVANT CODE OF PRACTICE. ACROW FORMWORK & SCAFFOLDING CAN PROVIDE DESIGN CERTIFICATION UPON REQUEST.
- UNLESS DRAWINGS HAVE BEEN ACCOMPANIED BY A DESIGN CERTIFICATE, IT IS TO BE ASSUMED THE DRAWINGS ARE NOT CERTIFIED AND CONSTRUCTION WORK SHOULD NOT PROCEED.

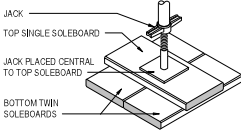


acrow.
ENGINEERING & CONSTRUCTION

DRAWING REFERENCE	
SHEETS START-END	DESCRIPTION
1 9	TITLE SHEETS & TYPICAL DETAILS
10 99	GENERAL ARRANGEMENTS
100 199	SECTIONS & ELEVATIONS
200 299	PERSPECTIVES
300 399	STANDARD DETAILS



SINGLE TIMBER SOLEBOARD
MAXIMUM DESIGN LEG LOAD - 4kN
MINIMUM REQUIRED GROUND BEARING CAPACITY - 4kPa



TRIPLE TIMBER SOLEBOARD
MAXIMUM DESIGN LEG LOAD - 4kN
MINIMUM REQUIRED GROUND BEARING CAPACITY - 4kPa

TIMBER SOLEBOARD SELECTION REQUIREMENTS:

- THE CLIENT MUST ENSURE THAT THE SUPPORTING FOUNDATION HAS ADEQUATE BEARING CAPACITY TO SUPPORT THE LOAD IMPOSED BY THE SCAFFOLD. THIS SHOULD BE SUPPLIED TO THE CLIENT BY MEANS OF A REPORT BY A ENGINEER COMPETENT IN GEOTECHNICAL ENGINEERING OR WRITTEN CONFIRMATION THAT THE SUPPORTING GROUND OR STRUCTURE IS ADEQUATE FOR THE LOADS APPLIED AS SHOWN IN THE DESIGN.
- THE CLIENT SHALL ENSURE THAT THE SCAFFOLD IS ERECTED ON A LEVEL AND STABLE BEARING SURFACE FOR ALL SOLEBOARD LOCATIONS.
- SOLEBOARDS SHALL BE MINIMUM 225mm WIDE x 38mm THICK x 500mm LONG PER SOLEBOARD.

TYPICAL DETAIL - SOLEBOARD LAYOUTS

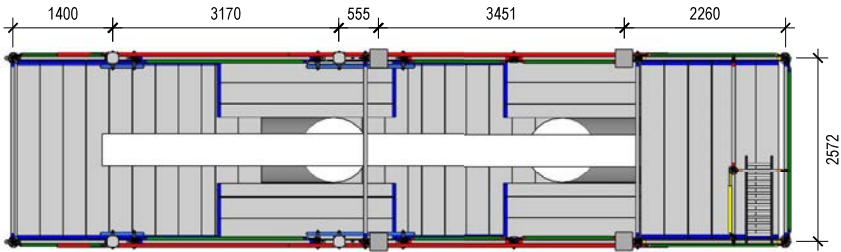
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CONCEPT ISSUE

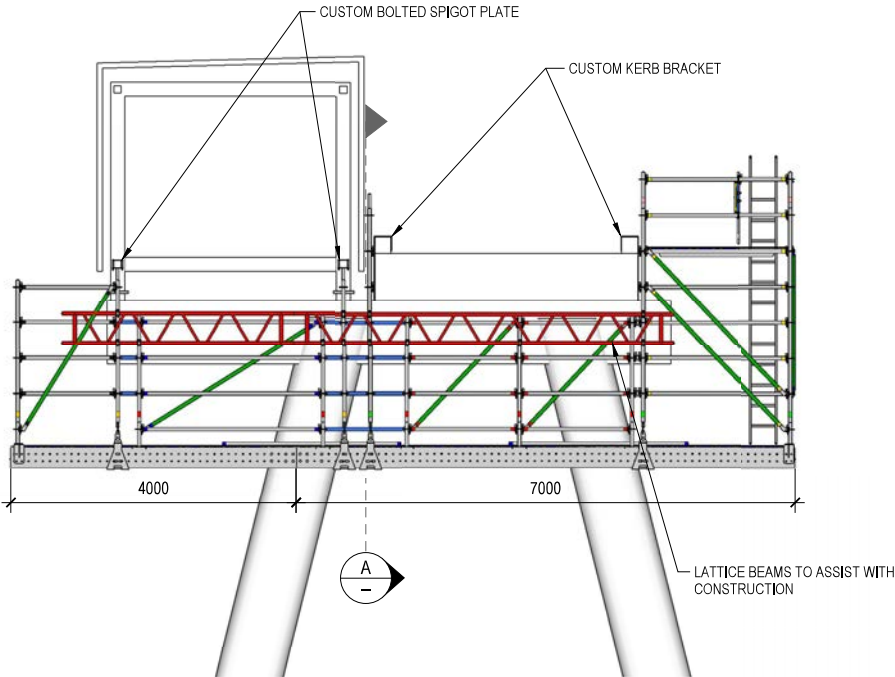
A	13.05.2025	CONCEPT ISSUE, NOT FOR CONSTRUCTION		M.F.	J.H.
REV	DATE	DESCRIPTION		DRAWN/CHKD	
CLIENT		DRAWING TITLE			
BENCHMARK		TITLE PAGE & GENERAL NOTES			
PROJECT		LBST TYPICALS	JOB NO. 250218	DWG. 01	
					
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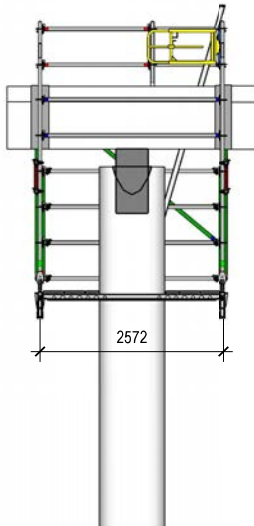
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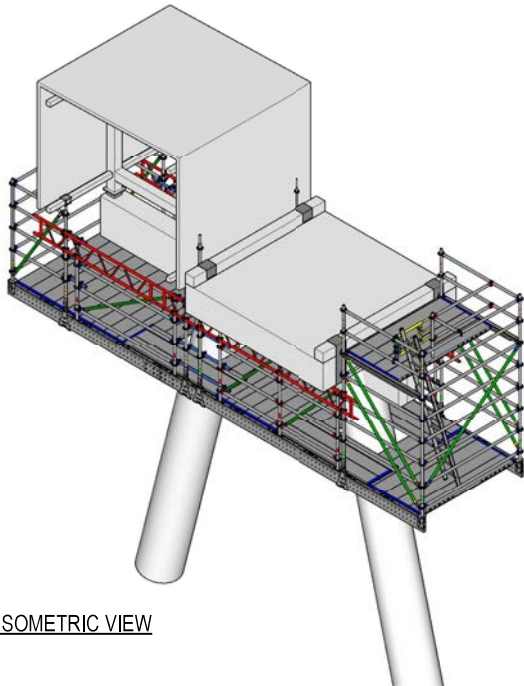
PLAN VIEW
SCALE: 1:75 @ A3



ELEVATION
SCALE: 1:75 @ A3



SECTION
SCALE: 1:75 @ A3



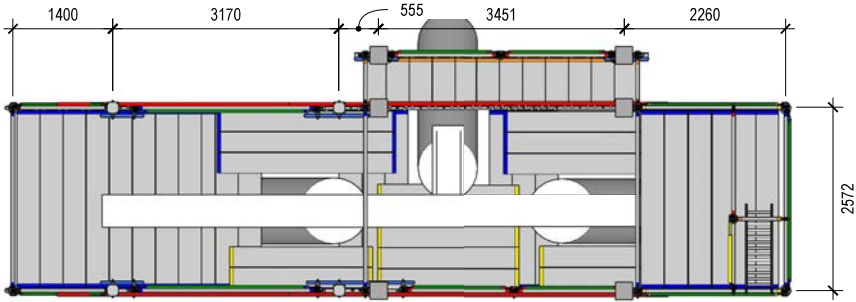
ISOMETRIC VIEW

ALL SCAFFOLD TO BE RINGLOCK

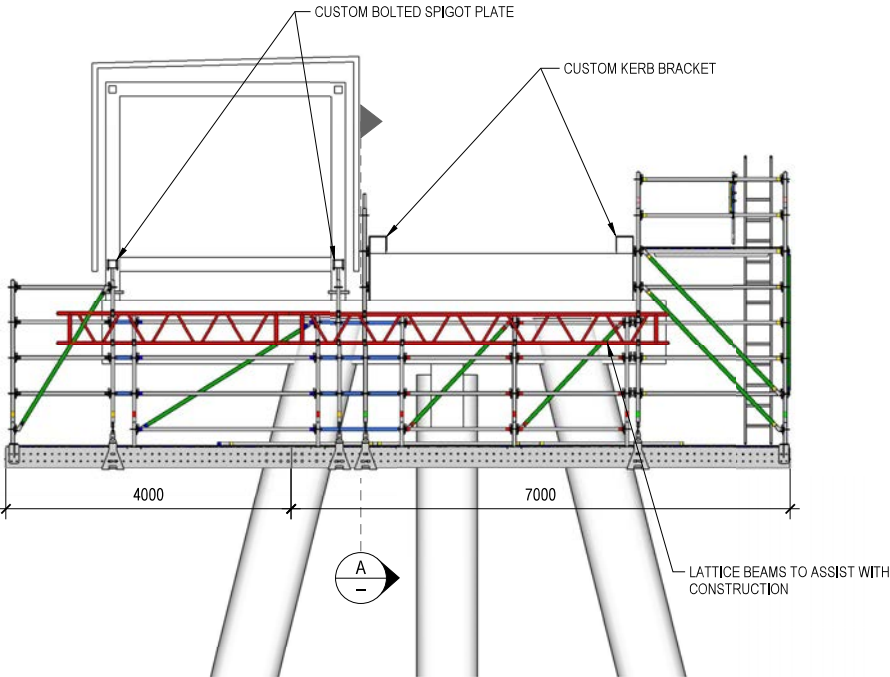
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A	13.05.2025	CONCEPT ISSUE, NOT FOR CONSTRUCTION		M.F	J.H
REV	DATE	DESCRIPTION		DRAWN/CHKD	
CLIENT		DRAWING TITLE			
BENCHMARK		TYPICAL BAY			
PROJECT		LBST TYPICALS		JOB NO.	DWG.
				250218	10
 acrow. CONSTRUCTION & SERVICES					
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GENERAL NOTES

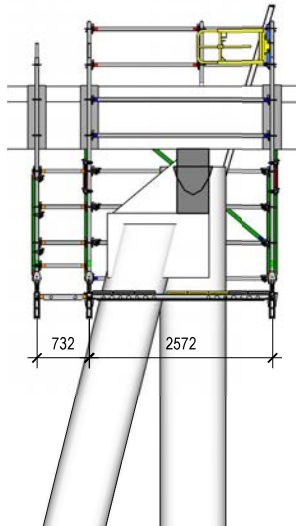
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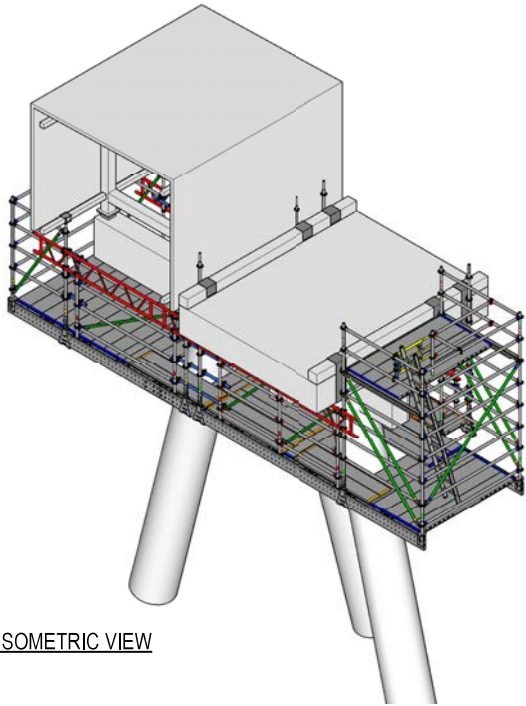
PLAN VIEW
SCALE: 1:75 @ A3



ELEVATION
SCALE: 1:75 @ A3



SECTION
SCALE: 1:75 @ A3



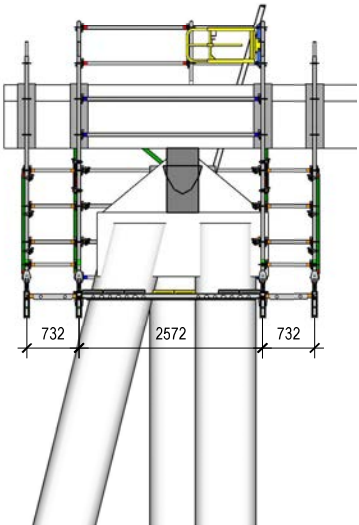
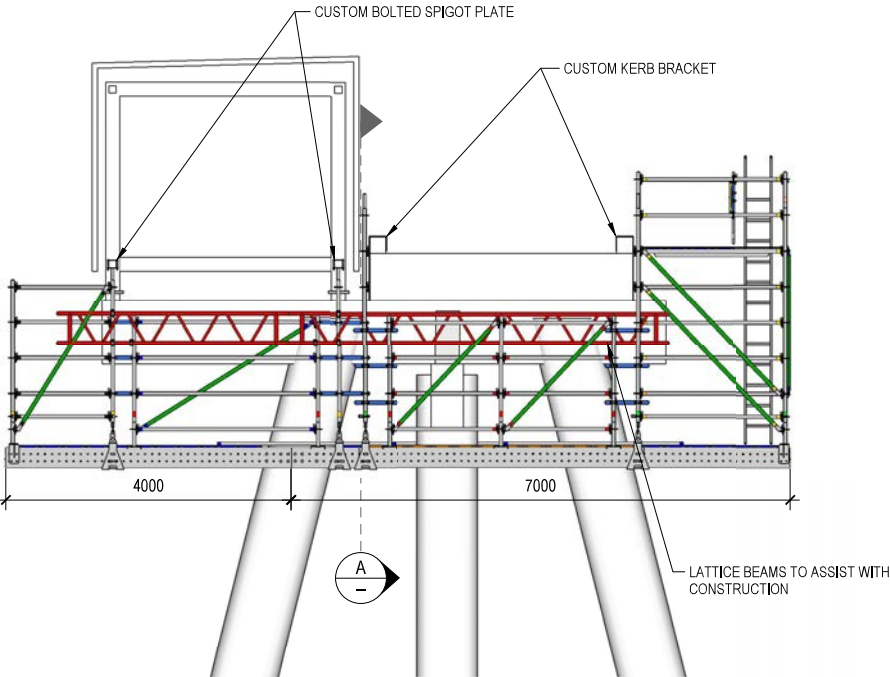
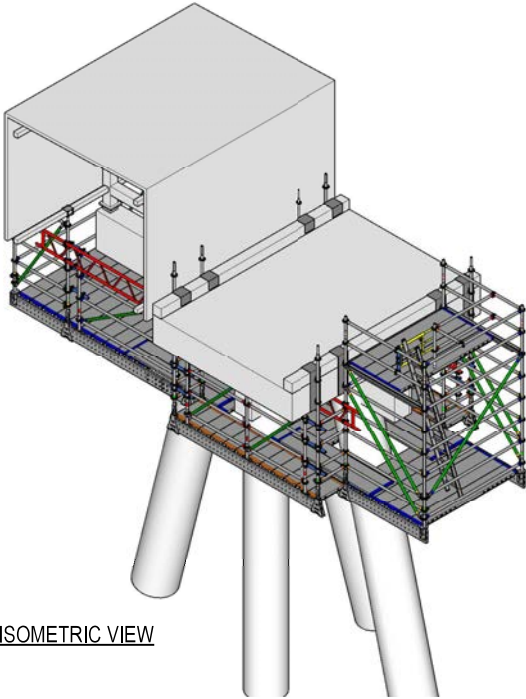
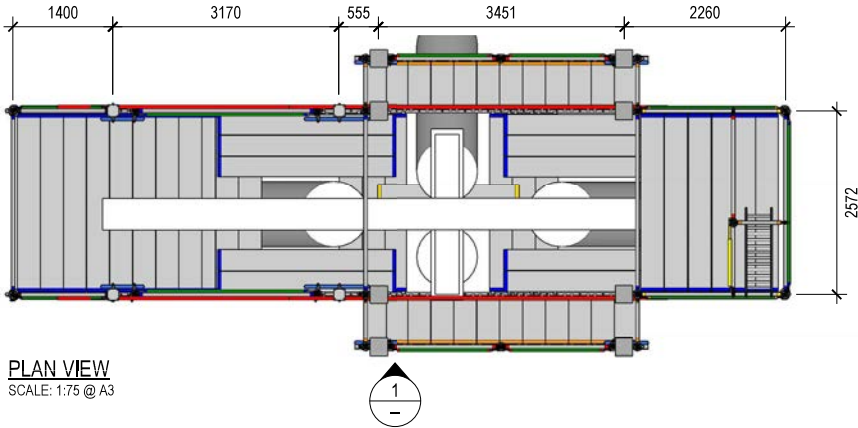
ISOMETRIC VIEW

ALL SCAFFOLD TO BE RINGLOCK

CONCEPT ISSUE				
A	13.05.2025	CONCEPT ISSUE, NOT FOR CONSTRUCTION		M.F. J.H.
REV	DATE	DESCRIPTION	DRAWN CHKD	
CLIENT		DRAWING TITLE		
BENCHMARK		1 PILE ANCHOR BENT		
PROJECT		JOB NO.	DWG.	
LBST TYPICALS		250218	11	
				
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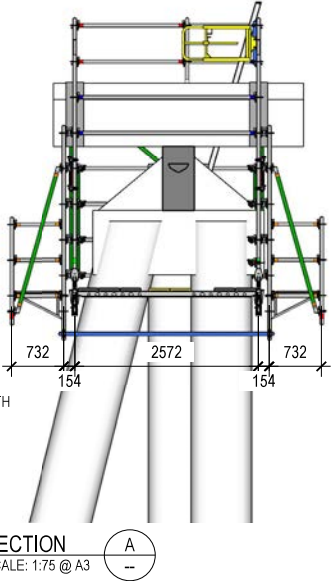
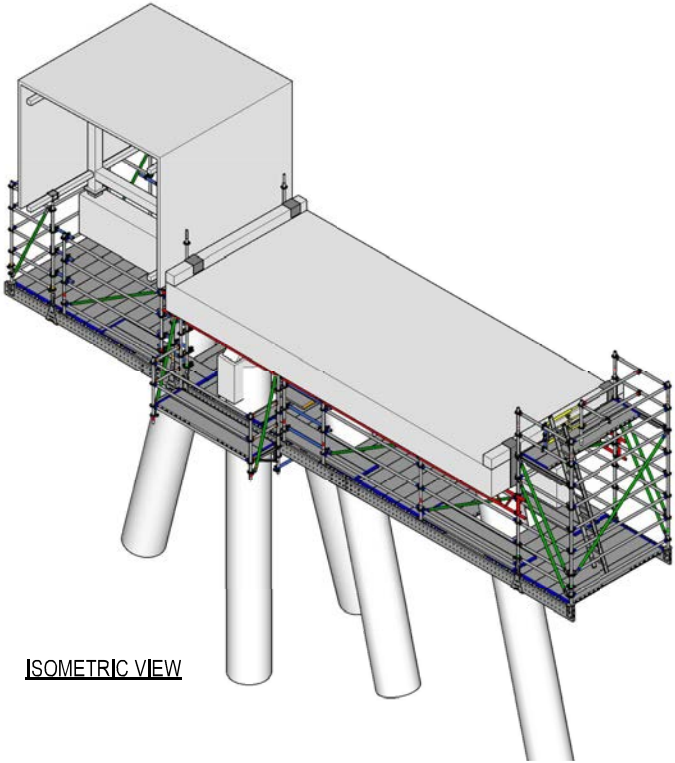
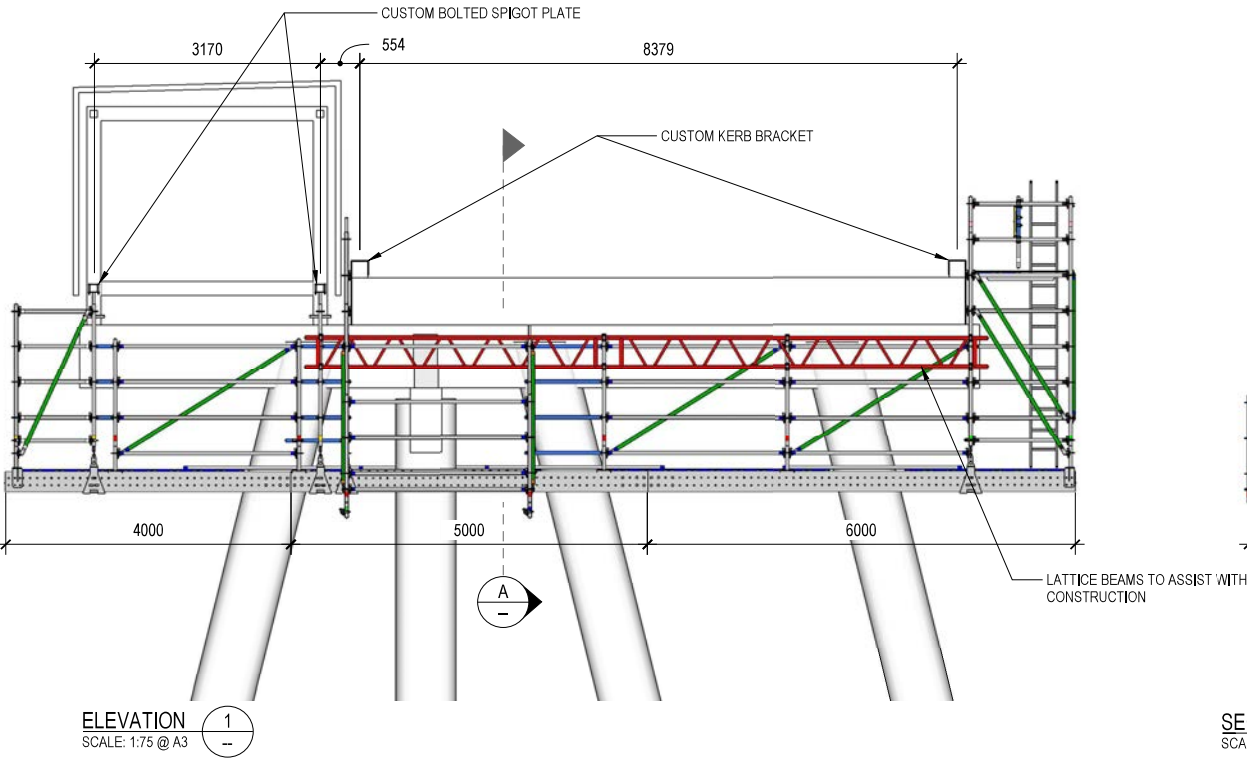
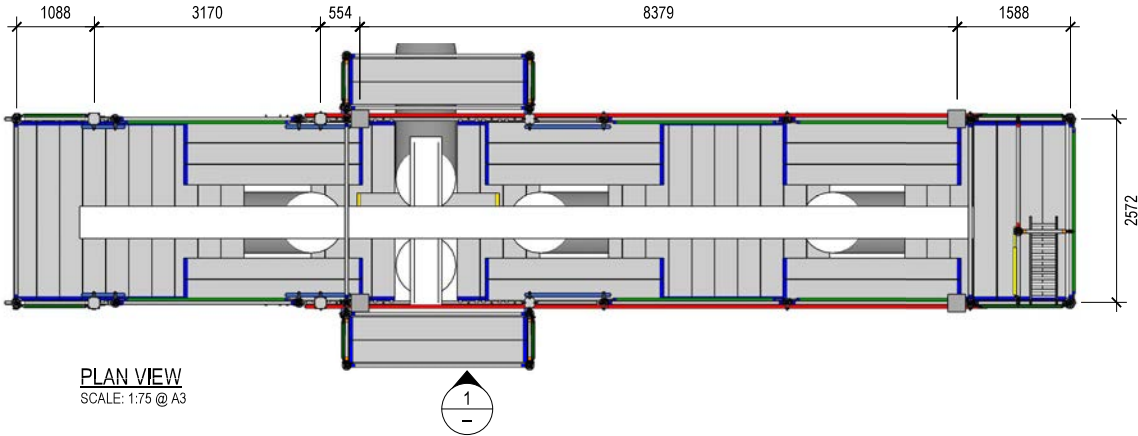


ALL SCAFFOLD TO BE RINGLOCK

CONCEPT ISSUE				
REV	DATE	DESCRIPTION	M.F.	J.H.
A	13.05.2025	CONCEPT ISSUE, NOT FOR CONSTRUCTION		
CLIENT		DRAWING TITLE		
BENCHMARK		2 PILE ANCHOR BENT		
PROJECT			JOB NO.	DWG.
LBST TYPICALS			250218	12
				
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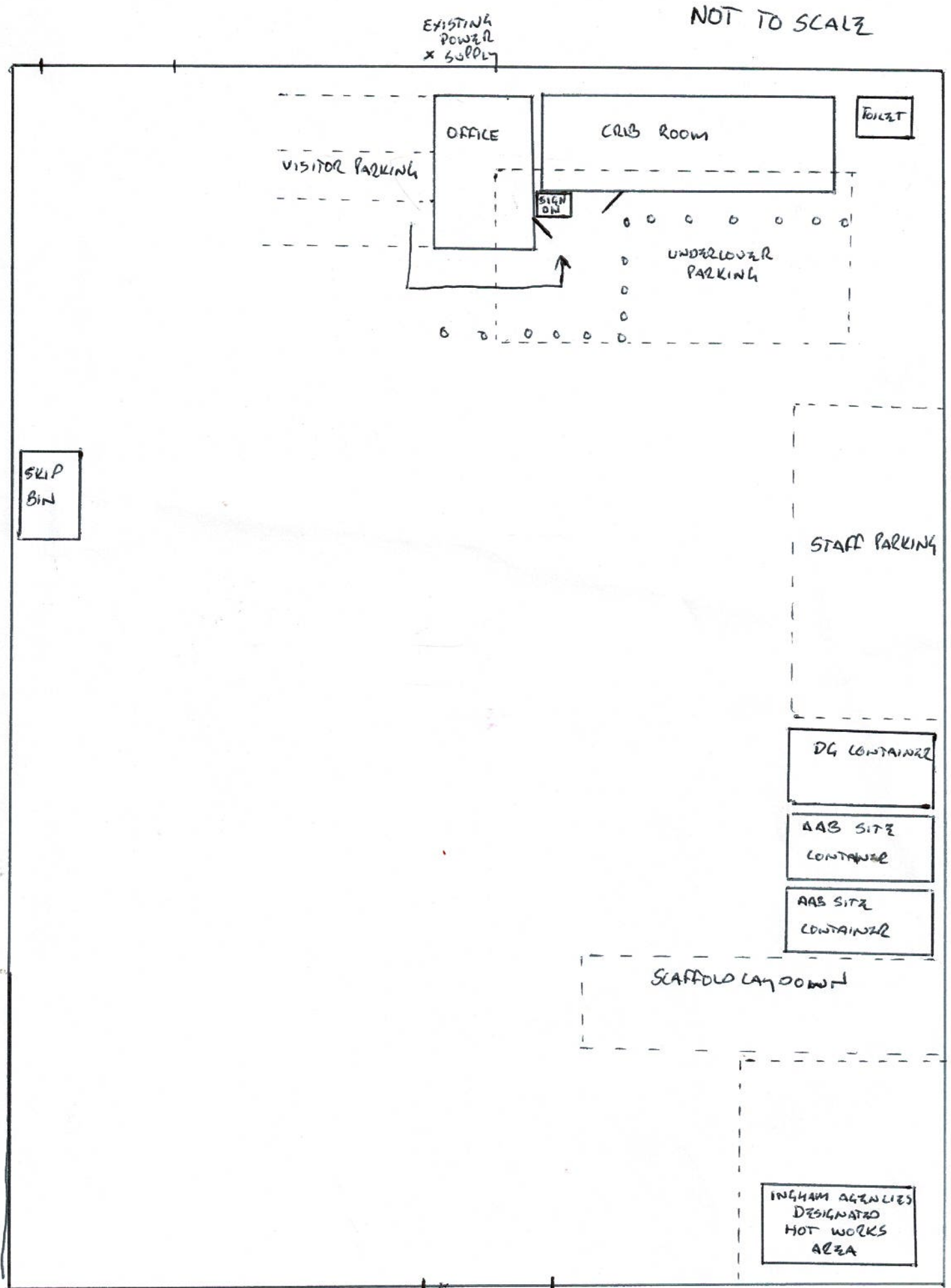
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ALL SCAFFOLD TO BE RINGLOCK

CONCEPT ISSUE				
A	13.05.2025	CONCEPT ISSUE, NOT FOR CONSTRUCTION		M.F. J.H.
REV	DATE	DESCRIPTION		DRAWN/CHKD
CLIENT		DRAWING TITLE		
BENCHMARK		PASSING BAY		
PROJECT		LBST TYPICALS	JOB NO. 250218	DWG. 13
				
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RESPONSE SCHEDULES SCHEDULE H – SAFETY, ENVIRONMENTAL and QUALITY

Schedule H1.1 – Work Health and Safety Methodology

Please provide details on how the Tenderer proposes to minimise workplace health and safety risks during the performance of the WUC including:

1. ***Brief description of the Tenderer's safety management systems proposed to be used in performance of the WUC.***

CONTRACTOR SAFETY MANAGEMENT PLAN:

This plan will: Addresses all the specific WHS hazards and issues relevant to the Contractor' obligations under the Contract which can be reasonably anticipated or ascertained from time to time during the performance of the Contractor's obligations under the Contract;

Documents the system and control methods to be implemented for the performance of its obligations under the Contract. **Listed below.**

INDUCTIONS:

Advanced Aqua Blasting (AAB) will hold a formal **SITE SPECIFIC** induction for all AAB, Benchmark Scaffolding and Ingham Agencies and others who will be working on the project. QSL and the safety audit company will be invited to give presentations. QSL Lucinda site inductions will also take place.

This will be held at the function room of Slip Ways Restaurant at Dungeness Lucinda

AAB will conduct inductions as the need arises during the projects first and subsequent years.

WEEKLY TOOL BOX MEETINGS:

AAB will conduct documented weekly tool box meetings for all staff on the project, usually on a Monday.

Safety will be an agenda item with each work group invited to give a "**safety share**".

One of the "**NEVER EVERS**" will be an agenda item at each Tool Box meeting.

DAILY PRE START MEETINGS:

AAB will conduct documented daily pre start for the project as a whole with each work group conducting their own prestart. QSL staff will be invited to attend or an AAB supervisor will attend the QSL prestart.

DAILY SIGN ON / OFF

AAB will have a sign off / on book for all staff and visitors at the onshore site office.

AAB will **use QSL Swipedon appt** for sign on / off to assist in emergency evacuation procedures.

TRAFFIC MANAGEMENT PLAN:

AAB will have a traffic management plan for the roadway work areas. It will outline communication protocols, speed limits, load limits and vehicle parking requirements. It will have warning signage at the start and end of the work area as well as identifying gallery work areas with additional warning signage.

SAFE WORK METHOD STATEMENTS (SWMS):

AAB will have SWMS for all work procedures but paying particular to High Risk Work such as

Operation of mobile plant eg forklift (Licence required)

Working at heights - scaffold erection / dismantle (Certificate of Competency required)

Working over water and near an exposed edge

Working near a roadway

AAB will require all contractors engaged on the project to submit their SWMS for approval.

All SWMS will be **reviewed weekly or if circumstances change.**

WHS RISK ASSESSMENT:

AAB has policies and procedures in place to manage workplace risks.

These are communicated to staff through inductions, tool box and pre start meeting.

THIRD PARTY SAFETY AUDIT

AAB will engage a third party to conduct a safety audit as per S11.3 of the Contract General Specifications.

2. ***Key identified safety risks and mitigations the Tenderer proposes to put in place***

ELECTRICAL: ELECTROCUTION

AAB will maintain an up to date **Electrical Equipment Register** that will document all electrical equipment on site with test due dates.

AAB and contractors will use Residual Current Devices (RCD) for all applications.

HIGH PRESSURE RELEASE: AIR / LIQUID

AAB will maintain a High Pressure Equipment Register that will record all pressure equipment on site with test due dates.

AAB and contractors have daily machinery pre start check books.

FIRE:

AAB will comply with QSL'S **Hot work permit** system and conditions.

AAB will have the appropriate indate fire extinguishers on site.

All staff will comply with QSL Lucinda's staff directions and Evacuation Plan.

WORKING OVER WATER: DROWNING

Harness and Fall arrest systems will be used when working near an exposed edge. (SWMS)

AAB will mark areas where Personal Flotation Devices (PFD) must be worn. (SWMS)

AAB will have rescue devices such as rope ladders, lifebuoys and throw ropes on site.

AAB will have a "MAN OVER BOARD RESCUE PLAN".

No working alone and AAB will have a list of trained first aiders, with 1 minimum always on site.

PEDESTRIAN AND VEHICLES ON ROADWAY:

AAB will have a **Traffic Management Plan** communicated to all Contractors and QSL staff.

FITNESS FOR WORK / FATIGUE MANAGEMENT:

AAB will conduct random drug / alcohol testing as per AAB Policy.

All staff working on the contract will submit to QSL random drug and alcohol testing when asked.

AAB and contractors will work 10 hour shifts, 6 days on and 1 day off. (24 hour) RDO.

Hydration, UV recognition, rest breaks will be a daily pre start agenda item.

STORAGE OF MATERIALS

AAB will have an approved Dangerous Goods Container (DG) for the storage of all coating material.

AAB will have in date Safety Data Sheets (SDS) for all material on site.

AAB will have an appropriate container on site for the storage of consumables, PPE etc.

Supervisors will conduct documented random site walk throughs to ensure safety standards are met.

REGULATED WASTE:

The removed coal tar epoxy coating is a regulated waste. This will be covered in the SWMS.

The regulated waste management is covered in the Environmental Management Methodology.

WEATHER: RAIN / STRONG WIND

Inclement weather, rain and or strong wind creates Health and Safety hazards for welding and scaffold erection and dismantling. Agreed stand downs will apply in these circumstances.

Work in the gallery applying Denso will be undertaken if feasible. The lock out procedure would apply.

SHIPPING:

No work in the C7B gallery during shipping.

HAZARD IDENTIFICATION AND INCIDENT REPORTING

Staff encouraged to report hazards / incidents immediately. Pre start agenda item for all work groups.

AAB's Workplace Compliance Officer (WCO) will be responsible for all hazards and incidents investigations and reporting.

RESPONSE SCHEDULES SCHEDULE H – SAFETY, ENVIRONMENTAL and QUALITY

Schedule H2.1 – Environmental Management Methodology

Please provide details on how the Tenderer proposes to manage obligations and minimise risks to the environment including:

1. ***Brief description of the Tenderer's environmental management systems proposed to be used in performance of the WUC.***

INTERGRATED ENVIRONMENTAL MANAGEMENT SYSTEM (IEMS) V1 10/2024

Advanced Aqua Blasting (AAB) has an Integrated Environmental Management System (v11 10/2024) that was submitted as a condition of AAB's Environmental Protection Agency's licencing conditions in April 2002.

Although changes in the EPA'S licencing conditions have occurred over the years ibn regards to Abrasive blasting and protective coating AAB still adheres to the conditions, particularly the "Abrasive blasting over water" section of the IEMS.

The IEMS deals in detail with Environmental Relevant Activities ERA 57 2(a) Regulated Waste Transport. LICENCE NUMBER ENRE00252505 for fixed and mobile activities from Mackay north to the Torres Straights and west to Mount Isa.

INTERGRATED ENVIRONMENTAL MANAGEMENT SYSTEM: Notification of Emergencies / Incidents

This is an extract from the IEMS.

As soon as practicable after becoming aware of any emergency or incident which results in the release of contaminates not in accordance with this environmental licence AAB will advise the administrating authority the following information. (EPA or HSC) **and in this case QSL Lucinda.**

- Holder of Environmental Licence. Advanced Aqua Blasting
- Location of emergency or incident.
- Environmental Licence number. ENRE00252505
- Name and contact details of contact person from Advanced Aqua Blasting.
- Time of release of the contaminates.
- Time Advanced Aqua Blasting became aware of the release.
- Suspected cause of the release.
- Possible extent of the environmental harm.
- Actions taken and to be undertaken.
- Any other requested information.

ENVIRONMENTAL MANAGEMENT PLAN:

AAB has an Environmental Management Plan for this contract QSL – LBST - JPHR2025.

SPILL PREVENTION AND RESPONSE PROCEDURE:

AAB has a "*Spill prevention and Response Procedure*" that in detail outlines the response to a spill and the preventive measures that need to be in place at the workplace.

HAZARDOUS CHEMICALS:

AAB has a "*Hazardous Chemicals Policy*" and a "*Hazardous chemicals Management Procedure*" that AAB supervisors implement when setting up the work site, such as the placement of Spill Kits, Dangerous Goods containers and bringing the minimum daily amount of chemicals to the off shore work site.

WASTE MANAGEMENT:

The removed coal tar epoxy regulated waste will be collected at the headstock, placed in sealable bulk bags and transported to the on shore contractors yard to be collected on an as need basis by MAMS Industrial Services Industrial Estate Challands Street Ingham and disposed of according to MAM's licence conditions.

All relevant EPA paper work will be completed and available for viewing.

Recyclable waste such as paint tins will be transported to AAB Challands Street depot for correct disposal.

All other non recyclable trade waste from the work site will be disposed of in an industrial bin at the contractor lay down yard collected by MAMS Industrial Services weekly.

2. Key identified environmental risks and mitigations the Tenderer proposes to put in place

CHEMICAL SPILL:

AAB will be conducting diesel refuelling of its plant, welder generators, compressors, forklift off shore. An appropriate approved Spills Kit will be kept on the refuelling vehicle. This activity has a SWMS that staff who do the refuelling will be familiar with.

Paint mixing will be in the original container which will be placed in a plastic bund dish before the paint tin is opened and mixing takes place. A Spills Kit will be in place. Only the minimum daily requirements of paint and thinners will be taken off shore and stored in the DG container.

Painting equipment cleaning will also take place in a plastic bund dish. All used cleaning solvents will be stored in the original container, marked as used (SLOPS) and returned onshore to the contractors lay down yard for disposal.

RELEASE OF REGULATED WASTE

The removed coating material from the headstock and piles is coal tar epoxy. This is classified as a **regulated waste** and handling and disposal need to be considered.

AAB will fully encapsulate the headstock (SWMS) for abrasive blasting so that the removed coal tar epoxy and the used garnet can be recovered. This will be done in a combination of manual and mechanical methods. (SWMS). The amount of recovered material will be approximately 1 tonne per headstock.

Due to the small amount of waste generated, the encapsulation is not negative pressure so a small amount of "nuisance dust" will escape. This is acceptable as it is used garnet.

If a large release of the regulated waste takes place, AAB's notification procedure will begin.

FAUNA

The Lucinda port limits which are defined in the regulations of the Transport Infrastructure Act 1994 extend between 8.5 to 10 kilometres from the shore. The whole port lies within the Great Barrier Reef World Heritage Area and borders the Hinchinbrook Island National Park and the Hinchinbrook Channel National Park.

Due to the site of the works, there is significant risks to the fauna – birds and fish that need to be addressed. This will happen in the Environmental Management Plan, Construction Management Plan, Inductions, Tool Box and pre start meetings.

The birds resting nesting on the jetty will be covered in the inductions" all birds have right of way" and what to do if an injured bird is encountered.

ACCIDENTAL NON REGULATED WASTE RELEASE:

The most common way for waste to enter the water is from the tray of a moving vehicle on the roadway. All vehicles used in the contract will have a cargo net to prevent this happening. Failure to use the net will be a disciplinary offence.

Rubbish bins will be at all work site and the contents taken on shore daily.

AAB will position a portable toilet on the passing bay closest to the work site.

There will be "no smoking or vaping" off shore. Failure to comply will be a disciplinary offence.

RESPONSE SCHEDULES SCHEDULE H – SAFETY, ENVIRONMENTAL and QUALITY

Schedule H3.1 – Quality Management Methodology

Please provide details on how the Tenderer proposes to achieve high levels of quality control and quality assurance during performance of the WUC including:

1. **A description of the Tenderer's proposed quality management systems to be used in performance of the WUC.**

Quality Management:

Advanced Aqua Blasting (AAB) if successful with this tender will engage **The APP Group Townsville** to conduct an audit review upgrade of all of AAB's QA practices and procedures to ensure tender compliance.

This audit review upgrade will look at electronic quality assurance management and ensure that all records such as photographic records, results of the Dilapidation survey, coating inspection reports and non compliance and rectification reports, Inspection and Test Plans (ITP) completed and underway are recorded on a work lot headstock number. This work lot containing all records will be available electronically to the Principal or 3rd party auditors engaged by the Principal.

AAB will have as part of its Key Personnel a full time Protective Coatings QA/QC Officer supplied by The APP Group. This QA/QC Officer will also train a NACE accredited (or booked to do a NACE Level 1) AAB staff member to cover for RDO's and other work absences.

AAB will submit a **Quality Project Management Plan** that complies with *Contract General Specification S9.2 Quality Management Plan*.

AAB has an ITP's that covers all the points outlined in the *Contract General Specification S9.3 Inspection and Test Plans*.

AAB has Painting Contractors Certification Programme (PCCP) accreditation. AAB has held this certification since 2010.

Certification is ☐ Class 1 Shop Application – Open Yard, ☐ Class 2 Shop Application – Enclosed Blast Cleaning Paint Application, ☐ Class 3 Field Application – Atmospheric Exposure Service, ☐ Class 4 Field Application – Immersion Service

2. **A description of the Tenderers proposed Quality Assurance testing and reporting methodology.**

Protective Coating Testing:

The full time QA/QC Officer will conduct or supervise all testing on the protective coatings works and complete the relevant section of the ITP.

The NACE 1 AAB Site Supervisor will be present for this testing. This is especially important if defects are identified.

Below is an extract from an ITP on the application of the first coat:

Remarks/Records (eg: Test Frequency, Reports, Certificates, Checklists, etc.)

Ensure coating application down the 500mm clean abrasive blasted section of the pile under the headstock. This is to ensure a clean surface for the application of the Denso Seashield jacket.

Remove all spent abrasive and residual dust by using dry compressed air, sweeping with a clean brush or vacuum cleaning prior to application of the coating. Avoid handling blasted steel with bare hands.

The surface must be inspected prior to coating application to ensure neither surface defects nor contamination exist, otherwise solvent clean rectification will be required before any coating is applied.

Apply initial coating within a four (4) hour period after blasting or before any surface deterioration occurs.

Wet Film Test daily on first area coated and then on each painted area coated as required. Record on AS3894.14 Inspection Report Daily Painting or similar. Dulux Protective Coatings Specification; PCQ-115-0121-4 Section 7.5.0 Records

All coating materials supplied is by an Australian Paint Approval Scheme (APAS) recognised manufacturer – Dulux Australia

Dulux Protective Coatings Specification; PCQ-115-0121-4 Section 7.3.1 Approved Wet Film Gauge is to be used during application to minimise the possibility of low dry film thicknesses and in accordance with AS3894.3.

Protective Coating Reporting Methodology.

All AS Inspection sheets used and the ITP will be recorded under the (headstock) work lot number as will any rectification repairs. This will initially be a paper recording until entered into the electronic data base with photos for each headstock work lot.

AS 3894.10 Inspection Report -Daily

AS 3894.11 Equipment Report.

AS 3894.12 Inspection Report- Coating

AS 3894.13 Inspection Report – Daily.

AS 3894.14 Inspection Report – Daily Painting

Daily Surface and Ambient conditions will be measured. The results will be recorded on **AS 3894.10 Inspection Report – Daily Surface and Ambient Conditions.**

An **AS 3894.11 Equipment Report** will be completed recording all testing equipment brand and calibration due date.

Paint Containers will be clearly labelled with manufacturer's name, product type, colour and batch number. This will be recorded on **AS 3894.13 Inspection Report – Daily Blast and Paint.**

The surface profile will be recorded on **AS 3894.13 Inspection Report – Daily Blast and Paint.**

Wet film thickness will be measured in accordance with **AS 3894.3** and recorded on **AS 3894.13** Inspection Report – Daily Blast and Paint.

Dry film thickness shall be measured in accordance with AS 3894.3 and recorded on AS 3894.14 Inspection Report – Daily Painting.

Rectification and Touch-up Painting.

Surfaces where the paint coating has been damaged, failed Dry Film Testing or marked in any way will be touched up or repainted in accordance with the instructions *Dulux Protective Coatings Specification; PCQ-115-0121-4 Section 8.1.0 Repair of Damaged Coating.*

Treatment will be: Pre-clean the damaged area. (2) Remove the damaged coating back to a sound adherent surface. (3) Clean the damaged area in accordance with Section 6.2 of this Specification. (4) Feather back the repair area by a minimum 50mm overlap by low pressure whip abrasive blasting, sanding or hand tools. (5) Clean surface to remove dust. (6) Build up the repair area using the same type of paints and same dry film thickness as specified.

Structural Steel – Testing and weld repairs.

The full time QA/QC Officer will conduct or supervise all testing on the steel inspection and welding works and complete the relevant section of the ITP.

AAB will use the Pile Condition Inspection Sheet for detailed recording so that weld patches can be cut. A similar Headstock Inspection Sheet will be designed and used.

AAB will use a calibrated ultrasonic thickness gauge and remove a small amount of corrosion from the surface with a grinder/flapper disc to achieve a thickness reading. The inspection procedure shall be as follows:

AAB will refer to Aurecon Drawing 1 Approach Jetty Typical Repair Details and Drawing 2 Structural Utilisation Assessment diagram/drawing for the particular pile or headstock.

Steel Repair and Identification Reporting Methodology.

All Pile and Headstock Inspection sheets used and the ITP will be recorded under the (headstock) work lot number as will all welding repairs both pad and patch. This will initially be a paper recording until entered into the electronic data base with photos for each headstock work lot.

3 *Key identified quality risks and mitigations the Tenderer proposes to put in place.*

Surface contamination.

Protective coating will NOT be performed in the following circumstances:

If the surface is wet, or likely to become wet after abrasive blasting and drying with compressed air or if the clean surface has failed the salt test.

On surface where the formation of condensation is likely i.e. where the surface temperature of steel is less than 3°C above dew point.

When the relative humidity of the surrounding air is above 85%.

When the ambient temperature, or the surface temperature, exceeds 50°C or is less than 5°C.

Incorrect product Part A and B quantities and insufficient mixing.

AAB will have available at the workplace Dulux Product Data Sheets for the products being applied. DUREMAX GFX and epoxy thinners as the use of thinner (solvents) other than nominated for each coating in the relevant Technical Data Sheet may impair performance of the coating.

Stripe coating.

Prior to the spray application of each coating all edges, bolts and boltholes (including slots) shall be stripe coated by brush method. Staff will pay particular attention to achieving adequate DFT at edges by Stripe Coating to ensure long service life

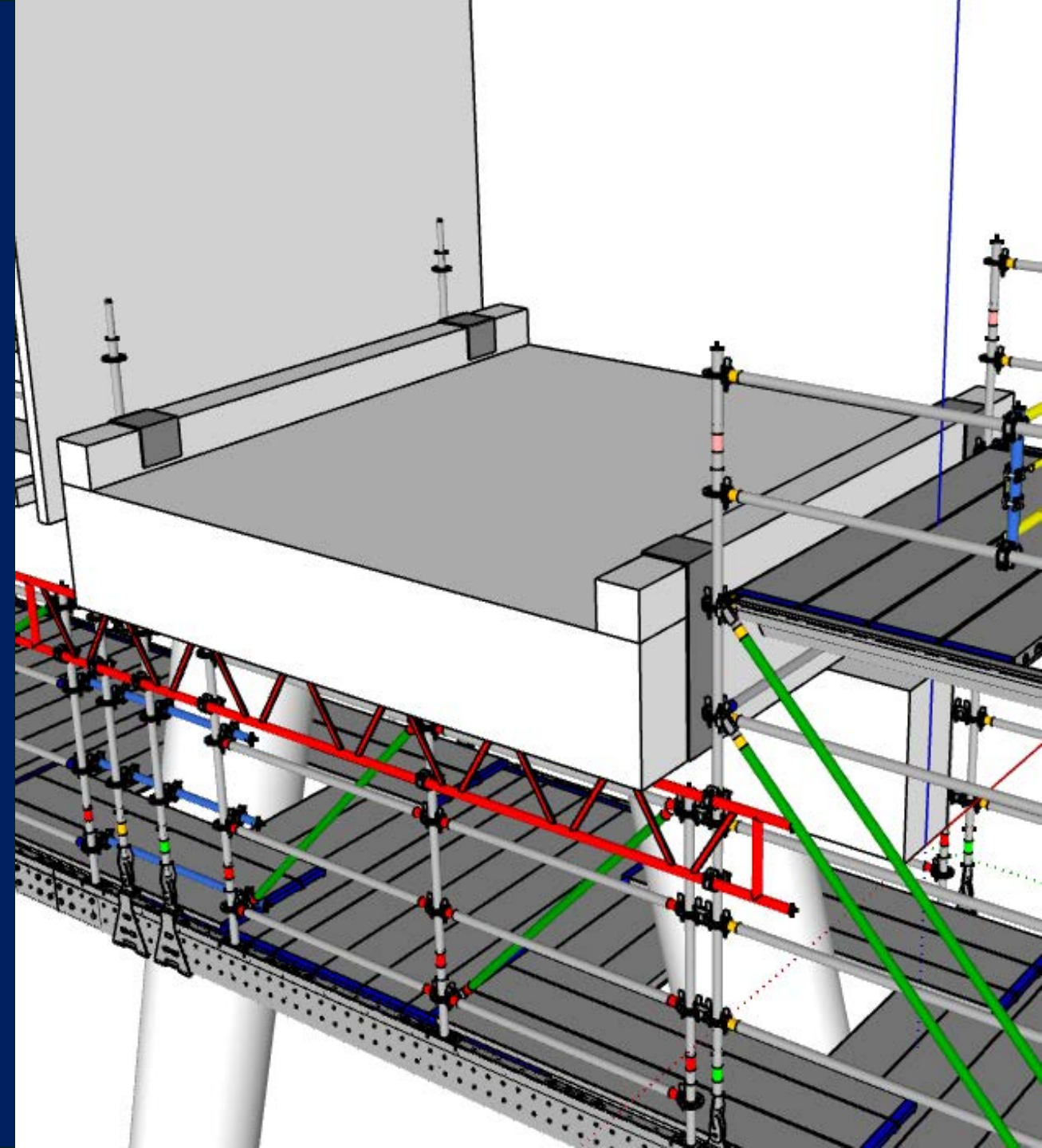
The coating should be protected from the elements and contamination during coating cure to achieve maximum life.

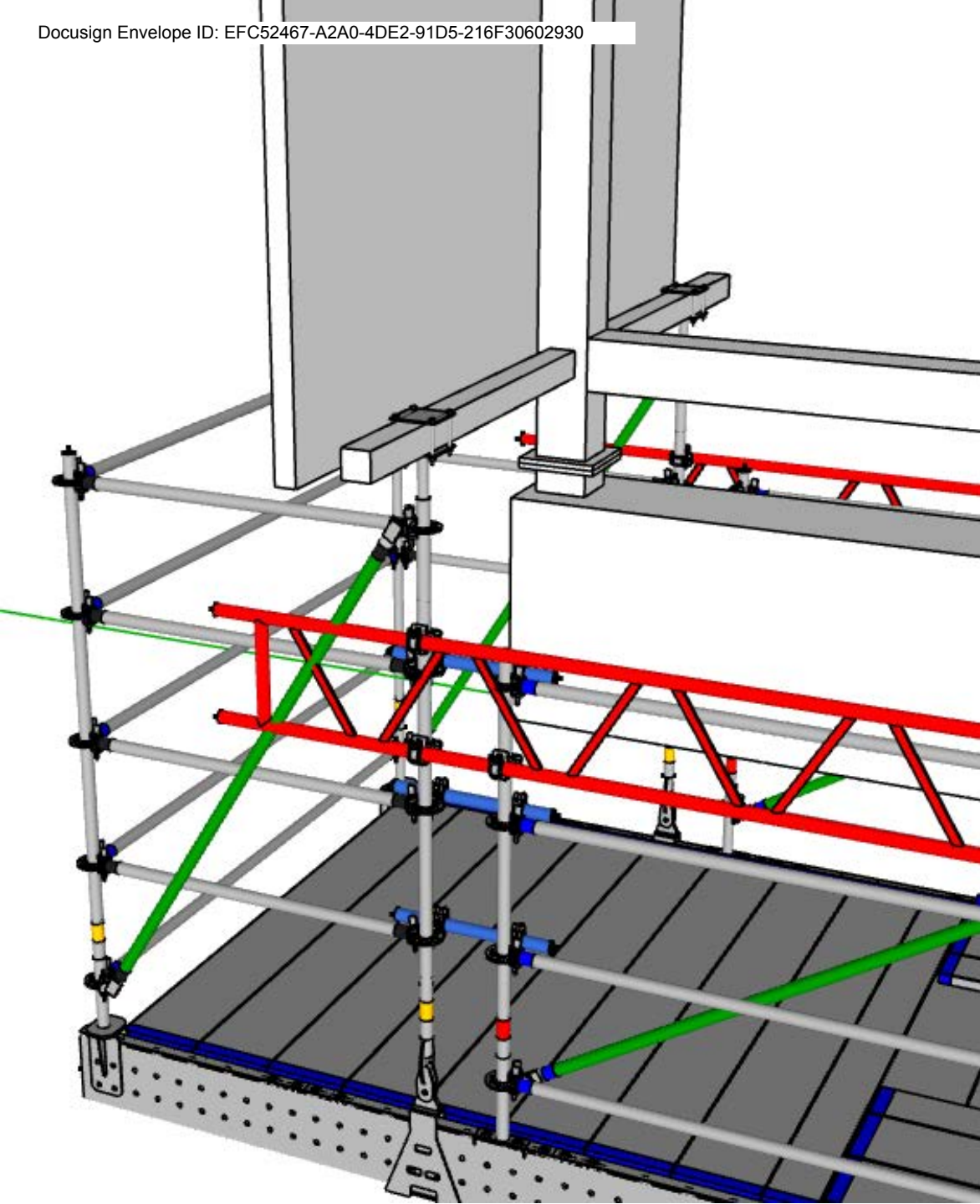
Lines of communication: For ITP Witness and Hold points.

The ABB Project Manager will maintain a 1-week estimated look ahead of expected activities, Hold Points and Witness Points for both welding activities and protective coating and will provide at least one Business Days' notice of all Hold Points and Witness Points.

Typical Scaffold Design

- Southern end of the scaffold to be supported from the roadway kerbs with engineered fabricated brackets.
- Access the northern end of the roadway, the scaffolders will need to remove the hood under the conveyor to access.
- Lattice Beam to be installed under roadway to installed additional components for the working deck.



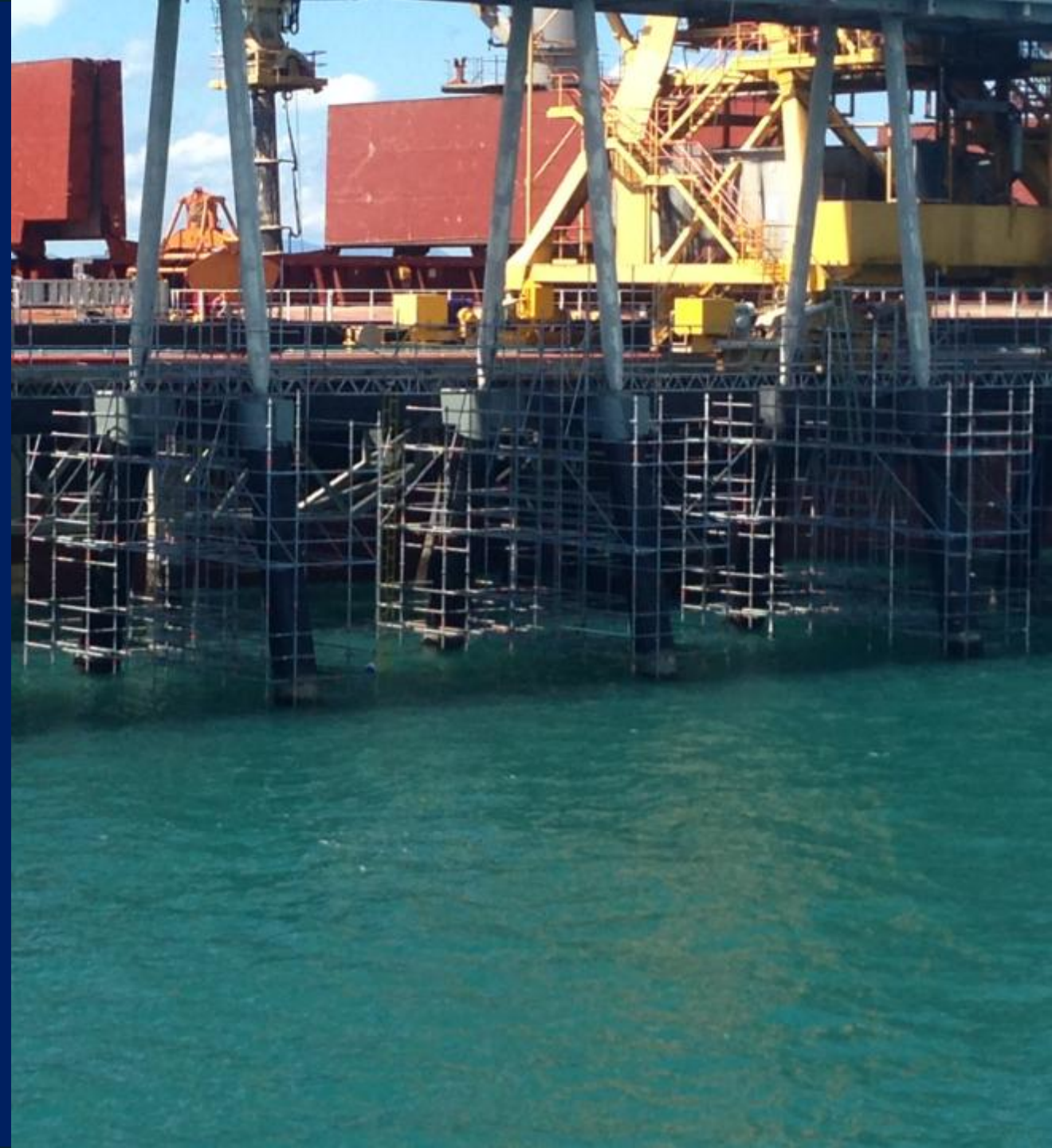


Typical Scaffold Design

- Northern end of the scaffold will be supported from another specific engineered bracket clamping to the conveyor frame underneath.
- Doing it this way, we can eliminate the need for AAB to come back after the scaffold is remove to do touch ups.
- Once in place, extend another Lattice beam to this connection and complete the installation of the working deck.

Typical Scaffold Design

- Six scaffolds will be built with 4 available for abrasive blasting, painting, welding and Denso jacket fitting at any one time.
- One scaffold will be in the process of dismantling with another in the process of erection.
- Scaffold will be built on Priority A Pilebent 257 first with work proceeding out to Pilebent 281.
- Removal of the scaffold will be the same steps but in reverse.
- The removed scaffold should be able to fit onto the back of the truck for relocation to the next pilebent site.

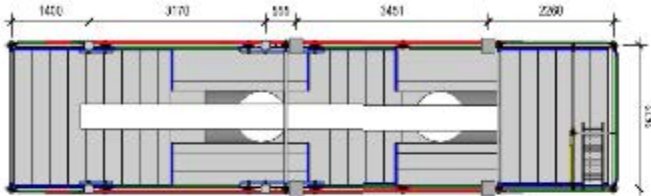


Scaffolding Design

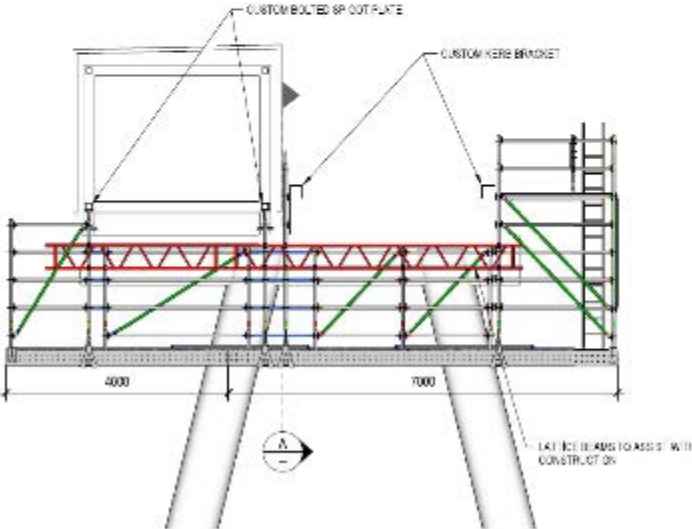
Typical Bay

GENERAL NOTES

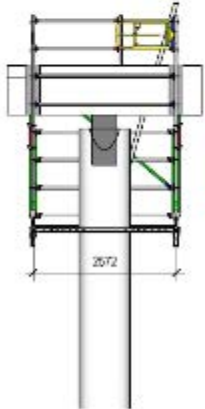
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- 2. THIS SCAFFOLDING IS TO BE CONSTRUCTED IN ACCORDANCE WITH THE FOLLOWING NOTES.



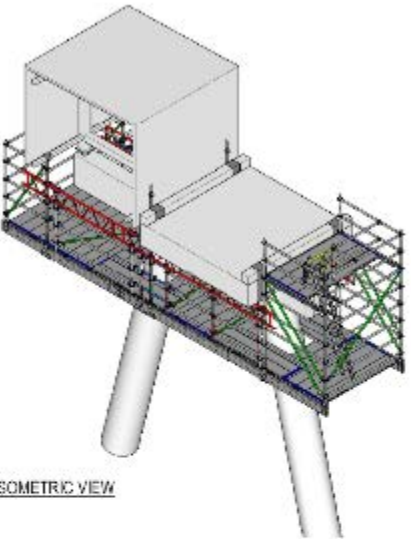
PLAN VIEW
SCALE: 1/16" = 1'-0"



ELEVATION
SCALE: 1/16" = 1'-0"



SECTION
SCALE: 1/16" = 1'-0"



ISOMETRIC VIEW

ALL SCAFFOLD TO BE RINGLOCK

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1 Pile Anchor Bent

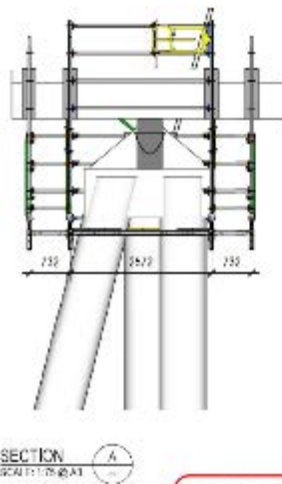
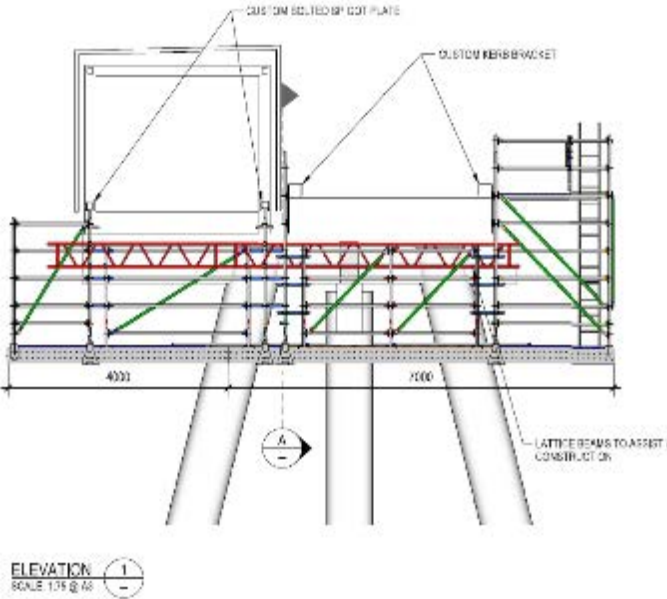
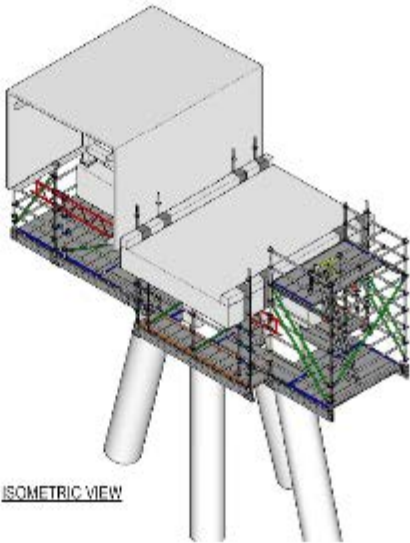
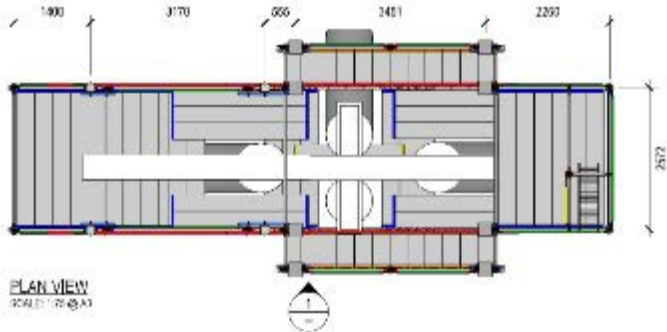


Scaffolding Design

2 Pile Anchor Bent

GENERAL NOTES

- 1. THIS DRAWING SHALL BE PRINTED IN COLOR.
- 2. THIS DRAWING IS TO BE USED FOR CONSTRUCTION OF THE ABOVE DESCRIBED SCAFFOLD.



ALL SCAFFOLD TO BE RINGLOCK

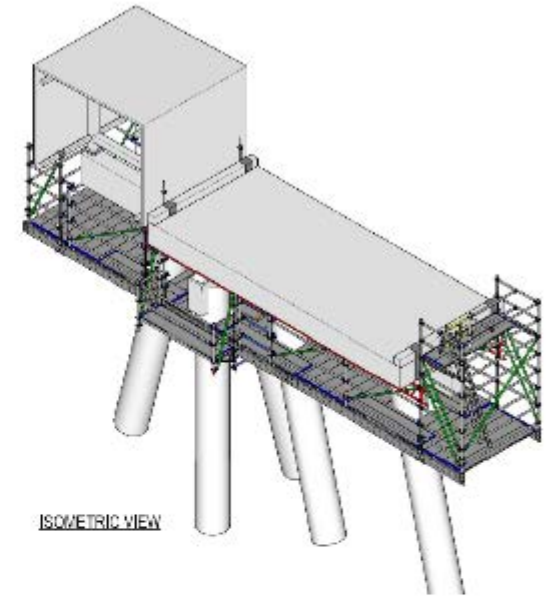
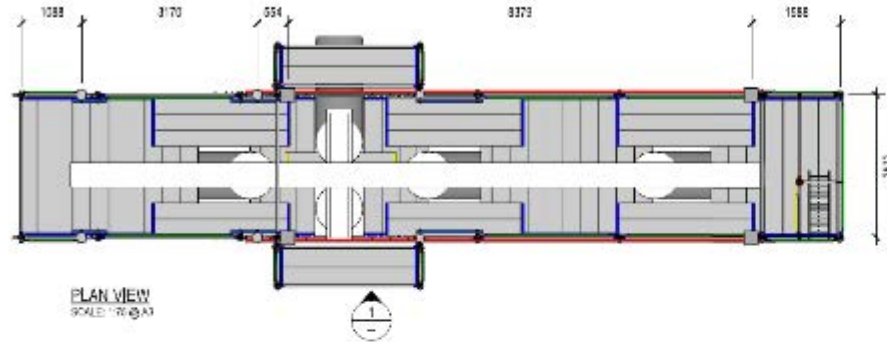
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Scaffolding Design

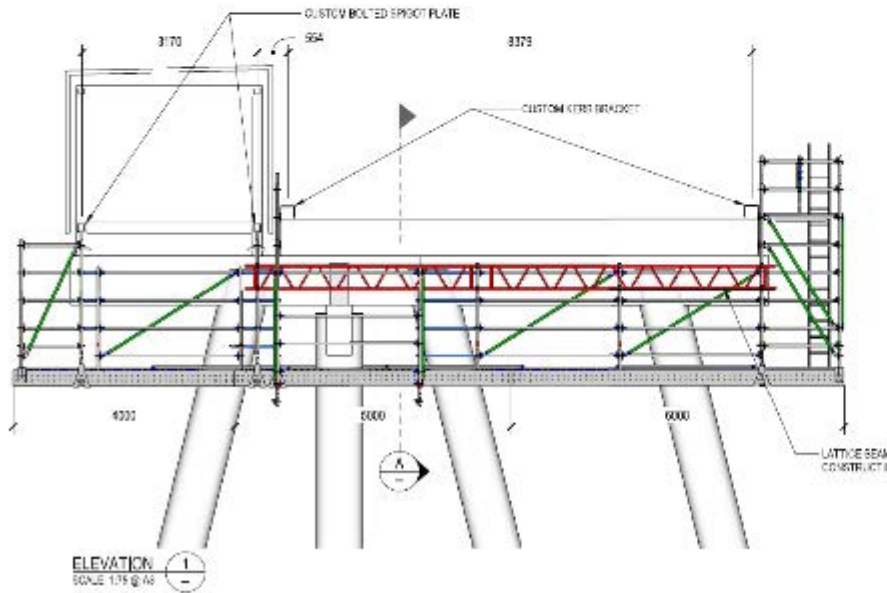
Passing Bay

GENERAL NOTES

1. THIS DRAWING MUST BE PRINTED IN COLOR.
2. THIS DRAWING TO BE USED IN CONJUNCTION WITH ALL DRAWINGS ACCOMPANYING THIS PROJECT.



ISOMETRIC VIEW



SECTION
SCALE: 1/8\"/>

ALL SCAFFOLD TO BE RINGLOCK

CONCEPT ISSUE			
DATE: 01/11/2018	PROJECT: 10000000000000000000	REV: 1	ISSUED FOR: 10000000000000000000
DATE: 01/11/2018	PROJECT: 10000000000000000000	REV: 2	ISSUED FOR: 10000000000000000000
DATE: 01/11/2018	PROJECT: 10000000000000000000	REV: 3	ISSUED FOR: 10000000000000000000
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DATE: 01/11/2018	PROJECT: 10000000000000000000	REV: 9	ISSUED FOR: 10000000000000000000
DATE: 01/11/2018	PROJECT: 10000000000000000000	REV: 10	ISSUED FOR: 10000000000000000000

acrow.

Scaffold on one
pilebent with
another scaffold
in use on another
pilebent



Pressure water washing

Pressure water washing will be used to remove salt on the headstock after welding repairs and before abrasive blasting as per specifications listed in Dulux PCQ-015-0121-4/A.

Testing for residual salt will be as per AS 3894.6 with a salt test of < 50ppm acceptable.



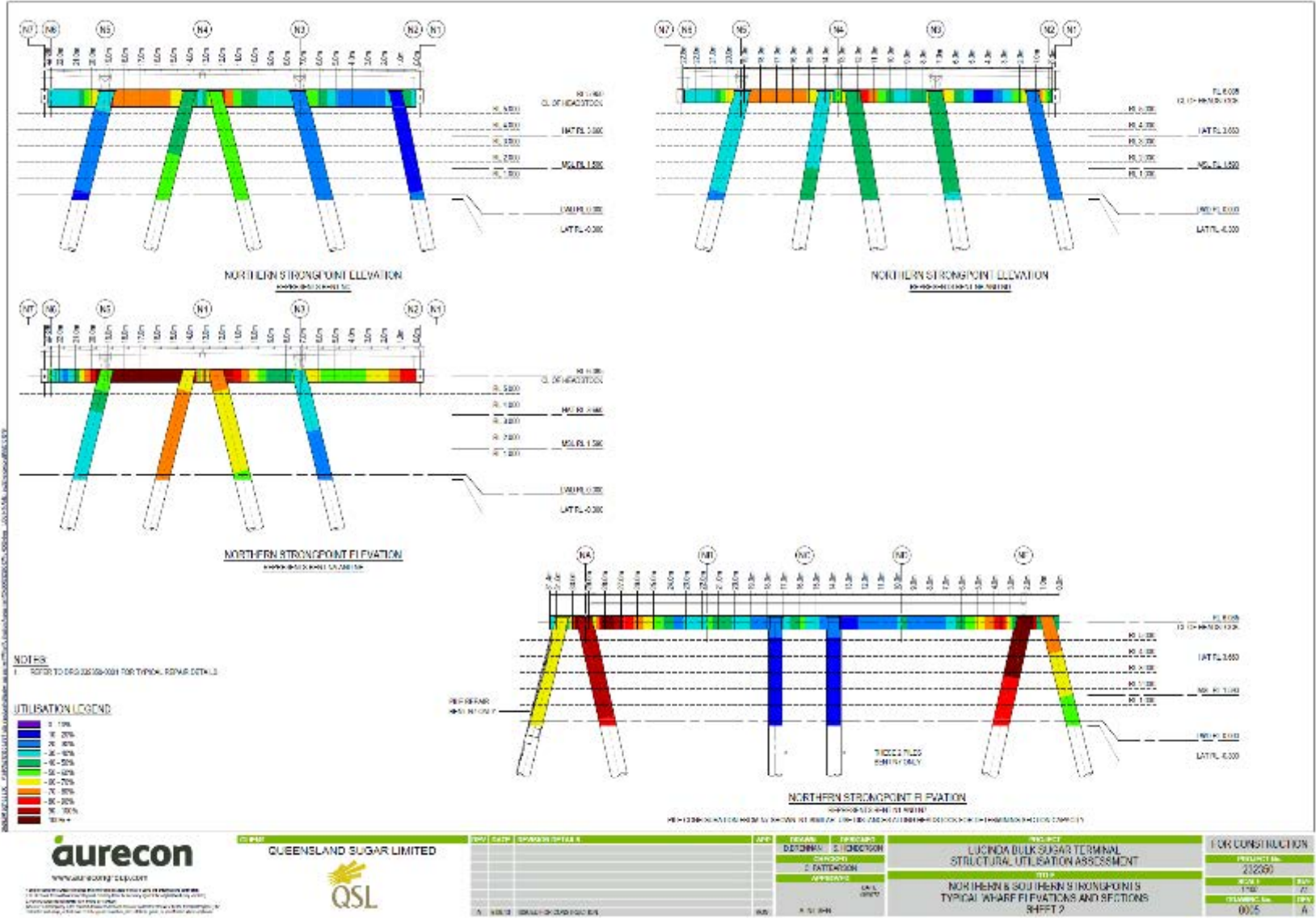
Welding repairs

All welders shall have the qualifications requirements of Clause 4.12.2.3 of AS/NZS 1554.1 and others requirements stated in the contract QSL-LBST-JPHR2025



AURECON

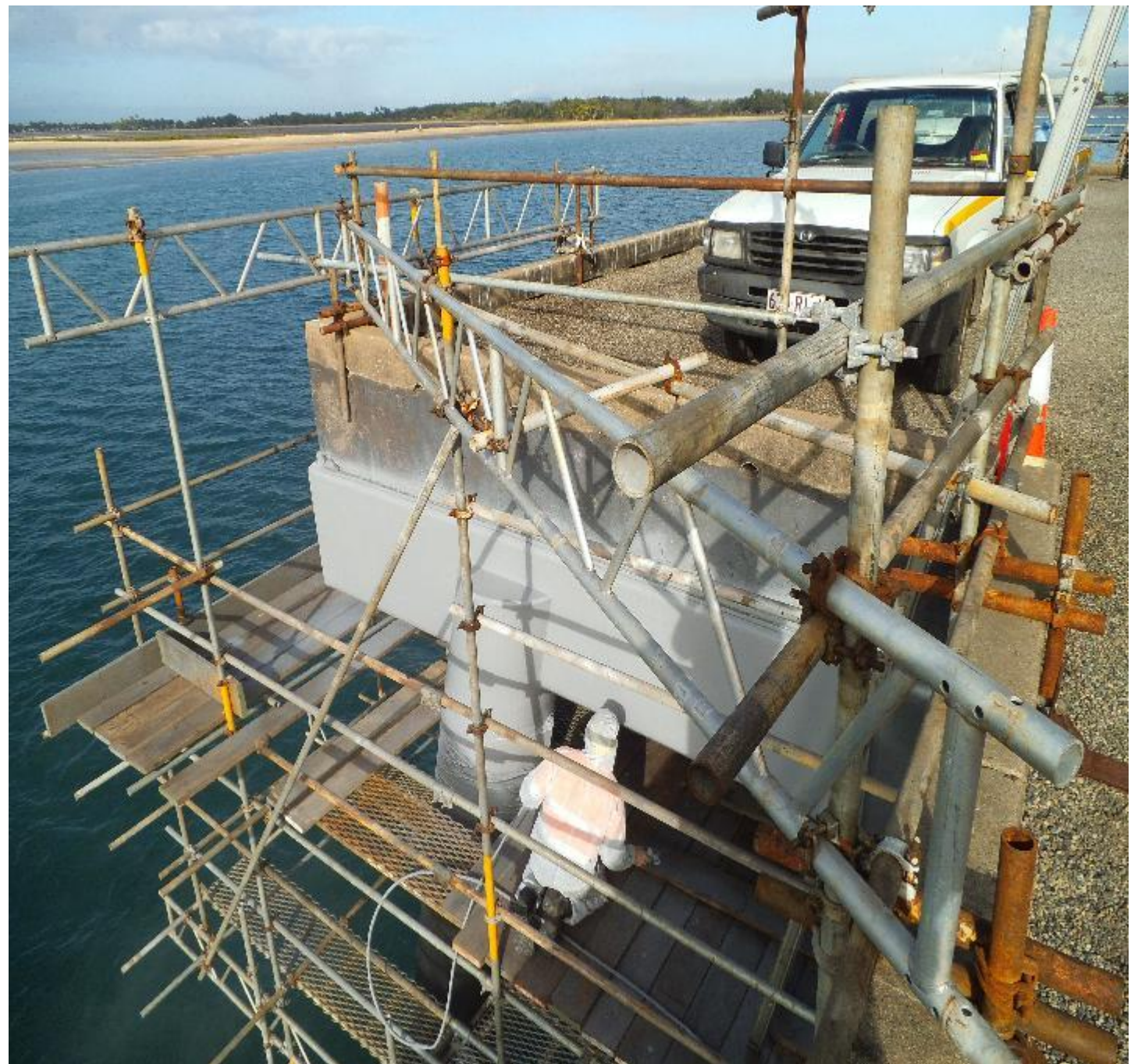
Structural
Utilisation



Protective coating application on the headstock

AAB applicators are familiar and experienced with the product being applied, Dulux DUREMAX GFX.

AAB will comply with all the product application QA specifications listed in Dulux PCQ-015-0121-4/A



Application of Denso Grease and Tape on steel piles

SPECIFICATION:
Denso Engineering
Specification Sea Shield series
2000
Grease and Tape application



Denso Seashield jacket fitting

SPECIFICATION:
Denso Engineering Specification
Sea Shield series 2000
Jacket fitting



Reporting works schedule

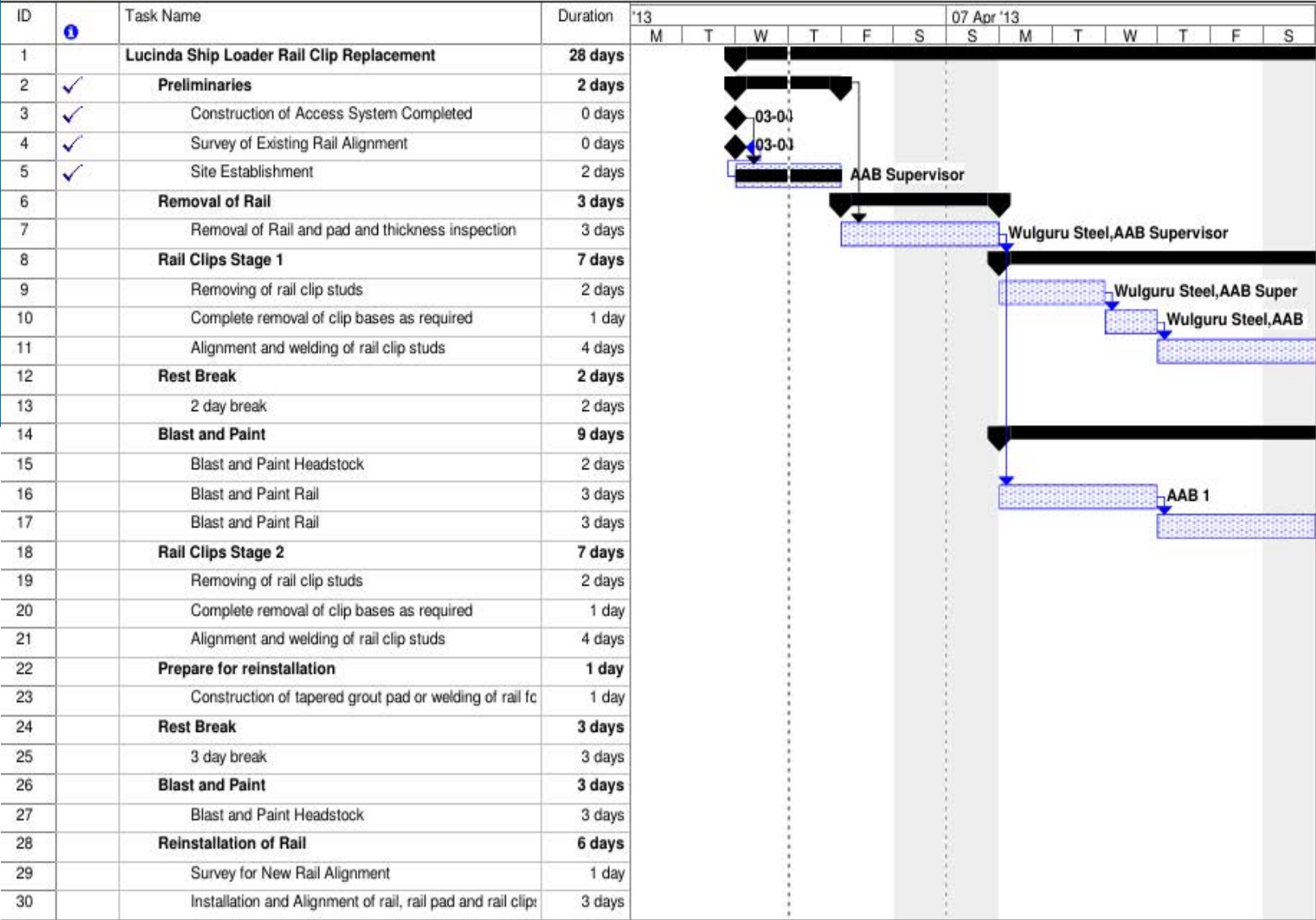
Pilebent 257
258 259 260
262 263



Reporting works schedule

LBST SR1213

04-04-2013



Project: LucindaShipLoaderRailClipRep
Date: Thu 04-04-13

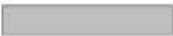
Task



Milestone



External Tasks



Split



Summary



External Milestone



Progress



Project Summary



Deadline

